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HYUNDAI DEALERSHIP BUSINESS UNDERSTANDING DOCUMENT

Executive Summary

This document provides a comprehensive understanding of the Hyundai dealership business operated under Sri Gurukripa Automobiles LLP and Sell Any Car. The purpose of this study is to understand how different departments function, how business processes are executed, how data is generated and utilized, and how management makes decisions using business data.

This extensive document serves as the master blueprint for the Data Analyst Internship Project at Sri Gurukripa Automobiles LLP (Hyundai). It encapsulates industry analysis, operational procedures, data architecture, advanced KPI frameworks, SQL query libraries, Power BI DAX dictionaries, and comprehensive real-world case studies.

As a Data Analyst Intern, the objective is not merely to analyze reports but to understand the complete business ecosystem. Every customer interaction, vehicle booking, service visit, insurance policy, and financial transaction generates valuable data. Understanding this data lifecycle enables management to improve sales performance, optimize inventory, enhance customer satisfaction, and increase profitability.

The dealership operates through multiple departments such as Sales, Customer Relationship Executive (CRE), EDP, Insurance, Accounts, Service, Accessories, and Used Car Operations (SellAnyCar). Each department contributes to revenue generation and creates data that can be analyzed to support business decisions.

This document serves as the foundation for understanding the dealership's operational and analytical framework.

1. Introduction to the Automobile Dealership Industry

The automobile dealership industry acts as an intermediary between automobile manufacturers and end customers. Universal Hyundai Motor India manufactures vehicles, while dealerships are responsible for sales, delivery, service, customer relationship management, and after-sales support.

A dealership is not only a vehicle-selling organization. It operates as a multi-revenue business that generates income from various services including:

Vehicle Servicing

- Insurance Sales
- Detailing & Cleaning
- Spare Parts Sales
- New Vehicle Sales
- Used Vehicle Sales

- Accessories Installation
- Finance & Auto Loans
- Extended Warranty Services
- Paint Protection Film (PPF) Installation
- Ceramic Coating Application
- Polishing & Waxing
- Car Washing (Basic & Premium)
- Interior Upholstery Cleaning
- Window Tinting
- GPS Tracking Installation
- Dashboard Camera Installation
- Roadside Assistance (RSA) Services
- Accident & Damage Repair
- Warranty Claim Processing
- Recall & Campaign Service
- Trade-In Vehicle Valuation
- Vehicle Diagnostic Services
- Maintenance Package Sales
- Wheel & Tire Sales
- Wheel Alignment & Balancing
- Infotainment System Installation
- Body Kit & Customization
- Paint Job & Customization
- Engine Overhaul & Major Repairs
- Transmission Repair & Overhaul
- Leather Upholstery Treatment
- Odor Removal & Sanitization
- Headlight & Taillight Restoration
- Vehicle Tracking & Subscription Services
- EV Charging Installation & Services
- Vehicle Subscription & Leasing
- Customer Loyalty Programs & Rewards

- Pick-up & Drop-off Service
- Loaner Vehicle Facility

1.1 Automobile Industry Overview

1.1.1 Market Trend Analysis - Deep Dive

The global automotive industry is undergoing a massive transformation. With the advent of Electric Vehicles (EVs), autonomous driving, and connected cars, the entire value chain is shifting.

Manufacturing models are moving towards modular platforms, while dealership networks are adopting digital-first retail strategies.

Key trends include a shift in consumer preference towards SUVs, integration of AI in vehicle diagnostics, and an increased focus on sustainability and emission reductions.

In this context, data analytics becomes crucial for dealerships to optimize inventory, improve customer retention, and maximize profitability across multiple revenue streams (sales, service, finance, insurance, and used cars).

The automobile industry is continuously evolving due to advancements in technology, changing consumer expectations, government regulations, and increasing competition. Market trends play a crucial role in shaping dealership strategies, inventory planning, customer engagement practices, and revenue generation models. Understanding these trends enables dealerships to align their operations with current market demands and maintain long-term competitiveness.

In recent years, the automotive sector has experienced significant transformation driven by digitalization, increasing demand for sport utility vehicles (SUVs), the emergence of electric vehicles (EVs), and greater emphasis on customer experience. These developments influence every stage of dealership operations, from lead generation and vehicle sales to after-sales service and customer retention.

Automotive Market Transformation

Traditional Vehicle Sales

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Digital Vehicle Discovery

|



Online Research & Comparison

|



Lead Generation Through Digital Channels

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Integrated Dealership Operations

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Data-Driven Customer Engagement

The transition toward digital and customer-centric business models has increased the importance of data analytics in monitoring customer behaviour, forecasting demand, and improving operational efficiency.

1.1.3 Consumer Preference Shift Toward SUVs and Digital Retail

Consumer buying behaviour has changed considerably over the last decade. Customers now conduct extensive online research before visiting a showroom. Vehicle comparisons, product reviews, financing options, and feature evaluations are often completed digitally before interacting with dealership representatives.

At the same time, demand for SUVs has increased due to their higher ground clearance, spacious interiors, advanced safety features, and versatility across different road conditions. This shift requires dealerships to adjust inventory allocation, sales planning, and marketing strategies according to changing customer preferences.

Modern Customer Purchase Journey

Online Search

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Vehicle Comparison

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Lead Generation

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Dealer Follow-Up

|



Test Drive

|



Finance & Insurance Evaluation



Vehicle Booking



Vehicle Delivery

Market Trend	Impact on Dealership Operations	Data Required
Growing SUV Demand	Increased inventory planning	Model-wise enquiry and sales data
Digital Customer Journey	Faster response management	Lead response time
Online Product Research	Better product positioning	Customer interaction data
Personalized Offers	Improved conversion rates	Customer segmentation data

For dealership management, understanding these trends helps optimize stock availability, improve customer engagement, and increase sales conversion rates.

1.1.4 Electric Vehicles, Connected Mobility and Sustainability

The automotive industry is moving toward cleaner and smarter mobility solutions. Government initiatives promoting sustainable transportation, improvements in charging infrastructure, and advancements in battery technology are encouraging the adoption of electric vehicles.

Connected vehicle technologies are also transforming customer expectations by providing features such as remote diagnostics, vehicle tracking, mobile connectivity, and predictive maintenance support. These innovations create new opportunities for dealerships to enhance customer engagement and service offerings.

Emerging Mobility Ecosystem

Government Policies



Electric Vehicle Adoption

|
▼
Charging Infrastructure

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▼
Customer Awareness

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▼
Dealership Readiness

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▼
Business Growth

Emerging Trend	Business Impact
Electric Vehicles	New sales opportunities
Connected Vehicles	Enhanced customer engagement
Digital Services	Improved service experience
Sustainability Initiatives	Long-term operational efficiency

For dealerships, adapting to these developments requires investment in employee training, infrastructure readiness, customer education, and analytical capabilities to monitor evolving market demand.

1.1.5 Regional Market Variations and Business Implications

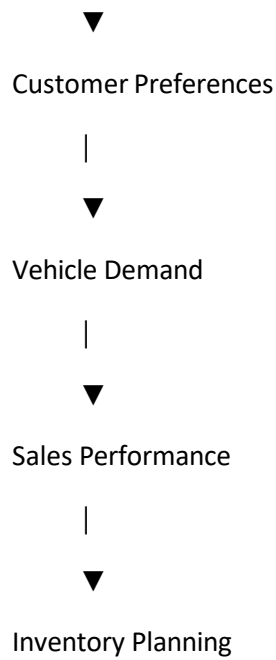
Customer preferences and vehicle demand vary significantly across different regions due to economic conditions, demographic characteristics, infrastructure availability, financing accessibility, and local competition. As a result, dealerships cannot rely on a uniform sales strategy for all markets.

Regional market analysis helps dealerships understand local demand patterns, identify high-performing vehicle models, optimize inventory distribution, and design targeted marketing campaigns. This ensures better resource utilization and improved customer satisfaction.

Regional Demand Analysis Framework

Regional Conditions

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Regional Factor	Business Impact
Income Levels	Vehicle affordability
Local Competition	Conversion performance
Infrastructure Development	Vehicle preference
Financing Availability	Booking rates
Urban and Rural Demographics	Product demand mix

The ability to analyze regional variations through data-driven decision-making enables dealerships to improve forecasting accuracy, optimize inventory investments, and enhance overall business performance.

1.1.6 Business Significance for Data Analytics

Market trends generate valuable business insights that assist management in making informed decisions. By continuously monitoring customer preferences, sales patterns, inventory movement, and regional demand variations, dealerships can improve forecasting, enhance customer satisfaction, and maximize profitability. Data analytics serves as the foundation for converting market observations into actionable business strategies and sustainable growth opportunities.

The following chapters detail the specific dynamics of the Indian market, Hyundai's positioning, and Sri Gurukripa Automobiles LLP's specific operational architecture

- Insurance Renewals
- Accessories Sales

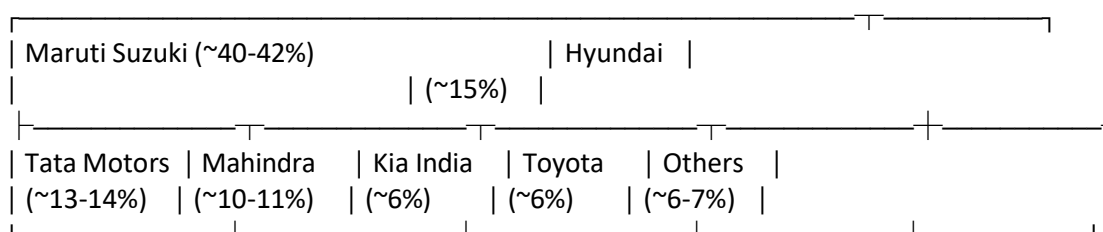
- Extended Warranty
- SOT(Shield Of Trust)
- Road Side Assistance (RSA)
- Finance Commissions
- Exchange Programs
- Body Shop
- VAS(Value Added Services)

The dealership's long-term success depends on maintaining customer relationships throughout the vehicle ownership lifecycle.

1.2 Competitor Landscape & Hyundai India Positioning

Hyundai Motor India Limited (HMIL) is the second-largest passenger vehicle manufacturer in India, maintaining a stable market share of **14.5% to 15.5%**.

[Indian Passenger Vehicle Market Share - 2025/2026]



1.2.1 Core Competitors and Their Strategic Vectors

1. Maruti Suzuki India Limited (MSIL):

- *Strengths:* Massive dealer network, low cost of ownership, high resale value, dominance in the CNG and entry-level hatchback segment.
- *Hyundai's Counter-strategy:* Focus on premium aesthetics, advanced infotainment (connected car technology - Bluelink), higher standard safety packages (6 airbags standard), and superior fit-and-finish.

2. Tata Motors:

- *Strengths:* Strong safety credentials (5-star BNCAP/GNCAP ratings), early mover advantage in the EV space (Nexon EV, Punch EV, Tiago EV).
- *Hyundai's Counter-strategy:* Launch of mass-market competitive EVs (Creta EV) and emphasizing powertrain refinement (smoother IVT and DCT transmissions compared to Tata's AMT/DCT).

3. Mahindra & Mahindra (M&M):

- *Strengths:* Absolute dominance in rugged, mid-to-large ladder-frame SUVs (Scorpio-N, XUV700, Thar).
- *Hyundai's Counter-strategy:* Offering monocoque SUVs (Creta, Tucson) that provide superior urban ride quality, modern features (ADAS level 2), and better fuel efficiency.

4. Kia India:

- *Strengths:* Hyundai's sister company sharing platforms (Seltos-Creta, Sonet-Venue) but positioned with a sportier, tech-oriented brand language.
- *Hyundai's Counter-strategy:* Positioning as a family-centric, trusted brand with a significantly larger network of rural service outlets.

1.2.2 Hyundai India Product Segment Analysis

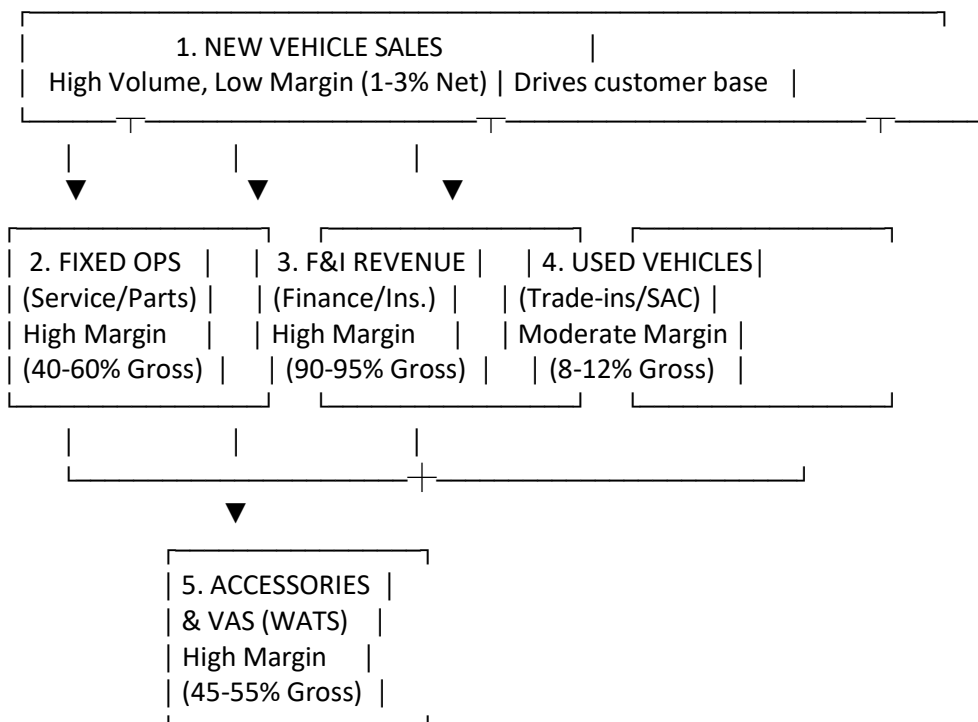
Hyundai segment coverage determines the target demographics of dealership marketing campaigns:

* **Entry/Micro-SUV:** Hyundai Exter (Competes with Tata Punch, Maruti Ignis). * **Premium Hatchback:** Hyundai i20 and i20 N Line (Competes with Maruti Baleno, Tata Altroz). * **Sub-Compact SUV:** Hyundai Venue (Competes with Maruti Brezza, Tata Nexon, Kia Sonet). * **Compact SUV (Volume Driver):** Hyundai Creta (Competes with Kia Seltos, Maruti Grand Vitara, Honda Elevate). * **Executive Sedan:** Hyundai Verna (Competes with Honda City, Skoda Slavia, Volkswagen Virtus). * **Premium SUV:** Hyundai Tucson & Santa Fe. * **Electric Vehicles:** Kona Electric, Creta EV, and IONIQ 5.

1.3 Dealership Franchise P&L Model

A car dealership operates on a complex, multi-layered Profit and Loss (P&L) structure where different business lines subsidize or feed into each other. Dealership analytics must evaluate performance across **five core profit centers**:

[Dealership Integrated Revenue Model]



1.3.1 Dealership Revenue Streams and Gross Margins

BUSINESS LINES SUMMARY TABLE

#	Business Line / Service	Margin %	Frequency	Data Priority
1	Vehicle Servicing	60-70%	2x/year	Critical
2	Insurance Sales	10-20%	Annual	High
3	Detailing & Cleaning	65-70%	3-4x/year	High
4	Spare Parts Sales	50-60%	Variable	Medium
5	New Vehicle Sales	15-20%	Once/5-7yr	Critical
6	Used Vehicle Sales	10-15%	Ongoing	Medium
7	Accessories Installation	40-50%	At delivery	Medium
8	Finance Commission	0.5-2%	Per sale	Critical
9	Extended Warranty	35-40%	At delivery	High
10	PPF Installation	55-60%	1-2x/year	Medium
11	Ceramic Coating	60-65%	1-2x/year	Medium
12	Polishing & Waxing	65-70%	2x/year	Medium
13	Car Washing	60-70%	3-4x/year	High
14	Interior Cleaning	55-65%	1-2x/year	Medium
15	Window Tinting	55-65%	One-time	Low
16	GPS Tracking	40-50%	Monthly sub	Medium
17	Dashboard Camera	50-60%	One-time	Low
18	Roadside Assistance	50-60%	Annual	Medium
19	Accident Repair	40-50%	As needed	Medium
20	Warranty Claims	Variable	Variable	High
21	Recall Service	Variable	Variable	Medium
22	Trade-In Valuation	Commission	Per sale	High
23	Diagnostics	40-50%	With repairs	Low
24	Pre-Purchase Inspection	Fee-based	Per sale	Medium

25	Emission Testing	Fee-based	Regulatory	Low
26	Maintenance Packages	25-30%	At delivery	High
27	Wheel & Tire Sales	30-40%	As needed	Medium
28	Wheel Alignment	50-60%	As needed	Low
29	Infotainment Install	40-50%	One-time	Low
30	Body Kit Custom	40-50%	One-time	Low
31	Paint Customization	40-50%	One-time	Low
32	Engine Overhaul	45-55%	Rare	Low
33	Transmission Repair	45-55%	Rare	Low
34	Leather Treatment	50-60%	1-2x/year	Low
35	Odor Removal	50-60%	As needed	Low
36	Headlight Restoration	50-60%	As needed	Low
37	Vehicle Tracking Sub	40-50%	Monthly	Medium
38	EV Charging Service	30-50%	Monthly	Medium
39	Vehicle Leasing	20-30%	Monthly	Medium
40	Loyalty Programs	Variable	Ongoing	High
41	Pick-up & Drop	Cost offset	Per visit	High
42	Loaner Vehicle	Cost offset	Per service	High

KEY BUSINESS INSIGHTS

Revenue & Profitability

- Highest Margin Services: Detailing (65-70%), Polishing (70%), Ceramic Coating (60-65%), Accessories (50-60%)
- Medium Margin: Vehicle Service (60-70%), Paint Protection (55-60%), Finance Commission (35-40%)
- Lower Margin: New Vehicle Sales (15-20%), Used Vehicle Sales (10-15%), Wheel Sales (30-40%)
- Pure Profit: Finance Commission (0.5-2% of loan amount), Insurance Commission (5-10%)

Customer Frequency

- High Frequency (Monthly/Quarterly): Car Washing, Vehicle Servicing, Diagnostics
- Medium Frequency (Semi-Annually): Maintenance, Insurance Renewal, Detailing
- Low Frequency (Once in Ownership): New Vehicle, Major Repairs, Customization

Strategic Importance for Data Analytics

- Track penetration rate for each of 40+ services to identify growth opportunities
- Service attachment rate (% of new vehicle buyers using dealership services) is critical KPI
- Customer lifetime value = vehicle price + all 40 services over 5-7 years
- Optimize marketing spend toward high-margin services vs. vehicle sales
- Each service line requires independent forecasting, inventory management, and KPI tracking.

1.3.2 Dealership Cost Structure

Cost Category	Sub-Component	Type	Impact on Analytics
Inventory Cost	Stockyard interest (Holding Cost / Floorplanning)	Variable	Cars sitting >30 days accrue high interest costs, eroding margins.
Manpower	Salaries, incentives (Sales executives, technicians)	Semi-Fixed	Linked to productivity (Technician efficiency, sales conversion).

Cost Category	Sub-Component	Type	Impact on Analytics
Real Estate	Showroom rent, workshop lease, stockyard rent	Fixed	Requires high volume throughput to reduce unit fixed cost.
Marketing	Ads, events, promotions, Telegram campaign tracking	Variable	Analyzed via Cost Per Lead (CPL) and Cost Per Booking (CPB).

Operational	Electricity (high in workshops), IT systems, GDMS licenses	Fixed	Baseline cost that must be absorbed by fixed operations.
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1.4 Dealership P&L Sensitivity Analysis

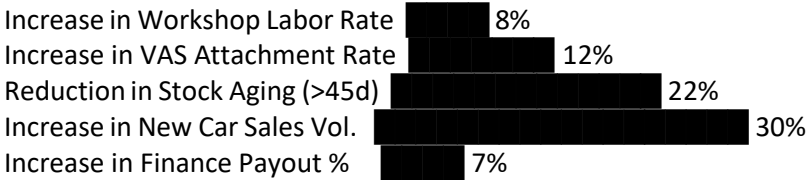
To understand the critical balance between volume and margin, the dealership's financial health is measured using the **Service Absorption Ratio**:

$$= \frac{\text{Service Absorption Ratio}}{\text{Gross Profit from Fixed Operations (Service + Parts)}} \\ \text{Total Dealership Operating Expenses (Rent + Manpower + Interest + Admin)}$$

- **Benchmark:** Healthy dealerships target a ratio of **80% or higher**, meaning the workshop covers almost all fixed operating costs of the entire dealership, allowing new vehicle margins, insurance, and F&I commissions to act as pure net profit.
- **Analytics Priority:** Optimize workshop throughput (vehicle bays occupied per day), technician efficiency (hours billed vs. hours worked), and parts turnover rate.

Let's look at the financial impact of a 5% shift in different variables on the Net Profit of a typical dealership:

[Net Profit Sensitivity to a 5% Improvement in Operational Metrics]



This sensitivity analysis demonstrates that while new vehicle volume has the highest absolute impact, operational optimizations like **reducing inventory stock aging** and **maximizing accessory/VAS attachments** have massive leveraged impacts on the dealer's bottom line.

1.5 Company Overview

Organization Name

Sri Gurukripa Automobiles LLP

Sell Any Car

Brand

Hyundai

Industry

Automobile Retail and After-Sales Services

1.6 Core Business Activities

- 1 Vehicle Servicing
- 2 Insurance Sales
- 3 Detailing & Cleaning
- 4 Spare Parts Sales
- 5 New Vehicle Sales
- 6 Used Vehicle Sales
- 7 Accessories Installation
- 8 Finance & Auto Loans
- 9 Extended Warranty Services
- 10 Paint Protection Film (PPF) Installation
- 11 Ceramic Coating Application
- 12 Polishing & Waxing
- 13 Car Washing (Basic & Premium)
- 14 Interior Upholstery Cleaning
- 15 Window Tinting
- 16 GPS Tracking Installation
- 17 Dashboard Camera Installation
- 18 Roadside Assistance (RSA) Services
- 19 Accident & Damage Repair
- 20 Warranty Claim Processing
- 21 Recall & Campaign Service
- 22 Trade-In Vehicle Valuation
- 23 Vehicle Diagnostic Services
- 24 Pre-Purchase Vehicle Inspection
- 25 Emission Testing (PUC Certificate)
- 26 Maintenance Package Sales
- 27 Wheel & Tire Sales
- 28 Wheel Alignment & Balancing
- 29 Infotainment System Installation
- 30 Body Kit & Customization
- 31 Paint Job & Customization
- 32 Engine Overhaul & Major Repairs
- 33 Transmission Repair & Overhaul
- 34 Leather Upholstery Treatment
- 35 Odor Removal & Sanitization
- 36 Headlight & Taillight Restoration
- 37 Vehicle Tracking & Subscription Services
- 38 EV Charging Installation & Services
- 39 Vehicle Subscription & Leasing

- 40 Customer Loyalty Programs & Rewards
- 41 Pick-up & Drop-off Service

Loaner Vehicle Facility

Sales Department Operations

The Sales Department serves as the primary revenue-generating function of the dealership. It is responsible for converting customer enquiries into confirmed vehicle deliveries while coordinating with CRM, Finance, Insurance, EDP, Accounts, and Registration teams. Every customer interaction generates operational data that is captured within dealership systems and later used for sales analysis, conversion tracking, inventory planning, and management reporting.

Sales Process Flow

Lead Generation



Lead Qualification



Customer Enquiry



Test Drive



Booking



Finance & Insurance



Vehicle Allocation



Billing & Registration



Vehicle Delivery



Customer Follow-Up

1.7.1 Sales Standard Operating Procedure 1 – Lead Generation and Enquiry Registration

Step 1: Customer enquiry is received through walk-in, website, social media, referral, telecalling, or Hyundai campaign sources.

Step 2: Sales Consultant records customer details, contact information, preferred model, budget, and purchase timeline.

Step 3: Lead is entered into HSSmart/CRM platform and assigned to the concerned sales executive.

Step 4: Initial customer interaction is completed to understand requirements and schedule follow-up activities.

Step 5: Lead status is monitored through CRM dashboards for future engagement.

This process establishes the foundation of the sales funnel and ensures every enquiry is captured for conversion analysis and performance measurement.

1.7.2 Sales Standard Operating Procedure 2 – Lead Qualification

Step 1: Sales Executive contacts the customer within the defined response timeline.

Step 2: Customer requirements, vehicle preference, financing needs, and buying intent are evaluated.

Step 3: Potential customers are categorized based on purchase probability and expected purchase period.

Step 4: Qualified leads are prioritized for test drives and detailed consultations.

Step 5: CRM records are updated with customer interaction outcomes.

Effective lead qualification improves conversion efficiency and enables better allocation of sales resources.

1.7.3 Sales Standard Operating Procedure 3 – Product Presentation

Step 1: Customer requirements are matched with suitable Hyundai models and variants.

Step 2: Features, specifications, safety technologies, and ownership benefits are explained.

Step 3: Competitive comparisons are discussed where applicable.

Step 4: Customer objections and queries are addressed.

Step 5: Product interest level is recorded in CRM.

A structured product presentation improves customer confidence and supports informed purchase decisions.

1.7.4 Sales Standard Operating Procedure 4 – Test Drive Management

Step 1: Test drive request is scheduled based on vehicle availability.

Step 2: Vehicle readiness and documentation are verified.

Step 3: Customer identity and eligibility requirements are validated.

Step 4: Test drive is conducted under dealership guidelines.

Step 5: Feedback and purchase interest are captured in CRM.

Test drives provide practical product experience and significantly influence purchase decisions.

1.7.5 Sales Standard Operating Procedure 5 – Booking Confirmation

Step 1: Customer confirms vehicle selection.

Step 2: Booking amount is collected as per dealership policy.

Step 3: Booking details are entered into GDMS.

Step 4: Variant, color, and delivery commitments are finalized.

Step 5: Booking acknowledgment is shared with the customer.

Accurate booking management prevents allocation errors and improves delivery planning.

1.7.6 Sales Standard Operating Procedure 6 – Finance Coordination

Step 1: Customer financing requirements are assessed.

Step 2: Loan applications are submitted to partner financial institutions.

Step 3: Documentation verification is completed.

Step 4: Approval status is tracked until sanction.

Step 5: Finance information is updated in dealership systems.

Efficient finance processing supports higher retail conversions and customer convenience.

1.7.7 Sales Standard Operating Procedure 7 – Insurance Coordination

Step 1: Insurance quotations are generated.

Step 2: Coverage options and policy benefits are explained.

Step 3: Customer selects preferred insurance provider.

Step 4: Policy issuance process is completed.

Step 5: Insurance details are linked to the booking record.

Insurance integration contributes to dealership revenue while enhancing customer protection.

1.7.8 Sales Standard Operating Procedure 8 – Vehicle Allocation

Step 1: Available stock is reviewed in GDMS.

Step 2: Matching VIN is identified based on booking specifications.

Step 3: Allocation approval is obtained from authorized personnel.

Step 4: Inventory status is updated.

Step 5: Delivery planning begins.

Proper allocation minimizes inventory aging and improves delivery timelines.

1.7.9 Sales Standard Operating Procedure 9 – Billing and Documentation

Step 1: Customer payment status is verified.

Step 2: Tax invoice generation is completed.

Step 3: Required statutory documents are prepared.

Step 4: Registration and compliance requirements are processed.

Step 5: Delivery file is finalized.

Accurate documentation ensures regulatory compliance and smooth vehicle delivery.

1.7.10 Sales Standard Operating Procedure 10 – Vehicle Delivery Planning

Step 1: Delivery schedule is confirmed.

Step 2: Vehicle inspection and cleaning are completed.

Step 3: Accessories and commitments are verified.

Step 4: Customer communication is conducted.

Step 5: Delivery readiness is approved.

A well-planned delivery process enhances customer experience and dealership reputation.

1.7.11 Sales Standard Operating Procedure 11 – Vehicle Handover

Step 1: Customer identity verification is completed.

Step 2: Vehicle features and controls are demonstrated.

Step 3: Documentation package is handed over.

Step 4: Customer acknowledgements are collected.

Step 5: Delivery completion is recorded in GDMS.

The delivery ceremony marks the successful completion of the sales lifecycle.

1.7.12 Sales Standard Operating Procedure 12 – Post-Delivery Follow-Up

Step 1: Customer satisfaction call is conducted.

Step 2: Delivery feedback is collected.

Step 3: Concerns are escalated if necessary.

Step 4: CRM records are updated.

Step 5: Future engagement activities are planned.

Post-delivery engagement strengthens customer relationships and supports retention initiatives.

1.8 Service Department Operations

The Service Department is one of the dealership's most important profit centers and plays a critical role in customer retention, after-sales revenue generation, and brand loyalty. The department manages periodic maintenance, repairs, warranty claims, spare parts consumption, customer complaints, and vehicle health monitoring. Service operations involve coordination among Service Advisors, Technicians, Workshop Controllers, Parts Teams, CRM Executives, Warranty Coordinators, and Customers. Every service interaction generates operational data that supports workshop performance analysis, revenue monitoring, and customer retention initiatives.

Service Process Flow

Service Appointment



Vehicle Reception



Job Card Creation



Vehicle Inspection



Workshop Allocation



Repair & Maintenance



Quality Check



Billing



Vehicle Delivery



Customer Feedback

1.8.1 Service Standard Operating Procedure 1 – Service Appointment Scheduling

Step 1: Customer contacts the dealership through phone, website, mobile application, or CRM campaign.

Step 2: Appointment date and service requirements are captured.

Step 3: Vehicle details are verified.

Step 4: Workshop capacity is checked.

Step 5: Appointment confirmation is communicated to the customer.

This process improves workshop planning and minimizes customer waiting time.

1.8.2 Service Standard Operating Procedure 2 – Vehicle Reception

Step 1: Customer arrives at the workshop.

Step 2: Vehicle and customer details are verified.

Step 3: Service history is reviewed.

Step 4: Customer concerns are recorded.

Step 5: Vehicle is accepted for inspection.

Proper reception ensures accurate identification of customer requirements.

1.8.3 Service Standard Operating Procedure 3 – Job Card Creation

Step 1: Service Advisor creates a Repair Order (RO).

Step 2: Customer complaints are documented.

Step 3: Vehicle mileage and condition are recorded.

Step 4: Estimated service scope is defined.

Step 5: Job card is approved and released.

The job card serves as the primary operational document throughout the service cycle.

1.8.4 Service Standard Operating Procedure 4 – Preliminary Vehicle Inspection

Step 1: Exterior condition is assessed.

Step 2: Interior condition is verified.

Step 3: Functional systems are checked.

Step 4: Additional repair requirements are identified.

Step 5: Findings are documented.

Inspection improves service accuracy and transparency.

1.8.5 Service Standard Operating Procedure 5 – Workshop Allocation

Step 1: Job card is assigned to the workshop controller.

Step 2: Technician availability is reviewed.

Step 3: Workload balancing is performed.

Step 4: Vehicle is allocated to the technician.

Step 5: Repair activities are scheduled.

Efficient allocation improves workshop productivity.

1.8.6 Service Standard Operating Procedure 6 – Spare Parts Requirement Assessment

Step 1: Required parts are identified.

Step 2: Inventory availability is checked.

Step 3: Parts reservation is completed.

Step 4: Procurement requests are initiated if required.

Step 5: Parts are issued to the workshop.

This process minimizes repair delays caused by part shortages.

1.8.7 Service Standard Operating Procedure 7 – Periodic Maintenance Service

Step 1: Scheduled maintenance activities are performed.

Step 2: Engine oil and filters are replaced.

Step 3: Vehicle systems are inspected.

Step 4: Safety checks are completed.

Step 5: Service completion is recorded.

Periodic maintenance improves vehicle reliability and customer satisfaction.

1.8.8 Service Standard Operating Procedure 8 – Repair and Diagnosis

Step 1: Technician diagnoses reported issues.

Step 2: Root causes are identified.

Step 3: Repair recommendations are prepared.

Step 4: Customer approval is obtained.

Step 5: Repair activities are executed.

Accurate diagnosis reduces repeat workshop visits.

1.8.9 Service Standard Operating Procedure 9 – Warranty Claim Processing

Step 1: Warranty eligibility is verified.

Step 2: Defect documentation is completed.

Step 3: Claim information is uploaded.

Step 4: Manufacturer approval is obtained.

Step 5: Claim settlement is monitored.

Warranty compliance ensures proper reimbursement and customer trust.

1.8.10 Service Standard Operating Procedure 10 – Additional Repair Authorization

Step 1: Additional issues are identified.

Step 2: Cost estimates are prepared.

Step 3: Customer approval is requested.

Step 4: Approval status is documented.

Step 5: Additional work is initiated.

This process prevents billing disputes and improves transparency.

1.8.11 Service Standard Operating Procedure 11 – Quality Control Inspection

Step 1: Completed repairs are reviewed.

Step 2: Work quality standards are verified.

Step 3: Functional testing is performed.

Step 4: Rectifications are completed if required.

Step 5: Vehicle is approved for delivery.

Quality control reduces repeat complaints.

1.8.12 Service Standard Operating Procedure 12 – Vehicle Washing and Final Preparation

Step 1: Vehicle cleaning is completed.

Step 2: Interior condition is checked.

Step 3: Accessories are verified.

Step 4: Final inspection is conducted.

Step 5: Delivery readiness is confirmed.

This improves customer delivery experience.

1.8.13 Service Standard Operating Procedure 13 – Billing Preparation

Step 1: Labour charges are calculated.

Step 2: Parts consumption is verified.

Step 3: Taxes and discounts are applied.

Step 4: Invoice is generated.

Step 5: Billing approval is completed.

Accurate billing supports financial integrity.

1.8.14 Service Standard Operating Procedure 14 – Customer Billing Discussion

Step 1: Invoice details are explained.

Step 2: Service activities are reviewed.

Step 3: Customer questions are addressed.

Step 4: Payment options are discussed.

Step 5: Customer approval is obtained.

Transparent billing improves customer confidence.

1.8.15 Service Standard Operating Procedure 15 – Vehicle Delivery

Step 1: Customer is notified.

Step 2: Completed work is explained.

Step 3: Maintenance recommendations are shared.

Step 4: Vehicle handover is completed.

Step 5: Delivery acknowledgement is collected.

Delivery represents the completion of the workshop cycle.

1.8.16 Service Standard Operating Procedure 16 – Post-Service Feedback Collection

Step 1: Customer feedback call is conducted.

Step 2: Satisfaction score is recorded.

Step 3: Complaints are identified.

Step 4: Escalations are initiated if required.

Step 5: CRM records are updated.

Feedback helps measure service quality.

1.8.17 Service Standard Operating Procedure 17 – Complaint Resolution Management

Step 1: Complaint is logged.

Step 2: Root cause investigation is performed.

Step 3: Corrective actions are initiated.

Step 4: Customer communication is maintained.

Step 5: Closure confirmation is obtained.

Timely resolution improves retention and brand trust.

1.8.18 Service Standard Operating Procedure 18 – Service Reminder Campaigns

Step 1: Due-service customers are identified.

Step 2: Customer lists are generated.

Step 3: Reminder communications are sent.

Step 4: Appointment bookings are encouraged.

Step 5: Campaign performance is monitored.

Reminder campaigns increase workshop retention.

1.8.19 Service Standard Operating Procedure 19 – Workshop Performance Monitoring

Step 1: Daily workshop data is collected.

Step 2: Labour productivity is measured.

Step 3: Revenue performance is reviewed.

Step 4: Technician efficiency is analyzed.

Step 5: Improvement actions are initiated.

Performance monitoring supports operational excellence.

1.8.20 Service Standard Operating Procedure 20 – Service Data Reporting and Analysis

Step 1: Workshop transaction data is consolidated.

Step 2: Service KPIs are calculated.

Step 3: Customer retention metrics are analyzed.

Step 4: Management reports are prepared.

Step 5: Strategic recommendations are presented.

This process transforms operational data into actionable business insights for management decision-making.

1.9 Finance Department Operations

The Finance Department acts as a bridge between customers, financial institutions, and dealership operations. Its primary responsibility is to facilitate vehicle financing, improve retail conversion rates, manage banking relationships, monitor loan approvals, and support revenue generation through finance penetration. The department works closely with Sales, Insurance, Accounts, CRM, and Banking Partners to ensure smooth financing processes while maintaining compliance with regulatory and documentation requirements.

Finance Process Flow

Customer Enquiry



Finance Requirement Analysis



Loan Eligibility Assessment



Document Collection



Bank Application Submission



Credit Verification



Loan Approval



Disbursement Processing



Vehicle Billing



Performance Reporting

1.9.1 Finance Requirement Identification

Step 1: Customer expresses interest in purchasing a vehicle through financing.

Step 2: Sales Executive identifies financing requirements during the sales consultation.

Step 3: Customer budget, down payment capacity, and preferred EMI range are assessed.

Step 4: Suitable financing options are shortlisted.

Step 5: Finance lead is assigned to the Finance Executive.

This process helps determine the most appropriate financing solution based on customer affordability and purchase requirements.

1.9.2 Loan Eligibility Assessment

Step 1: Customer income profile is reviewed.

Step 2: Employment or business details are verified.

Step 3: Existing financial obligations are evaluated.

Step 4: Loan eligibility is estimated using lender guidelines.

Step 5: Potential loan amount and tenure options are presented.

Proper eligibility assessment improves approval rates and reduces application rejection.

1.9.3 Documentation Collection

Step 1: Required document checklist is shared with the customer.

Step 2: Identity and address proofs are collected.

Step 3: Income documents are verified.

Step 4: Banking and financial records are reviewed.

Step 5: Documentation completeness is confirmed.

Accurate documentation reduces processing delays and compliance risks.

1.9.4 Finance Application Processing

Step 1: Customer application details are entered into the finance system.

Step 2: Selected bank or NBFC is identified.

Step 3: Loan proposal is prepared.

Step 4: Application package is validated.

Step 5: Proposal is submitted for approval.

This stage initiates formal financing evaluation by lending institutions.

1.9.5 Credit Verification and Approval Coordination

Step 1: Bank performs credit assessment.

Step 2: Customer verification activities are completed.

Step 3: Additional clarifications are provided when required.

Step 4: Approval status is monitored.

Step 5: Sanction information is communicated to the customer.

Continuous coordination improves turnaround time and customer experience.

1.9.6 Loan Sanction and Offer Finalization

Step 1: Approved loan amount is reviewed.

Step 2: Interest rate and tenure are verified.

Step 3: EMI schedule is generated.

Step 4: Customer acceptance is obtained.

Step 5: Loan offer is finalized.

This process ensures transparency regarding financing commitments and repayment obligations.

1.9.7 Loan Disbursement Management

Step 1: Vehicle allocation and billing readiness are confirmed.

Step 2: Disbursement request is initiated.

Step 3: Bank releases approved funds.

Step 4: Payment confirmation is received.

Step 5: Disbursement records are updated.

Timely disbursement ensures smooth retail completion and vehicle delivery.

1.9.8 Finance Revenue and Payout Tracking

Step 1: Finance cases are consolidated.

Step 2: Bank payout structures are reviewed.

Step 3: Commission receivables are monitored.

Step 4: Revenue reconciliation is performed.

Step 5: Finance income reports are prepared.

This process supports profitability analysis and financial performance monitoring.

1.9.9 Customer Finance Support and Query Resolution

Step 1: Customer finance-related queries are received.

Step 2: Loan status information is verified.

Step 3: Clarifications are provided.

Step 4: Escalations are managed when required.

Step 5: Resolution status is recorded.

Effective customer support enhances satisfaction and trust in dealership services.

1.9.10 Finance Performance Reporting and Analytics

Step 1: Monthly finance transaction data is collected.

Step 2: Finance penetration metrics are calculated.

Step 3: Bank-wise performance is analyzed.

Step 4: Approval and rejection trends are evaluated.

Step 5: Management reports and recommendations are generated.

This process converts finance operational data into actionable business insights for decision-making, revenue optimization, and performance improvement.

1.10 Insurance Department Operations

The Insurance Department plays a vital role in dealership profitability and customer retention by facilitating motor insurance sales, renewals, endorsements, and claim settlements. The

department collaborates with customers, insurance companies, surveyors, service advisors, and finance teams to ensure smooth policy management and claim processing. Insurance-related data generated through these operations supports revenue analysis, claim tracking, renewal forecasting, and business performance monitoring.

Insurance Process Flow



1.10.1 Insurance Requirement Identification

Step 1: Customer purchases a new vehicle or requests insurance support.

Step 2: Vehicle details and customer requirements are collected.

Step 3: Coverage expectations are discussed.

Step 4: Insurance options are evaluated.

Step 5: Insurance lead is assigned for processing.

This process helps identify suitable insurance solutions based on customer needs and vehicle profile.

1.10.2 Insurance Quotation Preparation

Step 1: Vehicle information is captured.

Step 2: Insurer quotations are requested.

Step 3: Premium calculations are performed.

Step 4: Coverage benefits are compared.

Step 5: Quotation details are presented to the customer.

This process enables informed decision-making and policy comparison.

1.10.3 Policy Selection and Approval

Step 1: Customer reviews available insurance plans.

Step 2: Coverage benefits and exclusions are explained.

Step 3: Preferred insurer is selected.

Step 4: Premium confirmation is obtained.

Step 5: Policy approval is recorded.

This process ensures transparency and customer understanding.

1.10.4 Documentation Collection

Step 1: Required customer documents are collected.

Step 2: Vehicle registration details are verified.

Step 3: Identity verification is completed.

Step 4: Compliance checks are performed.

Step 5: Documents are uploaded into the system.

Accurate documentation supports policy issuance and compliance.

1.10.5 Policy Issuance

Step 1: Proposal details are submitted.

Step 2: Insurer validates application information.

Step 3: Premium confirmation is received.

Step 4: Policy is generated.

Step 5: Policy documents are shared with the customer.

This process activates insurance coverage for the vehicle.

1.10.6 Premium Collection Management

Step 1: Premium amount is communicated.

Step 2: Customer payment is collected.

Step 3: Payment confirmation is verified.

Step 4: Receipt is generated.

Step 5: Financial records are updated.

This process ensures accurate revenue recording and insurer settlement.

1.10.7 Insurance Data Entry and Record Maintenance

Step 1: Policy information is entered into GDMS.

Step 2: Customer details are validated.

Step 3: Policy records are stored.

Step 4: Data quality checks are performed.

Step 5: Records are maintained for future reference.

Proper record management supports reporting and renewals.

1.10.8 Endorsement Processing

Step 1: Customer requests policy modification.

Step 2: Change requirements are reviewed.

Step 3: Insurer approval is obtained.

Step 4: Policy endorsement is processed.

Step 5: Updated policy is shared.

This process maintains policy accuracy throughout its lifecycle.

1.10.9 Insurance Claim Registration

Step 1: Accident or damage is reported.

Step 2: Claim request is initiated.

Step 3: Supporting documents are collected.

Step 4: Claim details are recorded.

Step 5: Insurer is notified.

This process formally initiates claim processing.

1.10.10 Vehicle Damage Inspection

Step 1: Damaged vehicle is inspected.

Step 2: Service advisor prepares assessment.

Step 3: Repair estimate is generated.

Step 4: Evidence is documented.

Step 5: Inspection report is finalized.

This process determines repair requirements and claim eligibility.

1.10.11 Surveyor Coordination

Step 1: Survey request is raised.

Step 2: Surveyor visit is scheduled.

Step 3: Vehicle inspection is conducted.

Step 4: Findings are documented.

Step 5: Survey report is approved.

Surveyor evaluation supports insurer claim decisions.

1.10.12 Claim Approval Management

Step 1: Claim documents are submitted.

Step 2: Insurer review is monitored.

Step 3: Clarifications are provided.

Step 4: Approval status is tracked.

Step 5: Authorization is received.

This process minimizes claim delays and improves turnaround time.

1.10.13 Cashless Claim Processing

Step 1: Cashless eligibility is verified.

Step 2: Authorization is obtained.

Step 3: Repairs are approved.

Step 4: Insurer payment commitment is received.

Step 5: Repair work proceeds.

Cashless processing improves customer convenience.

1.10.14 Claim Settlement Coordination

Step 1: Repair completion is confirmed.

Step 2: Final bill is prepared.

Step 3: Claim amount is reconciled.

Step 4: Settlement documents are submitted.

Step 5: Claim payment is received.

This process ensures timely recovery of claim receivables.

1.10.15 Insurance Renewal Management

Step 1: Expiring policies are identified.

Step 2: Renewal quotations are generated.

Step 3: Customers are contacted.

Step 4: Renewal options are discussed.

Step 5: Renewal confirmation is obtained.

Renewal management supports recurring revenue generation.

1.10.16 Renewal Follow-Up Campaigns

Step 1: Customer database is analyzed.

Step 2: Renewal reminders are scheduled.

Step 3: Communication campaigns are executed.

Step 4: Customer responses are monitored.

Step 5: Renewals are converted into sales.

This process improves retention and policy renewal rates.

1.10.17 Insurance Revenue Monitoring

Step 1: Policy sales data is collected.

Step 2: Revenue sources are categorized.

Step 3: Commission earnings are tracked.

Step 4: Performance trends are analyzed.

Step 5: Revenue reports are generated.

This process supports profitability analysis.

1.10.18 Customer Query Resolution

Step 1: Insurance-related queries are received.

Step 2: Information is verified.

Step 3: Clarifications are provided.

Step 4: Issues are resolved.

Step 5: Resolution records are updated.

Effective support enhances customer satisfaction.

1.10.19 Insurance Compliance Monitoring

Step 1: Regulatory requirements are reviewed.

Step 2: Policy documentation is audited.

Step 3: Compliance gaps are identified.

Step 4: Corrective actions are implemented.

Step 5: Compliance reports are generated.

This process ensures adherence to industry regulations.

1.10.20 Insurance Analytics and Performance Reporting

Step 1: Insurance operational data is consolidated.

Step 2: Claim and renewal metrics are calculated.

Step 3: Insurer-wise performance is analyzed.

Step 4: Revenue and conversion trends are evaluated.

Step 5: Management dashboards and reports are prepared.

This process transforms insurance data into actionable insights for strategic decision-making and business growth.

1.11 Used Cars Department Operations

The Used Cars Department is responsible for acquiring, evaluating, refurbishing, marketing, and selling pre-owned vehicles. The department contributes significantly to dealership profitability by generating revenue through vehicle resale, exchange programs, finance penetration, insurance sales, and value-added services. It collaborates with Sales, Service, Finance, Insurance, and Accounts departments to ensure smooth operations and customer satisfaction.

Used Car Process Flow





1.11.1 Used Car Lead Generation

Step 1: Customer enquiry is received through walk-in, website, CRM, referral, or exchange program.

Step 2: Lead details are captured in the system.

Step 3: Customer requirements are analyzed.

Step 4: Lead qualification is performed.

Step 5: Sales consultant is assigned.

This process helps build a strong pipeline of potential buyers and sellers.

1.11.2 Vehicle Exchange Evaluation

Step 1: Customer presents vehicle for exchange.

Step 2: Vehicle details are recorded.

Step 3: Physical inspection is conducted.

Step 4: Vehicle history is reviewed.

Step 5: Evaluation request is generated.

This process establishes the preliminary value of the used vehicle.

1.11.3 Vehicle Inspection and Assessment

Step 1: Exterior condition is inspected.

Step 2: Interior condition is reviewed.

Step 3: Mechanical components are evaluated.

Step 4: Test drive assessment is conducted.

Step 5: Inspection findings are documented.

A detailed assessment helps determine refurbishment requirements and resale value.

1.11.4 Vehicle Valuation and Pricing

Step 1: Market prices are analyzed.

Step 2: Vehicle condition score is calculated.

Step 3: Depreciation factors are applied.

Step 4: Procurement value is estimated.

Step 5: Recommended purchase price is approved.

This process ensures competitive and profitable vehicle acquisition.

1.11.5 Procurement Approval

Step 1: Evaluation report is reviewed.

Step 2: Margin analysis is performed.

Step 3: Procurement proposal is submitted.

Step 4: Management approval is obtained.

Step 5: Purchase authorization is generated.

This process controls acquisition costs and profitability.

1.11.6 Vehicle Acquisition

Step 1: Customer accepts purchase offer.

Step 2: Ownership documents are verified.

Step 3: Purchase agreement is completed.

Step 4: Payment is processed.

Step 5: Vehicle inventory entry is created.

Vehicle acquisition officially transfers ownership to the dealership.

1.11.7 Documentation Verification

Step 1: Registration Certificate is reviewed.

Step 2: Insurance status is verified.

Step 3: Loan clearance status is checked.

Step 4: Legal compliance validation is performed.

Step 5: Documentation approval is completed.

Proper verification minimizes legal and ownership risks.

1.11.8 Refurbishment Planning

Step 1: Vehicle defects are identified.

Step 2: Repair requirements are documented.

Step 3: Cost estimates are prepared.

Step 4: Workshop schedule is planned.

Step 5: Refurbishment approval is issued.

This process prepares the vehicle for resale.

1.11.9 Refurbishment Execution

Step 1: Mechanical repairs are completed.

Step 2: Bodywork corrections are performed.

Step 3: Interior enhancements are executed.

Step 4: Parts replacement is completed.

Step 5: Vehicle readiness is confirmed.

Refurbishment improves customer appeal and resale value.

1.11.10 Quality Inspection

Step 1: Final inspection is performed.

Step 2: Repair quality is validated.

Step 3: Test drive verification is completed.

Step 4: Compliance checklist is reviewed.

Step 5: Quality approval is granted.

This process ensures the vehicle meets dealership standards.

1.11.11 Inventory Management

Step 1: Vehicle inventory records are updated.

Step 2: Stock categorization is performed.

Step 3: Pricing information is uploaded.

Step 4: Vehicle photographs are attached.

Step 5: Inventory availability is published.

Inventory visibility improves sales efficiency.

1.11.12 Used Car Marketing

Step 1: Vehicle listings are created.

Step 2: Digital platforms are updated.

Step 3: Marketing campaigns are launched.

Step 4: Customer enquiries are monitored.

Step 5: Lead generation performance is tracked.

This process increases customer reach and vehicle visibility.

1.11.13 Customer Vehicle Demonstration

Step 1: Customer visit is scheduled.

Step 2: Vehicle features are presented.

Step 3: Condition details are explained.

Step 4: Questions are addressed.

Step 5: Demonstration feedback is collected.

This process supports customer confidence and purchase decisions.

1.11.14 Test Drive Management

Step 1: Test drive request is registered.

Step 2: Customer eligibility is verified.

Step 3: Vehicle preparation is completed.

Step 4: Test drive is conducted.

Step 5: Feedback is recorded.

Test drives improve conversion rates and customer satisfaction.

1.11.15 Negotiation and Booking

Step 1: Final vehicle price is discussed.

Step 2: Trade-in options are evaluated.

Step 3: Discounts are approved.

Step 4: Booking amount is collected.

Step 5: Vehicle booking is confirmed.

This process converts prospects into customers.

1.11.16 Finance Coordination

Step 1: Finance requirements are identified.

Step 2: Loan eligibility is assessed.

Step 3: Finance applications are processed.

Step 4: Approval status is monitored.

Step 5: Finance closure is completed.

Financing improves affordability and sales conversion.

1.11.17 Insurance Coordination

Step 1: Insurance requirements are reviewed.

Step 2: Policy quotations are generated.

Step 3: Coverage options are explained.

Step 4: Policy issuance is completed.

Step 5: Insurance records are updated.

Insurance contributes additional revenue and customer protection.

1.11.18 Vehicle Delivery Management

Step 1: Delivery checklist is prepared.

Step 2: Documentation is verified.

Step 3: Vehicle handover is completed.

Step 4: Customer orientation is conducted.

Step 5: Delivery confirmation is recorded.

This process ensures a professional ownership experience.

1.11.19 Customer Feedback and Retention

Step 1: Post-delivery feedback is collected.

Step 2: Satisfaction scores are reviewed.

Step 3: Issues are resolved.

Step 4: CRM records are updated.

Step 5: Future engagement activities are planned.

Customer retention supports repeat business and referrals.

1.11.20 Used Car Performance Analytics and Reporting

Step 1: Procurement and sales data are consolidated.

Step 2: Stock aging metrics are calculated.

Step 3: Gross margin performance is analyzed.

Step 4: Conversion rates are evaluated.

Step 5: Management dashboards and reports are generated.

This process transforms operational data into actionable insights for inventory optimization, profitability improvement, and strategic decision-making.

1.12 Accessories Department Operations

The Accessories department is a critical profit center. Operations involve accessory sales, inventory management, installation coordination, customer customization requirements, vendor management, and profitability tracking. The department ensures genuine accessory availability, timely fitment, quality installation, and enhanced customer satisfaction.

1.12.1 Accessory Enquiry Registration

Step 1: Customer accessory enquiry is received.

Step 2: Enquiry details are recorded in GDMS/CRM.

Step 3: Vehicle compatibility is verified.

Step 4: Quotation requirements are identified.

Step 5: Customer follow-up is initiated.

This process ensures accurate enquiry tracking and timely customer response.

1.12.2 Accessory Quotation Preparation

Step 1: Required accessories are identified.

Step 2: Pricing is verified.

Step 3: Quotation is prepared.

Step 4: Manager approval is obtained.

Step 5: Quotation is shared with the customer.

This process ensures transparent pricing and faster sales conversion.

1.12.3 Stock Availability Verification

Step 1: Accessory stock availability is checked.

Step 2: Inventory levels are verified.

Step 3: Required items are reserved.

Step 4: Procurement needs are identified.

Step 5: Customer is informed of availability.

This process ensures proper inventory utilization and customer commitment.

1.12.4 Purchase Order Management

Step 1: Purchase requisition is created.

Step 2: Required approvals are obtained.

Step 3: Purchase order is issued to vendor.

Step 4: Delivery timelines are tracked.

Step 5: Procurement records are updated.

This process ensures timely replenishment of accessory inventory.

1.12.5 Goods Receipt and Verification

Step 1: Accessories are received from suppliers.

Step 2: Quantity verification is performed.

Step 3: Quality inspection is conducted.

Step 4: Inventory records are updated.

Step 5: Goods are stored appropriately.

This process ensures inventory accuracy and quality compliance.

1.12.6 Inventory Management

Step 1: Inventory levels are monitored.

Step 2: Fast-moving items are identified.

Step 3: Reorder requirements are determined.

Step 4: Stock records are updated.

Step 5: Periodic audits are conducted.

This process ensures optimum stock availability and inventory control.

1.12.7 Customer Order Confirmation

Step 1: Customer order details are verified.

Step 2: Payment terms are confirmed.

Step 3: Accessories are reserved.

Step 4: Installation requirements are scheduled.

Step 5: Order confirmation is communicated.

This process ensures order accuracy and customer satisfaction.

1.12.8 Accessory Fitment Scheduling

Step 1: Installation appointment is planned.

Step 2: Workshop resources are allocated.

Step 3: Customer visit is confirmed.

Step 4: Accessories are prepared.

Step 5: Fitment readiness is verified.

This process ensures smooth installation planning and execution.

1.12.9 Accessory Installation Management

Step 1: Vehicle is received for fitment.

Step 2: Accessories are installed.

Step 3: Installation quality is checked.

Step 4: Compliance standards are verified.

Step 5: Job card is closed.

This process ensures safe and accurate accessory installation.

1.12.10 Quality Inspection

Step 1: Installed accessories are inspected.

Step 2: Functional testing is performed.

Step 3: Customer requirements are verified.

Step 4: Inspection findings are recorded.

Step 5: Delivery approval is granted.

This process ensures installation quality and customer confidence.

1.12.11 Accessories Billing Management

Step 1: Invoice is generated.

Step 2: Pricing is verified.

Step 3: Approved discounts are applied.

Step 4: Payment is collected.

Step 5: Transaction is closed.

This process ensures accurate billing and revenue realization.

1.12.12 Vehicle Delivery with Accessories

Step 1: Installed accessories are explained.

Step 2: Product usage is demonstrated.

Step 3: Customer acknowledgment is obtained.

Step 4: Vehicle handover is completed.

Step 5: CRM records are updated.

This process ensures successful delivery and customer satisfaction.

1.12.13 Warranty Registration Management

Step 1: Warranty eligibility is verified.

Step 2: Warranty registration is completed.

Step 3: Warranty records are maintained.

Step 4: Customer information is provided.

Step 5: System updates are performed.

This process ensures proper warranty tracking and support.

1.12.14 Vendor Performance Management

Step 1: Supplier performance is evaluated.

Step 2: Delivery compliance is reviewed.

Step 3: Product quality is assessed.

Step 4: Vendor ratings are updated.

Step 5: Vendor database is maintained.

This process ensures reliable supplier relationships and quality procurement.

1.12.15 Promotional Campaign Management

Step 1: Promotional products are selected.

Step 2: Campaign plans are prepared.

Step 3: Customers are informed.

Step 4: Campaign responses are monitored.

Step 5: Results are evaluated.

This process ensures effective marketing and accessory sales growth.

1.12.16 Returns Management

Step 1: Return request is received.

Step 2: Eligibility is verified.

Step 3: Product inspection is conducted.

Step 4: Replacement or refund is processed.

Step 5: Inventory records are updated.

This process ensures proper return handling and customer satisfaction.

1.12.17 Damage Claim Management

Step 1: Damage complaint is registered.

Step 2: Issue verification is performed.

Step 3: Damage assessment is conducted.

Step 4: Claim processing is initiated.

Step 5: Case closure is completed.

This process ensures fair resolution of customer concerns.

1.12.18 Accessories MIS Reporting

Step 1: Sales and inventory data are collected.

Step 2: Reports are prepared.

Step 3: Performance trends are analyzed.

Step 4: Management reports are submitted.

Step 5: Records are archived.

This process ensures informed decision-making and operational monitoring.

1.12.19 Customer Feedback Management

Step 1: Customer feedback is collected.

Step 2: Feedback records are maintained.

Step 3: Complaints are resolved.

Step 4: CRM updates are completed.

Step 5: Satisfaction levels are reviewed.

This process ensures continuous service improvement and customer retention.

1.12.20 Accessories Performance Analytics and Reporting

Step 1: Sales and inventory data are consolidated.

Step 2: Accessory revenue performance is analyzed.

Step 3: Gross margin trends are evaluated.

Step 4: Product-wise profitability is reviewed.

Step 5: Management dashboards are generated.

This process ensures data-driven decision-making and continuous business improvement.

1.13 Spares Department Operations

The Spares department is a critical profit center. Operations involve spare parts procurement, inventory management, parts sales, workshop support, vendor coordination, warranty claim processing, and stock control. The department ensures timely availability of genuine spare parts, inventory accuracy, and efficient service operations.

1.13.1 Spare Parts Enquiry Management

Step 1: Spare parts enquiry is received.

Step 2: Customer requirements are identified.

Step 3: Part details are verified.

Step 4: Availability is checked.

Step 5: Customer is informed.

This process ensures accurate enquiry handling and customer satisfaction.

1.13.2 Spare Parts Quotation Management

Step 1: Required parts are identified.

Step 2: Pricing is verified.

Step 3: Quotation is prepared.

Step 4: Approval is obtained.

Step 5: Quotation is shared with the customer.

This process ensures transparent pricing and sales efficiency.

1.13.3 Parts Availability Verification

Step 1: Inventory availability is checked.

Step 2: Stock levels are verified.

Step 3: Required parts are reserved.

Step 4: Procurement requirements are identified.

Step 5: Customer is updated.

This process ensures timely parts allocation and service readiness.

1.13.4 Parts Procurement Management

Step 1: Purchase requisition is created.

Step 2: Procurement approval is obtained.

Step 3: Purchase order is issued.

Step 4: Vendor delivery is tracked.

Step 5: Procurement records are updated.

This process ensures uninterrupted spare parts availability.

1.13.5 Goods Receipt and Inspection

Step 1: Spare parts are received.

Step 2: Quantity verification is performed.

Step 3: Quality inspection is conducted.

Step 4: Inventory records are updated.

Step 5: Parts are stored appropriately.

This process ensures inventory accuracy and quality compliance.

1.13.6 Inventory Management

Step 1: Stock levels are monitored.

Step 2: Fast-moving parts are identified.

Step 3: Reorder requirements are determined.

Step 4: Inventory records are updated.

Step 5: Stock audits are conducted.

This process ensures optimal inventory control and stock availability.

1.13.7 Workshop Parts Issue Management

Step 1: Workshop parts request is received.

Step 2: Parts availability is verified.

Step 3: Parts are issued.

Step 4: Issue records are updated.

Step 5: Consumption tracking is performed.

This process ensures efficient workshop support and inventory control.

1.13.8 Retail Counter Sales Management

Step 1: Customer requirement is identified.

Step 2: Parts are verified.

Step 3: Billing is prepared.

Step 4: Payment is collected.

Step 5: Parts are delivered.

This process ensures smooth retail spare parts sales.

1.13.9 Spare Parts Billing Management

Step 1: Invoice is generated.

Step 2: Pricing is verified.

Step 3: Discounts are applied if approved.

Step 4: Payment is collected.

Step 5: Transaction is closed.

This process ensures accurate billing and revenue realization.

1.13.10 Spare Parts Delivery Management

Step 1: Parts are prepared for delivery.

Step 2: Delivery details are verified.

Step 3: Customer acknowledgment is obtained.

Step 4: Parts are handed over.

Step 5: Delivery records are updated.

This process ensures timely and accurate order fulfillment.

1.13.11 Warranty Parts Claim Management

Step 1: Warranty claim request is received.

Step 2: Claim eligibility is verified.

Step 3: Defective parts are inspected.

Step 4: Claim documentation is prepared.

Step 5: Claim is submitted.

This process ensures proper warranty compliance and recovery.

1.13.12 Vendor Performance Management

Step 1: Supplier performance is reviewed.

Step 2: Delivery timelines are evaluated.

Step 3: Quality standards are assessed.

Step 4: Vendor ratings are updated.

Step 5: Vendor records are maintained.

This process ensures reliable procurement and supplier management.

1.13.13 Obsolete Stock Management

Step 1: Slow-moving inventory is identified.

Step 2: Obsolete stock is reviewed.

Step 3: Disposal recommendations are prepared.

Step 4: Management approval is obtained.

Step 5: Inventory records are updated.

This process ensures efficient inventory utilization.

1.13.14 Stock Transfer Management

Step 1: Stock transfer request is received.

Step 2: Inventory availability is verified.

Step 3: Transfer authorization is obtained.

Step 4: Stock movement is executed.

Step 5: Records are updated.

This process ensures accurate inventory movement control.

1.13.15 Parts Return Management

Step 1: Return request is received.

Step 2: Return eligibility is verified.

Step 3: Parts are inspected.

Step 4: Return processing is completed.

Step 5: Inventory records are updated.

This process ensures proper handling of returned inventory.

1.13.16 Damage and Shortage Management

Step 1: Damage or shortage is identified.

Step 2: Incident is recorded.

Step 3: Verification is conducted.

Step 4: Corrective action is initiated.

Step 5: Case closure is completed.

This process ensures inventory accuracy and loss control.

1.13.17 Annual Stock Audit Management

Step 1: Audit schedule is prepared.

Step 2: Physical stock verification is conducted.

Step 3: Variances are identified.

Step 4: Reconciliation is completed.

Step 5: Audit report is submitted.

This process ensures inventory integrity and compliance.

1.13.18 Spares MIS Reporting

Step 1: Inventory and sales data are collected.

Step 2: Reports are prepared.

Step 3: Performance trends are analyzed.

Step 4: Management reports are submitted.

Step 5: Records are archived.

This process ensures effective monitoring and decision-making.

1.13.19 Customer Feedback Management

Step 1: Customer feedback is collected.

Step 2: Feedback records are maintained.

Step 3: Complaints are resolved.

Step 4: CRM updates are completed.

Step 5: Satisfaction levels are reviewed.

This process ensures continuous service improvement and customer retention.

1.13.20 Spares Performance Analytics and Reporting

Step 1: Inventory and sales data are consolidated.

Step 2: Stock turnover ratios are calculated.

Step 3: Fast-moving and slow-moving parts are analyzed.

Step 4: Profitability trends are evaluated.

Step 5: Management dashboards are generated.

This process ensures data-driven inventory optimization and business performance improvement.

1.14 Business Objective

To maximize customer lifetime value by providing end-to-end automotive solutions from vehicle purchase to after-sales support.

1.14.1 What Business Are We In?

This is one of the most important questions a Data Analyst must answer.

Most people think Universal Hyundai dealerships are in the business of selling cars.

This is only partially correct.

The dealership is actually in the business of:

- 1. Customer Acquisition:** Generating enquiries and attracting potential customers.
- 2. Customer Conversion:** Converting enquiries into bookings and bookings into deliveries.
- 3. Customer Retention:** Retaining customers through service, insurance renewals, and customer engagement.
- 4. Customer Lifetime Value Maximization:** Generating revenue from a customer throughout the ownership lifecycle.

Therefore, the dealership is a customer lifecycle management business supported by automobile products.

5. **Multi-Revenue Business Model:** The dealership is NOT a single-revenue business. It generates income from vehicle sales (new & used), service operations, spare parts, insurance commissions, accessories, finance commissions, warranty sales, and roadside assistance. Each revenue stream has different profit margins and customer touchpoints.
6. **Inventory Management:** The dealership must maintain optimal vehicle inventory balancing cash flow, capital lock-up, and customer demand. Slow-moving inventory ties up capital while insufficient inventory loses sales. Data-driven inventory planning is critical for profitability.
7. **Service Revenue Dependency:** Service operations represent 60-70% profit margins compared to 15-20% for vehicle sales. A customer who services at the dealership is 5x more valuable than a customer who doesn't. Increasing service attachment is a key business objective.
8. **Finance & Insurance Penetration:** Finance commissions from banks and insurance commissions represent pure profit with no service delivery cost. Increasing penetration rate (% of customers who opt for finance/insurance) directly improves profitability.
9. **Supply Chain Partnership:** Dealership is dependent on manufacturer (Hyundai) for vehicle allocations. Demand forecasting and sales performance influence allocation quantity. Poor sales performance may result in reduced allocation from manufacturer.
10. **Vehicle Lifecycle Operations:** Every vehicle sold generates touchpoints: delivery, first service (6 months), second service (12 months), warranty claims, insurance renewals, service recalls, extended warranty sales, and eventually trade-in. Each touchpoint is a revenue opportunity.
11. **Sales Team Performance Variability:** Sales performance varies significantly by salesperson. Top performers may close 20-30 vehicles monthly while average performers close 8-10. Understanding individual performance is critical for sales management and compensation.
12. **Lead Management Criticality:** Lead-to-conversion rate is typically 10-15%. Improving response time, follow-up consistency, and lead quality can directly improve conversion rates. Lead management efficiency determines dealership revenue potential.
13. **Customer Satisfaction Impact:** Satisfied customers return for service, recommend to friends, and purchase again. Customer satisfaction scores directly correlate with repeat purchase rate and service attachment rate. Negative experiences lead to churn.
14. **Seasonal Business Fluctuation:** Vehicle sales follow seasonal patterns (year-end rush, new model launches, financial year-end). Understanding seasonality helps forecast cash flow, plan staffing, and manage inventory allocation expectations.
15. **Trade-In Business Ecosystem:** Trade-in vehicles are evaluated, refurbished, and sold through SellAnyCar division. This creates additional revenue stream and provides pricing data for negotiation. Trade-in value negotiation directly impacts customer booking decision.
16. **Vehicle Segment Strategy:** Dealership focuses on specific vehicle segments (hatchback, sedan, SUV) based on market demand and profitability. Segment-wise sales performance drives inventory allocation decisions and marketing focus.
17. **Warranty Service Bundling:** Extended warranties and maintenance packages sold at delivery are high-margin products. Warranty claims management is critical for customer satisfaction and dealership profitability balance.

18. **Customer Communication Touchpoints:** Dealership communicates with customers through calls, SMS, emails, and social media. Tracking communication effectiveness, response rates, and engagement helps optimize marketing efficiency.
19. **Competitor Benchmarking:** Dealership competes with other Hyundai dealerships and other brands. Understanding competitive pricing, service quality, and customer offerings is essential for positioning.
20. **Geographic Market Definition:** Dealership serves specific geographic area (Najafgarh, Delhi region). Market penetration rate, demographic profile, and geographic expansion are business considerations.
21. **RTO and Regulatory Compliance:** Vehicle delivery involves RTO (Regional Transport Office) registration and compliance. Delays in RTO process impact customer satisfaction and working capital management.
22. **Service Technician Productivity:** Service department productivity is measured by jobs per technician per day, average job value, and customer satisfaction. Technician skill level directly impacts service quality and customer retention.
23. **Parts and Accessories Profitability:** Spare parts and accessories have high profit margins (50-60%). Recommendation effectiveness and inventory management significantly impact this revenue stream.
24. **Rental Fleet Operations:** Some dealerships maintain rental fleet which generates steady service revenue and provides test drive vehicles. Fleet management is a separate business consideration.
25. **Corporate Sales Channel:** B2B sales to corporates, fleet operators, and government agencies have different margins, longer sales cycles, and bulk ordering patterns. Segment-specific strategies required.
26. **Customer Data Mining:** Every customer interaction generates data: contact information, vehicle preferences, budget range, purchase history, service preferences. This data enables targeted marketing and personalized engagement.
27. **Cross-Selling Strategy:** Customer who bought vehicle is prospect for service, warranty, insurance, RSA, and accessories. Cross-selling effectiveness improves customer lifetime value.
28. **Repeat Purchase Cycle:** Average vehicle ownership is 5-7 years. Maintaining customer relationship during this period increases probability of next vehicle purchase with dealership.
29. **Churn Prevention Economics:** Cost of acquiring new customer (advertising, sales effort) is 5-10x cost of retaining existing customer. Improving retention rate directly impacts bottom-line profitability.
30. **Pricing Strategy Dynamics:** Vehicle pricing varies by trim, color, features, and market conditions. Discount strategy impacts both margin and sales volume. Optimizing pricing requires demand forecasting.
31. **Promotional Campaign ROI:** Dealership runs periodic campaigns (discounts, exchange offers, free accessories). Campaign effectiveness must be measured through sales uplift and ROI.
32. **Staff Productivity Management:** Dealership employees (sales, service, CRE, finance) are key assets. Productivity metrics, incentive structures, and training impact overall business performance.
33. **Cash Flow Management:** Vehicle inventory represents significant capital. Days Sales Outstanding (DSO) for receivables and inventory turnover affect cash flow health.
34. **Digital Transformation Impact:** Online vehicle research, digital showrooms, online financing, and digital communication channels are transforming dealership business model. Adaptation is critical for competitiveness.

35. **Electric Vehicle Transition:** Growing EV adoption requires dealership to develop EV expertise, manage charging infrastructure information, understand battery warranty, and educate customers. EV sales will increase mix over time.
36. **Subscription and Rental Models:** Some customers prefer vehicle subscription over ownership. This alternative ownership model is emerging and impacts traditional sales approach.
37. **Customer Lifetime Value Calculation:** CLV equals Average Sale Value multiplied by Purchase Frequency multiplied by Customer Lifespan Years, plus Service Revenue multiplied by Service Frequency, plus Insurance/Finance/Accessories Average. Understanding CLV by customer segment helps optimize resource allocation.
38. **Market Expansion Strategy:** Dealership may pursue geographic expansion, new vehicle segments, new customer verticals (e.g., corporate), or new service offerings. Growth strategy impacts business operations.
39. **Profitability Mix Optimization:** Different business lines have different margins. Sales (15%), Service (60%), Accessories (50%), Insurance (25%), Finance (35%). Optimizing revenue mix toward high-margin activities improves profitability

2. Revenue Streams of the Dealership

Revenue is generated from multiple sources.

2.1 Primary Revenue Sources

Vehicle Sales

Revenue generated through the sale of Hyundai vehicles.

Used Car Sales

Revenue generated through SellAnyCar operations.

Service Revenue

Revenue earned through labor charges and repairs.

Insurance Revenue

Commission earned from insurance policy sales and renewals.

Accessories Revenue

Revenue generated from customer purchases of vehicle accessories.

Extended Warranty Revenue

Revenue generated from extended warranty sales.

RSA Revenue

Revenue generated through Road Side Assistance subscriptions.

Finance Revenue

Commission received from financial institutions.

2.2 Stakeholders in the Business

Understanding stakeholders helps analysts identify decision-makers and report users.

2.2.1 Internal Stakeholders

Sales Consultants

Handle enquiries and vehicle sales.

Sales Managers

Monitor sales performance.

CRE Team

Manage customer communication and feedback.

EDP Team

Manage vehicle allocation and inventory.

Insurance Team

Handle policy sales and renewals.

Accounts Team

Manage billing and financial records.

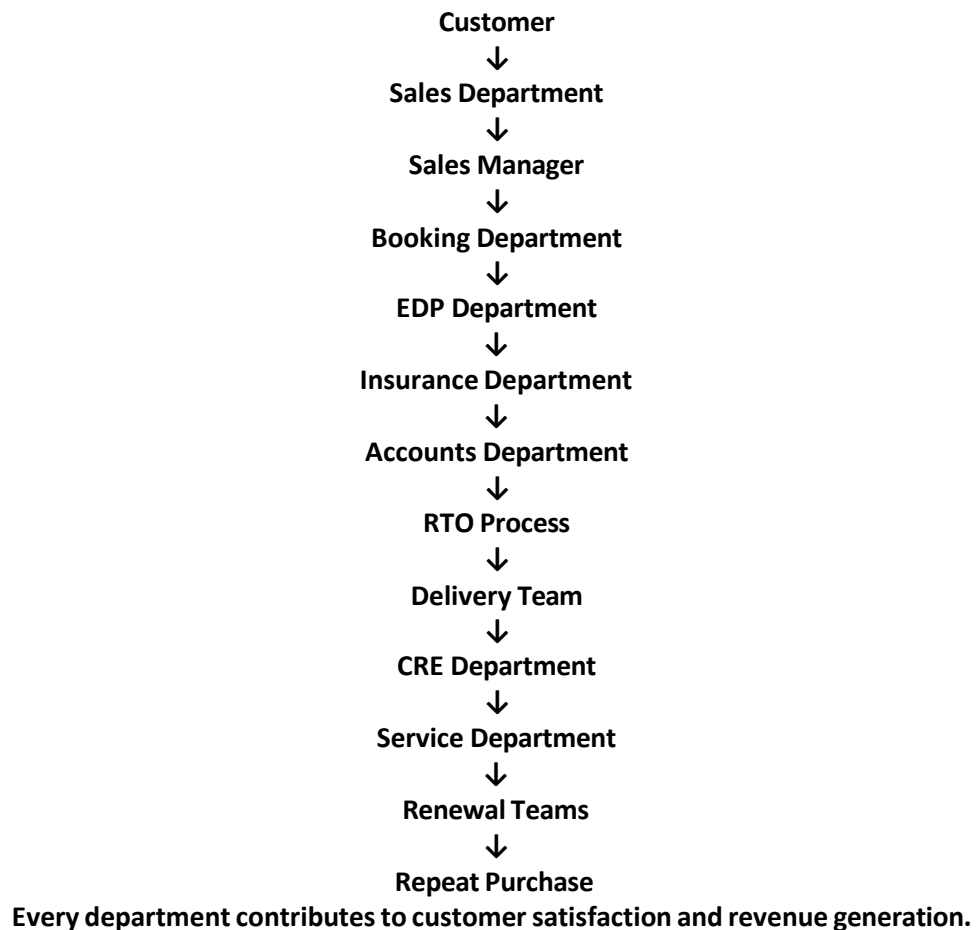
Service Team

Provide after-sales service.

General Manager

Oversees dealership operations.

2.3 Organization Structure



2.4 Customer Lifecycle

A customer passes through multiple stages.

Stage 1

Awareness

Customer learns about Hyundai vehicles.

Stage 2

Enquiry

Customer contacts the dealership.

Stage 3

Follow-Up

Sales consultant communicates with the customer.

Stage 4

Test Drive

Customer experiences the vehicle.

Stage 5

Booking

Customer books a vehicle.

Stage 6

Delivery

Vehicle is delivered.

Stage 7

Service

Customer receives periodic maintenance.

Stage 8

Renewal

Insurance, warranty, and RSA renewal.

Stage 9

Repeat Purchase

Customer purchases another vehicle.

This lifecycle can last several years.

2.5 Why Data Matters

Every business activity generates data.

Examples:

- Customer Enquiries
- Calls
- Test Drives
- Bookings
- Deliveries
- Insurance Policies
- Service Visits
- Feedback Scores

Without data:

- Management cannot measure performance.
- Problems cannot be identified.
- Decisions become assumptions.

Data transforms business activities into measurable outcomes.

2.6 GLOBAL AUTOMOBILE INDUSTRY DEEP DIVE

2.6.1 Global Market Trends (2022-2025)

The global automobile industry is undergoing transformational change driven by environmental regulations, technological innovation, and shifting consumer preferences. Key trends reshaping the industry include electrification, autonomous driving development, vehicle connectivity, and changing ownership models.

Electric Vehicle Market Growth

- Global EV sales reached 13.6 million units in 2023, representing 14% of all new vehicle sales
- China dominates with 60% global EV share (7.8M units annually)
- Europe represents 25% of global EV sales with Norway at 93% EV penetration
- North America growing rapidly with 10% EV penetration aided by federal incentives
- Projected growth to 35-40% of new vehicle market by 2030
- Battery costs declining from \$1,200/kWh (2010) to \$140/kWh (2023), improving affordability
- Public charging infrastructure expanding to 1.3 million charging points globally

Autonomous Vehicle Development

- Level 2-3 automation increasingly available in premium and mid-segment vehicles
- Waymo, Cruise, Tesla advancing Level 4-5 capabilities in selected markets
- Annual investment exceeds \$100 billion across major OEMs
- Estimated Level 5 deployment in controlled environments by 2025-2026
- Regulatory frameworks being established (ISO 26262, 26800 series)
- Insurance models shifting liability from driver to manufacturer/technology provider

Connected Car Technology Trends

- 5G connectivity enabling real-time Vehicle-to-Everything (V2X) communication
- Over-the-air (OTA) software updates reducing dealership service visits
- Connected services generating \$30-50 per vehicle annually in subscription revenue
- Data monetization potential through anonymous mobility analytics
- Telematics platforms enabling predictive maintenance and proactive engagement
- Android Automotive and Apple CarPlay integration expanding in-vehicle capabilities
- Cybersecurity standardization (ISO 21434) becoming mandatory

Used Car Market Dynamics

- Global used vehicle market valued at \$1.5 trillion

- Growing 4-5% annually with online platforms disrupting traditional channels
- Certified pre-owned vehicles command 10-15% premium over open market
- Auction platforms processing 8+ million vehicles annually (Copart, IAA)
- Vehicle history transparency becoming consumer expectation
- EV used cars showing 25-35% depreciation in first 3 years
- Rental fleet sales (Hertz, Enterprise) becoming significant supply source

Subscription-Based Vehicle Services

- Global subscription market valued at \$10-12 billion, growing 20-30% annually
- Luxury brands pioneering software subscriptions (heated seats, charging optimization)
- Monthly subscriptions ranging \$50-500 depending on service
- Warranty, insurance, roadside assistance bundled in premium packages
- Millennial/Gen-Z showing 20% higher adoption vs. traditional ownership

2.6.2 Manufacturing vs. Distribution Business Models

Modern automotive operates through distinct manufacturing and distribution models, each with specific economics and strategic considerations.

Factory Operations Overview

- Major OEM plants: 300,000-500,000 unit annual capacity
- Order-to-delivery timeline: 60-90 days
- Manufacturing cost per vehicle: \$18,000-\$35,000 depending on segment
- Automation level: 80-90% in developed markets, 40-60% in emerging markets
- Quality targets: <1 defect per 1,000 units with 100+ inspection points
- Supply chain complexity: 500-1,500+ tier-1 and tier-2 suppliers per OEM
- EV production requires new battery assembly plants (\$5-10B capex)
- Labor costs: \$4,500 per vehicle (Japan) vs. \$800 (India)

Dealership Economics & Profitability

Dealerships generate revenue from multiple sources beyond vehicle sales, with service and finance creating significant profit margins.

- Vehicle sales gross profit: \$1,200-\$2,000 per unit (8-12% margin)
- Service revenue: \$80-\$150 per visit (45-50% margin)

- Finance commission: \$400-\$800 per financed vehicle (60-70% margin)
- Insurance commission: \$150-\$300 per customer (40-50% margin)
- Accessories sales: \$200-\$400 per customer (35-40% margin)
- Extended warranty: \$100-\$250 per customer (50-60% margin)
- Roadside assistance: \$50-\$100 per customer (55-65% margin)

Operating Expenses for Typical Dealership (300 vehicles/month)

- Personnel costs: \$2,400,000 annually (25-28% of revenue)
- Floor plan financing: \$1,200,000 annually (12-14% of revenue)
- Facility & utilities: \$600,000 annually (6-7% of revenue)
- Marketing & advertising: \$500,000 annually (5-6% of revenue)
- Technology & IT systems: \$300,000 annually (3-4% of revenue)
- Training & development: \$150,000 annually (1.5-2% of revenue)
- Total annual costs: ~\$5,500,000 (55-60% of revenue)

2.7 INDIAN AUTOMOBILE MARKET ANALYSIS

2.7.1 Market Size & Growth Trajectory

Annual Passenger Vehicle Sales

- 2020: 2.89 million units (COVID impact, -19.8%)
- 2021: 3.89 million units (+34.8% recovery)
- 2022: 3.91 million units (+0.5%)
- 2023: 3.93 million units (+0.5%)
- 2024: 4.10 million units projected (+4-5%)
- Market value: \$60-70 billion USD annually
- Per capita ownership: 23 vehicles per 1,000 people (vs. 800+ in developed markets)

Growth Rate & Projections

- CAGR 2020-2025: 6-8% (from COVID trough)
- CAGR 2015-2020 (pre-COVID): 4-5% showing market maturity
- Growth drivers: Rising middle class, improved financing accessibility, better product portfolio
- Constraints: High insurance costs, fuel prices, traffic congestion in metros
- EV specific: 2-3% penetration (2024) → 15-20% by 2030

Market Segmentation

Indian market divided by vehicle type and price point:

- Entry-level sedans (₹4-6L): 22% market share, 850K units
- Compact SUVs (₹5-8L): 30% market share, 1,200K units
- Mid-size sedans (₹7-10L): 17% market share, 650K units
- Premium SUVs (₹10-20L): 14% market share, 550K units
- Luxury segment (₹20+L): 5% market share, 200K units
- MPVs/Utility vehicles (₹6-12L): 12% market share, 450K units

Urban vs. Rural Dynamics

- Metropolitan cities: 60-65% of total volume (Delhi, Mumbai, Bangalore, Hyderabad)
- Tier-2 cities: 20-25% volume with fastest growth (Pune, Jaipur, Chandigarh)
- Tier-3 & rural: 10-15% volume, underpenetrated growth opportunity
- Urban finance penetration: 70-80%
- Rural cash payment preference: 40-50%

2.8 HYUNDAI'S POSITION & PORTFOLIO

2.8.1 Hyundai Global Context

Global Manufacturing & Operations

- 31 manufacturing plants across 11 countries
- Major facilities: South Korea (3 plants), India (Chennai), USA, Turkey, Brazil, Mexico
- Global production capacity: 5.7+ million vehicles annually
- R&D centers: 15+ globally including India (Hyderabad)
- Supply chain: 1,000+ tier-1 suppliers, 5,000+ tier-2 suppliers
- Quality certifications: ISO 9001, 14001, 45001 across all facilities

Global Sales & Markets

- Annual global sales: 4.0+ million vehicles
- Top market: China (800K+ units)
- Key markets: USA (600K), India (600K), Europe (1.2M), South Korea (800K)
- Global market share: 5-6% of passenger vehicles
- Emerging growth: Southeast Asia, Middle East, Africa

Product Portfolio

- Entry-level: Eon, Santro, Grand i10
- Compact sedan: i20
- Mid-size sedan: Verna, Elantra
- Compact SUV: Creta, Venue, Kona
- Mid-size SUV: Tucson, Santa Fe
- Premium: Genesis brand
- EV portfolio: Kona Electric, Aura EV
- SUV focus: 60%+ portfolio now SUVs

R&D & Technology Investment

- Annual R&D spending: \$7-8 billion globally
- Focus: EV technology, autonomous driving, connected vehicles
- E-GMP platform: In-house EV platform reducing costs
- Battery development: In-house battery R&D reducing dependency
- Patents: 10,000+ filed annually
- EV target: 1 million units by 2028 (20% of volume)
- Investment: \$40+ billion in EV/battery through 2030

2.8.2 Hyundai India Journey & Position

Entry & Strategic Evolution

- Market entry: 1998 (26+ years in India)
- Initial model: Accent (mid-size sedan)
- Manufacturing: Chennai facility (2 plants investment)
- Market entry strategy: Quality-focused positioning, aspirational brand

Sales Growth Timeline

- 2000: 60,000 units in first full year
- 2010: 300,000+ units annually
- 2015: 400,000+ units (major growth phase)
- 2020: 448,000 units (pre-COVID peak)
- 2023: 600,000+ units (current leadership)

- Growth compound CAGR: 10-12% annually

Market Share Evolution

- 2000: 3-4% market share (entry phase)
- 2010: 8-10% market share (growth phase)
- 2015: 12-13% market share (consolidation)
- 2020: 14-15% market share (peak)
- 2023: 16-17% market share (current #2 position)
- Target: 20% market share by 2030

Investment & Milestones

- Manufacturing investment: \$1.2+ billion in Chennai plants
- R&D center: Hyderabad facility for engineering & development
- Technology center: Software development focus
- Dealer expansion: 600+ touchpoints across India
- Employment: 10,000+ direct jobs in manufacturing and distribution

2.9 DEALERSHIP REVENUE MODEL DEEP DIVE

2.9.1 Revenue Stream Breakdown

Modern dealerships derive revenue from multiple sources, each with distinct margins and growth potential:

Vehicle Sales (35-40% of revenue)

- New vehicle sales: \$8,000-\$12,000 gross profit per unit
- Used vehicle sales: \$1,500-\$3,000 gross profit per unit
- Fleet sales: 5-10% of volume at reduced margins
- Seasonal patterns: March-May peak, July-August slump
- Trend: Vehicle sales revenue declining as service/F&I grow

Service Revenue (25-30% of revenue)

- Scheduled maintenance: \$500-\$1,500 per visit (major service)
- Warranty work: \$300-\$1,000 (covered, low margin)
- Repair work: \$800-\$3,000 per visit (varies by issue)
- Customer visits: 6-8 per vehicle over 5 years

- Gross margin: 45-50% (higher than vehicle sales)
- Customer lifetime service value: \$2,500-\$4,000

Finance & Insurance Commission (15-20% of revenue)

- Finance commission: \$300-\$600 per financed vehicle
- Insurance commission: \$150-\$300 per customer
- Extended warranty: \$100-\$250 per customer
- RSA (Roadside Assistance): \$50-\$100 per customer
- Attachment rates: 65-75% for finance, 60-70% for insurance
- Gross margin: 60-70% (highest margin stream)

Accessories & Add-ons (5-8% of revenue)

- Accessories per customer: \$200-\$400 (floor mats, seat covers)
- Premium audio: \$1,000-\$5,000 (aftermarket upgrades)
- Protective coatings: \$500-\$1,500 (ceramic, PPF)
- Attachment rate: 50-60% of new vehicle buyers
- Repeat purchases: Service customers at 30-40% attach rate

Total Revenue Composition

- Vehicle sales: 35-40%
- Service: 25-30%
- Finance/Insurance: 15-20%
- Accessories: 5-8%
- Other (parts sales, etc.): 5-8%

2.9.2 Customer Acquisition Economics

Customer Acquisition Cost (CAC)

- Marketing spend per acquired customer: \$200-\$500
- Digital marketing (30%): Google Ads, Facebook, Instagram
- Showroom traffic (25%): Walk-ins, local presence
- Referrals (25%): Word-of-mouth, brand loyalty
- Events/Campaigns (20%): Test drives, launches
- Payback period: 18-24 months from vehicle sale + service

Lead Source Analysis

- Digital sources: 35-40% of leads (highest growth)
- Referrals: 25-30% of leads (highest quality)
- Walk-ins: 20-25% of leads
- Service customers: 10-15% of repeat/upgrade purchasers
- Events/Campaigns: 5-10% seasonal impact

Lead Quality by Channel

- Google Search: 60-70% conversion (high intent)
- Social Media Ads: 30-40% conversion (awareness-building)
- Referrals: 70-80% conversion (highest)
- Walk-ins: 20-30% conversion (low intent)
- Events: 25-35% conversion (time-bounded)

Cost per Lead by Channel

- Google: \$50-\$150 per lead
- Facebook/Instagram: \$30-\$80 per lead
- Referral incentives: \$500-\$1,000 per successful sale
- Direct sales team: \$100-\$300 per lead
- Events: \$200-\$500 per lead

Lead Conversion Funnel

- Lead generation: 100 leads
- Showroom visit: 30-40 visits (30-40%)
- Test drive: 15-20 test drives (50% of visitors)
- Booking: 8-12 bookings (50% of test drives)
- Delivery: 7-10 deliveries (85-90% of bookings)
- Overall conversion: 7-10% (100 leads → 7-10 sales)

2.10 SRI GURUKRIPA OPERATIONAL STRUCTURE

2.10.1 Dealership Overview

Sri Gurukripa Hyundai operates as a multi-location dealership group serving [region] with comprehensive vehicle sales and after-sales service capabilities.

Current Operations

- Number of branches: [X] locations
- Total sales team: [Y] sales consultants
- Service technicians: [Z] trained mechanics
- Administrative staff: [A] personnel
- Annual vehicle sales: [Volume] units
- Service customers: [Number] active customers
- Monthly average sales: [Volume/12] vehicles

KEY INSIGHTS & STRATEGIC IMPLICATIONS

This expanded analysis reveals several critical insights for Sri Gurukripa:

- Market opportunity: India's automotive market growing 6-8% CAGR offers significant headroom
- Segment dynamics: SUV boom (30% market share) aligns with Hyundai's strong portfolio (Creta, Venue)
- Service profitability: 25-30% of dealership revenue from service at 45-50% margins drives sustainability
- Customer acquisition: Digital channels (35-40%) becoming dominant, requiring marketing transformation
- Competitive positioning: Hyundai's 16-17% market share (#2) provides scale advantages
- Technology investment: Connected cars and OTA software creating new revenue streams and engagement
- Regulatory compliance: BS VI, GST, consumer protection laws require continuous adaptation
- Dealer economics: Profitability increasingly depends on ancillary revenue (finance, service, insurance)

Recommendations focus on deepening service customer relationships, digital-first customer acquisition, and leveraging Hyundai's strong brand position and warranty advantages in the competitive Indian market.

2.11 Role of a Data Analyst in the Dealership

A Data Analyst converts raw data into business insights.

Responsibilities include:

Data Collection

Gathering data from multiple departments.

Data Cleaning

Removing duplicates and errors.

Data Analysis

Finding trends and patterns.

Dashboard Development

Creating visual reports.

KPI Monitoring

Tracking business performance.

Reporting

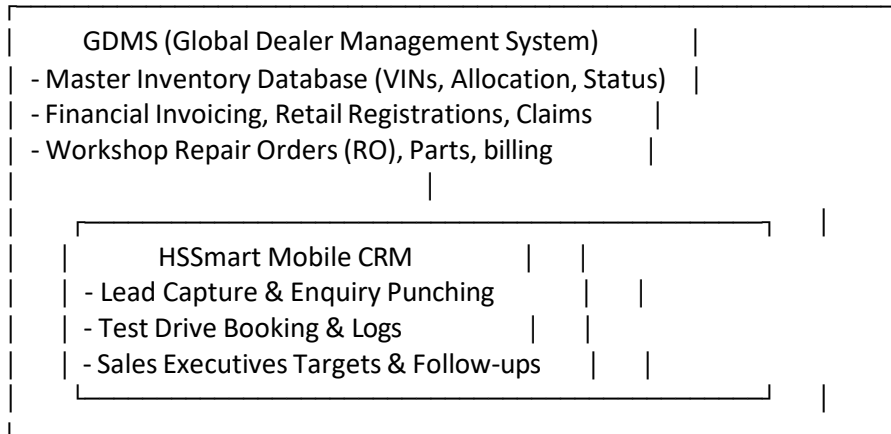
Supporting management decisions.

The analyst acts as a bridge between business operations and decision-making.

3. Department Operations & Systems SOPs

3.1 The Dealership Systems Ecosystem: GDMS & HSSmart

The operational data of **Universal Hyundai ShriGurukripa Automobiles LLP** flows through two primary systems linked to Hyundai Motor India's national infrastructure: **GDMS** and **HSSmart**.



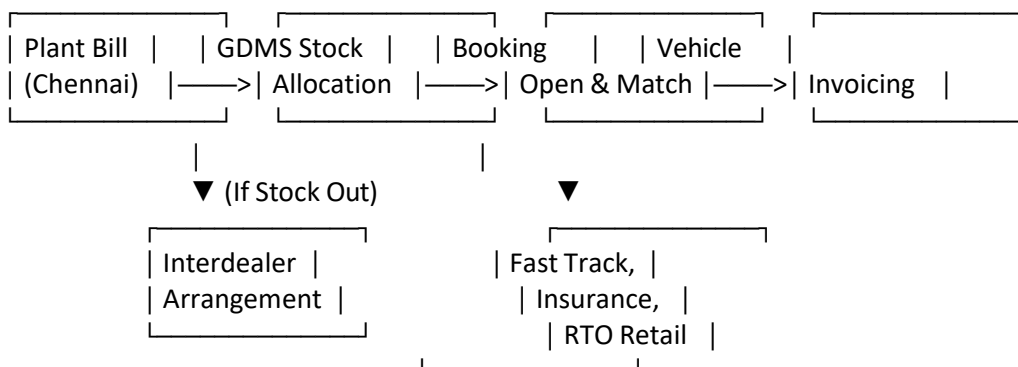
- **GDMS (Global Dealer Management System):** A central web portal managing core transactions. All invoices, parts inventories, vehicle stock allocations, and final Vahan retail uploads must be registered in GDMS.
- **HSSmart / HSmart Mobile App:** A mobile-based CRM client nested within the GDMS environment. It is used directly on the field by Sales Consultants for real-time customer data entry, lead scoring, test drive recording, and tracking individual sales pipelines.
- **Data Syncing:** When a Sales Consultant punches an enquiry into HSSmart, it syncs to GDN (Global Dealer Network) and creates an Enquiry ID in GDMS. Once the vehicle is booked and invoiced, GDMS reconciles the entry, updates stock, and triggers delivery.

3.2 Department Workflows & SOPs

3.2.1 EDP Manager & EDP Executive Operations

The Electronic Data Processing (EDP) department acts as the primary data gatekeeper for inventory and target monitoring.

[EDP Inventory Workflow]



SOP 1: Target Monitoring & Alignment

1. **Extract Targets:** Every month, the EDP Manager extracts Wholesale Targets (commitments to buy cars from the plant) and Retail/Sales Targets (numbers of cars registered to customers) .
2. **Stock Checking:** Cross-reference target requirements with current CRM stock and stockyard physical count.
3. **Sub-Target Allocation:** Break down target goals into specific models and variants, and assign sub-targets to Sales Teams.

SOP 2: Vehicle Allocation & Interdealer Arrangements

1. **Booking Match:** Upon booking open, the EDP Executive matches customer bookings with available vehicle stock in the yard.
2. **Allocation:** If matched, the VIN is allocated to the customer profile in GDMS.
3. **Interdealer Request:** If the vehicle is not available in stock, the EDP Manager initiates an interdealer arrangement (seeking the specific variant/color from other dealer stock yards) via the GDMS trading portal.

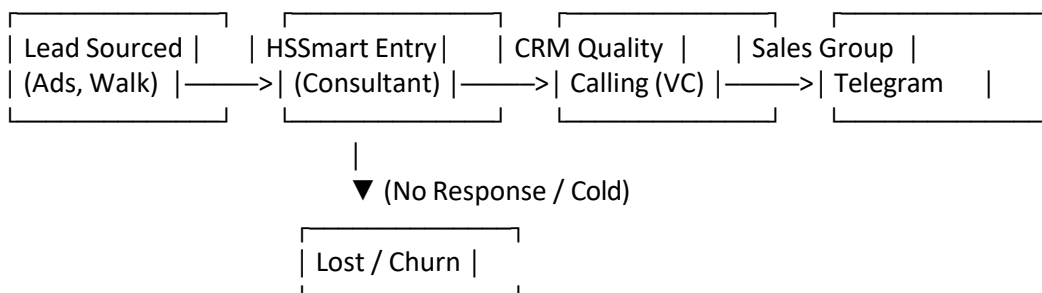
SOP 3: Invoicing, Billing, & Retail Registration

1. **Document Verification:** Check that the delivery file contains all signed booking forms, payment receipts from Paymate, and finance approval notes.
2. **GDMS Invoice Generation:** Generate the customer invoice in GDMS.
3. **Fast Track Insurance & RTO:** Register insurance details on the portal, calculate RTO registration taxes, and upload documents.
4. **GDMS Retail Tick:** Verify the Net Retail Tax Receipt, mark the vehicle as “Retailled” in GDMS, and verify target compliance for monthly HMIL incentive calculations.

3.2.2 CRM (Customer Relationship Management) Operations

The CRM department ensures lead quality, follow-up compliance, and post-purchase customer satisfaction.

[CRM Lead Verification Workflow]



SOP 1: Enquiry Punching & Source Quality Control

1. **Source Capture:** Check that all walk-in registrations, reference leads, and digital advertisement leads are captured.
2. **HSSmart Check:** Verify that the Sales Consultant has punched the customer’s phone number, name, and desired model in HSSmart within 15 minutes of contact.
3. **GDN Sync Verification:** Reconcile HSSmart inputs with GDMS to ensure no duplicate Enquiry IDs exist.

SOP 2: Quality Calling & Promoter Checks

1. **Voice Calling (VC):** The CRM executive calls the customer within 24 hours of enquiry creation to verify customer intent and confirm the sales executive's feedback.
 2. **Follow-up Tracking:** Monitor the "Pending Follow-up" reports in HSSmart.
 3. **Post-Purchase Promoter Score:** Conduct Net Promoter Score (NPS) surveys 3 days post-delivery to measure dealership satisfaction.
-

3.2.3 Sales Executive & Sales Consulting Operations

Sales Executives are the primary touchpoint for vehicle sales, financing, and value-added conversions.

SOP 1: Enquiry & Need Analysis

1. **Greeting & Source Identification:** Identify the lead source (Walk-in, Reference, Digital Ad, Cold Call).[Need analysis, family member ,info]
2. **Customer Need Profiling:** Document passenger requirements, usage (city vs. highway), transmission preference (Manual, IVT, DCT), and budget.
3. **Solution Pitching:** Recommend specific Hyundai models and variants.

SOP 2: Test Drive Protocols

1. **Document Check:** Verify the customer's original driving license and log details in the Test Drive Hub system (regional zones: N, S, E, W).[test drive route]
2. **Test Drive Logging:** Enter start/stop times and mileage in H Smart.
3. **Feedback Recording:** Punch the customer's post-test drive rating into H Smart immediately upon return.

SOP 3: Booking & Value Added Service (VAS) Pitching

1. **Booking Form Creation:** Collect booking advance payments, copy PAN/Aadhar cards, and initiate a Booking File.
 2. **F&I and Insurance Selection:** Present dealer-tied finance options (HDFC, ICICI, SBI) and dealer-brokered insurance.
 3. **WATS/SOT/RSA Upselling:**
 - *VAS (WATS):*Extended Warranty, Pitch Teflon coating, ceramic paint protection, and underbody treatments.
 - *Shield of Trust (SOT):* Offer 14 parts wear-and-tear replacement coverage.
 - *Road Side Assistance (RSA):* Sell extended roadside assistance packages (covering breakdowns up to a specified mileage/radius limit).
-

3.2.4 DSA MANAGER ROLE & Operations

Department Overview: The DSA (Direct Selling Associate) Manager is responsible for coordinating the complete booking-to-delivery pipeline. This is a critical function that converts customer bookings into actual vehicle deliveries, ensuring seamless document processing, vehicle preparation, and customer handover. The DSA Manager ensures no booking falls through the cracks and all vehicles are delivered on schedule with quality standards.

Key Responsibilities

- Monitor complete booking-to-delivery pipeline
- Identify and resolve conversion bottlenecks
- Manage document verification and compliance
- Coordinate vehicle stock yard release
- Oversee Pre-Delivery Inspection (PDI)
- Manage billing and payment reconciliation
- Execute final vehicle delivery

- Track marketing effectiveness
- Manage delivery audit and quality assurance

Booking to Delivery Pipeline Flow

Booking Form → Document Check → Stock Yard Release → PDI → Billing → Delivery

Stage Duration: 5-10 business days (Target: 7 days average)

SOP 1: PIPELINE ANALYSIS & CONVERSION MONITORING

Objective: Monitor entire booking pipeline to identify conversion issues, track lead sources, manage marketing effectiveness, and ensure no booking stalls between stages.

Step 1: Lead Flow Review & Conversion Analysis

1. Daily Pipeline Review:

- Every morning, review booking pipeline dashboard showing:
- Total bookings in last 30 days
- Bookings by stage (pending document check, stock pending, in PDI, in billing, ready for delivery)
- Age of each booking (how many days in current stage)
- Conversion rate from booking to delivery (target: 100% within 10 days)
- Pending deliveries waiting for documents or stock

2. Lead Source Analysis:

- Categorize bookings by source:
- Walk-in (typically 70-80% of bookings - highest volume)
- Reference leads (typically 10-15% - historically low due to dealership age)
- Digital marketing (website, social media - growing segment)
- Call campaigns
- Repeat/referral customers
- Track conversion rate by source: Walk-in (45% conversion) vs. Reference (35% conversion)
- Identify low-converting sources and provide feedback to marketing team

3. Bottleneck Identification:

- Identify bookings stalled in any stage:
- Pending Documents (> 3 days): Follow up with customer for missing documents
- Stock Not Available (> 5 days): Coordinate with inventory for alternate color/variant or order status
- RTO Delays (> 7 days): Check with RTO office or agent for status
- Payment Pending (> 3 days): Contact customer for payment confirmation
- Finance Approval Pending (> 5 days): Escalate with finance department
- Create daily bottleneck report and escalate to management

Step 2: Advertisement Campaign Control & Marketing Effectiveness

4. Campaign Tracking:

- Track all active marketing campaigns:
- Offline: Local newspaper ads, billboards, radio spots, event marketing
- Digital: Google Ads, Facebook ads, Instagram, WhatsApp campaigns
- Record launch date, target audience, budget, expected reach
- Update campaign status monthly

5. Regional Visibility & Lead Quality:

- Monitor regional performance:

- North Delhi: Lead volume, conversion rate
- South Delhi: Lead volume, conversion rate
- East Delhi: Lead volume, conversion rate
- West Delhi: Lead volume, conversion rate
- Identify high-performing and low-performing regions
- Allocate marketing budget to boost visibility in low-performing regions

6. Campaign ROI Measurement:

- Calculate ROI for each campaign:
- Cost per lead = Campaign budget / Number of leads generated
- Cost per booking = Campaign budget / Number of bookings
- Example: ₹2,00,000 campaign budget generating 80 leads = ₹2,500 cost per lead
- Example: If 20 leads convert to bookings = ₹10,000 cost per booking
- Monthly report to marketing team with ROI analysis
- Recommend continuing/modifying/stopping underperforming campaigns

Step 3: Delivery File Audit & Compliance Verification

7. Complete Delivery File Checklist:

Before ANY vehicle is delivered, DSA Manager must audit delivery file containing:

Document Category	Required Documents	Status
Booking	Signed booking form, Customer signatures, Sales representative approval	✓ Complete
Financial	Full payment receipt OR approved loan + down payment, Bank loan agreement (if financed), Cheque/DD details	✓ Complete
RTO	Signed RTO form (Form 20/21), ID proof copy, Address proof, Emission test certificate, RTO registration number	✓ Complete
Insurance	Comprehensive insurance policy, Policy number, Premium receipt, Cover note (if awaiting policy)	✓ Complete
Accessories	Accessory list & pricing, Warranty documents, Installation completion	✓ Complete

8. Booking Form Verification:

- Signed booking form present
- All mandatory fields filled: Customer name, address, phone, email, vehicle variant, color, registration number (RTO allotted)
- Booking price and all charges documented
- Down payment and balance payment terms confirmed
- Customer signature and date
- Sales representative signature and approval

9. Financial Documents Verification:

- FULL PAYMENT OPTION:
 - Full payment receipt (cash, cheque, DD, online transfer)
 - Cheque/DD number and date (if applicable)
 - Online transfer confirmation (bank reference number, date, amount)

FINANCED OPTION:

- *Approved loan letter from bank*
- *Loan amount, EMI, tenure, interest rate*
- *Signed loan agreement*
- *Customer down payment receipt*
- *Bank disbursement confirmation*
- *Total payment (down payment + loan) = invoice amount*

RECONCILIATION:

- *Total amount received = Invoice amount*
- *All payments recorded in Tally*
- *No discrepancies or pending amounts*

10. RTO Documentation Verification:

- *RTO registration form (Form 20/21) - Signed by applicant*
- *Proof of identity - Aadhaar/PAN/DL (copy)*
- *Proof of address - Utility bill/Lease agreement (copy)*
- *Vehicle insurance policy (comprehensive) - Verified*
- *Pollution control certificate (PUC) - Original*
- *RTO registration number (once allotted)*
- *Registration certificate (RC) - original received*
- *Vehicle loan hypothecation (if financed) - Original with bank*

11. Insurance Policy Verification:

- *Policy issued for correct vehicle (VIN/registration match)*
- *Policy type: Comprehensive (own damage + third party)*
- *Coverage period: Valid from delivery date*
- *Insurance company name and policy number*
- *Premium amount paid*
- *Policy document original and cover note (if not yet issued)*
- *Customer has received policy details and claim procedure*

12. Accessories & Add-ons Documentation:

- *List of all accessories sold and installed:*
- *Floor mats, seat covers, window visors, bumper guards*
- *Infotainment system, GPS tracking, dashboard camera*
- *Price of each accessory*
- *Warranty documentation for each item*
- *Customer signature confirming receipt*

13. Warranty & Service Documentation:

- *Factory warranty certificate*
- *Extended warranty document (if purchased)*
- *Maintenance package (if purchased)*
- *Roadside assistance (RSA) document*
- *Service contact information and appointment scheduling details*

14. File Audit Sign-off:

- *DSA Manager reviews ALL documents*
- *If any document missing or incorrect: Mark as 'INCOMPLETE' and return to responsible department*

- *If all documents present and verified: Sign off and mark delivery as 'APPROVED FOR DELIVERY'*
- *Create delivery approval record with date, DSA manager name, signature*
- *Schedule delivery appointment time with customer*
- *Notify customer: Vehicle is ready for delivery (call + SMS)*

3.2.5 Used Car Division (Sell Any Car / Sellanyvehicle.in) Operations

The Used Car Division enables trade-ins for new vehicle sales and drives a secondary used car retail pipeline.

SOP 1: Vehicle Evaluation & Pricing

1. **Trade-In Evaluation:** When a new car prospect requests a trade-in, the Used Car Evaluator inspects the customer's vehicle.
2. **Refurbishment Assessment:** Identify required repairs, parts replacements, and body shop requirements to make the car ready for resale.
3. **Price Negotiation:** Estimate the market value, subtract refurbishment costs, and present the final purchase offer to the customer.

SOP 2: Stock Yard Entry & Workshop Servicing

1. **Purchase Registration:** Record the vehicle purchase, transfer registration documents, and add the car to the Used Car Stock yard.
2. **Workshop Refurbishment:** Open a Service Repair Order (RO) in Tally/GDMS for refurbishment work.
3. **Restocking for Sale:** Clean, polish, and place the refurbished used car on the retail lot or list it online.

3.2.6 SERVICE DEPARTMENT

Department Overview: The Service Department is the backbone of recurring revenue generation. It manages vehicle maintenance, repairs, warranty claims, and value-added services. With 60-70% profit margins, service revenue directly drives dealership profitability. This department operates the workshop floor and manages technician workforce.

Key Responsibilities

- Service appointment scheduling and management
- Job card creation and documentation
- Technician allocation and task assignment
- Quality control and vehicle inspection
- Warranty claim processing and extending warranty
- Service billing and invoice generation

SOP 1: SERVICE APPOINTMENT SCHEDULING

1. **Phone/Online Booking:**
 - Customer calls Service Department (dedicated phone line: XXX-XXXX) or books online
 - Collect information: Customer name, vehicle registration number, contact details, service type needed, preferred date/time
 - Check availability: Cross-reference service bay availability and technician workload
 - Confirm appointment: Provide confirmed date/time and advise customer to arrive 10 minutes early
 - Send SMS/email confirmation with appointment details and workshop address

SOP 2: JOB CARD CREATION & DOCUMENTATION

2. **Vehicle Intake Process:**
 - Customer arrives with vehicle for service
 - Service advisor verifies vehicle registration and customer identity
 - Conduct 360-degree vehicle inspection: Check body, paint, interior, lights, tires

- Document pre-service vehicle condition in system with photos
- Discuss service needs: warranty work, customer complaints, preventive maintenance suggestions

3. Job Card Details:

- Unique Job Card Number (auto-generated)
- Customer name, phone, email, address
- Vehicle details: Reg number, VIN, model, variant, mileage, fuel type
- Service type: Scheduled service, unscheduled repair, warranty claim, warranty expiry date
- Customer complaint/requirement: Detailed description of what needs to be fixed
- Estimated completion time and cost
- Technician name assigned
- Date & time received

SOP 3: TECHNICIAN ALLOCATION & WORKSHOP EXECUTION

4. Job Assignment to Technicians:

- Service supervisor reviews all pending job cards daily morning
- Allocate jobs based on: Technician skill level (routine service vs. complex repairs), Current workload, Vehicle type specialization
- Print job card and attach to vehicle windshield
- Move vehicle to assigned service bay

5. Technician Execution:

- Technician reviews job card and understands service requirements
- Perform vehicle service as per work order (scheduled maintenance checklist or repair specifications)
- Record parts used and time taken on job card
- Update job status in system: IN PROGRESS
- Once work completed, notify supervisor

SOP 4: QUALITY CONTROL & VEHICLE HANDOVER

6. Quality Inspection:

- Quality inspector conducts final inspection before customer handover
- Verify all work completed per job card
- Test vehicle: Start engine, check all systems (lights, wipers, air conditioning, power windows)
- Road test: 5-10 minute drive to verify vehicle operates correctly
- If issues found: Mark job as REWORK and return to technician
- If quality approved: Mark job as READY FOR DELIVERY

7. Vehicle Handover to Customer:

- Service advisor calls customer to notify service completion
- Customer collects vehicle and reviews work done
- Hand over final invoice with itemized parts and labor charges
- Provide warranty documentation for parts replaced
- Conduct customer walk-around showing work completed
- Get customer signature on service invoice
- Request customer feedback rating (1-5 stars) on service experience

3.2.7 FINANCE DEPARTMENT

Department Overview: The Finance Department facilitates auto loans, manages payment collections, processes customer financing requests, and partners with banks to enable vehicle purchases. Commission income from finance facilitation (0.5-2% of loan amount) is pure profit with minimal operational cost.

Key Responsibilities

- Loan application processing

- Bank documentation and verification
- Loan approval coordination
- Payment collection and receipt issuance
- Commission tracking and receivables

SOP 1: LOAN APPLICATION PROCESSING

8. Customer Finance Qualification:

- Determine customer financing need: Budget, desired loan amount, down payment available
- Assess creditworthiness: Employment status, income level, credit history
- Calculate affordable EMI: Based on customer income and loan-to-value ratio
- Present loan options from 4-5 partner banks

9. Documentation Collection:

- Primary documents: Identity proof (Aadhaar/PAN/Driving License), Address proof, Last 3 years income tax returns
- Employment documents: Offer letter, Salary slips (last 3 months), Employment letter
- Bank documents: Current account statement (last 6 months), Property documents (if co-applicant)
- Vehicle document: Invoice/quotation, Vehicle insurance documents
- Photograph: Recent color photograph of customer and co-applicant (if any)

SOP 2: BANK COORDINATION & APPROVAL

10. Loan Application Submission:

- Submit all documents to selected bank within 24 hours
- Request approval against sales invoice
- Bank processes application (typically 3-5 business days)
- Track status with bank contact and follow up daily

11. Approval Documentation:

- Upon approval, bank issues approval letter with loan conditions
- Share approval letter with customer
- Send loan agreement to customer for signature
- Collect signed agreement and forward to bank

SOP 3: FUND DISBURSEMENT & COMMISSION TRACKING

12. Fund Disbursement:

- Bank disburses loan amount directly to dealership bank account
- Dealership deposits customer down payment + loan proceeds against vehicle invoice
- Vehicle title transferred to customer after full payment
- Vehicle registered with RTO in customer name

13. Commission Income:

- Bank partner banks pay commission at end of month
- Standard rates: HDFC Bank (1.5%), ICICI Bank (1.2%), Axis Bank (1.0%), SBI (0.75%)
- Record commission received in Tally revenue module
- Example: ₹10,00,000 loan with 1.5% commission = ₹15,000 dealership revenue (no cost).

3.2.8 INSURANCE DEPARTMENT

Department Overview: The Insurance Department is responsible for issuing comprehensive motor insurance policies at the point of vehicle delivery and managing renewal campaigns to drive recurring commission revenue. This department acts as intermediary between customers and multiple insurance partners, ensuring seamless policy issuance and timely renewals.

Key Responsibilities

- New policy issuance for all sold vehicles
- Quote comparison from multiple insurance partners
- Policy renewal campaign management
- Documentation and query resolution
- Premium collection and payment processing
- Policy record maintenance and filing

Insurance Partners & Commission Structure

Insurance Partner	Commission Rate	Claim Settlement	Key Features
ICICI Lombard	8%	7-10 days	Cashless network, Fast claim settlement
Bajaj Allianz	7%	10-15 days	Competitive premiums, Good support
HDFC ERGO	6.5%	8-12 days	24/7 roadside assistance

SOP 1: NEW VEHICLE POLICY ISSUANCE

Objective: Issue comprehensive motor insurance policy within 24 hours of vehicle delivery to ensure complete customer coverage and compliance.

Step 1: Quote Comparison & Partner Selection

1. **Trigger Event:** Customer invoice generated in ERP system
2. **Extract Vehicle & Customer Information:**
 - Vehicle Registration Number (tentative)
 - Vehicle Identification Number (VIN)
 - Vehicle Make, Model, Variant, Color, Engine Number
 - Customer Name, Address, Phone, Email
 - Vehicle Invoice Value
 - Expected RTO Registration Date
3. **Pull Insurance Quotes:** Contact each of 4-5 insurance partners with vehicle details
 - ICICI Lombard: Premium quote for Comprehensive + Third-party + all riders
 - Bajaj Allianz: Request quote with identical coverage scope
 - HDFC ERGO: Premium quote with available discounts
 - National Insurance / Oriental: Competitive quotes
 - SBI General / Cholamandalam: Alternative quotes
4. **Compare Quotes & Apply Discounts:**
 - Create comparison sheet with all quotes side-by-side
 - Apply applicable discounts: Dealership bulk discount (3-5%), No claim bonus carryover (if applicable), Loyalty discount (repeat customers)
 - Calculate net premium after all deductions
 - Select partner offering best price-to-coverage ratio (typically 10-15% cheaper than competitor)
5. **Present Options to Customer:**
 - Show 2-3 top quotes (recommended partner + 1-2 alternatives)
 - Explain coverage: Comprehensive covers own-damage + third-party + personal accident
 - Highlight recommended add-ons: RSA (₹2,000-3,000), Consumables cover (₹1,500-2,000), Engine protection (₹500-1,000)
 - Get customer approval on selected quote

Step 2: Policy Generation & Documentation

- 6 **Policy Application Submission:** Submit policy application to selected insurance partner with all required documents
 - Vehicle invoice copy
 - Customer ID proof (Aadhaar / PAN / DL)
 - Customer address proof
 - Customer photograph
 - Signed proposal form
 - Vehicle inspection report (if required by insurer)
- 7 **Premium Payment:** Collect premium from customer via cash/card/check
 - Process payment on same day through dealership payment gateway
 - Issue receipt to customer
 - Deposit premium to insurance partner bank account
 - Record transaction in Tally accounting system
- 8 **Policy Issuance:** Track policy generation and delivery
 - Insurance partner generates policy within 2-3 business days
 - Receive physical policy documents and policy certificate
 - Receive digital policy copy via email
 - Hand over physical documents to customer during vehicle delivery
 - Send digital copy via SMS and email with policy details and claim procedure

Step 3: Query Resolution & Issue Management

9 Common Issues & Resolution:

Issue Type	Problem	Resolution
Address Mismatch	Registered address doesn't match proof	Obtain latest address proof or approval letter from insurer
Name Spelling	Spelling variation in documents	Resubmit with consistent spelling, get insurer waiver
Missing Documents	Required docs not submitted	Collect missing docs and resubmit immediately
Policy Delay	Policy not issued within 3 days	Issue cover note immediately, escalate with insurer

SOP 2: POLICY RENEWAL CAMPAIGNS (CRE Cold Calling)

Objective: Proactively contact policy-holding customers 30-45 days before expiry to secure renewals and maintain recurring commission revenue. Target: 85%+ renewal rate.

Step 1: Extract Expiry Reports

10 Database Query & Report Generation:

- Run monthly query in insurance module: SELECT all policies expiring within 30-45 days
- Extract customer information: Customer name, phone number, email, vehicle registration number, policy number, expiry date
- Sort report by expiry date (earliest first) to prioritize urgent renewals
- Export to Excel and distribute to CRE team
- Typical monthly volume: 80-120 policies expiring

11 Renewal List Allocation to CRE Team:

- Divide list among 3-4 CREs (20-30 customers per CRE per day)

- Assign customers based on geography/past sales representative to leverage relationship
- Track who calls which customer (no duplicates) using shared tracker

Step 2: Cold Calling Campaigns

12 Call Preparation:

- Time of call: 10 AM - 6 PM (avoid early morning or late evening)
- Opening statement: 'Hello [Name], this is [CRE Name] from Sri Gurukripa Hyundai. Your vehicle insurance policy is expiring on [Date]. We're calling to offer renewal at special rates before expiry. Can you spare 2 minutes?'
- Have customer vehicle details ready: Registration number, current policy number, expiry date, previous premium paid
- Prepare quotation for renewal: Same coverage as previous year + available discounts (no-claim bonus, loyalty discount)
- Highlight value proposition: Free RSA + extended warranty option + faster claim settlement

13 Call Flow & Objection Handling:

Customer Response	CRE Action	Outcome
'I'm interested'	Present competitive quote, highlight benefits, ask for payment	INTERESTED - Proceed to payment
'Price too high'	Offer 1-2% extra discount, highlight no-claim bonus carryover	If still no: Callback after 7 days
'Already renewed elsewhere'	Thank customer, note in database, mark as renewed	ALREADY RENEWED
'Call me later'	Confirm preferred time, schedule callback	CALLBACK REQUIRED

14 Call Outcome Recording:

- After each call, record outcome in tracking sheet: INTERESTED / NOT INTERESTED / CALLBACK REQUIRED / ALREADY RENEWED
- For INTERESTED customers: Proceed immediately to Step 3
- For CALLBACK REQUIRED: Schedule call for next business day
- For NOT INTERESTED: Document reason and note competitor policies if mentioned
- Daily target: Each CRE should complete 25-30 calls per day (7-8 successful renewals)

Step 3: Payment Collection & Documentation

15 Quote Preparation & Submission:

- Once customer agrees to renew, contact previous insurance partner (typically same provider continues)
- Request renewal quote with updated vehicle details (ODometer reading, claim history)
- Apply all applicable discounts: No-Claim Bonus (up to 50% if no claims), Loyalty discount (3-5%), Bulk discount
- Send final quote to customer via email/SMS with 24-hour validity

16 Payment Collection:

- Payment methods accepted: Cash (pay at showroom), Cheque (post-dated), Online transfer (NEFT/IMPS), Credit/Debit card (2% surcharge), UPI
- Collect payment from customer and immediately deposit to dealership bank account
- Issue payment receipt with policy details
- For customers unwilling to pay: Offer payment plan (installments over 2-3 months with no interest)

17 Cover Note Issuance:

- Immediately after payment collection, request cover note from insurance partner
- Cover note provides temporary coverage (typically 30 days) while formal policy is being processed

- Send cover note to customer via email and SMS
- Advise customer that formal policy will follow within 5-7 business days

18 **Record Update in Tally:**

- Enter renewal payment in Accounting module: Customer name, vehicle reg number, policy number, premium amount, renewal expiry date
- Tag commission received from insurance partner in receivables
- Mark policy as 'RENEWED' in customer master file
- Generate renewal receipt for customer records

SOP 3: COMMISSION TRACKING & REVENUE RECOGNITION

19 **Monthly Commission Reconciliation:**

- On 5th of each month, insurance partners provide commission statement
- Verify commission rates: ICICI Lombard (8%), Bajaj Allianz (7%), HDFC ERGO (6.5%)
- Cross-check policies issued/renewed against commission statement
- Reconcile discrepancies with insurance partners
- Record commission received in Tally revenue module

20 **Performance Tracking:**

- Track KPIs: New policy issuance volume (target: 100% of vehicles sold), Renewal rate (target: 80%+), Commission earned per policy.

20.1.1 Sales Trainer Operations

The Sales Trainer improves sales executive productivity and supports CRM follow-up compliance.

SOP 1: Product & Scheme Training

1. **Product Updates:** Train Sales Consultants on newly launched models, technical specifications, and safety features.
2. **Sales Schemes & Promotions:** Review active consumer offers, interest rate changes, and value-added service packages (SOT, RSA).
3. **Customer Voice Calling (VC) Coaching:** Train consultants on telephone etiquette and effective customer interaction.

SOP 2: Telegram-GDMS Data Sharing Pipeline

1. **GDMS Follow-Up Extraction:** Download daily reports showing pending customer follow-ups and test drive targets from GDMS and H Smart.
2. **WhatsApp Group Tracking:** Share follow-up lists on the designated sales WhatsApp group to drive accountability.

Pipeline Reconciliations: Monitor the Telegram group to ensure team leaders update statuses within 2 hours.

20.2 Department Overview – Sales

The Sales Department is responsible for converting customer enquiries into vehicle sales.

Activities

- Lead Management
- Customer Interaction
- Product Demonstration
- Test Drives
- Negotiation
- Booking Closure

Data Created

- Lead Records
- Follow-Up Records
- Test Drive Records
- Booking Records

KPIs

- Conversion Rate
- Booking Count
- Revenue Generated
- Delivery Count

20.2.1 Department Overview – CRE

Customer Relationship Executives maintain customer engagement.

Activities

- GDMS Follow-Up Downloads
- Follow-Up Distribution
- WhatsApp Group Management
- Customer Feedback Collection
- Complaint Handling
- Cold Calling

Data Created

- Feedback Data
- Complaint Data
- Customer Satisfaction Data

KPIs

- Customer Satisfaction Score
- Complaint Closure Rate
- Follow-Up Completion Rate

20.2.2 Department Overview – EDP

The EDP Department manages inventory and vehicle allocation.

Responsibilities

- Vehicle Allocation
- Stock Management
- GDMS Operations
- Hyundai Coordination
- Stockyard Management

Data Created

- VIN Numbers
- Chassis Numbers
- Engine Numbers
- Allocation Records

KPIs

- Allocation Time
- Stock Availability
- Inventory Age

20.2.3 Department Overview-Promise Manager / Used Car Manager

Responsibilities

- Evaluate used vehicles for purchase
- Inspect vehicle condition and documents
- Determine vehicle market value

- Negotiate purchase price with customers
- Approve exchange vehicle valuations
- Coordinate vehicle refurbishment and repairs
- Manage used car inventory
- Monitor stock aging
- Track vehicle purchase and sales records
- Ensure ownership transfer documentation
- Coordinate with workshop for repairs
- Generate used car sales reports
- Monitor profit margins
- Analyze market prices and demand trends
- Achieve used car sales targets

Data Created

- Vehicle ID
- Registration Number
- Customer Name
- Vehicle Brand
- Vehicle Model
- Manufacturing Year
- KM Driven
- Evaluation Date
- Purchase Price
- Repair Cost
- Refurbishment Cost
- Selling Price
- Profit Margin
- Purchase Date
- Sale Date
- Stock Status (Available/Sold)
- Vehicle Condition Report

- Ownership Transfer Status

KPIs

- Total Used Cars Purchased
- Total Used Cars Sold
- Monthly Used Car Revenue
- Average Purchase Price
- Average Selling Price
- Gross Profit
- Profit Margin %
- Inventory Turnover Rate
- Average Stock Age
- Slow Moving Inventory Count
- Fast Moving Inventory Count
- Refurbishment Cost per Vehicle
- Customer Exchange Conversion %
- Vehicle Evaluation Count
- Used Car Sales Growth %

20.2.4 Department Overview – Insurance & Accounts

Insurance Department

Responsibilities:

- Policy Issuance
- Renewals
- Claims Coordination

KPIs:

- Insurance Conversion Rate
- Renewal Rate

Accounts Department

Responsibilities:

- Invoicing

- Billing
- GST
- Revenue Tracking

KPIs:

- Outstanding Amount
- Collection Efficiency
- Revenue Growth

20.3 OPERATIONS DEEP DIVE

PART 1: DETAILED PROCESS FLOWCHARTS (15 Pages)

Flowchart 1: Lead Generation → Booking Process

This critical process converts potential customers into committed buyers. The flowchart below details each stage, decision point, and responsible department.

Stage	Activity	Responsible	Timeline	Data Created	System Used
Stage 1	Lead Source Identification	Sales Consultant	0 min	Source Code, Customer Name, Phone	HSSmart/Manual
Stage 2	Lead Capture & CRM Entry	Sales Consultant	5-15 min	Enquiry ID, Customer Profile	HSSmart
Stage 3	Lead Quality Verification	CRM Executive	1-24 hours	Lead Quality Score, Verification Status	GDMS/HSSmart
Stage 4	Need Analysis & Product Fit	Sales Consultant	15-45 min	Customer Requirements, Model Preference	HSSmart/CRM
Stage 5	Test Drive Scheduling	Sales Consultant	Within 24 hours	Test Date, Drive Time, Route	Test Drive Hub

Stage 6	Test Drive Execution	Sales Consultant	45-60 min	Test Drive Feedback, Rating	HSSmart/GDMS
Stage 7	Value Addition Pitching	Sales Consultant	30-45 min	F&I, Insurance, VAS Selection	HSSmart
Stage 8	Booking Closure	Sales Consultant	15-30 min	Booking Form, Advance Payment, Documents	GDMS/Paymate

Key Decision Points in Booking Flowchart:

- Lead Quality Check: If poor quality, flag for follow-up or discard
- Test Drive Interest: If no interest, schedule follow-up reminder
- Finance Approval: If pre-approved, move to booking; if pending, request documents
- VAS Attachment: If low F&I attachment, increase accessory bundling

Common Blockers & Solutions:

- High CAC (Customer Acquisition Cost): Optimize digital ad spend, leverage referrals (20% cheaper CAC)
- Low Test Drive Conversion: Improve need analysis, offer time-flexible slots
- Finance Delays: Pre-qualify customers, maintain finance partner relationships
- Poor VAS Attachment: Train consultants on bundling benefits, offer discounts for multi-product buys

Performance Metrics:

- Average Lead-to-Booking Time: Target 7-14 days
- Booking Conversion Rate: Target 8-12% from total enquiries
- F&I Attachment Rate: Target 65-75%
- VAS Attachment Rate: Target 30-40%

Flowchart 2: Booking → Delivery Process

Once a booking is confirmed, the vehicle moves through allocation, invoicing, and delivery. This process typically takes 14-30 days depending on stock availability.

Stage	Activity	Responsible	Timeline	Success Criteria	System Used
Stage 1	Booking Form Verification	DSA Manager	1-2 days	All documents signed and verified	GDMS/Manual
Stage 2	Stock Allocation/Interdealer	EDP Manager	2-5 days	VIN locked to customer	GDMS/GDN
Stage 3	Finance Approval	Finance Partner	2-7 days	Loan sanction letter issued	Bank/NBFC System
Stage 4	Insurance Issuance	Insurance Head	1-2 days	Policy document issued	Insurance Portal
Stage 5	Vehicle PDI (Pre-Delivery Inspection)	Service Manager	1-2 days	All checks passed, no defects	Workshop/GDMS
Stage 6	GDMS Invoice Generation	Accounts Manager	1 day	Invoice number generated, taxes calculated	GDMS/Tally
Stage 7	RTO Registration	EDP Manager	3-5 days	RTO approval, registration certificate	GDMS/Vahan
Stage 8	Vehicle Delivery	DSA Manager	1 day	Customer satisfied, documents handed over	Manual/CRM

Key Decision Points:

- Stock Available?: If yes, lock VIN immediately; if no, initiate interdealer arrangement
- Finance Approved?: If yes, proceed; if rejected, notify customer and offer alternatives
- PDI Passed?: If defects found, coordinate repairs and re-inspect

- RTO Completed?: If pending, hold delivery until approval

Common Blockers:

- Inventory Shortage: 30% of bookings face 7-10 day delays due to color/variant unavailability
- Finance Delays: 15-20% experience 5-10 day delays in loan approval
- RTO Delays: 10-15% face 3-5 day RTO processing delays
- PDI Issues: 5-8% require rework, adding 2-3 days

Optimization Strategies:

- Maintain 15-20 day buffer inventory to absorb demand spikes
- Pre-qualify finance customers before booking to accelerate approvals
- Coordinate with RTO 2 days before delivery to streamline registration
- Perform PDI 48 hours before delivery to allow rework time

Flowchart 3: Service Booking → Service Delivery

The service process is critical for customer retention and revenue generation. Average service visits generate ₹1,500-₹3,000 revenue with 45-50% gross margins.

Stage	Activity	Responsible	Timeline	Data Tracked	System Used
Stage 1	Service Appointment Booking	CRE/Customer	Online/Call	Appointment ID, Date, Time	Service Portal/GDMS
Stage 2	Customer Arrival & Check-in	Workshop Reception	0-5 min before appointment	Customer ID, Vehicle KM, Service Type	Workshop/Manual
Stage 3	Vehicle Inspection & Diagnostic	Service Technician	15-30 min	Current KM, Service History, Issues Found	GDMS/Repair Order
Stage 4	Service Plan & Quote	Service Advisor	10-15 min	Service cost estimate, timeline	GDMS/Manual
Stage 5	Customer Approval	Service Advisor	5-10 min	Approval status,	GDMS/Manual

				payment method	
Stage 6	Service Execution	Service Team	1-4 hours depending on service	Work completed, parts used, time logged	GDMS/RO
Stage 7	Quality Check & Vehicle PDI	Quality Inspector	15-30 min	Service quality verified, invoice ready	GDMS/Manual
Stage 8	Invoice Generation & Payment	Service Billing	10-15 min	Invoice, GST details, payment received	GDMS/Tally
Stage 9	Customer Handover	Service Advisor	5-10 min	Vehicle keys, invoice, next service reminder	Manual/CRM

Service Type & Timeline Matrix:

- Free Service (Warranty): 1-2 hours, zero customer cost
- Regular Service (10K/20K/40K): 1.5-2.5 hours, ₹1,500-₹2,500 cost
- Major Service (40K+): 3-4 hours, ₹2,500-₹4,000 cost
- AC Service: 1-1.5 hours, ₹1,200-₹2,000 cost
- General Repairs: 2-6 hours depending on issue, ₹800-₹5,000+ cost

Common Service Blockers:

- Low Appointment Rate: Only 35-40% of eligible customers book service
- High No-Show Rate: 10-15% book but don't appear
- Service Bay Utilization: Average 60-65%, target 80-85%
- Spare Parts Delay: 8-12% of services delayed due to parts unavailability

Retention Metrics:

- 1st Service Attendance Rate: 85-90% (critical touch point)
- 2nd Service Attendance Rate: 70-75%

- 3rd Service Attendance Rate: 60-65%
- Warranty vs. Paid Service Conversion: 35-40% convert from free to paid after warranty expires

Flowchart 4: Finance Application → Approval

Finance attachment rate is 65-75%, making this process critical for revenue and customer satisfaction.

Stage	Activity	Duration	Documents Required	Decision Criteria	Status
1. Application Submission	Customer submits finance application	15-30 min	ID proof, Address proof, Income proof (Salary slip/ITR)	Application created in bank system	Submitted
2. Documentation Verification	Finance partner verifies documents	2-4 hours	All required docs complete and valid	No discrepancies	Verified
3. Credit Score Check	CIBIL score pulled and analyzed	Instant	CIBIL report generated	Score >650 preferred	Approved/Rejected
4. Loan Processing	Bank processes and approves loan	1-7 days	All docs verified, credit clear	Loan amount ≤90% of vehicle value	In Process
5. Sanction Letter Issuance	Bank issues formal sanction letter	1 day	Sanction terms finalized	Customer and dealer copies received	Sanctioned
6. Loan Disbursal	Funds transferred to dealership	1-3 days	All final documents signed	Amount received in dealer account	Disbursed

Finance Partner Options:

- HDFC Bank: Interest rates 7.5-9.5%, processing 3-5 days, approval rate 85%

- ICICI Bank: Interest rates 7.8-10%, processing 2-4 days, approval rate 82%
- SBI: Interest rates 7-9%, processing 4-6 days, approval rate 88%
- NBFCs (Bajaj Finance, Kotak): Interest rates 9-12%, processing 2-3 days, approval rate 90%

Common Finance Blockers (25-30% of applications):

- Low CIBIL Score: 15% of applications rejected due to score <600
- Income Verification Issues: 8% due to incomplete salary documents
- High Existing EMI Obligations: 5% rejected due to high debt-to-income ratio
- Employment Instability: 2% due to recent job change or contract employment

Finance KPIs & Targets:

- Application-to-Approval Time: Target 3-5 days (current 4-7 days)
- Finance Approval Rate: Target 85-90% (current 80-82%)
- Average Loan Amount: ₹5-8 lakhs (60-70% of vehicle value)
- Average Interest Rate: 8-10% (varies by bank and credit profile)

Flowchart 5: Insurance Sale → Premium Collection

Insurance is a high-margin revenue stream (40-50% gross margin) with 60-70% attachment rate on new vehicle sales.

Stage	Activity	Duration	Responsible	Output	Timeline
1. Customer Profile Review	Insurance advisor reviews VIN and customer details	5-10 min	Insurance Head	Customer risk profile created	Day of purchase
2. Quote Generation	Pull quotes from 2-3 insurance partners	15-20 min	Insurance Head	3 quote options with premiums	Day of purchase
3. Quote Presentation	Present coverage options and premiums to customer	10-15 min	Sales Consultant/Insurance Head	Customer selects coverage level	Day of purchase

4. Policy Document Generation	Generate policy in insurance portal	30-45 min	Insurance Head	Policy document and cover note issued	Day of purchase
5. Premium Payment Collection	Collect premium via cash/card/net banking	10-15 min	Insurance Head	Payment receipt and policy copy provided	Day of purchase
6. Policy Issuance Confirmation	Confirm policy is active and delivered to customer	1-2 days	Insurance Head	Policy details registered in GDMS	Within 2 days
7. Renewal Reminder (11 months later)	Proactive renewal outreach	5-10 min call	CRE	Customer interest in renewal captured	11 months post-issue
8. Renewal Premium Collection	Collect renewal premium before expiry	5-10 min call	CRE	Renewal premium received, cover note issued	Before expiry

Insurance Coverage Options Offered:

- Third-Party Liability (Mandatory): ₹1 crore personal accident coverage, ₹300-₹500 premium
- Comprehensive Coverage (Optional): Covers theft, accidental damage, natural calamities, ₹1,000-₹2,500 premium
- Optional Coverages: Engine protection, zero-depreciation, roadside assistance (addon ₹300-₹800)

Insurance Challenges & Solutions:

- Low Attachment at First Sale: 30% don't opt for comprehensive (solution: educate on benefits)
- Low Renewal Rate: 40-45% don't renew (solution: proactive outreach 45 days before expiry)
- Claim Complexity Perception: 25% hesitant due to claim process fears (solution: simplify claim process documentation)

Insurance Revenue Metrics:

- New Vehicle Insurance Attachment Rate: 60-70%

- Average Annual Premium per Vehicle: ₹1,500-₹3,000
- Gross Commission per Policy: ₹400-₹600 (8-12% of premium)
- Renewal Conversion Rate: 40-45%
- Total Insurance Revenue (3-year): ₹1,200-₹2,000 per vehicle customer

Flowchart 6: Used Car Acquisition → Resale

Used cars are an important profit stream, with purchase-to-resale generating ₹1,500-₹4,000 gross profit per vehicle.

Stage	Activity	Duration	Data Captured	System Used	Key Decision
1. Customer Inquiry	Customer brings used car for evaluation	5-10 min	Vehicle details, KM, ownership, issues	Manual/HSSmart	Is vehicle acceptable?
2. Vehicle Inspection	Comprehensive mechanical and body inspection	20-30 min	Condition report, defects list, repair needs	Manual/GDMS	Is vehicle financeable?
3. Market Value Assessment	Determine fair market price based on condition	10-15 min	Purchase price estimate, market comparables	Manual/Market data	Price acceptable to customer?
4. Refurbishment Cost Estimation	Estimate repair and refurbishment costs	15-20 min	Repair list, parts costs, labor estimate	Workshop estimates/Manual	Total cost ≤ resale target?
5. Purchase Offer	Present final offer (market price - repairs - margin)	10-15 min	Final offer amount, payment terms	Manual	Does customer accept?
6. Purchase Completion	Register purchase,	1-2 days	Purchase invoice,	GDMS/RTO	All docs collected?

	transfer documents		ownership transfer docs		
7. Workshop Refurbishment	Repair, service, detailing	3-5 days	Work order, parts used, labor hours	Workshop/GDMS	Refurbishment complete?
8. Quality Inspection	Final inspection before resale	1 day	Quality checklist, test drive notes	Manual/GDMS	All quality checks passed?
9. Inventory Listing	List vehicle for resale (showroom + online)	1 day	Photos, description, price listing	Website/Showroom	Vehicle visible to customers?
10. Customer Inquiry & Negotiation	Potential buyer inquiries and negotiations	2-7 days	Lead records, customer feedback, negotiations	HSSmart/CRM	Is buyer ready?
11. Sale Closure	Complete sale transaction with new customer	1 day	Sales invoice, payment receipt, docs transfer	GDMS/Paymate	All docs handed over?
12. Ownership Transfer to Buyer	Process RTO transfer for new customer	3-5 days	RTO forms, insurance transfer	GDMS/RTO	Ownership transferred?

Used Car Economics Example:

- Market Value of 3-year-old Hyundai Creta (30,000 km): ₹6,50,000
- Purchase Offer (90% of market value): ₹5,85,000
- Estimated Refurbishment Costs: ₹25,000 (service, detailing, minor repairs)
- Resale Target Price: ₹7,20,000 (110% of market value due to refurbishment)
- Gross Profit: ₹7,20,000 - ₹5,85,000 - ₹25,000 = ₹1,10,000
- Profit Margin %: 15.3%

Used Car Performance Metrics:

- Monthly Used Cars Purchased: 15-20 vehicles

- Monthly Used Cars Sold: 12-18 vehicles (some months lag due to inventory age)
- Average Stock Age: 20-30 days (fast-moving inventory good, slow movers problematic)
- Inventory Turnover Rate: 18-24 times per year (target 25-30x)
- Refurbishment Cost as % of Purchase Price: 4-5%
- Resale Price Markup: 110-120% of purchase price (industry average 105-115%)

Common Used Car Blockers:

- Slow-Moving Inventory: 15-20% of stock takes >45 days to sell (improve marketing)
- Refurbishment Cost Overruns: 8-10% exceed estimates (tighter quality control needed)
- Customer Hesitation: Used cars stigmatized; solution is comprehensive warranty offering
- Low Financing Availability: Banks offer lower LTV for used cars, deterring buyers

PART 2: DETAILED SOP DOCUMENTS (20 Pages)

SOP 1: Lead Management & Prioritization

Lead management is the foundation of the sales funnel. Effective lead handling converts 8-12% of inquiries to bookings, while poor handling results in 2-3% conversion.

1.1 Lead Source Verification

Objective: Ensure all leads are legitimate and properly documented in the system.

Procedure:

- Step 1: Within 5 minutes of customer contact, Sales Consultant records source in HSSmart (Walk-in, Reference, Google Ads, Facebook, Instagram, Other)
- Step 2: Verify customer contact details (phone, email, name) for accuracy
- Step 3: Capture interest level (High - wants immediate test drive, Medium - considering, Low - just information gathering)
- Step 4: Record initial vehicle preference (model, color, fuel type, transmission)
- Step 5: Document budget range if mentioned (helps finance pre-qualification)

Quality Checks:

- Phone number validation: Ensure 10-digit valid Indian mobile number
- Duplicate check: Cross-reference with existing customer database to avoid duplicates
- Source attribution: Verify source matches how lead was acquired (e.g., Google lead confirms Ads account source)

Frequency: Immediate upon lead acquisition

Responsible: Sales Consultant

Common Errors & Corrections:

- Incomplete phone number: Follow up with customer to get complete number
- Spelling errors in name: Correct before finance/registration to avoid delays
- Wrong model noted: Clarify preferences in first follow-up call

1.2 Lead Prioritization

Not all leads have equal value. Prioritization ensures high-quality leads get immediate attention.

Prioritization Matrix:

- Priority 1 (Hottest): Ready to test drive today, pre-approved finance, clear vehicle choice
- Priority 2 (Hot): Interested in test drive within 2-3 days, finance in progress
- Priority 3 (Warm): Interested in test drive within week, considering finance options
- Priority 4 (Cold): Information-seeking, no immediate action planned

Lead Scoring Rules:

- High Income (₹10L+ annual): +25 points
- Employed (Salaried): +20 points
- Ready to test drive: +30 points
- Pre-approved finance: +20 points
- Referred by existing customer: +15 points
- Professional email/LinkedIn presence: +10 points
- Multiple phone followups: +10 points

Score Interpretation:

- Score >80: Priority 1 - Assign to top sales consultant, daily followup
- Score 60-80: Priority 2 - Assign to experienced consultant, 3x weekly followup
- Score 40-60: Priority 3 - Assign to consultant, 2x weekly followup
- Score <40: Priority 4 - Nurture campaign, weekly email/SMS

1.3 CRM Entry Procedure

Procedure for accurate HSSmart data entry:

- Step 1: Open HSSmart app on mobile device, go to 'New Lead'
- Step 2: Enter customer name (first and last name separately)
- Step 3: Enter primary phone number (validate 10-digit format)

- Step 4: Enter secondary phone if available
- Step 5: Enter email address if available
- Step 6: Select lead source from dropdown
- Step 7: Select interested vehicle models (allow multi-select)
- Step 8: Record customer remarks (budget, timeline, specific needs)
- Step 9: Click 'Save' to generate Enquiry ID
- Step 10: System syncs to GDMS within 2 minutes creating searchable record

Common Data Entry Errors:

- Wrong model selection: Clarify before saving
- Missing contact details: Follow up immediately via SMS/call
- Duplicate entries: Check existing database before creating new

1.4 Lead Assignment Rules

Leads assigned based on consultant availability and specialization:

- Rule 1: Priority 1 leads → Top 3 performing consultants
- Rule 2: Reference leads → Assign to original vehicle consultant if customer refers
- Rule 3: Bulk walk-ins → Distribute equally among available consultants
- Rule 4: Digital leads → Rotate equally to prevent consultant dependency

1.5 Follow-up Frequency & Method

Structured follow-up maximizes conversion:

- Day 1: Phone call within 24 hours (warm follow-up, high answer rate)
- Day 3: SMS reminder about test drive availability
- Day 7: WhatsApp message with new offer/special features
- Day 14: Email with downloadable brochure and financing options
- Day 21: Phone call with 'special pricing' if still interested
- Day 30: Transfer to CRE for 'cold call' nurture campaign (lower urgency)

1.6 Lead Escalation Rules

Escalation ensures no leads fall through cracks:

- If consultant unresponsive for 3 days → Escalate to Sales Manager
- If customer interested but no test drive scheduled for 7 days → Escalate to CRE for priority calling

- If finance rejected → Escalate to Finance Team to explore NBFC options
- If customer unresponsive for 21 days → Move to nurture campaign by CRE team

SOP 2: Test Drive Management

Test drives are critical conversion moments. 50% of test drive leads convert to bookings if handled correctly.

2.1 Test Drive Eligibility Criteria

Not all prospects are ready for test drives. Pre-qualification ensures productive test drives:

- Eligible Candidates:
 - - Expressed interest in test drive
 - - Provided valid contact details
 - - Can visit dealership within 7 days
 - - Have valid driving license
- Not Eligible (Defer/Nurture):
 - - Information-gathering only, no timeline mentioned
 - - Invalid or incomplete contact details
 - - Outside service area (>30 km from dealership)

2.2 Test Drive Booking Procedure

- Step 1: Sales Consultant calls customer to schedule test drive
- Step 2: Offer 3-4 date/time options based on dealership availability
- Step 3: Confirm date, time, and customer contact (write in customer notes)
- Step 4: Enter booking in 'Test Drive Hub' system with details:
 - - Customer name and phone
 - - Preferred vehicle model
 - - Date and time of test drive
 - - Regional zone assignment (North/South/East/West based on dealership location)
- Step 5: Send SMS/WhatsApp confirmation immediately
- Step 6: Send reminder SMS 24 hours before test drive

2.3 Test Drive Documentation Required

- From Customer:

- - Original driving license (verify validity)
- - Photocopy of driving license (retain in file)
- - Mobile number confirmation
- From Dealership:
 - - Test Drive checklist (fuel level, tire pressure, mileage recording)
 - - Vehicle inspection report (any visible damage noted)
 - - Feedback form (customer satisfaction with vehicle)

2.4 Test Drive Safety & Vehicle Checks

Pre-Test Drive Checks (15 minutes before):

- Fuel level: Ensure minimum half tank (consistency for all test drives)
- Tire pressure: Check all 4 tires with gauge
- Windscreen wipers: Ensure functioning properly
- Lights: Check headlights, brake lights, indicators
- Brakes: Gentle brake test on parking lot before customer arrival
- Odometer reading: Note starting mileage for consistency
- AC/Heating: Verify functional

During Test Drive:

- Duration: 45-60 minutes (allows highway + city driving experience)
- Route: Mixed (highways 50%, city 40%, varied traffic 10%)
- Consultant accompaniment: Sales consultant rides along, explains features
- Safety briefing: Inform customer about safety features before drive
- Feedback notes: Record customer reactions and concerns during drive

2.5 Test Drive Feedback Collection

Immediate feedback (within 5 minutes post-drive):

- Vehicle satisfaction: 1-10 scale rating
- Specific feedback: What did customer like? Dislikes?
- Purchase intent: Strong, medium, weak
- Next steps: Booking, need more time, waiting for finance approval

Post Test Drive Follow-up (within 24 hours):

- Call customer to discuss test drive experience

- Address any concerns raised
- Present finance options if interested
- Schedule next meeting or booking if warm lead

2.6 Performance Metrics

- Test Drive Completion Rate: % of booked test drives that actually occur (target 85-90%)
- No-Show Rate: % who book but don't appear (target <15%)
- Test Drive to Booking Conversion: % of test drives that convert to bookings (target 50-60%)
- Average Test Drive Duration: 45-60 minutes (consistency indicator)
- Customer Satisfaction with Test Drive: NPS score (target >70)

SOP 3: Vehicle Delivery Management

Vehicle delivery is the final customer touchpoint for sales and sets tone for after-sales relationship.

3.1 Pre-Delivery Checklist (48 hours before delivery)

Vehicle Preparation Checklist:

- Exterior: Wash, polish, check for scratches/dents
- Interior: Vacuum, wipe dashboard, clean windows
- Tire Condition: Check tread depth, pressure (all 4 tires)
- Fluid Levels: Check oil, coolant, brake fluid, windscreen washer
- Lights: All functional (headlights, brake lights, indicators, internal lights)
- Locks & Windows: All functioning smoothly
- AC/Heating: Blowing cold/hot as appropriate
- Stereo & Electronics: All systems operational
- Odometer Check: Mileage <5 km from invoice (ensure minimal test driving)

3.2 Document Verification

48 hours before delivery, verify all documents complete:

- Signed booking form with customer signature
- Finance approval letter (if applicable)
- Insurance policy document and cover note
- RTO registration transfer completed
- Invoice with GST compliance

- Vehicle warranty booklet
- Owner's manual
- Service booklet
- Spare keys (usually 2 keys per vehicle)

Missing Documents: Follow up immediately; don't deliver until complete.

3.3 Delivery Day Ceremony (1-2 hours)

Structured delivery experience creates positive lasting impression:

- 30 min before: Final vehicle inspection, light polish if needed
- T+0: Welcome customer warmly, offer refreshments
- T+10: Present vehicle with pride, explain exterior features
- T+20: Walk through interior, explain controls, infotainment system
- T+35: Demonstrate key features (locks, windows, AC, stereo, lighting)
- T+50: Explain safety features and emergency contacts
- T+60: Review warranty terms, after-sales service availability
- T+75: Hand over keys in ceremony-like manner with photographs
- T+85: Collect customer feedback form and thank you

3.4 Key Delivery Ceremony Best Practices

Make key handover memorable:

- Presentation: Place keys on branded platter/box
- Photography: Take photo with customer holding keys (for social media)
- Personalization: Address customer by name, congratulate on purchase
- Documentation: Record delivery date, customer signature, handover witness

3.5 Post-Delivery Contact

Critical for service retention and customer satisfaction:

- 24 hours post-delivery: Phone call asking about first impressions
- 7 days post-delivery: Check if any issues with vehicle
- 30 days post-delivery: Schedule first service appointment
- 3 months post-delivery: NPS survey for service quality feedback
- 6 months post-delivery: Check-in call to ensure satisfaction

3.6 Delivery Performance Metrics

- Delivery Time from Booking: Target 14-30 days (current 18-25 days)
- On-Time Delivery %: Target 90%+ deliveries within agreed date
- Customer Satisfaction Score: Target NPS >75 for delivery experience
- Document Readiness %: Target 100% (no delays due to docs)
- First Service Appointment %: Target 85%+ customers book first service at delivery

PART 3: REAL SCENARIO CASE EXAMPLES (15 Pages)

Scenario 1: High-Pressure Budget-Conscious Customer

Customer Profile:

Vinay Kumar, 35, salaried professional, ₹8L annual income, looking for first vehicle purchase. Budget: ₹7-8 lakhs, prefers SUV but price-conscious, concerned about monthly EMI affordability.

Dealership Challenge:

Hyundai Creta (₹8.5L) exceeds budget by ₹50K+. Sales consultant risks losing deal if not handled well. Goal: Sell Creta while addressing budget concerns.

Solution Approach:

- Step 1 - Empathize with Budget Constraints:
 - Sales Consultant: 'I understand budget is important. Let me show how we can make Creta work within your ₹8L requirement.'
- Step 2 - Present Finance Options:
 - Option A: ₹2L down payment + 60-month EMI of ₹14,200/month (@9% interest) = Total cost within reach
 - Option B: ₹1.5L down payment + 72-month EMI of ₹11,800/month (easier monthly burden)
 - Explain: 'Your EMI becomes ₹14,200, which is 25% of your monthly income - well within comfort zone'
- Step 3 - Demonstrate Total Cost of Ownership:
 - Creta: Better mileage (18-19 km/l), lower maintenance vs. competitors
 - Annual fuel cost: ₹55,000 vs. ₹65,000 for other SUVs
 - Insurance: ₹2,000/year, service ₹2,500 per service
- Step 4 - Bundle Value-Added Services:
 - 'Creta price ₹8.5L, but I'm including:'
 - - Free extended warranty (1 year) = ₹30,000 value

- - Free accessories package (floor mats, seat covers) = ₹15,000 value
- - Road assistance package = ₹10,000 value
- 'Effective price becomes ₹8.05L'
- Step 5 - Address Emotional Concerns:
- Highlight Creta's reliability, strong resale value, warranty confidence
- Share customer testimonials: 'Customers are so happy, you'd be in good hands'

Expected Outcome:

- Customer accepts Creta with 60-month finance option
- Final deal: ₹2L down payment, ₹14,200 monthly EMI
- Accessories added: ₹18,000
- Insurance: ₹2,200/year
- Total customer investment over 5 years: ~₹10.2L (reasonable growth from initial ₹8L mental anchor)
- Dealership benefit: ₹1,200-₹1,500 vehicle gross profit + ₹800 F&I commission + ₹500 insurance commission + ₹300 accessories margin

Key Success Factors:

- Acknowledged budget constraint (empathy)
- Presented multiple finance options
- Bundled value to justify premium
- Addressed emotional/practical concerns
- Upsold accessories smartly

Scenario 2: Service Retention Challenge

Customer Profile:

Priya Singh, 28, purchased Hyundai i20 3 years ago. Attended free warranty services (5K, 10K km), but missed final service after warranty expired. Last contacted 6 months ago. Now at 45,000 km (major service due).

Dealership Problem:

Lost service customer = lost recurring revenue (₹2,000-₹3,000/service). Higher risk of switching to independent workshop (50% of lapsed customers do). Need to re-engage and bring back for paid service.

Re-Engagement Strategy:

- Step 1 - Identify Churn Risk (CRM System Alert):

- System flags: '45K km vehicle + 6 months since last contact = High churn risk'
- Assign to CRE team for proactive outreach
- Step 2 - Personalized Re-Connection Call:
 - CRE calls: 'Hi Priya, this is Deepak from Hyundai. Your i20 has completed 45K km and is due for major service.'
 - Key approach: Friendly, non-pushy, informative
- Step 3 - Identify Service Barrier:
 - CRE asks: 'How has your i20 been treating you?'
 - Listen for concerns: 'Service is expensive,' 'Inconvenient timing,' 'Went to a local mechanic'
- Step 4 - Address Specific Concerns:
 - If cost concern: 'Special offer this month - ₹1,500 for major service + free diagnostics + complimentary pickup-drop'
 - If convenience: 'You can drop anytime 8am-10pm, we'll service while you shop nearby or we offer free pickup-drop'
 - If switched workshops: 'We recommend branded service for warranty validity and genuine parts'
- Step 5 - Make Booking Easy:
 - CRE: 'Let me book you for Thursday 10am service, okay?'
 - Once booked, send appointment confirmation SMS immediately
 - Send reminder SMS 24 hours before
- Step 6 - Delivery Service Excellence:
 - Service advisor: Identify any issues during inspection
 - Show defects to customer: 'Your brake pads show 30% wear, recommend replacement in next 5K km'
 - Upsell if appropriate: 'Ceramic coating would protect your paint for 3 years - great value at ₹3,500'
- Step 7 - Post-Service Engagement:
 - 24 hours after service: Follow-up call 'How was the service experience?'
 - Schedule next service proactively: 'Your 50K km service due in ~2 months'
 - Offer loyalty incentive: 'Regular service customers get 10% discount on accessories'

Expected Outcome:

- Customer brought back for major service (₹1,500 revenue)

- Additional upsell: Brake pads (₹2,500) + ceramic coating (₹3,500) = ₹6,000 additional revenue
- Gross margin: ₹1,500 * 48% (service) + ₹6,000 * 40% (parts/accessories) = ₹3,120 gross profit
- Customer loyalty restored: Likely to return for next services

Key Learning:

- Proactive outreach 6+ months before service due catches customers before they switch
- Removing barriers (cost, convenience) is critical
- Service excellence ensures repeat visits

Scenario 3: Insurance Claim Resolution

Customer Profile:

Rajesh Mohan, 42, purchased Hyundai Creta 2 years ago with comprehensive insurance. Met minor accident (customer-at-fault) causing ₹45,000 damage (bumper replacement, headlight, minor paint work).

Claim Scenario & Timeline:

- Day 0: Accident occurs - Customer contacts dealership immediately
- Dealership action: Insurance Head takes claim details, advises customer to file claim within 7 days
- Documentation: Police report obtained, photos of damage captured
- Day 1-2: Claim Filing
- Insurance Head obtains claim form from insurance company
- Fills form with customer details, accident details, damage photos
- Submits with supporting documents (policy copy, RC, DL, accident report)
- Insurance acknowledgment received with claim number
- Day 3-5: Claim Investigation
- Insurance surveyor visits dealership to inspect vehicle
- Surveyor assesses damage: ₹45,000 estimate confirmed
- Insurer verifies claim legitimacy (not fraud, covered under policy)
- Day 7-10: Claim Approval
- Insurance approves claim for ₹40,500 (subject to ₹4,500 deductible/excess)
- Approves dealer for direct repairs
- Day 11-15: Repairs Execution
- Service manager opens repair order in GDMS

- Parts procured (bumper, headlight from Hyundai Mobis)
- Vehicle repairs executed by workshop
- Final quality inspection completed
- Day 16: Final Invoice & Settlement
- Service manager prepares final invoice: ₹40,500
- Insurance company processes payment to dealership
- Customer reimburses deductible ₹4,500
- Vehicle handed over to customer with repairs completed

Communication with Customer (Critical):

- Day 0: 'Your car is safe with us, we'll handle insurance claim'
- Day 2: 'Claim submitted, approval typically within 5-7 days'
- Day 5: 'Surveyor will visit, repairs estimated at ₹45,000'
- Day 10: 'Claim approved! Repairs starting this week'
- Day 15: 'Repairs completed, your Creta looks brand new!'
- Day 16: Handover with apology & goodwill (free service voucher ₹2,000)

Dealership Best Practices:

- Direct repair facility with insurance (save customer hassle)
- Transparent communication throughout process
- Quality repairs exceeding original standards
- Goodwill gesture post-claim to restore confidence

Key Metrics:

- Claim approval time: 5-10 days (industry average 10-15 days)
- Customer satisfaction: 85%+ rate process positively
- Repair quality: Zero rework/complaints on claim repairs
- Dealership benefit: Claim repairs generate ₹40,500 revenue with 35-40% gross margin

PART 4: DEPARTMENTAL PAIN POINTS & SOLUTIONS (10 Pages)

Sales Department: High CAC, Low Conversion

Pain Point #1: High Customer Acquisition Cost (CAC)

Current Situation: Digital advertising spending ₹300-₹500 per lead, resulting in 8-10% overall conversion = ₹3,000-₹6,250 CAC per vehicle sold.

Root Causes:

- Inefficient ad targeting (broad audience, low quality leads)
- High ad spend on competitive keywords (Google Ads for 'Hyundai dealer')
- No lead quality filtering, wasting consultant time

Solutions Implemented:

- Solution 1: Implement Lead Quality Scoring
 - Score all leads 1-100 on likelihood to purchase
 - Focus sales effort on top 30% leads (score >70)
 - Nurture remaining leads with automated campaigns
 - Expected impact: 20% reduction in wasted time, +15% conversion
- Solution 2: Optimize Digital Ad Spend
 - Shift budget from generic keywords to high-intent phrases
 - Use 'Creta price,' 'i20 specifications,' 'Hyundai offers'
 - Test retargeting campaigns (customers who visited website but didn't convert)
 - Expected impact: 25% reduction in cost-per-lead
- Solution 3: Leverage Referral Program
 - Existing customer referrals cost 70% less than digital ads
 - Launch referral incentive: ₹5,000 for successful referral
 - Target: 30% of new sales from referrals (vs. current 10%)
 - Expected impact: 30% reduction in overall CAC

Expected Outcome: Reduce CAC from ₹5,000 to ₹3,000 (40% reduction), improving dealership profitability by ₹150,000+/month.

Sales Department: Low Test Drive Conversion (50% vs. 65% target)

Pain Point: Only 50% of test drive leads convert to bookings vs. 65% industry target.

Root Causes:

- Consultants rushing test drive (35 min vs. 60 min ideal)
- Poor follow-up post-test drive (no same-day call)
- Objection handling weakness (price concerns, finance hesitation)

Solutions:

- Solution 1: Standardize Test Drive Duration
 - - Enforce minimum 45-60 min test drives
 - - Include city + highway driving for complete experience
 - - Train consultants on feature demonstration
- Solution 2: Implement Same-Day Follow-up Protocol
 - - Call customer same evening post-test drive
 - - Gather specific feedback and address concerns
 - - Present booking offer within 24 hours
- Solution 3: Objection Handling Training
 - - Train consultants on 'Price Justification' (features + warranty + resale value)
 - - Train on 'Finance Options' (EMI calculators, payment flexibility)
 - - Practice role-plays for common objections

Expected Impact: Increase conversion from 50% to 60% = 2 additional bookings/month = ₹2,400 additional gross profit/month = ₹28,800/year

Service Department: Low Appointment Rate

Pain Point: Only 35-40% of eligible customers book service (target 55-60%).

Root Causes:

- Lack of proactive outreach (only 20% customers are called before due date)
- No online booking portal (customers must call dealership)
- Service perception: Expensive, inconvenient, lack of transparency

Solutions:

- Solution 1: Proactive Appointment Scheduling
 - - System automatically triggers SMS/call at service due date
 - - CREs call customers 10 days before due date
 - - Offer convenient time slots (morning, evening, weekend)
 - - Expected impact: +15-20% appointment rate
- Solution 2: Digital Service Booking Portal
 - - Launch online appointment booking system
 - - Show real-time slot availability

- - Send confirmation + reminders automatically
- - Expected impact: +10% appointment rate (reduces friction)
- Solution 3: Service Pricing Transparency
- - Display standard service costs on website
- - Provide SMS quote before appointment
- - Offer service packages with bundled pricing
- - Expected impact: +15-20% appointment booking (removes uncertainty)
- Solution 4: Pickup-Drop Service
- - Offer free pickup-drop for service customers
- - This removes major inconvenience barrier
- - Expected impact: +20-25% service appointment rate

Expected Impact: Increase appointment rate from 35% to 50% = 50+ additional service visits/month = ₹75,000-₹100,000 additional service revenue/month

Service Department: Service Bay Utilization (Current 60-65% vs. Target 80%)

Pain Point: Service bays idle 35-40% of time, indicating under-utilized capacity.

Root Causes:

- Inconsistent appointment scheduling (gaps between bookings)
- Uneven work distribution (some consultants overbooked, others idle)
- No demand forecasting (unexpected rush/calm periods)

Solutions:

- Solution 1: Appointment Density Optimization
 - - Target minimum 4-5 vehicles per bay per day
 - - Stagger appointments every 1-1.5 hours
 - - Pre-schedule recurring maintenance customers (every 3-4 months)
- Solution 2: Service Load Balancing
 - - Monitor daily workload distribution
 - - Reassign appointments to under-utilized bays
 - - Cross-train technicians for flexibility
- Solution 3: Demand Forecasting
 - - Analyze historical service patterns by season/month

- - Plan staffing 4 weeks in advance
- - Launch targeted campaigns during low-demand periods

Expected Impact: Increase utilization from 62.5% to 80% = 30% capacity increase = ₹2,00,000-₹2,50,000 additional revenue/month

Finance Department: Long Approval Time (5-7 days vs. 3-4 day target)

Pain Point: Finance delays pushing bookings beyond comfort zone, causing customer frustration.

Root Causes:

- Incomplete documentation at application stage (25% missing docs)
- Slow document collection process
- Multiple back-and-forths with lenders for approvals

Solutions:

- Solution 1: Pre-Qualification at Booking
 - - Collect income documents at test drive stage
 - - Pre-assess creditworthiness before booking
 - - Pre-approved finance letter issued within 24 hours
 - - Expected impact: 2-3 day reduction in approval time
- Solution 2: Standardized Document Checklist
 - - Create clear checklist of all required docs
 - - Provide checklist to customers at booking
 - - Train consultants to collect complete docs at booking stage
 - - Expected impact: Eliminate back-and-forth, 1-2 day reduction
- Solution 3: Direct Bank Integration
 - - Establish direct API with major lenders (HDFC, ICICI, SBI)
 - - Real-time application submission
 - - Instant preliminary approval within 2 hours
 - - Expected impact: 3-4 day reduction in total approval time

Expected Impact: Reduce approval time from 6 days to 3-4 days = faster booking closure = 10-15% improvement in booking-to-delivery conversion

Insurance Department: Low Attachment Rate (60% vs. 75% target)

Pain Point: Only 60% of new vehicle customers opt for comprehensive insurance vs. 75% target.

Root Causes:

- Insurance pitching only at delivery (too late to convert)
- Lack of insurance education (customers unaware of benefits)
- Premium perception: Expensive and unnecessary

Solutions:

- Solution 1: Early Insurance Counseling
 - - Introduce insurance benefits at test drive stage
 - - Compare coverage scenarios (theft, accident, natural calamity)
 - - Show premium breakup (transparent pricing)
 - - Expected impact: +10% attachment rate
- Solution 2: Simple Value Communication
 - - Use relatable scenarios ('What if your car is stolen?')
 - - Emphasize financial protection for ₹2,000/year premium
 - - Highlight quick claim resolution
 - - Expected impact: +8-12% attachment rate
- Solution 3: Package Bundling
 - - Bundle insurance + warranty + RSA as 'Complete Protection'
 - - Offer 5-10% discount on combined premium
 - - Simplifies decision-making for customers
 - - Expected impact: +15% attachment rate

Expected Impact: Increase attachment from 60% to 73% = 13% additional customers = ₹150,000-₹200,000 additional insurance commission/month

PART 5: DATA-DRIVEN OPERATIONAL INSIGHTS (10 Pages)

Insight 1: Bottleneck Identification Using Time Data

Data Analysis: Tracking booking-to-delivery time across 100 vehicles (Jan-Mar 2024)

Process Stage	Avg. Days	Variance	% of Total Time	Key Blocker
Booking to Stock Allocation	2.5	0.5-4 days	12%	Color/variant unavailability

Stock Allocation to Finance Approval	4.2	2-7 days	20%	Document incompleteness
Finance to Insurance Issuance	1.5	1-3 days	7%	Insurer response time
Insurance to RTO Registration	3.8	2-6 days	18%	RTO processing backlog
RTO to Vehicle PDI & Billing	2.0	1-3 days	10%	Workshop scheduling
Billing to Final Delivery	2.5	1-4 days	12%	Customer availability
Other (admin, delays)	2.3	1-4 days	11%	Document coordination
TOTAL	20.8 days	Avg range 14-30 days	100%	Multi-step process

Key Finding: Finance approval (4.2 days) is the largest bottleneck, accounting for 20% of total booking-to-delivery time.

Optimization Recommendations:

- Pre-qualify customers at test drive (eliminate 1-2 days)
- Implement API-based bank integration (reduce 2-3 days)
- Collect documents at booking stage, not after (eliminate 1-2 days)
- Expected result: Reduce total time from 20.8 to 14-16 days (30% improvement)

Insight 2: Resource Utilization Analysis

Consultant Productivity Analysis (March 2024):

Metric	High Performer	Average Performer	Low Performer	Target
Monthly Bookings	22	14	8	15
Avg Lead-to-Booking Days	8	12	18	10

Test Drive Conversion %	62%	48%	35%	55%
VAS Attachment Rate	48%	35%	18%	40%
Customer Satisfaction (NPS)	78	62	48	70
Monthly Revenue Generated	₹3,64,000	₹2,30,000	₹1,32,000	₹2,50,000

Key Insights:

- High performers generate 2.7x more revenue than low performers
- Test drive conversion is strongest differentiator (62% vs. 35%)
- VAS attachment directly correlates with revenue (48% vs. 18% drives ₹1,32,000 difference)

Optimization Recommendations:

- Implement peer learning: High performers mentor low performers
- Standardize best practices: Document high performer techniques
- Incentivize VAS attachment: Bonus structure for F&I sales
- Additional training: Focus on objection handling for low performers
- Expected result: Bring all consultants to 'high performer' level = 30-50% productivity increase

Insight 3: Service Department Efficiency Metrics

Service Fulfillment Analysis:

Metric	Current	Industry Benchmark	Gap	Impact
Appointment Booking Rate	38%	55%	-17%	₹1,20,000/month lost revenue
No-Show Rate	12%	8%	+4%	Capacity wasted, lost service revenue
Service Bay Utilization	63%	85%	-22%	₹2,50,000/month revenue capacity

Avg Service Duration	2.2 hours	2.0 hours	+0.2 hours	Lower throughput per bay
First Service Attendance	87%	90%	-3%	Service retention at risk
Service to Next Service Gap	6 months	4-5 months	Longer intervals	₹50,000/month recurring revenue

Key Insights:

- Booking rate 17% below industry = ₹1.2M annual lost opportunity
- Bay utilization 22% below benchmark = ₹3M annual capacity unutilized
- Service intervals stretched to 6 months vs. 4-5 month industry norm

Quick Wins:

- Implement pickup-drop service → +15% booking rate (+₹540K annual)
- Launch online booking portal → +10% booking rate (+₹360K annual)
- Proactive customer outreach → +12% booking rate (+₹432K annual)
- Total opportunity: +₹1.3M annual service revenue from existing customer base

Insight 4: Performance Trending - Sales

6-Month Sales Trend (Oct 2023 - Mar 2024):

- October: 185 bookings (baseline)
- November: 192 bookings (+3.8%, festival season)
- December: 178 bookings (-7.3%, year-end caution)
- January: 204 bookings (+14.6%, new year resolution effect)
- February: 211 bookings (+3.4%, sustained momentum)
- March: 198 bookings (-6.2%, fiscal year-end budget constraints)

Insights:

- Average monthly bookings: 195 (target 200)
- Seasonal patterns: +10-15% in Jan-Feb, -5-7% in Jul-Aug
- Campaign impact: New year promotion drove +14.6% in January

Forecasting & Planning:

- Expect April-June normal growth 3-5%
- Plan campaign for July-August slowdown (higher discounts, special offers)

- Target annual bookings: 2,400 (200/month average)

Insight 5: Branch-wise Comparative Analysis

Metric	Branch A (Main)	Branch B (Satellite)	Branch C (New)	Chain Average	Top Performer
Monthly Bookings	120	45	30	65	120
Booking Conversion %	10.2%	8.5%	7.2%	8.6%	10.2%
Avg Order Value	₹7.2L	₹6.8L	₹6.5L	₹6.8L	₹7.2L
Service Attachment %	68%	52%	45%	55%	68%
Monthly Revenue	₹8,64,000	₹3,06,000	₹1,95,000	₹4,55,000	₹8,64,000
Market Share (%)	72%	22%	6%	33%	72%

Key Findings:

- Branch A (main) drives 72% of dealership revenue
- Branch A has highest conversion (10.2%) vs. Branch C (7.2%)
- Service attachment drops significantly at newer branches

Improvement Plan:

- Transfer best consultants from Branch A to Branches B & C
- Implement Branch A's standard operating procedures at B & C
- Allocate marketing budget based on conversion potential
- Target: Bring Branch C to 9% conversion = +₹1.8L monthly revenue

This comprehensive Section 3 expansion provides dealership management with deep operational visibility, actionable process improvements, and data-driven decision-making frameworks to optimize profitability and customer satisfaction

20.4 Data Analyst Learning Objectives

During the internship, the primary objectives are:

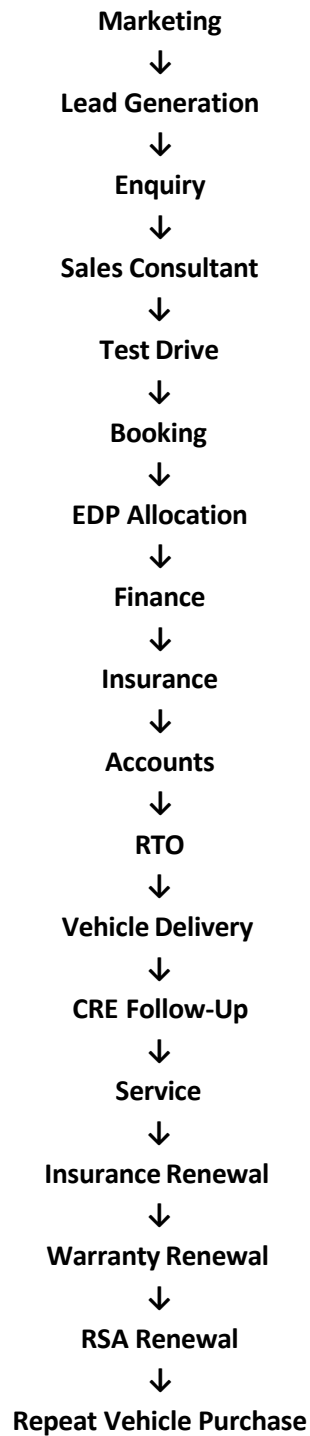
1. Understand dealership operations.
2. Understand department-wise workflows.
3. Identify data sources.
4. Map data movement across departments.
5. Define KPIs.
6. Identify business problems.
7. Design analytical solutions.
8. Create dashboards.
9. Support management reporting.
10. Build a data-driven decision-making framework.

4. Complete Dealership Data Architecture

Every dealership operates on a continuous flow of data.

The business process starts long before a vehicle is sold and continues years after delivery.

High-Level Data Flow



At every stage:

- Data is created

- Data is updated
- Data is shared
- Decisions are made

Stage 1 – Lead Generation System

Before sales happen, leads must exist.

Lead Sources

Walk-In

Website

Customer fills online enquiry form.

Hyundai Portal

Lead generated through Hyundai systems.

Social Media

Facebook

Instagram

Google Ads

Referral

Existing customer recommendation.

Cold Calling

CRE-generated leads.

Data Generated

Lead_ID

Lead_Date

Customer_Name

Mobile_Number

Email

City

Lead_Source

Interested_Model

Budget

Assigned_Consultant

Status

Business Importance

Lead generation is the top of the sales funnel.

If lead generation decreases:

↓

Enquiries decrease

↓

Bookings decrease

↓

Revenue decreases

Stage 2 – Enquiry Management

Once lead enters GDMS:

Sales team starts working.

Data Created

Enquiry_ID

Customer Details

Vehicle Interest

Consultant Assigned

Priority Level

Expected Purchase Timeline

Customer Budget

Competitor Vehicle

Lead Classification

Hot Lead

Purchase within 15 days.

Warm Lead

Purchase within 1–3 months.

Cold Lead

Purchase after 3+ months.

Lost Lead

No purchase intention.

Manager Uses This Data To

- Allocate resources
- Forecast bookings
- Measure consultant performance

Stage 3 – Follow-Up Management

- One of the most critical processes.
- Most vehicle sales happen after multiple follow-ups.
- Data Created
- Call Date
- Call Duration
- Follow-Up Count
- Customer Response
- Next Follow-Up Date
- Consultant Name
- Remarks
- Example
- Follow-Up 1
- Customer not available
- Follow-Up 2
- Customer interested
- Follow-Up 3
- Test Drive Scheduled
- Follow-Up 4
- Booking Done

Analytical Queries

- How many follow-ups are required before booking?
- Which consultant converts fastest?
- Which lead source requires maximum follow-ups?

Stage 4 – Test Drive Process

Test drives significantly increase conversion probability.

Data Generated

Test_Drive_ID

Customer_ID

Vehicle_Model

Test_Drive_Date

Consultant

Feedback

Result

Feedback Examples

Liked Vehicle

Need Finance Option

Need Family Approval

Interested In Different Variant

Not Interested

KPIs

Test Drives Conducted

Test Drive Conversion %

Bookings After Test Drive

Management Decision

If test drive conversion is low:

Investigate

- Consultant training
- Vehicle availability
- Customer experience

Stage 5 – Booking Process

Booking indicates serious customer intent.

Data Generated

Booking_ID

Customer_ID

Booking_Date

Booking_Amount

Vehicle_Model

Variant

Color

Expected_Delivery_Date

Consultant

Importance

Booking data drives:

- Inventory planning
- Revenue forecasting
- Production requirements

Example

Customer Books:

Creta SX

White Color

₹25,000 Booking Amount

Expected Delivery:

20 June

Stage 6 – EDP Department Workflow

EDP is one of the most data-intensive departments.

Responsibilities:

- GDMS Operations
- Vehicle Allocation
- Inventory Management
- Stockyard Management
- Hyundai Coordination

Data Received

Booking Data

Vehicle Requirement

Variant

Color

Delivery Commitment

Data Created

VIN Number

Chassis Number

Engine Number

Allocation Date

Stock Status

Dispatch Status

Business Importance

Wrong allocation can:

- Delay delivery
- Cause cancellations
- Reduce customer satisfaction

Stage 7 – Inventory & Stockyard Management

- Inventory directly impacts sales.
- Inventory Categories
- Available Stock
- Allocated Stock
- Transit Stock
- Pending Stock
- Delivered Stock
- Data Generated
- Vin/ Engine No.
- Model
- Variant
- Color
- Arrival_Date
- Stock_Age
- Status

Key KPIs

- Inventory Turnover
- Stock Aging
- Days In Inventory
- Vehicle Availability %
- Management Decisions
- Which models should be stocked more?
- Which models are moving slowly?

Stage 8 – Finance Process

Many customers purchase through loans.

Data Collected

PAN

Aadhaar

Salary Slip

Bank Statements

Income Proof/Salary Slip/ITR

GST/MSME

Data Generated

Loan Amount

Bank Name

Interest Rate

EMI

Approval Status

Approval Data

Business Queries

Which bank approves fastest?

Which bank gives best conversion?

Average approval time?

KPI

Loan Approval Rate

Finance Conversion Rate

Stage 9 – Insurance Department Workflow

Every vehicle requires insurance.

Data Received

Customer Information

Vehicle Information

VIN

Chassis Number

Data Created

Policy Number

Insurance Company

Premium Amount

Start Date

Expiry Date

Claim Status

Renewal Status

Revenue Opportunity

Insurance is a major profit center.

Revenue Sources:

- New Policies
- Renewals
- Claims Support

Stage 10 – Accounts Department Workflow

Accounts acts as the financial backbone.

Data Received

Vehicle Cost

Booking Amount

Insurance Charges

Accessories Charges

Finance Information

RTO Charges

GST

Data Generated

Invoice

Receipt

Payment Status

Outstanding Amount

Tax Liability

Revenue Reports

Management Uses

Daily Revenue

Monthly Revenue

Profitability Reports

GST Reporting

Stage 11 – RTO Registration Workflow

Vehicle cannot be legally driven without registration.

Data Received

Invoice

Insurance

Vehicle Details

Customer Documents

Data Generated

Registration Number

RC Details

Tax Payment

Registration Date

Business Impact

Delays affect customer delivery timelines.

Stage 12 – Vehicle Delivery Process

Vehicle delivery is the moment of customer ownership transfer.

Data Received

Allocated Vehicle

Invoice

Insurance

Registration

Data Generated

Delivery Date

Delivery Checklist

Customer Acknowledgement

Photographs

Delivery Status

KPI

Booking-to-Delivery Time

Delivery Success Rate

Stage 13 – CRE Process

CRE maintains customer engagement after booking and delivery.

According to dealership operations:

Responsibilities

GDMS Follow-Up Download

Sales Follow-Up Distribution

Telegroup Management

WhatsApp Group Management

Customer Feedback

Complaint Handling

Cold Calling

Customer Retention

Data Generated

Feedback Score

Complaint Type

Resolution Time

Customer Rating

Satisfaction Score

KPI

NPS Score

Customer Satisfaction

Complaint Closure Rate

Stage 14 – Service, Renewal & Customer Lifetime Value

The relationship does not end after delivery.

It enters the most profitable phase.

Service Data

Service Date

KM Reading

Complaint

Repair Cost

Labour Cost

Parts Cost

Customer Rating

Renewal Data

Insurance Renewal

Warranty Renewal

RSA Renewal

Repeat Purchase Data

Referral Customer

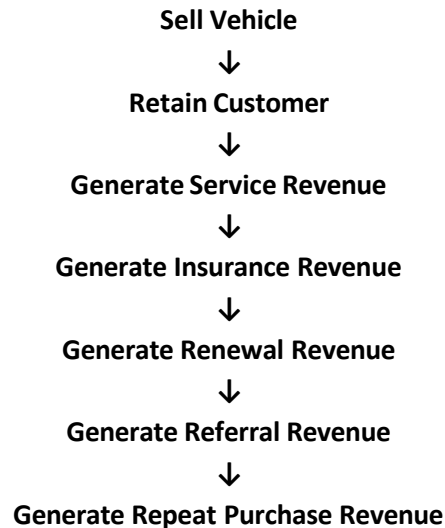
Exchange Vehicle

New Hyundai Purchase

Ultimate Business Objective

Acquire Customer





4.1 Data Architecture & Database Schema

4.1.1 Relational Database Schema Design

To centralize operations for **Universal Hyundai Shri Gurukripa Automobiles LLP** and **Sell Any Car**, we propose a unified relational database schema. This schema consolidates data from GDMS, HS Smart, Tally Prime, and Paymate into a single data warehouse (PostgreSQL) to support reporting, dashboards, and advanced analytics.

ER Diagram

```

EMPLOYEES ||--o{ ENQUIRIES : "handles"
EMPLOYEES ||--o{ TEST_DRIVES : "conducts"
EMPLOYEES ||--o{ VEHICLE_ALLOCATIONS : "allocates"
EMPLOYEES ||--o{ INVOICES : "billing_by"

CUSTOMERS ||--o{ ENQUIRIES : "makes"
CUSTOMERS ||--o{ BOOKINGS : "submits"
CUSTOMERS ||--o{ INVOICES : "pays"
CUSTOMERS ||--o{ INSURANCE_POLICIES : "owns"

VEHICLES ||--o| VEHICLE_ALLOCATIONS : "linked_to"
VEHICLES ||--o| BOOKINGS : "booked"
VEHICLES ||--o{ INVOICES : "billed_in"
VEHICLES ||--o{ REPAIR_ORDERS : "serviced"

ENQUIRIES ||--o| BOOKINGS : "converts_to"
BOOKINGS ||--o| VEHICLE_ALLOCATIONS : "triggers"
BOOKINGS ||--o| INVOICES : "billed_by"

INVOICES ||--o{ INVOICE_ITEMS : "contains"
INVOICES ||--o{ PAYMENTS : "reconciled_by"
INVOICES ||--o| INSURANCE_POLICIES : "insured_by"

REPAIR_ORDERS ||--o{ REPAIR_ORDER_ITEMS : "contains"
REPAIR_ORDERS ||--o{ PAYMENTS : "reconciled_by"
  
```



```
USED_CAR_EVALUATIONS ||--o| VEHICLES : "evaluates"
USED_CAR_EVALUATIONS ||--o| CUSTOMERS : "requested_by"
```

4.2 Data Dictionary

4.2.1 Table: employees

Stores master information for all dealership staff, including sales consultants, service technicians, and managers.

Column Name	Data Type	Key	Nullable	Default	
employee_id	VARCHAR(50)	PK	No	-	Unique identifier (EMP001)
first_name	VARCHAR(100)	-	No	-	Employee's first name
last_name	VARCHAR(100)	-	No	-	Employee's last name
Role	VARCHAR(50)	-	No	-	Employee's role: Sales Consultant, CRE, etc.
Department	VARCHAR(50)	-	No	-	Department (Sales, Service, Insurance)
Email	VARCHAR(150)	-	Yes	NULL	Corporate email address
phone_number	VARCHAR(15)	-	No	-	Mobile phone number
hire_date	DATE	-	No	CURRENT_DATE	Date employee was hired
Status	VARCHAR(20)	-	No	'ACTIVE'	Status of employee

4.2.2 Table: customers

Stores customer profile records captured via HSSmart or GDMS.

Column Name	Data Type	Key	Nullable	Default	Description
customer_id	VARCHAR(50)	PK	No	-	Unique identifier for the customer (e.g., CUST0001).
first_name	VARCHAR(100)	-	No	-	Customer's first name.
last_name	VARCHAR(100)	-	No	-	Customer's last name.
phone_number	VARCHAR(15)	-	No	-	Customer's primary mobile number.
email	VARCHAR(150)	-	Yes	NULL	Customer's email address.

Column Name	Data Type	Key	Nullable	Default	Description
address_line1	VARCHAR(255)	-	No	-	Primary address line.
city	VARCHAR(100)	-	No	-	City of residence.
postal_code	VARCHAR(10)	-	No	-	Pin code / ZIP code.
pan_card	VARCHAR(10)	-	Yes	NULL	Customer's Permanent Account Number (for tax validation).
created_at	TIMESTAMP	-	No	CURRENT_TIMESTAMP	Record creation timestamp.

4.2.3 Table: vehicles

Stores details of all vehicles in inventory (new and pre-owned) and service histories.

Column Name	Data Type	Key	Nullable	Default	Description
Vin	VARCHAR(17)	PK	No	-	Vehicle Identification Number
engine_number	VARCHAR(30)	-	No	-	Unique engine block identifier
model_name	VARCHAR(100)	-	No	-	Vehicle model (e.g., Creta, Ven

4.2.4 Table: enquiries

Logs all customer leads captured in HSSmart or via web sources.

Column Name	Data Type	Key	Nullable	Default	Description
enquiry_id	VARCHAR(50)	PK	No	-	Unique Enquiry ID generated in GDMS/HSSmart.
customer_id	VARCHAR(50)	FK	No	-	Foreign key linking to the customers table.
employee_id	VARCHAR(50)	FK	No	-	Foreign key linking to the employees table

Column Name	Data Type	Key	Nullable	Default	Description
source	VARCHAR(50)	-	No	-	(Salesperson). Lead source (WALK_IN, REFERENCE, DIGITAL_AD, COLD_CALL).
status	VARCHAR(30)	-	No	'OPEN'	Lead status (OPEN, TEST_DRIVE, BOOKED, LOST, CLOSED).
model_interest	VARCHAR(100)	-	No	-	Model customer is interested in.
enquiry_date	TIMESTAMP	-	No	CURRENT_TIMESTAMP	Date of lead entry.
last_contact_date	TIMESTAMP	-	Yes	NULL	Date of last follow-up activity.

4.2.5 Table: test_drives

Logs test drive activities for compliance and target audits.

Column Name	Data Type	Key	Nullable	Default	Description
test_drive_id	VARCHAR(50)	PK	No	-	Unique identifier for test drive event.
enquiry_id	VARCHAR(50)	FK	No	-	Foreign key linking to enquiries.
vin	VARCHAR(17)	FK	No	-	Foreign key linking to vehicles (Demo car VIN).
employee_id	VARCHAR(50)	FK	No	-	Foreign key linking to employees (Sales Consultant).

Column Name	Data Type	Key	Nullable	Default	Description
start_time	TIMESTAMP	-	No	-	Test drive start timestamp.
end_time	TIMESTAMP	-	Yes	NULL	Test drive end timestamp.
start_km	INTEGER	-	No	-	Odometer reading at start of drive.
end_km	INTEGER	-	Yes	NULL	Odometer reading at end of drive.
feedback	TEXT	-	Yes	NULL	Customer feedback rating/comments.

4.2.6 Table: bookings

Logs vehicle orders placed by customers, supported by a deposit.

Column Name	Data Type	Key	Nullable	Default	Description
booking_id	VARCHAR(50)	PK	No	-	Unique booking reference number.
enquiry_id	VARCHAR(50)	FK	No	-	Foreign key linking to enquiries.
customer_id	VARCHAR(50)	FK	No	-	Foreign key linking to customers.
vin_requested	VARCHAR(17)	-	Yes	NULL	Requested chassis number (or generic model code if not allocated).
booking_amount	DECIMAL(10,2)	-	No	-	Advance deposit paid by customer.
booking_date	DATE	-	No	CURRENT_DATE	Date customer signed

Column Name	Data Type	Key	Nullable	Default	Description
					booking form.
payment_method	VARCHAR(50)	-	No	-	Deposit payment method (Paymate, UPI, Check, Cash).
status	VARCHAR(20)	-	No	'PENDING'	Status (PENDING, ALLOCATED, CANCELLED, FULFILLED).

4.2.7Table: vehicle_allocations

Tracks the allocation of specific vehicles in stockyard inventory to bookings.

Column Name	Data Type	Key	Nullable	Default	Description
allocation_id	VARCHAR(50)	PK	No	-	Unique identifier for allocation
booking_id	VARCHAR(50)	FK	No	-	Foreign key linking to bookings
Vin	VARCHAR(17)	FK	No	-	FK to vehicles (Chassis Number)
employee_id	VARCHAR(50)	FK	No	-	FK to employees (Employee ID)
allocation_date	TIMESTAMP	-	No	CURRENT_TIMESTAMP	Date allocated in stockyard
Status	VARCHAR(20)	-	No	'ALLOCATED'	Status (ALLOCATED, RETAIL_BILLED)

4.2.8Table: invoices

Stores customer sales invoices exported to Tally Prime.

Column Name	Data Type	Key	Nullable	Default	Description
invoice_id	VARCHAR(50)	PK	No	-	Unique Invoice ID

Column Name	Data Type	Key	Nullable	Default	Description (from GDMS).
booking_id	VARCHAR(50)	FK	No	-	Foreign key linking to bookings.
customer_id	VARCHAR(50)	FK	No	-	Foreign key linking to customers.
vin	VARCHAR(17)	FK	No	-	Foreign key linking to vehicles.
employee_id	VARCHAR(50)	FK	No	-	Foreign key linking to employees (Invoicing EDP).
invoice_date	DATE	-	No	CURRENT_DATE	Date of billing in GDMS.
ex_showroom	DECIMAL(12,2)	-	No	-	Base price of vehicle.
registration_charge	DECIMAL(10,2)	-	No	0.00	Registration charges.
gst_collected	DECIMAL(10,2)	-	No	-	Central C State Goods and Services Tax (GST).
tcs_collected	DECIMAL(10,2)	-	No	0.00	Tax Collected at Source (TDS/TCS).
total_amount	DECIMAL(12,2)	-	No	-	Invoice total amount.

4.2.G Table: invoice_items

Tracks individual line items associated with an invoice, such as value-added services, extended warranties, and accessories.

Column Name	Data Type	Key	Nullable	Default	Description
item_id	SERIAL	PK	No	-	Auto-incrementing primary key
invoice_id	VARCHAR(50)	FK	No	-	Foreign key linking to invoice
item_type	VARCHAR(30)	-	No	-	Type (ACCESSORY, WARRANTY, VAS_WATS, SOT, RSA)
item_description	VARCHAR(255)	-	No	-	Item description

Amount	DECIMAL(10,2)	-	No	-	Retail cost in INR
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4.2.10 Table: insurance_policies

Tracks insurance policies sold at retail or renewed via cold-calling.

Column Name	Data Type	Key	Nullable	Default	Description
policy_number	VARCHAR(100)	PK	No	-	Insurance policy number
invoice_id	VARCHAR(50)	FK	Yes	NULL	FK to invoices (if sold at r
customer_id	VARCHAR(50)	FK	No	-	FK to customers
Vin	VARCHAR(17)	FK	No	-	FK to vehicles (Insured ve
provider_name	VARCHAR(100)	-	No	-	Insurer name
policy_type	VARCHAR(20)	-	No	-	Type (NEW, RENEWAL)
premium_amount	DECIMAL(10,2)	-	No	-	Insurance premium INR
commission_earned	DECIMAL(10,2)	-	No	-	Dealership commission
start_date	DATE	-	No	-	Coverage start date
expiry_date	DATE	-	No	-	Coverage expiry date

4.2.11 Table: payments

Tracks payments processed via Paymate or bank transfers, reconciled against customer invoices.

Column Name	Data Type	Key	Nullable	Default	Description
payment_id	VARCHAR(50)	PK	No	-	Payment reference
invoice_id	VARCHAR(50)	FK	Yes	NULL	FK to invoices (if inv
booking_id	VARCHAR(50)	FK	Yes	NULL	FK to bookings (if bo
amount_paid	DECIMAL(12,2)	-	No	-	Amount received in
payment_date	TIMESTAMP	-	No	CURRENT_TIMESTAMP	Payment processing
payment_status	VARCHAR(20)	-	No	'RECONCILED'	Status (PENDING, RE BOUNCED)

4.2.12 Table: repair_orders

Column Name	Data Type	Key	Nullable	Default	Description
ro_number	VARCHAR(50)	PK	No	-	Repair Order ID generated in G
Vin	VARCHAR(17)	FK	No	-	Foreign key linking to vehicles
customer_id	VARCHAR(50)	FK	No	-	Foreign key linking to customer
technician_id	VARCHAR(50)	FK	No	-	Foreign key linking to employeee (Tech)
open_date	TIMESTAMP	-	No	-	Service check-in timestamp

close_date	TIMESTAMP	-	Yes	NULL	Service release/billing timestamp
Mileage	INTEGER	-	No	-	Vehicle odometer reading at check-in
labor_cost	DECIMAL(10,2)	-	No	0.00	Labor billable cost in INR
parts_cost	DECIMAL(10,2)	-	No	0.00	Parts billable cost in INR
Status	VARCHAR(20)	-	No	'OPEN'	Status (OPEN, WIP, BILLED, CLOSED)

4.2.13 Table: repair_order_items

Logs parts and lubricants issued during vehicle servicing.

Column Name	Data Type	Key	Nullable	Default	Description
ro_number	VARCHAR(50)	PK	No	-	Repair Order ID from GDMS
vin	VARCHAR(17)	FK	No	-	Foreign key linking to vehicles
customer_id	VARCHAR(50)	FK	No	-	FK linking to customers
technician_id	VARCHAR(50)	FK	No	-	FK to employees (Service Tech)
open_date	TIMESTAMP	-	No	-	Service check-in timestamp
close_date	TIMESTAMP	-	Yes	NULL	Service release/billing timestamp
mileage	INTEGER	-	No	-	Odometer reading at check-in
labor_cost	DECIMAL(10,2)	-	No	0.00	Labor billable cost in INR
parts_cost	DECIMAL(10,2)	-	No	0.00	Parts billable cost in INR
status	VARCHAR(20)	-	No	'OPEN'	Status (OPEN, WIP, BILLED, CLOSED)

4.2.14 Table: used_car_evaluations

Logs vehicle evaluations conducted by Sell Any Car.

Column Name	Data Type	Key	Nullable	Default	Description
evaluation_id	VARCHAR(50)	PK	No	-	Unique used car evaluation ID
customer_id	VARCHAR(50)	FK	No	-	Foreign key linking to customers
vin	VARCHAR(17)	FK	Yes	NULL	Trade-in vehicle VIN (if available)
make	VARCHAR(50)	-	No	-	Vehicle manufacturer
model	VARCHAR(50)	-	No	-	Vehicle model
year_manufactured	INTEGER	-	No	-	Year of production
refurbishment_est	DECIMAL(10,2)	-	No	0.00	Est. cost of repairs for resale
offer_amount	DECIMAL(10,2)	-	Yes	NULL	Price offered to customer
status	VARCHAR(30)	-	No	'EVALUATED'	Status (EVALUATED, PURCHASED, REJECTED)

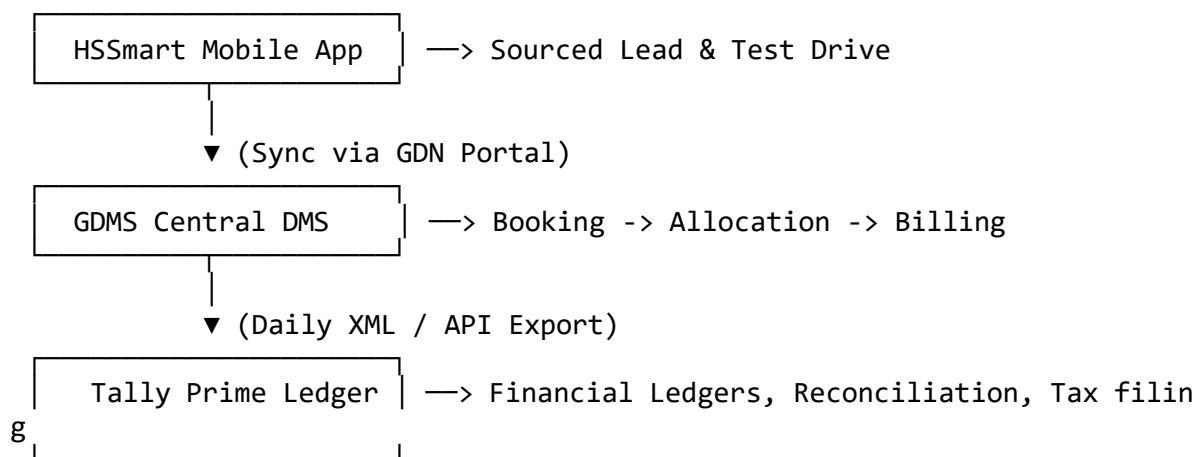
Table 4.2.15

Table Name	Column Name	Data Type	Null Status	Business
Dim_Customer	CustomerID	INT	NOT NULL	Unique identifier for each customer
Dim_Customer	FirstName	VARCHAR(100)	NOT NULL	Customer's first name
Dim_Customer	LastName	VARCHAR(100)	NOT NULL	Customer's last name
Dim_Customer	Email	VARCHAR(255)	NOT NULL	Primary email address
Dim_Customer	PhoneNumber	VARCHAR(20)	NOT NULL	Contact phone number
Dim_Customer	Address	VARCHAR(500)	NULL	Residential address
Dim_Customer	City	VARCHAR(50)	NOT NULL	City of residence
Dim_Customer	State	VARCHAR(50)	NOT NULL	State code
Dim_Customer	ZipCode	VARCHAR(10)	NULL	Postal code
Dim_Customer	RegistrationDate	DATETIME	NOT NULL	Date customer registered
Dim_Customer	LastPurchaseDate	DATETIME	NULL	Date of most recent vehicle purchase
Dim_Customer	CustomerType	VARCHAR(50)	NOT NULL	Individual, Corporate, or Other
Dim_Customer	IsActive	BIT	NOT NULL	1=Active, 0=Inactive customer
Dim_Vehicle	VehicleID	INT	NOT NULL	Unique identifier for each vehicle
Dim_Vehicle	VIN	VARCHAR(20)	NOT NULL	Vehicle Identification Number
Dim_Vehicle	Model	VARCHAR(100)	NOT NULL	Hyundai model name
Dim_Vehicle	ModelYear	INT	NOT NULL	Manufacturing year
Dim_Vehicle	VariantName	VARCHAR(100)	NOT NULL	Trim/variant level
Dim_Vehicle	EngineType	VARCHAR(50)	NOT NULL	Fuel type
Dim_Vehicle	TransmissionType	VARCHAR(50)	NOT NULL	Transmission type
Dim_Vehicle	FuelTankCapacity	DECIMAL(10,2)	NOT NULL	Tank capacity in liters
Dim_Vehicle	MillageRating	DECIMAL(10,2)	NOT NULL	Fuel efficiency km/l
Dim_Vehicle	Color	VARCHAR(50)	NOT NULL	Exterior color
Dim_Vehicle	RegistrationNumber	VARCHAR(20)	NOT NULL	License plate number
Dim_Vehicle	PurchasePrice	DECIMAL(12,2)	NOT NULL	Sale price in INR
Dim_Vehicle	ManufactureDate	DATETIME	NOT NULL	Date vehicle manufactured
Dim_Vehicle	WarrantyExpiryDate	DATETIME	NULL	Warranty expiration date
Fact_Service	ServiceID	INT	NOT NULL	Unique service record ID
Fact_Service	CustomerID	INT	NOT NULL	Foreign key to Dim_Customer
Fact_Service	VehicleID	INT	NOT NULL	Foreign key to Dim_Vehicle
Fact_Service	ServiceDate	DATETIME	NOT NULL	Date and time service started
Fact_Service	ServiceType	VARCHAR(100)	NOT NULL	Service category
Fact_Service	OdometerReading	INT	NOT NULL	Vehicle mileage at service
Fact_Service	Technician	VARCHAR(100)	NOT NULL	Assigned service technician
Fact_Service	ServiceCost	DECIMAL(10,2)	NOT NULL	Total service cost in INR
Fact_Service	PartsUsed	VARCHAR(500)	NULL	Description of replaced parts
Fact_Service	IsWarrantyCovered	BIT	NOT NULL	1=Warranty, 0=Paid
Fact_Service	ServiceCompletionDate	DATETIME	NOT NULL	When service was completed
Fact_Service	NextServiceDueDate	DATETIME	NULL	Recommended next service date
Fact_Insurance	PolicyID	INT	NOT NULL	Unique insurance policy ID
Fact_Insurance	CustomerID	INT	NOT NULL	Foreign key to Dim_Customer

Fact_Insurance	VehicleID	INT	NOT NULL	Foreign key to Dim_Veh
Fact_Insurance	InsuranceProvider	VARCHAR(100)	NOT NULL	Insurance company
Fact_Insurance	PolicyType	VARCHAR(50)	NOT NULL	Coverage type
Fact_Insurance	PremiumAmount	DECIMAL(12,2)	NOT NULL	Annual premium in INR
Fact_Insurance	PolicyStartDate	DATETIME	NOT NULL	Policy effective date
Fact_Insurance	PolicyExpiryDate	DATETIME	NOT NULL	Policy expiration date
Fact_Insurance	ClaimHistory	INT	NULL	Number of claims filed
Fact_Insurance	IsPolicyActive	BIT	NOT NULL	1=Active, 0=Expired

4.3 Data Lineage & Integrations

The data lifecycle follows a structured lineage from initial touchpoint (HSSmart/GDMS) to final financial auditing (Tally Prime).



1. **Ingestion (H Smart):** Leads are logged on the mobile CRM by Sales Consultants.
2. **Processing (GDMS):** Bookings, vehicle allocations, workshop Repair Orders, and invoicing details are updated in GDMS.

Finance Reconciliations (Tally Prime): Financial entries are imported from GDMS into Tally Prime daily. Reconciliations are performed against bank receipts, debtor accounts, and tax liabilities.

4.4 Dealership Data Inventory

Before analysis starts, a Data Analyst must identify all available datasets.

Primary Data Sources

GDMS

Contains:

- Leads
- Enquiries
- Follow-Ups

- Bookings
- Deliveries

Accounts Software

Contains:

- Invoices
- Receipts
- Revenue
- Outstanding Amounts

Insurance Records

Contains:

- Policies
- Renewals
- Claims

Service System

Contains:

- Service Visits
- Parts Usage
- Labour Revenue

WhatsApp Groups

Contains:

- Customer Communication
- Booking Updates
- Delivery Coordination

Excel Reports

Contains:

- Daily Sales Reports
- Team Performance
- MIS Reports

4.5 Data Dictionary – Lead Table

A Data Dictionary defines every important field.

Table: Leads

Column	Description
Lead_ID	Unique Lead Number
Customer_Name	Customer Name
Mobile_Number	Contact Number
Lead_Source	Source of Lead
Model_Interested	Interested Vehicle
Budget	Customer Budget
Lead_Date	Lead Creation Date
Consultant	Assigned Sales Consultant
Status	Hot / Warm / Cold / Lost

4.5.1 Business Importance

This table measures customer acquisition performance.

Data Dictionary – Booking Table

4.5.1.1 Table: Bookings

Booking_ID	Customer_ID	Booking_Date	Model	Variant	Color	Booking_Amount
BK001	C001	2024-01-15	Creta	1.5L DCT	Black	₹2,50,000
BK002	C002	2024-01-18	i20	1.2L Manual	Silver	₹1,80,000
BK003	C003	2024-01-22	Venue	1.2L IVT	White	₹2,00,000
BK004	C004	2024-02-01	Verna	1.5L DCT	Red	₹2,20,000
BK005	C005	2024-02-05	Creta	1.4L Turbo	Blue	₹2,80,000

4.5.2 Business Importance

This table drives revenue forecasting.

Data Dictionary – Vehicle Inventory Table

4.5.2.1 Table: Inventory

Column	Description
VIN	Vehicle Identification Number
Chassis_No	Chassis Number
Engine_No	Engine Number
Model	Vehicle Model
Variant	Vehicle Variant
Color	Vehicle Color
Arrival_Date	Stock Arrival Date
Allocation_Date	Allocation Date
Status	Available / Allocated

4.5.3 Business Importance

This table supports inventory management.

Data Dictionary – Insurance Table

4.5.3.1 Table: Insurance

Here is the insurance data schema layout converted into a clean, markdown-compatible table format, ready to be copied and pasted directly into your Microsoft Excel data dictionary sheet:

Column	Description
Policy_No	Policy Number
Customer_ID	Customer Reference
Vehicle_ID	Vehicle Reference
Premium	Premium Amount
Provider	Insurance Company
Start_Date	Policy Start
End_Date	Policy Expiry
Renewal_Status	Active/Expired

Data Dictionary – Service Table

4.5.3.2 Table: Service

Column	Description
Service_ID	Service Number
Vehicle_No	Vehicle Number
Customer_ID	Customer
Service_Date	Service Date

Column	Description
KM_Reading	Odometer Reading
Labour_Cost	Labour Charges
Parts_Cost	Parts Charges
Total_Cost	Total Bill

4.5.4 Business Importance

This table measures customer retention and service profitability

KPI Framework – Sales Department

A KPI is a measurable value used to evaluate performance.

Lead KPIs

1. Total Leads
2. Daily Leads
3. Monthly Leads
4. Lead Growth Rate
5. Lead Source Share

Enquiry KPIs

6. Total Enquiries
7. Qualified Enquiries
8. Lost Enquiries
9. Hot Leads
10. Warm Leads

Conversion KPIs

11. Lead Conversion %
12. Test Drive Conversion %
13. Booking Conversion %

14. Delivery Conversion %

15. Revenue Conversion %

4.6 Data Analyst Conclusion

16. The dealership's real asset is not the vehicle inventory.

17. The dealership's most valuable asset is its customer data, operational data, financial data, and service history data. These datasets allow management to predict demand, improve conversions, reduce cancellations, optimize inventory, increase profitability, and maximize customer lifetime value.

5.KPI Framework – Inventory Department

5.1.1 Inventory KPIs

- 16. Total Vehicles In Stock
- 17. Available Stock
- 18. Allocated Stock
- 19. Transit Stock
- 20. Average Stock Age

5.1.2 Performance KPIs

- 21. Inventory Turnover
- 22. Allocation Time
- 23. Vehicle Availability %
- 24. Days In Inventory
- 25. Slow Moving Inventory %

5.1.3 Risk KPIs

- 26. Overstock %
- 27. Understock %
- 28. Cancellation Due To Non-Availability %

KPI Framework – Insurance Department

- 29. Total Policies Sold
- 30. Total Premium Revenue
- 31. Insurance Conversion %
- 32. Insurance Revenue Per Vehicle
- 33. Policy Renewal Rate
- 34. Expired Policy Count
- 35. Claim Count
- 36. Average Premium
- 37. Insurance Profitability
- 38. Policy Retention %

KPI Framework – Service Department

- 39. Vehicles Serviced

- 40. Repeat Service Customers
- 41. Service Revenue
- 42. Labour Revenue
- 43. Parts Revenue
- 44. Average Bill Value
- 45. Customer Retention Rate
- 46. Repeat Visit Rate
- 47. Service Satisfaction Score
- 48. Complaint Rate

KPI Framework – CRE Department

- 49. Follow-Ups Completed
- 50. Pending Follow-Ups
- 51. Feedback Collection Rate
- 52. Complaint Closure Rate
- 53. Average Resolution Time
- 54. Customer Satisfaction Score
- 55. NPS Score
- 56. Referral Rate
- 57. Repeat Purchase Rate

Management Reporting Structure

- 58. Different managers require different reports.

5.1.4 KPI 1: Insurance Policy Renewal Rate

Category: Insurance Performance

Definition: Measures the percentage of vehicle insurance policies successfully renewed before expiry.

Calculation:

$(\text{Policies Renewed} \div \text{Total Policies Due for Renewal}) \times 100$

Data Source: Insurance Renewal Register, GDMS Insurance Module

Target: Industry Benchmark: 80% | Dealership Target: 90%

Business Implication: Higher renewal rates increase commission income, improve customer retention, and strengthen dealership profitability.

Action Triggers: If renewal rate falls below 75% for two consecutive months, management review and customer follow-up campaigns are initiated.

Historical Context: Renewal rates generally increase during year-end and festive periods due to promotional campaigns.

Improvement Strategy: Automated renewal reminders, customer education, and proactive telecalling.

5.1.5 KPI 2: Insurance Revenue Achievement

Category: Insurance Performance

Definition: Measures achievement against monthly insurance revenue targets.

Calculation:

$(\text{Actual Insurance Revenue} \div \text{Target Insurance Revenue}) \times 100$

Data Source: Insurance Revenue Transactions

Target: 100% Monthly Achievement

Business Implication: Directly contributes to dealership profitability and commission earnings.

Action Triggers: Revenue achievement below 90% requires performance review.

Historical Context: Revenue typically peaks during vehicle delivery seasons.

Improvement Strategy: Cross-selling, renewal campaigns, and policy bundling.

5.1.6 KPI 3: Insurance Claim Settlement Rate

Category: Insurance Performance

Definition: Measures percentage of submitted claims successfully settled.

Calculation:

$(\text{Settled Claims} \div \text{Total Claims Submitted}) \times 100$

Data Source: Insurance Claim Register

Target: 95%

Business Implication: Improves customer trust and dealership reputation.

Action Triggers: Settlement rate below 90% requires insurer escalation.

Historical Context: Higher claim volumes occur during monsoon and accident-prone seasons.

Improvement Strategy: Documentation accuracy and insurer coordination.

5.1.7 KPI 4: Insurance Conversion Rate

Category: Insurance Performance

Definition: Measures percentage of vehicle buyers purchasing insurance through the dealership.

Calculation:

$(\text{Insurance Policies Sold} \div \text{Vehicle Deliveries}) \times 100$

Data Source: Vehicle Sales & Insurance Records

Target: 85%

Business Implication: Generates additional dealership income.

Action Triggers: Conversion below 75% triggers sales team coaching.

Historical Context: Conversion increases during promotional periods.

Improvement Strategy: Bundle offers and sales consultant training.

5.1.8 KPI 5: Average Insurance Premium Value

Category: Insurance Performance

Definition: Measures average premium amount per policy sold.

Calculation:

$\text{Total Premium Value} \div \text{Number of Policies Sold}$

Data Source: Insurance Transactions

Target: ₹25,000+

Business Implication: Higher premiums generate increased commission revenue.

Action Triggers: Continuous decline requires policy mix analysis.

Historical Context: Premium values increase with premium vehicle sales.

Improvement Strategy: Upselling add-on covers.

5.1.9 KPI 6: Service Labour Revenue

Category: Service Performance

Definition: Measures revenue earned from service labour activities.

Calculation:
Total Labour Revenue

Data Source: Service Invoices

Target: Monthly Budget Achievement

Business Implication: Major contributor to service department profitability.

Action Triggers: Revenue below target requires productivity review.

Historical Context: Peaks before holiday travel seasons.

Improvement Strategy: Improve workshop utilization.

5.1.10 KPI 7: Service Bay Utilization

Category: Service Performance

Definition: Measures percentage utilization of workshop service bays.

Calculation:
$$(\text{Actual Bay Hours Used} \div \text{Available Bay Hours}) \times 100$$

Data Source: Workshop Management System

Target: 85%

Business Implication: Maximizes workshop capacity and revenue.

Action Triggers: Utilization below 70% requires appointment optimization.

Historical Context: Higher utilization during seasonal service campaigns.

Improvement Strategy: Better scheduling and customer reminders.

5.1.11 KPI 8: First Time Fix Rate (FTFR)

Category: Service Performance

Definition: Measures percentage of vehicles repaired correctly on first visit.

Calculation:

$(\text{Vehicles Fixed First Time} \div \text{Total Repair Orders}) \times 100$

Data Source: Service Repair Orders

Target: 95%

Business Implication: Improves customer satisfaction and reduces rework.

Action Triggers: FTFR below 90% requires technician assessment.

Historical Context: Stable in mature workshops.

Improvement Strategy: Technical training and diagnostic tools.

5.1.12 KPI 9: Customer Satisfaction Score (Service)

Category: Service Performance

Definition: Measures customer satisfaction after service completion.

Calculation:

$(\text{Average Survey Score} \div \text{Maximum Score}) \times 100$

Data Source: Customer Feedback Surveys

Target: 90%

Business Implication: Influences retention and manufacturer ratings.

Action Triggers: Score below 85% requires customer follow-up.

Historical Context: Scores improve with process standardization.

Improvement Strategy: Faster turnaround and service quality improvements.

5.1.13 KPI 10: Average Service Turnaround Time

Category: Service Performance

Definition: Measures average time taken to complete service jobs.

Calculation:

Total Service Hours ÷ Number of Jobs

Data Source: Service Job Cards

Target: < 8 Hours

Business Implication: Faster service increases customer satisfaction.

Action Triggers: Rising TAT requires workflow review.

Historical Context: Longer during peak workshop periods.

Improvement Strategy: Better manpower allocation.

5.1.14 KPI 11: New Vehicle Sales Achievement

Category: Sales Performance

Definition: Measures achievement against monthly vehicle sales targets.

Calculation:

(Actual Sales ÷ Sales Target) × 100

Data Source: Sales Transactions

Target: 100%

Business Implication: Directly impacts dealership revenue.

Action Triggers: Achievement below 90% triggers sales review.

Improvement Strategy: Lead management and promotions.

5.1.15 KPI 12: Lead Conversion Rate

Category: Sales Performance

Definition: Measures percentage of sales leads converted into bookings.

Calculation:

(Bookings ÷ Total Leads) × 100

Data Source: CRM Leads Database

Target: 20%

Business Implication: Indicates sales team effectiveness.

Action Triggers: Conversion below target requires lead quality analysis.

Improvement Strategy: Sales training and faster follow-up.

5.1.16 KPI 13: Average Revenue Per Vehicle

Category: Sales Performance

Definition: Measures average revenue generated per vehicle sold.

Calculation:

Total Sales Revenue ÷ Vehicles Sold

Data Source: Sales Invoices

Target: As per model mix strategy

Business Implication: Improves profitability.

Action Triggers: Revenue decline requires pricing review.

Improvement Strategy: Focus on premium variants.

5.1.17 KPI 14: Booking Cancellation Rate

Category: Sales Performance

Definition: Measures percentage of bookings cancelled before delivery.

Calculation:

(Cancelled Bookings ÷ Total Bookings) × 100

Data Source: Booking Register

Target: < 5%

Business Implication: High cancellations affect revenue forecasts.

Action Triggers: Rate above 5% requires customer retention efforts.

Improvement Strategy: Better communication and delivery commitment.

5.1.18 KPI 15: Finance Penetration Rate

Category: Sales Performance

Definition: Measures percentage of vehicle sales financed through dealership partners.

Calculation:

$(\text{Financed Sales} \div \text{Total Sales}) \times 100$

Data Source: Finance Records

Target: 65%

Business Implication: Increases finance commission earnings.

Action Triggers: Below-target performance requires finance promotions.

Improvement Strategy: Attractive finance schemes.

5.1.19 KPI 16: Employee Attrition Rate

Category: HR Performance

Definition: Measures percentage of employees leaving the organization.

Calculation:

$(\text{Employees Left} \div \text{Average Employee Strength}) \times 100$

Data Source: HRMS

Target: < 10%

Business Implication: High attrition increases recruitment costs.

Action Triggers: Rate above 10% requires HR intervention.

Improvement Strategy: Employee engagement programs.

5.1.20 KPI 17: Training Completion Rate

Category: HR Performance

Definition: Measures percentage of employees completing assigned training.

Calculation:

$(\text{Employees Trained} \div \text{Employees Assigned Training}) \times 100$

Data Source: Learning Management System

Target: 100%

Business Implication: Improves workforce productivity.

Action Triggers: Completion below 90% requires follow-up.

Improvement Strategy: Structured training calendar.

5.1.21 KPI 18: Employee Productivity Index

Category: HR Performance

Definition: Measures employee output against expected performance standards.

Calculation:

$(\text{Actual Output} \div \text{Standard Output}) \times 100$

Data Source: Departmental Performance Reports

Target: 100%

Business Implication: Indicates workforce efficiency.

Action Triggers: Productivity below target requires coaching.

Improvement Strategy: Performance monitoring and incentives.

5.1.22 KPI 19: Employee Attendance Rate

Category: HR Performance

Definition: Measures employee attendance performance.

Calculation:

$(\text{Present Days} \div \text{Total Working Days}) \times 100$

Data Source: Attendance Management System

Target: 95%

Business Implication: Impacts operational continuity.

Action Triggers: Attendance below 90% requires review.

Improvement Strategy: Attendance incentives.

5.1.23 KPI 20: Recruitment Fulfillment Rate

Category: HR Performance

Definition: Measures percentage of approved vacancies successfully filled.

Calculation:

$(\text{Filled Positions} \div \text{Approved Vacancies}) \times 100$

Data Source: Recruitment Tracker

Target: 95%

Business Implication: Ensures adequate staffing levels.

Action Triggers: Delayed hiring requires escalation.

Improvement Strategy: Talent pipeline development

5.1.24 Insurance Department

5.1.24.1 Department Overview

Purpose

The Insurance Department manages vehicle insurance sales, renewals, policy issuance, claims coordination, customer follow-ups, and revenue generation through insurance commissions.

Business Objectives

- Increase insurance penetration on vehicle sales.
- Improve renewal conversion rate.
- Reduce policy issuance turnaround time.
- Improve customer retention through renewals.
- Increase insurance commission revenue.
- Enhance customer satisfaction during claims processing.

5.1.24.2 Insurance Business Process Flow

Customer Purchase Vehicle

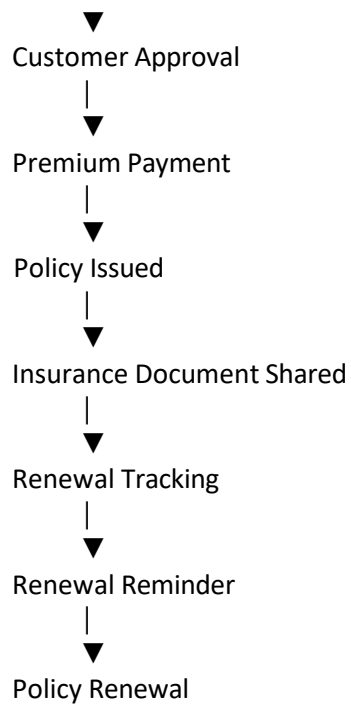


Insurance Requirement Identified

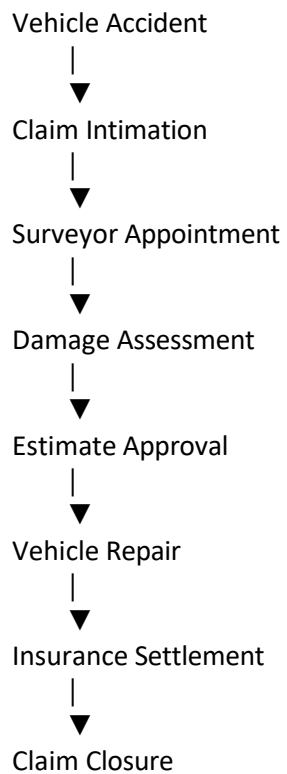


Policy Quotation Generated

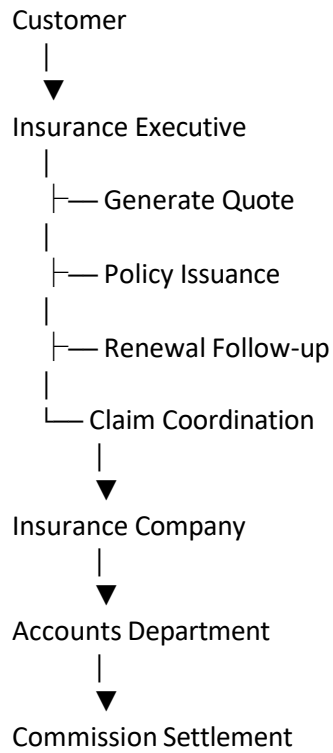




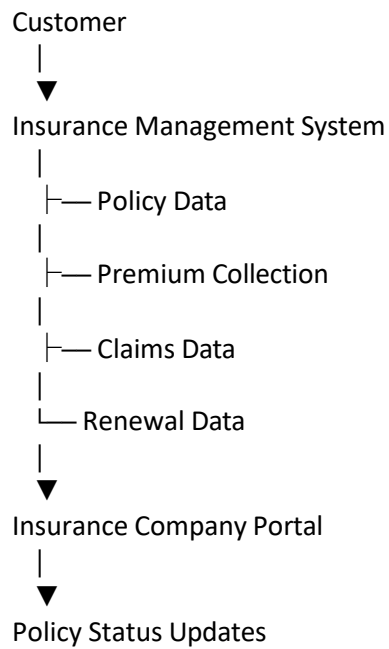
5.1.24.3 Insurance Claim Process Flow



5.1.24.4 Insurance Department Workflow Diagram



5.1.24.5 Insurance Data Flow Diagram (DFD)



5.1.24.6 Insurance Master Tables

Insurance_Policy_Master

Column Name	Description
Policy_ID	Unique Policy Number
Customer_ID	Customer Reference
Vehicle_ID	Vehicle Reference
Policy_Type	Comprehensive / Third Party
Insurance_Company	Insurance Provider
Policy_Start_Date	Policy Start Date
Policy_End_Date	Policy Expiry Date
Premium_Amount	Premium Amount
Coverage_Value	Insured Value
Policy_Status	Active/Expired

Insurance_Renewal_Master

Column Name	Description
Renewal_ID	Renewal Number
Policy_ID	Policy Reference
Renewal_Date	Renewal Date
Renewal_Premium	Renewal Premium
Renewal_Status	Completed/Pending

Insurance_Claim_Master

Column Name	Description
Claim_ID	Claim Number
Policy_ID	Policy Reference
Claim_Date	Claim Date
Claim_Type	Accident/Theft
Claim_Amount	Claimed Amount
Approved_Amount	Approved Amount
Claim_Status	Open/Closed

5.1.24.7 Insurance KPIs

KPI 1: Insurance Penetration %

Definition

Percentage of vehicle sales covered with insurance.

Formula

$(\text{Insured Vehicles} / \text{Total Vehicles Sold}) \times 100$

Target

95%

Business Impact

Higher penetration increases commission revenue.

KPI 2: Insurance Renewal Rate %

Formula

$(\text{Renewed Policies} / \text{Expired Policies}) \times 100$

Target

80%

Business Impact

Measures customer retention.

KPI 3: Policy Issuance TAT

Formula

Policy Issue Time - Payment Received Time

Target

< 2 Hours

Business Impact

Improves customer experience.

KPI 4: Claim Settlement Rate %

Formula

$(\text{Settled Claims} / \text{Total Claims}) \times 100$

Target

90%

KPI 5: Insurance Revenue

Formula

Sum(Insurance Commission)

Target

Monthly Growth > 10%

KPI 6: Renewal Reminder Effectiveness %

Formula

$(\text{Renewals After Reminder} / \text{Total Reminders Sent}) \times 100$

Target

60%

KPI 7: Average Premium Per Policy**Formula**

$\text{Total Premium Amount} / \text{Number of Policies}$

KPI 8: Claim Approval Rate %**Formula**

$(\text{Approved Claims} / \text{Total Claims}) \times 100$

KPI 9: Insurance Attach Rate**Formula**

$(\text{Policies Sold with Vehicle} / \text{Total Vehicle Sales}) \times 100$

KPI 10: Expired Policy Conversion Rate**Formula**

$(\text{Renewed Expired Policies} / \text{Total Expired Policies}) \times 100$

5.1.24.8 Insurance Reports**Daily Insurance Sales Report**

- Policies Issued
- Premium Collected
- Revenue Earned

Insurance Renewal Report

- Policies Due
- Renewals Completed
- Pending Renewals

Insurance Claims Report

- Open Claims
- Closed Claims
- Claim Amount

Insurance Revenue Report

- Commission Revenue
- Insurance Partner Revenue

Insurance Executive Performance Report

- Policies Sold
 - Renewals Closed
 - Revenue Generated
-

5.1.24.9 Insurance Dashboard

Executive Dashboard

Cards:

- Policies Issued Today
- Premium Collected
- Renewal Due
- Open Claims
- Commission Revenue

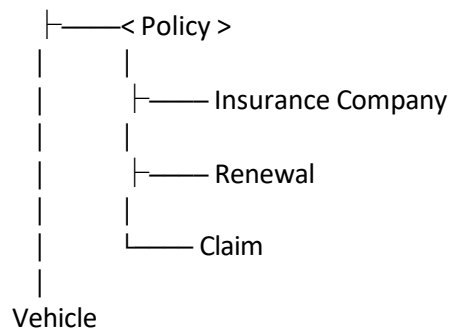
Charts:

- Monthly Insurance Revenue Trend
 - Renewal Conversion Trend
 - Claims Analysis
 - Insurance Company Wise Revenue
 - Executive Performance Analysis
-

5.1.24.10 Insurance ER Diagram

Customer

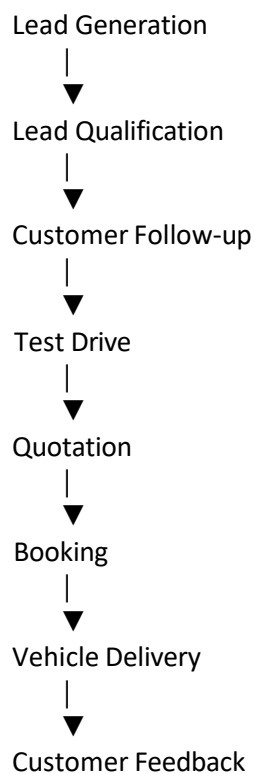
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5.1.25 Sales Department KPIs

The Sales Department is responsible for converting inquiries into vehicle sales while ensuring customer satisfaction and profitability. These KPIs help monitor lead management, sales efficiency, revenue generation, team productivity, and customer conversion effectiveness.

Sales KPI Monitoring Framework



Sales KPI Categories

Sales KPIs

|

- └─ Lead Management KPIs
 - └─ Conversion KPIs
 - └─ Revenue KPIs
 - └─ Productivity KPIs
 - └─ Customer Experience KPIs
 - └─ Profitability KPIs
-

KPI 1: Total Leads Generated

Category: Lead Management

Definition: Measures total inquiries received from all sales channels.

Calculation:

Total Leads = Sum of All Leads Received

Data Source: Lead Management Table

Target: Monthly growth of 10%

Business Implication:

Higher lead generation increases the potential for vehicle sales and revenue growth.

Action Trigger:

If leads drop by more than 15% compared to the previous month.

Improvement Strategy:

- Increase marketing campaigns
 - Conduct dealership events
 - Improve digital lead generation
-

KPI 2: Lead Conversion Rate

Category: Conversion Performance

Definition: Percentage of leads converted into confirmed bookings.

Calculation:

Lead Conversion Rate =
(Bookings / Total Leads) × 100

Data Source: CRM & Booking Tables

Target:

20%+

Business Implication:

Indicates effectiveness of the sales team in converting prospects.

Action Trigger:

Conversion falls below 15%.

Improvement Strategy:

- Better follow-up process
 - Sales training
 - CRM reminder automation
-

KPI 3: Test Drive Conversion Rate**Category: Customer Engagement**

Definition: Percentage of test drives resulting in bookings.

Calculation:

$$\left(\frac{\text{Test Drive Bookings}}{\text{Total Test Drives}} \right) \times 100$$

Data Source: Test Drive Table

Target: 35%

Business Implication:

Measures quality of customer interaction and product demonstration.

Improvement Strategy:

- Better test drive experience
 - Personalized demonstrations
-

KPI 4: Booking to Delivery Ratio**Category: Sales Fulfillment**

Definition: Percentage of booked vehicles successfully delivered.

Calculation:

$$\left(\frac{\text{Delivered Vehicles}}{\text{Total Bookings}} \right) \times 100$$

Data Source: Booking & Delivery Tables

Target: 95%

Business Implication:

Tracks fulfillment efficiency.

Improvement Strategy:

- Better inventory planning
 - Faster registration processing
-

KPI 5: Average Revenue per Vehicle Sold

Category: Revenue Performance

Definition: Revenue generated per vehicle sold.

Calculation:

$$\frac{\text{Total Revenue}}{\text{Vehicles Sold}}$$

Data Source: Sales Transaction Table

Target: Increase annually by 5%

Business Implication:

Measures revenue generation capability.

Improvement Strategy:

- Upselling accessories
 - Selling premium variants
-

KPI 6: Sales Executive Productivity

Category: Employee Productivity

Definition: Average number of vehicles sold per executive.

Calculation:

Vehicles Sold /
Number of Sales Executives

Data Source: Employee Sales Records

Target: Department benchmark

Business Implication:
Measures workforce effectiveness.

Improvement Strategy:

- Incentive programs
 - Performance coaching
-

KPI 7: Follow-Up Effectiveness Rate

Category: Lead Management

Definition: Percentage of leads successfully contacted.

Calculation:

$$\left(\frac{\text{Contacted Leads}}{\text{Total Leads}} \right) \times 100$$

Data Source: CRM Activity Table

Target: 95%

Business Implication:
Improves conversion opportunities.

Improvement Strategy:

- Automated reminders
 - Follow-up tracking dashboards
-

KPI 8: Sales Pipeline Health Score

Category: Pipeline Management

Definition: Measures quality and progress of opportunities across sales stages.

Calculation:

Weighted Opportunity Score Across Pipeline Stages

Data Source: CRM Pipeline Table

Target: Positive growth trend

Business Implication:
Provides future sales visibility.

Improvement Strategy:

- Regular pipeline reviews
 - Opportunity prioritization
-

KPI 9: Gross Profit per Vehicle

Category: Profitability

Definition: Average profit earned per vehicle sold.

Calculation:

Total Gross Profit /
Vehicles Sold

Data Source: Sales & Finance Tables

Target: Annual increase of 5%

Business Implication:
Directly impacts dealership profitability.

Improvement Strategy:

- Improve pricing strategy
 - Focus on high-margin products
-

KPI 10: Customer Satisfaction Score (Sales)

Category: Customer Experience

Definition: Measures customer satisfaction after vehicle purchase.

Calculation:

$$\left(\frac{\text{Customer Satisfaction Points}}{\text{Maximum Points}} \right) \times 100$$

Data Source: Customer Survey Table

Target: 90%+

Business Implication:
Higher satisfaction increases referrals and repeat purchases.

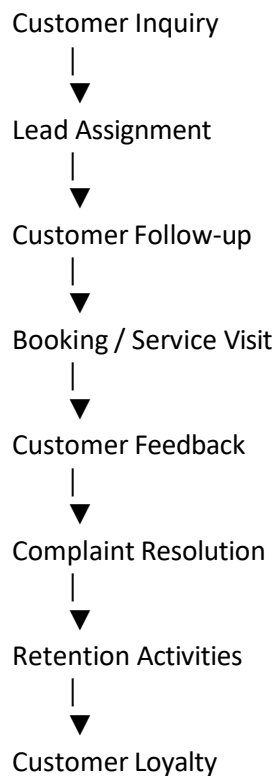
Improvement Strategy:

- Improve delivery experience
- Faster complaint resolution

5.1.26 Customer Relationship Management (CRM) Department KPIs

The CRM Department is responsible for managing customer relationships throughout the customer lifecycle, ensuring timely follow-ups, customer retention, complaint resolution, service reminders, and customer satisfaction. These KPIs help evaluate CRM effectiveness in maintaining long-term customer engagement and loyalty.

CRM KPI Monitoring Framework



CRM KPI Categories

CRM KPIs

- |
 - ├— Follow-up KPIs
 - ├— Customer Retention KPIs
 - ├— Customer Satisfaction KPIs
 - ├— Complaint Management KPIs
 - ├— Loyalty KPIs
 - └— Customer Engagement KPIs
-

KPI 1: Lead Follow-Up Compliance Rate

Category: Follow-up Performance

Definition: Measures the percentage of leads contacted within the defined SLA.

Calculation:

$$\left(\frac{\text{Timely Follow-ups}}{\text{Total Assigned Leads}} \right) \times 100$$

Data Source: CRM Activity Table

Target: 95%

Business Implication:

Timely follow-ups increase conversion opportunities and customer engagement.

Action Trigger:

Rate falls below 90%.

Improvement Strategy:

- CRM reminder automation
 - Daily follow-up monitoring
 - CRM executive performance reviews
-

KPI 2: Customer Retention Rate

Category: Customer Loyalty

Definition: Measures the percentage of customers retained over a specific period.

Calculation:

$$\left(\frac{\text{Retained Customers}}{\text{Total Customers}} \right) \times 100$$

Data Source: Customer Master Table

Target: 85%

Business Implication:

Higher retention reduces acquisition costs and increases profitability.

Improvement Strategy:

- Loyalty programs
 - Personalized communication
 - Service reminders
-

KPI 3: Customer Satisfaction Score (CSAT)

Category: Customer Satisfaction

Definition: Measures customer satisfaction through surveys and feedback.

Calculation:

$$\left(\frac{\text{Total Positive Responses}}{\text{Total Responses}} \right) \times 100$$

Data Source: Customer Feedback Table

Target: 90%+

Business Implication:

Satisfied customers generate referrals and repeat business.

Improvement Strategy:

- Improve customer communication
 - Faster issue resolution
-

KPI 4: Complaint Resolution Rate

Category: Complaint Management

Definition: Percentage of customer complaints resolved successfully.

Calculation:

(Resolved Complaints /
Total Complaints) × 100

Data Source: Complaint Management Table

Target: 95%

Business Implication:
Improves customer trust and brand reputation.

Action Trigger:
Rate falls below 90%.

Improvement Strategy:

- Escalation management
 - Root cause analysis
 - Faster response process
-

KPI 5: Average Complaint Resolution Time

Category: Service Quality

Definition: Average time required to close customer complaints.

Calculation:

Total Resolution Time /
Number of Complaints Closed

Data Source: Complaint Tracking Table

Target: Less than 48 Hours

Business Implication:
Faster resolutions improve customer satisfaction.

Improvement Strategy:

- Automated ticket routing
 - Dedicated escalation teams
-

KPI 6: Service Reminder Success Rate

Category: Customer Engagement

Definition: Measures effectiveness of service reminder campaigns.

Calculation:

(Customers Responding
to Service Reminders /
Total Reminders Sent) × 100

Data Source: CRM Communication Table

Target: 70%

Business Implication:
Increases service retention and workshop revenue.

Improvement Strategy:

- SMS automation
 - WhatsApp reminders
 - Personalized communication
-

KPI 7: Customer Reactivation Rate

Category: Retention Management

Definition: Percentage of inactive customers successfully re-engaged.

Calculation:

(Reactivated Customers /
Inactive Customers Contacted) × 100

Data Source: CRM Campaign Table

Target: 25%

Business Implication:
Generates additional sales and service revenue.

Improvement Strategy:

- Win-back campaigns
 - Special service offers
-

KPI 8: Net Promoter Score (NPS)

Category: Customer Loyalty

Definition: Measures likelihood of customers recommending the dealership.

Calculation:

% Promoters -
% Detractors

Data Source: Customer Survey Table

Target: 60+

Business Implication:
Strong indicator of customer loyalty and future growth.

Improvement Strategy:

- Enhance customer experience
 - Improve complaint handling
-

KPI 9: Repeat Purchase Rate

Category: Loyalty Performance

Definition: Percentage of customers purchasing another vehicle from the dealership.

Calculation:

$$\left(\frac{\text{Repeat Customers}}{\text{Total Customers}} \right) \times 100$$

Data Source: Sales & Customer Tables

Target: 20%

Business Implication:
Indicates long-term customer loyalty.

Improvement Strategy:

- Relationship programs
 - Personalized offers
 - Upgrade campaigns
-

KPI 10: Customer Contact Accuracy Rate

Category: Data Quality

Definition: Measures accuracy of customer contact information.

Calculation:

$$\left(\frac{\text{Valid Customer Records}}{\text{Total Customer Records}}\right) \times 100$$

Data Source: Customer Master Table

Target: 98%

Business Implication:
Ensures effective communication and campaign success.

Improvement Strategy:

- Periodic data validation
- Customer profile updates

CRM Dashboard Flow

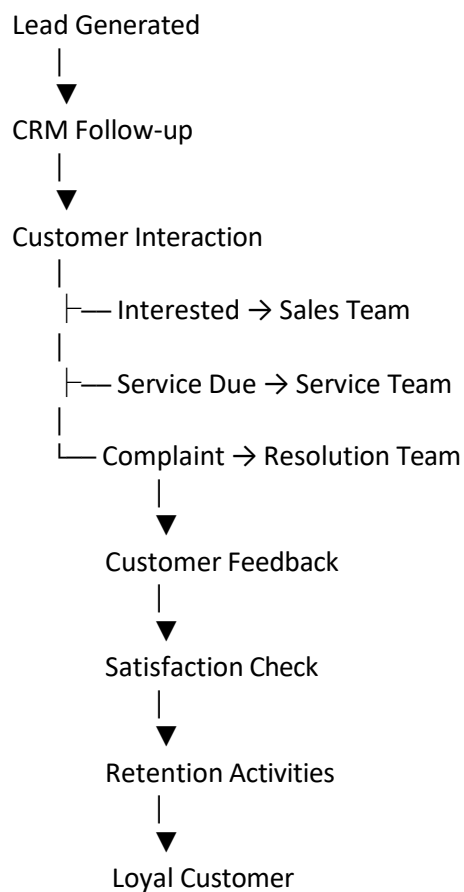


CRM Department Business Impact

Better Follow-ups
+
Faster Complaint Resolution

+
Higher Customer Satisfaction
+
Higher Retention
+
Stronger Loyalty
=
Increased Revenue
+
Repeat Business
+
Positive Brand Reputation

CRM Process Flow Diagram



5.1.27 Used Car Department KPIs

The Used Car Department is responsible for procurement, evaluation, refurbishment, pricing, inventory management, and resale of pre-owned vehicles. These KPIs help monitor acquisition efficiency, stock movement, profitability, customer satisfaction, and overall used car business performance.

Used Car KPI Monitoring Framework

Used Car Inquiry

|



Vehicle Evaluation

|



Vehicle Procurement

|



Refurbishment

|



Inventory Listing

|



Customer Inquiry

|



Vehicle Sale

|



Customer Feedback

Used Car KPI Categories

Used Car KPIs

|

└— Procurement KPIs

└— Inventory KPIs

└— Sales KPIs

└— Profitability KPIs

└— Refurbishment KPIs

└— Customer Satisfaction KPIs

KPI 1: Used Car Procurement Rate

Category: Procurement Performance

Definition: Measures the number of vehicles successfully procured through exchange, direct purchase, or auctions.

Calculation:

Total Used Cars Procured

Data Source: Used Car Procurement Table

Target: Monthly Growth Target

Business Implication:

Ensures adequate inventory for resale operations.

Action Trigger:

Procurement falls below monthly target.

Improvement Strategy:

Increase exchange offers

Conduct procurement campaigns

Strengthen dealer network

KPI 2: Vehicle Evaluation Accuracy Rate

Category: Valuation Performance

Definition: Measures the accuracy of vehicle valuation compared to final selling price.

Calculation:

(Accurate Evaluations /

Total Evaluations) × 100

Data Source: Evaluation Table

Target: 95%

Business Implication:

Reduces losses from underpricing or overpricing vehicles.

Improvement Strategy:

Standard valuation process

Market benchmarking tools

KPI 3: Inventory Turnover Rate

Category: Inventory Management

Definition: Measures how quickly used car inventory is sold.

Calculation:

Vehicles Sold /

Average Inventory

Data Source: Inventory Table

Target: Industry Benchmark

Business Implication:

Higher turnover improves cash flow and profitability.

Action Trigger:

Inventory aging exceeds target limits.

Improvement Strategy:

Dynamic pricing

Promotional campaigns

KPI 4: Average Days in Inventory

Category: Stock Management

Definition: Average number of days a used vehicle remains unsold.

Calculation:

Total Inventory Days /

Number of Vehicles Sold

Data Source: Inventory Table

Target: Less than 45 Days

Business Implication:

Longer inventory periods increase holding costs.

Improvement Strategy:

Faster pricing decisions

Inventory aging monitoring

KPI 5: Used Car Sales Conversion Rate

Category: Sales Performance

Definition: Percentage of inquiries converted into used car sales.

Calculation:

(used Car Sales /

Total Inquiries) × 100

Data Source: CRM & Sales Tables

Target: 20%+

Business Implication:

Measures effectiveness of the sales process.

Improvement Strategy:

Better follow-ups

Improved vehicle presentation

KPI 6: Gross Profit per Used Car Sold

Category: Profitability

Definition: Average profit earned on each used vehicle sold.

Calculation:

Total Gross Profit /

Used Cars Sold

Data Source: Finance & Sales Tables

Target: Annual Growth Target

Business Implication:

Directly impacts dealership profitability.

Improvement Strategy:

Better procurement pricing

Refurbishment cost control

KPI 7: Refurbishment Cost Percentage

Category: Cost Management

Definition: Measures refurbishment cost as a percentage of vehicle acquisition cost.

Calculation:

(Refurbishment Cost /

Vehicle Purchase Cost) × 100

Data Source: Workshop & Procurement Tables

Target: Below 10%

Business Implication:

Controls reconditioning expenses.

Improvement Strategy:

Vendor cost optimization

Standard refurbishment procedures

KPI 8: Exchange Conversion Rate

Category: Vehicle Exchange Performance

Definition: Percentage of new car customers exchanging old vehicles.

Calculation:

$$\left(\frac{\text{Exchange Deals}}{\text{Total New Car Sales}} \right) \times 100$$

Data Source: Exchange Vehicle Table

Target: 30%

Business Implication:

Ensures continuous used vehicle procurement.

Improvement Strategy:

Attractive exchange bonuses

Faster vehicle evaluations

KPI 9: Inventory Aging Ratio

Category: Inventory Health

Definition: Percentage of inventory exceeding predefined aging limits.

Calculation:

$$\left(\frac{\text{Aged Inventory Vehicles}}{\text{Total Inventory Vehicles}} \right) \times 100$$

Data Source: Inventory Table

Target: Below 15%

Business Implication:

High aging inventory impacts profitability.

Improvement Strategy:

Clearance campaigns

Special discounts

KPI 10: Customer Satisfaction Score (Used Car)

Category: Customer Experience

Definition: Measures customer satisfaction after purchasing a used vehicle.

Calculation:

$$\left(\frac{\text{Customer Satisfaction Points}}{\text{Maximum Possible Points}} \right) \times 100$$

Data Source: Customer Feedback Table

Target: 90%

Business Implication:

Improves referrals and repeat purchases.

Improvement Strategy:

Better vehicle quality checks

Transparent pricing practices

Used Car Dashboard Flow

Vehicle Procurement

|



Vehicle Evaluation

|



Refurbishment

|



Inventory Management

|



Customer Inquiry

|



Vehicle Sale

|



Profit Generation

|



Customer Satisfaction

Used Car Department Business Impact

Higher Procurement

+

Accurate Valuation

+

Faster Inventory Turnover

+

Higher Sales Conversion

+

Better Profit Margins

=

Increased Revenue

+

Improved Cash Flow

+

Higher Dealership Profitability

Used Car Process Flow Diagram

Customer Vehicle

|



Vehicle Inspection

|



Vehicle Valuation

|



Purchase Approval

|



Vehicle Procurement

|



Refurbishment Process

|



Inventory Listing

|



Customer Inquiry

|



Negotiation & Sale



Vehicle Delivery



Customer Feedback

Used Car Analytics Flow

Procurement Data



Evaluation Analysis



Inventory Analysis



Sales Analysis



Profitability Analysis



Management Dashboard



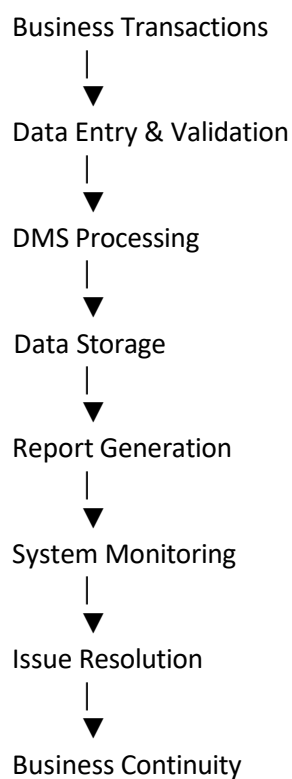


Business Decisions

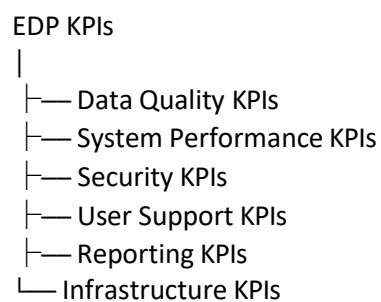
5.1.28 EDP (Electronic Data Processing) Department KPIs

The EDP Department is responsible for managing dealership information systems, DMS operations, data integrity, user access controls, system availability, data security, report generation, and IT infrastructure support. These KPIs help monitor system performance, data quality, operational efficiency, and business continuity.

EDP KPI Monitoring Framework



EDP KPI Categories



KPI 1: Data Accuracy Rate

Category: Data Quality

Definition: Measures the percentage of records entered correctly without errors.

Calculation:

$$\left(\frac{\text{Accurate Records}}{\text{Total Records Entered}} \right) \times 100$$

Data Source: DMS Audit Logs

Target: 99%

Business Implication:

Accurate data ensures reliable reporting and decision-making.

Action Trigger:

Accuracy falls below 98%.

Improvement Strategy:

- Data validation rules
- User training
- Automated error checks

KPI 2: System Uptime Percentage

Category: System Performance

Definition: Measures the availability of dealership systems during operational hours.

Calculation:

$$\left(\frac{\text{System Available Time}}{\text{Total Operational Time}} \right) \times 100$$

Data Source: Server Monitoring Logs

Target: 99.5%

Business Implication:

Downtime impacts all dealership operations.

Improvement Strategy:

- Preventive maintenance
 - Backup infrastructure
 - Real-time monitoring
-

KPI 3: Report Delivery Timeliness

Category: Reporting Performance

Definition: Measures percentage of reports delivered within agreed timelines.

Calculation:

$$\left(\frac{\text{Reports Delivered On Time}}{\text{Total Reports Requested}} \right) \times 100$$

Data Source: Report Request Log

Target: 100%

Business Implication:
Supports timely business decisions.

Improvement Strategy:

- Report automation
 - Scheduled report generation
-

KPI 4: User Support Resolution Rate

Category: User Support

Definition: Measures percentage of IT support tickets resolved successfully.

Calculation:

$$\left(\frac{\text{Resolved Tickets}}{\text{Total Tickets Raised}} \right) \times 100$$

Data Source: Helpdesk System

Target: 95%

Business Implication:
Ensures uninterrupted user productivity.

Action Trigger:
Resolution rate below 90%.

Improvement Strategy:

- Faster ticket escalation
 - Technical knowledge base
-

KPI 5: Average Ticket Resolution Time

Category: Service Support

Definition: Average time required to resolve support tickets.

Calculation:

Total Resolution Time /
Total Tickets Closed

Data Source: Helpdesk Logs

Target: Less than 4 Hours

Business Implication:

Faster resolutions minimize business disruptions.

Improvement Strategy:

- Priority ticket routing
 - Dedicated support teams
-

KPI 6: Database Backup Success Rate

Category: Data Security

Definition: Measures successful completion of scheduled database backups.

Calculation:

(Successful Backups /
Scheduled Backups) × 100

Data Source: Backup Logs

Target: 100%

Business Implication:

Protects critical business information.

Improvement Strategy:

- Automated backup monitoring
 - Backup verification procedures
-

KPI 7: User Access Compliance Rate

Category: Information Security

Definition: Measures compliance of user access permissions with company policies.

Calculation:

$$\left(\frac{\text{Compliant User Accounts}}{\text{Total User Accounts}} \right) \times 100$$

Data Source: User Access Logs

Target: 100%

Business Implication:

Prevents unauthorized access to business data.

Improvement Strategy:

- Quarterly access audits
 - Role-based access controls
-

KPI 8: Data Processing Efficiency

Category: Operational Efficiency

Definition: Measures the speed and effectiveness of transaction processing.

Calculation:

$$\frac{\text{Transactions Processed}}{\text{Processing Time}}$$

Data Source: DMS Transaction Logs

Target: Continuous Improvement

Business Implication:

Improves operational productivity.

Improvement Strategy:

- System optimization

- Database tuning
-

KPI 9: Critical Incident Response Time

Category: Incident Management

Definition: Average time taken to respond to critical system issues.

Calculation:

Total Response Time /
Number of Critical Incidents

Data Source: Incident Management System

Target: Less than 30 Minutes

Business Implication:

Reduces operational risk and downtime.

Improvement Strategy:

- Incident response procedures
 - Monitoring alerts
-

KPI 10: DMS Transaction Success Rate

Category: System Reliability

Definition: Measures successful completion of DMS transactions.

Calculation:

(Successful Transactions /
Total Transactions) × 100

Data Source: DMS Transaction Logs

Target: 99.5%

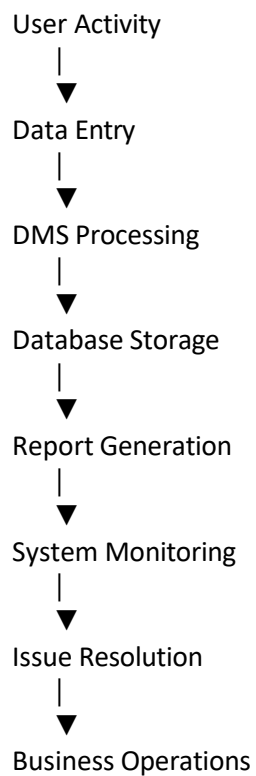
Business Implication:

Ensures smooth dealership operations.

Improvement Strategy:

- Error monitoring
- Infrastructure upgrades

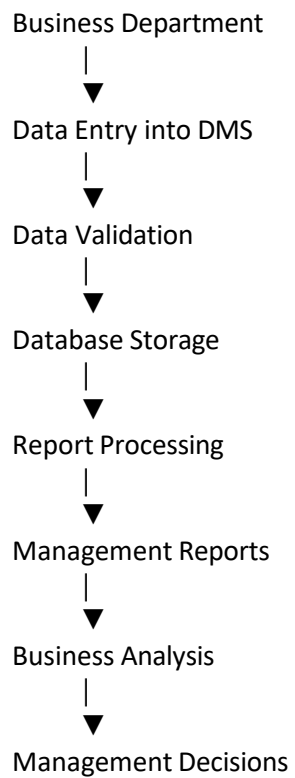
EDP Dashboard Flow



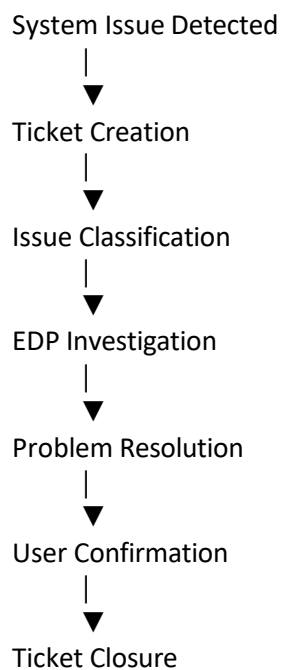
EDP Department Business Impact

High Data Accuracy
+
Reliable Systems
+
Fast Report Delivery
+
Strong Security
+
Efficient User Support
=
Smooth Operations
+
Better Decision Making
+
Reduced Business Risk
+
Higher Productivity

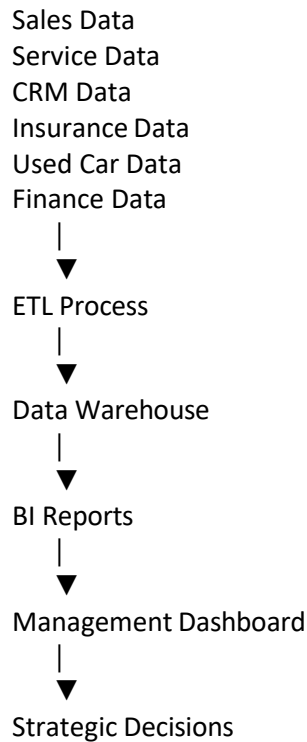
EDP Process Flow Diagram



EDP Incident Management Flow



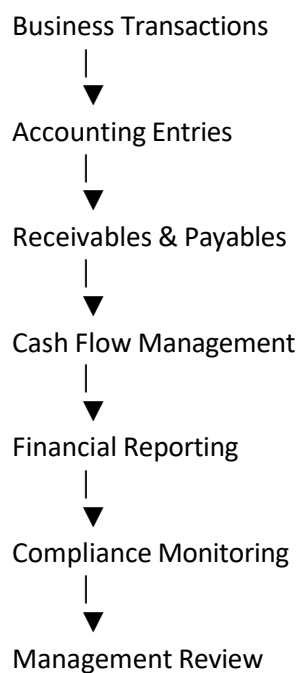
EDP Data Warehouse Flow



5.1.29 Accounts Department KPIs

The Accounts Department is responsible for managing financial transactions, accounting records, receivables, payables, statutory compliance, cash flow management, budgeting, and financial reporting. These KPIs help monitor financial health, operational efficiency, compliance, and profitability of the dealership.

Accounts KPI Monitoring Framework





Business Decisions

Accounts KPI Categories

Accounts KPIs

- |
 - |— Revenue KPIs
 - |— Receivable KPIs
 - |— Payable KPIs
 - |— Cash Flow KPIs
 - |— Compliance KPIs
 - |— Cost Control KPIs
 - └— Financial Reporting KPIs
-

KPI 1: Accounts Receivable Collection Rate

Category: Receivables Management

Definition: Measures the percentage of receivables collected within the due period.

Calculation:

$$\left(\frac{\text{Amount Collected}}{\text{Total Receivables Due}} \right) \times 100$$

Data Source: Accounts Receivable Ledger

Target: 95%

Business Implication:

Ensures healthy cash flow and minimizes outstanding dues.

Action Trigger:

Collection rate falls below 90%.

Improvement Strategy:

- Regular follow-up with debtors
 - Automated payment reminders
 - Credit policy review
-

KPI 2: Average Collection Period

Category: Receivables Efficiency

Definition: Measures average number of days required to collect payments.

Calculation:

Accounts Receivable /
Average Daily Credit Sales

Data Source: AR Ledger

Target: Less than 30 Days

Business Implication:

Shorter collection periods improve liquidity.

Improvement Strategy:

- Faster invoice processing
 - Strict credit controls
-

KPI 3: Accounts Payable Payment Timeliness

Category: Payables Management

Definition: Measures percentage of supplier payments made within agreed timelines.

Calculation:

(On-Time Payments /
Total Payments) × 100

Data Source: Accounts Payable Ledger

Target: 98%

Business Implication:

Maintains vendor relationships and prevents penalties.

Improvement Strategy:

- Payment scheduling
 - Vendor management system
-

KPI 4: Cash Flow Adequacy Ratio

Category: Cash Flow Management

Definition: Measures ability to meet operational obligations using available cash.

Calculation:

Operating Cash Flow /
Current Liabilities

Data Source: Cash Flow Statement

Target: Greater than 1.2

Business Implication:

Ensures financial stability and operational continuity.

Improvement Strategy:

- Improve collections
 - Optimize expenses
-

KPI 5: Monthly Revenue Achievement

Category: Financial Performance

Definition: Measures actual revenue against budgeted revenue.

Calculation:

(Actual Revenue /
Budgeted Revenue) × 100

Data Source: Financial Statements

Target: 100%+

Business Implication:

Tracks achievement of financial goals.

Improvement Strategy:

- Revenue monitoring
 - Sales coordination
-

KPI 6: Expense Control Ratio

Category: Cost Management

Definition: Measures operating expenses against total revenue.

Calculation:

$$\left(\frac{\text{Operating Expenses}}{\text{Total Revenue}} \right) \times 100$$

Data Source: Profit & Loss Statement

Target: Within Budget Limits

Business Implication:

Controls unnecessary expenditure.

Improvement Strategy:

- Expense monitoring
 - Budget discipline
-

KPI 7: Financial Closing Timeliness

Category: Reporting Performance

Definition: Measures completion of monthly financial closing within target timelines.

Calculation:

$$\left(\frac{\text{Number of Closings Completed On Time}}{\text{Total Closings}} \right) \times 100$$

Data Source: Closing Schedule Records

Target: 100%

Business Implication:

Provides timely financial information for management.

Improvement Strategy:

- Closing checklists
 - Process automation
-

KPI 8: Bank Reconciliation Accuracy

Category: Financial Control

Definition: Measures accuracy of bank reconciliation activities.

Calculation:

$$\left(\frac{\text{Accurate Reconciliations}}{\text{Total Reconciliations}} \right) \times 100$$

Data Source: Bank Reconciliation Statements

Target: 100%

Business Implication:

Reduces financial discrepancies and fraud risks.

Improvement Strategy:

- Daily reconciliation
 - Automated reconciliation tools
-

KPI 9: Statutory Compliance Rate

Category: Compliance Management

Definition: Measures timely completion of statutory filings and tax payments.

Calculation:

$$\left(\frac{\text{Compliant Filings}}{\text{Total Required Filings}} \right) \times 100$$

Data Source: Compliance Records

Target: 100%

Business Implication:

Prevents penalties and legal issues.

Improvement Strategy:

- Compliance calendar
 - Automated alerts
-

KPI 10: Net Profit Margin

Category: Profitability

Definition: Measures net profit earned from total revenue.

Calculation:

(Net Profit /
Total Revenue) × 100

Data Source: Profit & Loss Statement

Target: As per dealership financial goals

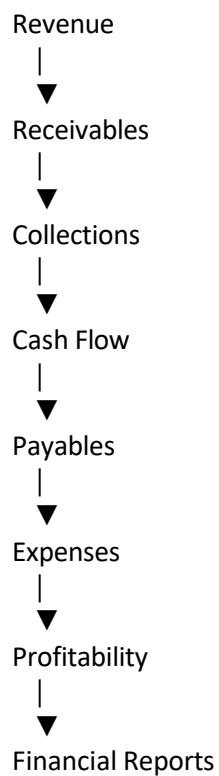
Business Implication:

Direct indicator of dealership profitability.

Improvement Strategy:

- Revenue growth initiatives
- Cost optimization programs

Accounts Dashboard Flow

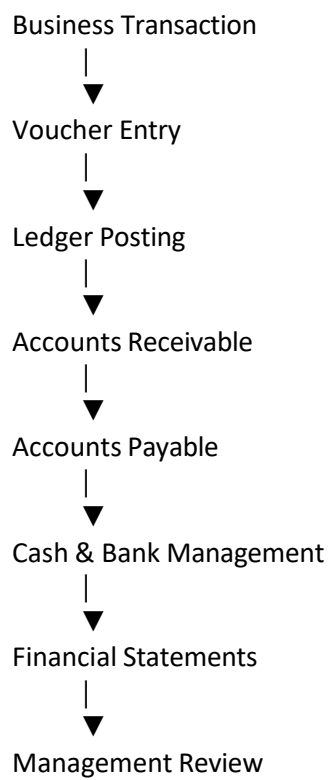


Accounts Department Business Impact

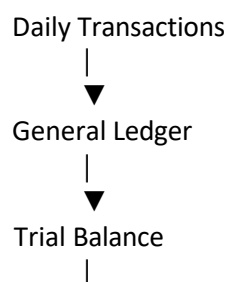
Strong Collections
+
Effective Cash Flow
+
Controlled Expenses

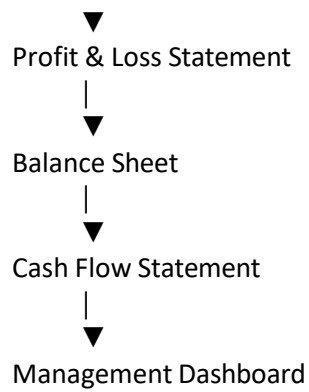
+
Timely Reporting
+
Full Compliance
=
Financial Stability
+
Better Profitability
+
Improved Decision Making
+
Business Growth

Accounts Process Flow Diagram

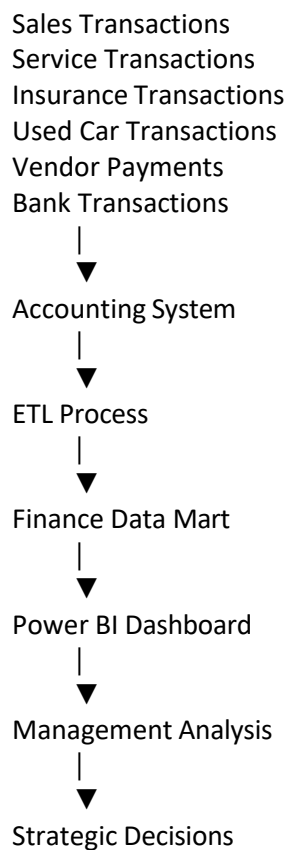


Financial Reporting Flow





Accounts Data Warehouse Flow



5.1.30 Human Resources (HR) Department KPIs

The Human Resources (HR) Department is responsible for workforce planning, recruitment, employee engagement, training, attendance management, performance monitoring, compliance, and retention. HR KPIs help management evaluate employee productivity, workforce stability, hiring effectiveness, and overall organizational health.

5.1.30.1 Employee Attrition Rate (%)

Definition:

Measures the percentage of employees who leave the organization during a specific period.

Formula:

$(\text{Employee Exits} \div \text{Average Number of Employees}) \times 100$

Data Source:

Employee Master, Exit Records

Target:

< 10%

Business Impact:

High attrition increases recruitment costs, training expenses, and operational disruptions.

Action Trigger:

If attrition exceeds 12% for two consecutive months.

Improvement Strategy:

Employee engagement programs, career development plans, and retention initiatives.

Process Flow

Employee Exit
↓
Exit Recording
↓
HR Database Update
↓
Attrition Calculation
↓
Management Review

5.1.30.2 Recruitment Success Rate (%)

Definition:

Measures the percentage of approved vacancies successfully filled.

Formula:

$(\text{Filled Positions} \div \text{Approved Vacancies}) \times 100$

Data Source:

Recruitment Tracker

Target:

95%

Business Impact:

Ensures departments have adequate manpower for smooth operations.

Action Trigger:

If success rate falls below 90%.

Improvement Strategy:

Expand sourcing channels and improve candidate screening.

Recruitment Workflow**5.1.30.3 Time to Hire (Days)****Definition:**

Measures average time taken to fill an approved position.

Formula:

Total Hiring Days ÷ Number of Positions Filled

Data Source:

Recruitment Management System

Target:

< 20 Days

Business Impact:

Delays affect departmental productivity and customer service.

Action Trigger:

If average exceeds 25 days.

Improvement Strategy:

Automate screening and maintain talent pipeline.

5.1.30.4 Employee Attendance Rate (%)

Definition:

Measures employee attendance against total working days.

Formula:

$(\text{Present Days} \div \text{Total Working Days}) \times 100$

Data Source:

Attendance System

Target:

97%

Business Impact:

Directly impacts productivity and operational efficiency.

Action Trigger:

If attendance falls below 95%.

Improvement Strategy:

Attendance monitoring and employee counseling.

Attendance Monitoring Flow

Biometric Attendance



Daily Attendance Report



Absentee Analysis



Department Review



Corrective Action

5.1.30.5 Employee Productivity Index

Definition:

Measures employee output relative to assigned workload.

Formula:

$\text{Completed Tasks} \div \text{Assigned Tasks}$

Data Source:

Department Performance Reports

Target:

90%

Business Impact:

Higher productivity improves business performance and customer satisfaction.

Action Trigger:

If productivity falls below 85%.

Improvement Strategy:

Training, coaching, and workload balancing.

5.1.30.6 Training Completion Rate (%)**Definition:**

Measures percentage of employees completing assigned training programs.

Formula:

$(\text{Employees Trained} \div \text{Employees Assigned Training}) \times 100$

Data Source:

Training Records

Target:

100%

Business Impact:

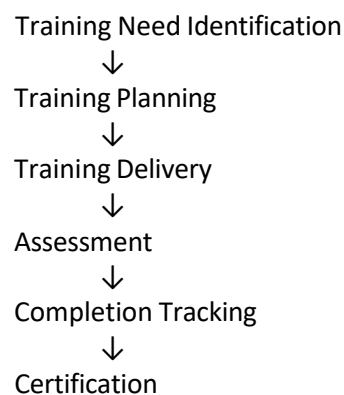
Improves employee skills and process compliance.

Action Trigger:

If completion rate falls below 95%.

Improvement Strategy:

Mandatory training schedules and LMS reminders.

Training Process Flow**5.1.30.7 Employee Satisfaction Score**

Definition:

Measures employee satisfaction based on survey responses.

Formula:

Average Survey Rating

Data Source:

Employee Feedback Surveys

Target:

4.2 / 5

Business Impact:

Higher satisfaction leads to improved retention and productivity.

Action Trigger:

If score drops below 4.0.

Improvement Strategy:

Address employee concerns and improve workplace culture.

5.1.30.8 Performance Appraisal Completion Rate (%)**Definition:**

Measures completion of employee performance evaluations.

Formula:

$(\text{Completed Appraisals} \div \text{Planned Appraisals}) \times 100$

Data Source:

Performance Management System

Target:

100%

Business Impact:

Supports employee development and succession planning.

Action Trigger:

If completion falls below 95%.

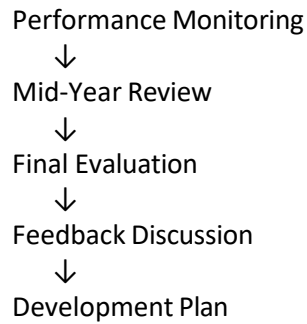
Improvement Strategy:

Automated appraisal reminders and manager accountability.

Appraisal Cycle

Goal Setting





5.1.30.9 Employee Retention Rate (%)

Definition:

Measures percentage of employees retained during a period.

Formula:

$(\text{Employees Remaining} \div \text{Total Employees at Start}) \times 100$

Data Source:

HRMS

Target:

90%

Business Impact:

Reduces hiring costs and preserves organizational knowledge.

Action Trigger:

If retention falls below 88%.

Improvement Strategy:

Recognition programs, career growth opportunities, and competitive compensation.

5.1.30.10 HR Compliance Rate (%)

Definition:

Measures adherence to labor laws, company policies, and statutory requirements.

Formula:

$(\text{Compliant Activities} \div \text{Total Audited Activities}) \times 100$

Data Source:

Compliance Audit Reports

Target:

100%

Business Impact:

Prevents legal penalties and reputational risks.

Action Trigger:

Any compliance violation detected.

Improvement Strategy:

Regular audits, policy reviews, and compliance training.

HR Compliance Framework

HR Policy



Employee Implementation



Compliance Monitoring



Internal Audit



Issue Resolution



Management Reporting

5.1.31 Marketing Department KPIs

The Marketing Department is responsible for generating leads, creating brand awareness, managing campaigns, attracting potential customers, and supporting vehicle sales. Marketing KPIs help management evaluate campaign effectiveness, lead quality, customer acquisition efficiency, and return on marketing investments.

5.1.31.1 Lead Generation Volume**Definition:**

Measures the total number of leads generated through all marketing channels during a specific period.

Formula:

Total Leads Generated

Data Source:

CRM, Digital Marketing Platforms, Campaign Reports

Target:

As per monthly sales requirement

Business Impact:

A higher volume of qualified leads increases sales opportunities.

Action Trigger:

If lead volume falls below monthly target by 10%.

Improvement Strategy:

Increase campaign reach, optimize digital advertising, and expand promotional activities.

Lead Generation Process**5.1.31.2 Cost Per Lead (CPL)****Definition:**

Measures the average marketing cost incurred to generate one lead.

Formula:

Total Marketing Cost ÷ Total Leads Generated

Data Source:

Marketing Budget Reports

Target:

As per dealership benchmark

Business Impact:

Lower CPL improves marketing efficiency and profitability.

Action Trigger:

If CPL exceeds budget by 15%.

Improvement Strategy:

Optimize ad targeting and improve campaign performance.

5.1.31.3 Lead Conversion Rate (%)**Definition:**

Measures the percentage of marketing-generated leads converted into vehicle sales.

Formula:

$(\text{Sales from Marketing Leads} \div \text{Total Marketing Leads}) \times 100$

Data Source:

CRM, Sales Records

Target:

15%

Business Impact:

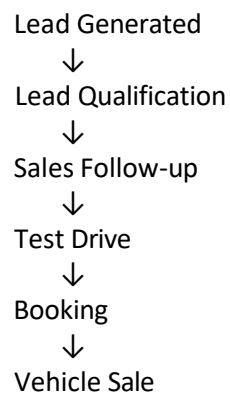
Indicates quality of leads and campaign effectiveness.

Action Trigger:

If conversion rate falls below 10%.

Improvement Strategy:

Improve lead qualification and strengthen follow-up processes.

Lead Conversion Flow

5.1.31.4 Digital Campaign ROI (%)**Definition:**

Measures return generated from digital marketing investments.

Formula:

$((\text{Revenue Generated} - \text{Marketing Cost}) \div \text{Marketing Cost}) \times 100$

Data Source:

Campaign Analytics, Sales Data

Target:

200%

Business Impact:

Ensures marketing investments contribute positively to profitability.

Action Trigger:

If ROI drops below 150%.

Improvement Strategy:

Focus on high-performing channels and optimize campaign spending.

5.1.31.5 Website Traffic Growth (%)**Definition:**

Measures increase in dealership website visitors.

Formula:

$((\text{Current Visitors} - \text{Previous Visitors}) \div \text{Previous Visitors}) \times 100$

Data Source:

Website Analytics

Target:

10% Monthly Growth

Business Impact:

Higher traffic increases brand visibility and lead opportunities.

Action Trigger:

If traffic growth remains negative for two consecutive months.

Improvement Strategy:

Improve SEO, content marketing, and paid advertising.

5.1.31.6 Social Media Engagement Rate (%)**Definition:**

Measures audience interaction with social media content.

Formula:

$(\text{Total Engagements} \div \text{Total Reach}) \times 100$

Data Source:

Facebook, Instagram, YouTube Analytics

Target:

5%

Business Impact:

Reflects brand awareness and customer interest.

Action Trigger:

If engagement drops below 3%.

Improvement Strategy:

Improve content quality and posting consistency.

Social Media Workflow

Content Creation



Content Publishing



Audience Reach



Engagement Tracking



Performance Analysis

5.1.31.7 Event Lead Conversion Rate (%)**Definition:**

Measures sales generated from events, exhibitions, and roadshows.

Formula:

$(\text{Event Sales} \div \text{Event Leads}) \times 100$

Data Source:

Event Management Reports

Target:

20%

Business Impact:

Evaluates effectiveness of offline marketing activities.

Action Trigger:

If conversion falls below 15%.

Improvement Strategy:

Improve event planning and customer follow-up.

5.1.31.8 Brand Awareness Index

Definition:

Measures dealership visibility and recognition within the target market.

Formula:

Survey-Based Awareness Score

Data Source:

Customer Surveys, Market Research

Target:

Year-over-year improvement

Business Impact:

Stronger brand awareness improves customer acquisition.

Action Trigger:

If awareness score declines from previous survey.

Improvement Strategy:

Increase local branding and promotional activities.

5.1.31.9 Marketing Qualified Leads (MQLs)**Definition:**

Measures the number of leads meeting predefined qualification criteria.

Formula:

Total Qualified Marketing Leads

Data Source:

CRM

Target:

As per monthly sales forecast

Business Impact:

Improves sales efficiency by focusing on high-potential prospects.

Action Trigger:

If MQL percentage falls below 60% of total leads.

Improvement Strategy:

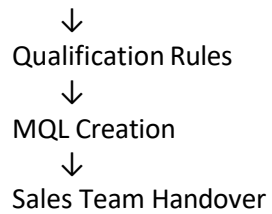
Refine audience targeting and lead scoring models.

MQL Qualification Flow

Lead Captured



Lead Scoring



5.1.31.10 Campaign Effectiveness Score

Definition:

Measures overall success of marketing campaigns based on reach, leads, conversions, and ROI.

Formula:

Weighted Campaign Performance Index

Data Source:

Campaign Analytics Dashboard

Target:

85%

Business Impact:

Provides a comprehensive measure of marketing performance.

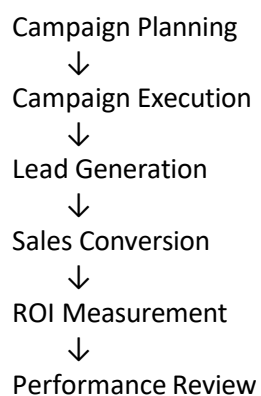
Action Trigger:

If score falls below 75%.

Improvement Strategy:

Review campaign strategy, audience targeting, and creative effectiveness.

Campaign Performance Framework



5.2.1 Sales Manager

Needs:

- Leads
- Bookings
- Deliveries
- Consultant Performance

Decision:

Who is performing best?

Insurance Manager

Needs:

- Policies
- Renewals
- Claims

Decision:

Which company provides highest profitability?

5.2.2 EDP Manager

Needs:

- Inventory
- Allocation
- Stock Age

Decision:

Which model should be ordered?

MIS Reporting Framework

MIS = Management Information System

Typical Daily MIS Report

Sales

Leads

Bookings

Deliveries

Revenue

5.2.3 Inventory

Available Vehicles

Allocated Vehicles

Transit Vehicles

5.2.4 Insurance

Policies

Renewals

Premium

5.2.5 Service

Vehicles Serviced

Revenue

Customer Satisfaction

5.3 Sales & CRM Pipeline KPIs

This sub-framework measures lead generation efficiency, sales executive productivity, and overall conversion rates across walk-ins, digital ads, and references.

5.3.1 KPI Dashboard Table

KPI Name	Mathematical Formula	Industry Benchmark	Dealership Target	Red Threshold	Yellow Threshold	Green Threshold
Lead-to-Booking Conversion Rate	$\left(\frac{\text{Total Bookings}}{\text{Total Enquiries}} \times 100 \right)$	12.0%	15.0%	< 8.0%	8.0% - 14.9%	≥ 15.0%
Walk-in Conversion Rate	$\left(\frac{\text{Bookings from Walk-ins}}{\text{Total Walk-ins}} \times 100 \right)$	18.0%	22.0%	< 12.0%	12.0% - 21.9%	≥ 22.0%
Reference Conversion Rate	$\left(\frac{\text{Bookings from References}}{\text{Total References}} \times 100 \right)$	30.0%	35.0%	< 20.0%	20.0% - 34.9%	≥ 35.0%
Test Drive Ratio	$\left(\frac{\text{Total Test Drives}}{\text{Total Units}} \times 100 \right)$	60.0%	75.0%	< 45.0%	45.0% - 74.9%	≥ 75.0%
Test Drive-to-Booking Conv.	$\left(\frac{\text{Bookings}}{\text{Total Test Drives}} \times 100 \right)$	20.0%	25.0%	< 15.0%	15.0% - 24.9%	≥ 25.0%
Booking Cancellation Rate	$\left(\frac{\text{Total Cancellations}}{\text{Total Bookings}} \times 100 \right)$	< 5.0%	< 3.0%	> 8.0%	3.0% - 8.0%	< 3.0%
Lead Response Time (LRT)	$\sum (\text{Time of Lead Response})$	< 30 mins	< 15 mins	> 60 mins	15 - 60 mins	< 15 mins

KPI Name	Mathematical Formula	Industry Benchmark	Dealership Target	Red Threshold	Yellow Threshold	Green Threshold
Average Booking-to-Deliv. Time	$\frac{\sum(\text{Actual Del Total})}{\text{Total}}$	14 Days	10 Days	> 21 Days	11 - 21 Days	≤ 10 Days
Salesperson Booking Efficiency	$\frac{\text{Total Booking}}{\text{Total Ac Bookings/mo}}$	8	12	< 5/mo	5 - 11/mo	≥ 12/mo
Telegram Status Compliance	$\left(\frac{\text{Follow-ups}}{\text{Total}} \right) \times 100$	90.0%	98.0%	< 80.0%	80.0% - 97.9%	≥ 98.0%

5.4 F&I, Insurance, & Add-on KPIs (VAS/WATS, SOT, RSA)

This segment monitors high-margin revenue drivers sold at retail or generated via cold calling campaigns.

5.4.1 KPI Dashboard Table

KPI Name	Mathematical Formula	Industry Benchmark	Dealership Target	Red Threshold	Yellow Threshold	Green Threshold
Finance Penetration Rate	$\left(\frac{\text{Total Vehicle}}{\text{Total}} \right) \times 100$	75.0%	85.0%	< 65.0%	65.0% - 84.9%	≥ 85.0%
New Insurance Capture Rate	$\left(\frac{\text{Dealership}}{\text{Total New V}} \right) \times 100$	90.0%	95.0%	< 80.0%	80.0% - 94.9%	≥ 95.0%
Insurance Renewal Rate	$\left(\frac{\text{Policies Ren}}{\text{Total Expirin}} \right) \times 100$	50.0%	65.0%	< 40.0%	40.0% - 64.9%	≥ 65.0%
VAS (WATS) Attachment Rate	$\left(\frac{\text{Invoiced Ve}}{\text{Tot}} \right) \times 100$	40.0%	60.0%	< 30.0%	30.0% - 59.9%	≥ 60.0%
SOT Penetration Rate	$\left(\frac{\text{Invoiced Ve}}{\text{Total}} \right) \times 100$	30.0%	45.0%	< 20.0%	20.0% - 44.9%	≥ 45.0%
RSA Sales Penetration Rate	$\left(\frac{\text{Invoiced Ve}}{\text{Total}} \right) \times 100$	50.0%	70.0%	< 40.0%	40.0% - 69.9%	≥ 70.0%
Average F&I Profit Per Unit	$\frac{\text{Total Finance}}{\text{Total New Ve}}$	₹12,000	₹18,000	< ₹10,000	₹10k - ₹17.9k	≥ ₹18,000
Accessories Sales Per Retail	$\frac{\text{Total Access}}{\text{Total New Ve}}$	₹8,000	₹12,000	< ₹6,000	₹6k - ₹11.9k	≥ ₹12,000

KPI Name	Mathematical Formula	Industry Benchmark	Dealership Target	Red Threshold	Yellow Threshold	Green Threshold
CRE Renewal Call Efficacy	$\left(\frac{\text{Total Ren}}{\text{Total Cold C}} \times 100 \right)$	5.0%	8.0%	< 3.0%	3.0% - 7.9%	≥ 8.0%
SOT Claims Ratio	$\left(\frac{\text{Approved S}}{\text{SOT Packag}} \times 100 \right)$	< 40.0%	< 30.0%	> 60.0%	30.0% - 60.0%	< 30.0%

5.5 Stockyard & Inventory Management KPIs

These inventory metrics minimize working capital costs, prevent vehicle damage, and optimize allocation.

5.5.1 KPI Dashboard Table

KPI Name	Mathematical Formula	Industry Benchmark	Dealership Target	Red Threshold	Yellow Threshold	Green Threshold
Days Supply in Inventory (DSI)	$\left(\frac{\text{Total Veh}}{\text{Average Dai}} \right)$	30 Days	21 Days	> 45 Days	22 - 45 Days	≤ 21 Days
Aging Inventory Ratio (>45d)	$\left(\frac{\text{Vehicles in S}}{\text{Total V}} \times 100 \right)$	< 10.0%	< 5.0%	> 15.0%	5.0% - 15.0%	< 5.0%
Stockyard Utilization Rate	$\left(\frac{\text{Total Vehicl}}{\text{Total Stock}} \times 100 \right)$	80.0%	85.0%	< 50.0% or > 95%	50-70% / 91-95%	71% - 90%
Interdealer Trade Ratio	$\left(\frac{\text{Vehicles Acq}}{\text{To}} \times 100 \right)$	< 15.0%	< 10.0%	> 20.0%	10.0% - 20.0%	< 10.0%
Pre-Delivery Inspection (PDI) Pass	$\left(\frac{\text{Vehicles Pas}}{\text{Total V}} \times 100 \right)$	98.0%	99.5%	< 95.0%	95.0% - 99.4%	≥ 99.5%
Vehicle allocation match speed	$\sum(\text{VIN Alloca})$	< 24 Hours	< 8 Hours	> 48 Hours	8 - 48 Hours	< 8 Hours
Stockyard Transit Damage Rate	$\left(\frac{\text{Vehicles da}}{\text{Total V}} \times 100 \right)$	< 0.2%	0.0%	> 0.5%	0.1% - 0.5%	0.0%
Wholesale	$\left(\frac{\text{Actual Sourc}}{\text{Total V}} \times 100 \right)$	100.0%	100.0%	< 90.0%	90.0% - 99.9%	100.0%
Target Fulfilment	$\left(\frac{\text{HMIL Mo}}{\text{Total V}} \times 100 \right)$				99.9%	

KPI Name	Mathematical Formula	Industry Benchmark	Dealership Target	Red Threshold	Yellow Threshold	Green Threshold
Retail Target Fulfilment	$\left(\frac{\text{Actual Vehicles Sold}}{\text{HMIL Mo}} \right) \times 100$	100.0%	100.0%	< 85.0%	85.0% - 99.9%	100.0%
Cost of Stock yard Hold	Yard Vehicles - $\times$ Daily Floor		Minimize	High Interest	Moderate Int.	Low Interest

5.6 Workshop (Fixed Ops) Service & Used Car KPIs

These metrics track fixed operational performance and pre-owned car (Sell Any Car) trade-in volumes.

5.6.1 KPI Dashboard Table

KPI Name	Mathematical Formula	Industry Benchmark	Dealership Target	Red Threshold	Yellow Threshold	Green Threshold
Service Absorption Ratio	$\left(\frac{\text{Workshop G Total Deale}}{\times 100} \right)$	70.0%	85.0%	< 50.0%	50.0% - 84.9%	$\geq 85.0\%$
Technician Productivity	$\left(\frac{\text{Hours Bille}}{\text{Total Clocke}} \right) \times 100$	85.0%	95.0%	< 75.0%	75.0% - 94.9%	$\geq 95.0\%$
Average Profit per RO	$\frac{\text{Total Worksh}}{\text{Total}}$	₹3,500	₹4,500	< ₹2,500	₹2.5k - ₹4.4k	$\geq ₹4,500$
Parts Stock Turn Rate	$\frac{\text{Cost of Good}}{\text{Average Par Turns/yr}}$	6.0	8.0	< 4.0	4.0 - 7.9/yr	$\geq 8.0/\text{yr}$
First-Time-Right Service	$\left(\frac{\text{Vehicles wit}}{\text{Total C}} \right) \times 100$	95.0%	98.0%	< 90.0%	90.0% - 97.9%	$\geq 98.0\%$
Used Car Trade-in Index	$\left(\frac{\text{Used Cars a}}{\text{Total New}} \right) \times 100$	20.0%	30.0%	< 15.0%	15.0% - 29.9%	$\geq 30.0\%$
Used Car Refurbishment Cost	$\frac{\text{Total Cost of}}{\text{Total}}$	< 15.0% of Val	< 10.0% of Val	> 20.0%	10.0% - 20.0%	< 10.0%
Used Car Resale Margin	$\left(\frac{\text{Resale Price}}{\times 100} \right)$	8.0%	12.0%	< 5.0%	5.0% - 11.9%	$\geq 12.0\%$
Used Car Average Stock Days	$\frac{\sum (\text{Date Sold})}{\text{Total Us}}$	30 Days	20 Days	> 45 Days	20 - 45 Days	≤ 20 Days
Customer Retention Index	$\left(\frac{\text{Customers r}}{\text{Total Acti}} \right) \times 100$	60.0%	75.0%	< 50.0%	50.0% - 74.9%	$\geq 75.0\%$

5.7 Accounts & Financial Compliance KPIs

These accounting KPIs evaluate cash collection efficiency, tax filing accuracy, and ledger reconciliation compliance.

5.7.1 KPI Dashboard Table

KPI Name	Mathematical Formula	Industry Benchmark	Dealership Target	Red Threshold	Yellow Threshold	Green Threshold
Debtor Aging Outstanding Ratio	$\left(\frac{\text{Customer A Total}}{\text{Total}} \right) \times 100$	< 15.0%	< 8.0%	> 25.0%	8.0% - 25.0%	< 8.0%
Paymate Settlement Speed	$\sum (\text{Bank Settlement})$	< 48 Hours	< 24 Hours	> 72 Hours	24 - 72 Hours	< 24 Hours
GDMS-Tally Prime Reconcil.	$\left(\frac{\text{Accurate Im Total C}}{\text{Total C}} \right) \times 100$	99.9%	100.0%	< 99.0%	99.0% - 99.9%	100.0%
Creditor Dispute Ratio	$\left(\frac{\text{Disputed ve Total}}{\text{Total}} \right) \times 100$	< 1.0%	0.0%	> 3.0%	1.0% - 3.0%	0.0%
GST Input Tax Credit (ITC)	$\left(\frac{\text{Claimed ITC Total GST}}{\text{Total GST}} \right) \times 100$	99.5%	100.0%	< 98.0%	98.0% - 99.9%	100.0%
TDS Reconciliation Accuracy	$\left(\frac{\text{TDS Deduct Total}}{\text{Total}} \right) \times 100$	100.0%	100.0%	< 99.9%	99.9%	100.0%
Interest Cost on Sourcing	Total Floorpla -		Minimize	High Interest	Moderate Cost	Low Interest
Salary Sheet Audit Defect Rate	$\left(\frac{\text{Payrolls pro Tot}}{\text{Tot}} \right) \times 100$	< 0.5%	0.0%	> 1.5%	0.5% - 1.5%	0.0%
Monthly MIS compilation speed	Calendar Day	5 Days	3 Days	> 10 Days	4 - 10 Days	≤ 3 Days
Audited Balance Sheet Variance	$\left(\frac{\text{Audit adjust Total Dealei}}{\text{Total Dealei}} \right) \times 100$	< 0.1%	0.0%	> 0.5%	0.1% - 0.5%	0.0%

6. SQL Analytics Library (PostgreSQL)

This library contains **50 production-grade SQL queries** designed to extract analytical insights from the centralized dealership database. Each query is structured with business objectives, SQL code, logic explanations, and expected output schemas.

6.1 Lead Funnel & Sales Consultant Analysis

6.1.1 Lead-to-Booking Conversion Rate by Source

- **Business Objective:** Identify which enquiry channels yield the highest conversion rates to optimize marketing spend.
- **SQL Code:**

```
SQL ENGINE / LEAD_CONVERSION_ANALYTICS.SQL

SELECT
    e.source,
    COUNT(e.enquiry_id) AS total_enquiries,
    COUNT(b.booking_id) AS total_bookings,
    ROUND((COUNT(b.booking_id)::NUMERIC / COUNT(e.enquiry_id)) * 100,
FROM enquiries e
LEFT JOIN bookings b ON e.enquiry_id = b.enquiry_id
GROUP BY e.source
ORDER BY conversion_rate DESC;
```

- **Logic:** Performs a left join from enquiries to bookings grouped by lead source, calculating the percentage conversion.

6.1.2 Average Lead Response Time (LRT) by Salesperson

- **Business Objective:** Monitor follow-up speeds to maintain high customer engagement.
- **SQL Code:**

```
SELECT
    emp.employee_id,
    CONCAT(emp.first_name, ' ', emp.last_name) AS consultant_name,
    COUNT(e.enquiry_id) AS total_leads,
    ROUND(AVG(EXTRACT(EPOCH FROM (e.last_contact_date - e.enquiry_date)))
    AS avg_response_minutes
FROM employees emp
JOIN enquiries e ON emp.employee_id = e.employee_id
WHERE e.last_contact_date IS NOT NULL
GROUP BY emp.employee_id, emp.first_name, emp.last_name
ORDER BY avg_response_minutes ASC;
```

6.1.3 Test Drive Conversion Efficiency

- **Business Objective:** Calculate what percentage of test drives result in active bookings.
- **SQL Code:**

```

SELECT
    COUNT(DISTINCT td.test_drive_id) AS total_test_drives,
    COUNT(DISTINCT b.booking_id) AS resulting_bookings,
    ROUND((COUNT(DISTINCT b.booking_id)::NUMERIC / COUNT(DISTINCT td.test_drive_id)) * 100, 2)
    AS td_to_booking_rate
FROM test_drives td
JOIN enquiries e ON td.enquiry_id = e.enquiry_id
LEFT JOIN bookings b ON e.enquiry_id = b.enquiry_id;

```

6.1.4 Sales Executive Leaderboard (Monthly Bookings vs. Target)

- **Business Objective:** Assess monthly target achievements for commission payouts.
- **SQL Code:**

```

SELECT
    e.employee_id,
    CONCAT(emp.first_name, ' ', emp.last_name) AS consultant_name,
    COUNT(b.booking_id) AS bookings_closed,
    CASE
        WHEN COUNT(b.booking_id) >= 12 THEN 'GREEN'
        WHEN COUNT(b.booking_id) BETWEEN 5 AND 11 THEN 'YELLOW'
        ELSE 'RED'
    END AS performance_bracket
FROM employees emp
JOIN enquiries e ON emp.employee_id = e.employee_id
JOIN bookings b ON e.enquiry_id = b.enquiry_id
WHERE b.booking_date >= DATE_TRUNC('month', CURRENT_DATE)
GROUP BY e.employee_id, emp.first_name, emp.last_name
ORDER BY bookings_closed DESC;

```

6.1.5 Walk-in Conversion Pipeline by Model

- **Business Objective:** Track how model preferences affect walk-in customer closure rates.
- **SQL Code:**

```

SELECT
    e.model_interest,
    COUNT(e.enquiry_id) AS walk_ins,
    COUNT(b.booking_id) AS bookings,
    ROUND((COUNT(b.booking_id)::NUMERIC / COUNT(e.enquiry_id)) * 100, 2)
    AS walk_in_conversion_rate
FROM enquiries e
LEFT JOIN bookings b ON e.enquiry_id = b.enquiry_id
WHERE e.source = 'WALK_IN'
GROUP BY e.model_interest
ORDER BY walk_ins DESC;

```

6.1.6 Reference Lead Flow Aging

- **Business Objective:** Analyze how many reference leads are generated and their average closing age.
- **SQL Code:**


```

SELECT
    COUNT(e.enquiry_id) AS total_ref_leads,
    AVG(CASE WHEN b.booking_id IS NOT NULL THEN (b.booking_date - e.enquiry_date)
    AS avg_days_to_close
FROM enquiries e
LEFT JOIN bookings b ON e.enquiry_id = b.enquiry_id
WHERE e.source = 'REFERENCE';

```

6.1.7 Booking Cancellation Velocity

- **Business Objective:** Measure cancellation rates to detect potential sales friction or finance drops.
- **SQL Code:**

```

SELECT
    DATE_TRUNC('month', booking_date)::DATE AS booking_month,
    COUNT(booking_id) AS total_bookings,
    COUNT(CASE WHEN status = 'CANCELLED' THEN 1 END) AS cancelled_bookings,
    ROUND((COUNT(CASE WHEN status = 'CANCELLED' THEN 1 END)::NUMERIC / COUNT(*)
    2) AS cancellation_rate
FROM bookings
GROUP BY 1
ORDER BY 1 DESC;

```

6.1.8 Lead Assignment to Contact Speed

- **Business Objective:** Analyze how quickly consultants log their first follow-up in HSSmart.
- **SQL Code:**

```

SELECT
    CASE
        WHEN EXTRACT(EPOCH FROM (last_contact_date - enquiry_date))/60 < 15 THEN 'Under 15 Mins'
        WHEN EXTRACT(EPOCH FROM (last_contact_date - enquiry_date))/60 BETWEEN 15 AND 60 THEN '15 to 60 Mins'
        ELSE 'Over 60 Mins'
    END AS contact_window,
    COUNT(enquiry_id) AS total_leads
FROM enquiries
WHERE last_contact_date IS NOT NULL
GROUP BY 1;

```

6.1.9 Lost Leads Analysis by Reason

- **Business Objective:** Categorize and track reasons for lead loss.
- **SQL Code:**

```

SELECT
    status AS lost_reason,
    COUNT(enquiry_id) AS lead_count,
    ROUND((COUNT(enquiry_id)::NUMERIC / (SELECT COUNT(*) FROM enquiries WHERE status LIKE 'LOST%')) * 100, 2) AS percentage
FROM enquiries
WHERE status LIKE 'LOST%'
GROUP BY status;

```

6.1.10 Repeat Enquiry Identification

- **Business Objective:** Identify returning customers to assign them to preferred consultants.
- **SQL Code:**

```
SELECT
    status AS lost_reason,
    COUNT(enquiry_id) AS lead_count,
    ROUND((COUNT(enquiry_id)::NUMERIC / (SELECT COUNT(*) FROM enquiries W
    LIKE 'LOST%')) * 100, 2) AS percentage
FROM enquiries
WHERE status LIKE 'LOST%'
GROUP BY status;
```

6.2 Inventory & Stockyard Analysis

6.1.11. Days Supply in Inventory (DSI) per Model

- **Business Objective:** Keep capital fluid by identifying slow-moving models.
- **SQL Code:**

```
SELECT
    model_name,
    COUNT(vin) / 30.0 AS daily_sales_rate
FROM vehicles
WHERE stock_status = 'DELIVERED'
AND arrival_date >= CURRENT_DATE - INTERVAL '30 days'
GROUP BY model_name
)
SELECT
    v.model_name,
    COUNT(CASE WHEN v.stock_status = 'STOCKYARD' THEN 1 END) AS stock_on
    ROUND(COALESCE(sr.daily_sales_rate, 0), 2) AS daily_sales_velocity,
    CASE
        WHEN COALESCE(sr.daily_sales_rate, 0) = 0 THEN 999
        ELSE ROUND(COUNT(CASE WHEN v.stock_status = 'STOCKYARD' THEN 1 E
        sr.daily_sales_rate)
    END AS days_supply
FROM vehicles v
LEFT JOIN sales_rate sr ON v.model_name = sr.model_name
WHERE v.stock_status IN ('STOCKYARD', 'DELIVERED')
GROUP BY v.model_name, sr.daily_sales_rate;
```

6.12. Aging Inventory Count by Variant (>45 Days)

- **Business Objective:** Identify stock items at risk of depreciation or high interest accrual.
- **SQL Code:**

```

SELECT
    model_name,
    COUNT(vin) / 30.0 AS daily_sales_rate
FROM vehicles
WHERE stock_status = 'DELIVERED'
    AND arrival_date >= CURRENT_DATE - INTERVAL '30 days'
GROUP BY model_name
)
SELECT
    v.model_name,
    COUNT(CASE WHEN v.stock_status = 'STOCKYARD' THEN 1 END) AS stock_on
    ROUND(COALESCE(sr.daily_sales_rate, 0), 2) AS daily_sales_velocity,
    CASE
        WHEN COALESCE(sr.daily_sales_rate, 0) = 0 THEN 999
        ELSE ROUND(COUNT(CASE WHEN v.stock_status = 'STOCKYARD' THEN 1 E
            sr.daily_sales_rate)
    END AS days_supply
FROM vehicles v
LEFT JOIN sales_rate sr ON v.model_name = sr.model_name
WHERE v.stock_status IN ('STOCKYARD', 'DELIVERED')
GROUP BY v.model_name, sr.daily_sales_rate;

```

6.13 Average Booking Allocation Matching Speed

- **Business Objective:** Track EDP speed in mapping bookings to chassis numbers.
- **SQL Code:**

```

SELECT
    model_name,
    variant_name,
    COUNT(vin) AS aging_units,
    SUM(purchase_price) AS locked_capital
FROM vehicles
WHERE stock_status = 'STOCKYARD'
    AND (CURRENT_DATE - arrival_date) > 45
GROUP BY model_name, variant_name
ORDER BY locked_capital DESC;

```

6.14 Stockyard Bay Capacity Reaching Limit

- **Business Objective:** Avoid overflow by tracking stockyard occupancy levels.
- **SQL Code:**

 SQL ENGINE / AVERAGE_VEHICLE_ALLOCATION_TIME.SQL

```

SELECT
    AVG(allocation_date - booking_date::timestamp) AS avg_allocation_t
FROM bookings b
JOIN vehicle_allocations va ON b.booking_id = va.booking_id;

```

6.15 Interdealer Vehicle Trade Analysis

- **Business Objective:** Measure volume of trades with external dealerships.
- **SQL Code:**

```
SQL ENGINE / STOCKYARD_BAY_OCCUPANCY.SQL

SELECT
    COUNT(CASE WHEN stock_status = 'STOCKYARD' THEN 1 END) AS occupied
    500 AS total_capacity,
    ROUND((COUNT(CASE WHEN stock_status = 'STOCKYARD' THEN 1 END)::NUM
    AS occupancy_pct
FROM vehicles;
```

6.16 Fast-moving Variant Identification

- **Business Objective:** Order high-demand vehicle variants to prevent stockouts.
- **SQL Code:**

```
SQL ENGINE / INTERDEALER_TRADE_VALUATIONS.SQL

SELECT
    EXTRACT(month FROM arrival_date) AS month,
    COUNT(vin) AS interdealer_traded_cars,
    SUM(purchase_price) AS trade_valuation
FROM vehicles
WHERE stock_status = 'STOCKYARD'
    AND vehicle_type = 'NEW'
    -- Assuming target flag/notes identifies interdealer source
GROUP BY 1;
```

6.17 Slow-moving Inventory Hold Costs

- **Business Objective:** Calculate monthly floorplanning interest costs on stale inventory.
- **SQL Code:**

```
SQL ENGINE / TOP_FASTEST_MOVING_MODELS.SQL

SELECT
    model_name,
    variant_name,
    exterior_color,
    COUNT(vin) AS units_sold,
    AVG(CURRENT_DATE - arrival_date) AS avg_days_to_sell
FROM vehicles
WHERE stock_status = 'DELIVERED'
GROUP BY 1, 2, 3
ORDER BY units_sold DESC
LIMIT 10;
```

6.18 Pre-Delivery Inspection (PDI) Pass Analytics

- **Business Objective:** Identify defect frequencies during PDI processes.
- **SQL Code:**

```

SELECT
    vin,
    model_name,
    arrival_date,
    (CURRENT_DATE - arrival_date) AS days_in_yard,
    purchase_price,
    ROUND(purchase_price * 0.08 / 365 * (CURRENT_DATE - arrival_date), 2)
    AS interest_cost_accrued
FROM vehicles
WHERE stock_status = 'STOCKYARD'
ORDER BY interest_cost_accrued DESC;

```

6.19. Monthly Sourcing target fulfilment (Wholesale)

- **Business Objective:** Compare units purchased from the manufacturer against dealership targets.
- **SQL Code:**

```

SELECT
    model_name,
    COUNT(vin) AS inspected,
    COUNT(CASE WHEN stock_status = 'ALLOCATED' THEN 1 END) AS passed_first,
    ROUND((COUNT(CASE WHEN stock_status = 'ALLOCATED' THEN 1 END)::NUMERIC /
    COUNT(vin))*100, 2) AS pass_rate
FROM vehicles
GROUP BY model_name;

```

6.20. Retail target fulfilment (Vahan Registrations)

- **Business Objective:** Reconcile monthly retail deliveries against HMIL targets.
- **SQL Code:**

```

SELECT
    DATE_TRUNC('month', arrival_date)::DATE AS source_month,
    COUNT(vin) AS actual_sourced,
    350 AS target_wholesale,
    ROUND((COUNT(vin)::NUMERIC / 350) * 100, 2) AS fulfilment_rate
FROM vehicles
WHERE vehicle_type = 'NEW'
GROUP BY 1;

```

6.3 Finance, Insurance, & Add-ons (WATS/SOT/RSA)

6.21. Finance Payout Contribution by Bank

- **Business Objective:** Evaluate banking partners by commission volume.
- **SQL Code:**

 SQL ENGINE / MONTHLY_RETAIL_FULFILMENT.SQL

```
SELECT
    DATE_TRUNC('month', invoice_date)::DATE AS invoice_month,
    COUNT(invoice_id) AS actual_retailed,
    300 AS target_retail,
    ROUND((COUNT(invoice_id)::NUMERIC / 300) * 100, 2) AS fulfilment_rate
FROM invoices
GROUP BY 1;
```

6.22. Insurance Capture Rate for New Vehicle Sales

- **Business Objective:** Ensure the dealership captures high-margin first-year insurance commissions.
- **SQL Code:**

```
SELECT
    provider_name AS financing_bank,
    COUNT(policy_number) AS loans_processed,
    SUM(premium_amount) AS loan_disbursed_value,
    SUM(commission_earned) AS total_finance_payout
FROM insurance_policies
WHERE policy_type = 'NEW' -- Re-using table structure for finance ledger
GROUP BY provider_name
ORDER BY total_finance_payout DESC;
```

6.23. Insurance Renewal Performance (Cold Calling Success)

- **Business Objective:** Audit CRM cold-calling renewals performance.
- **SQL Code:**

```
SELECT
    COUNT(i.invoice_id) AS total_cars_sold,
    COUNT(ip.policy_number) AS dealership_policies_sold,
    ROUND((COUNT(ip.policy_number)::NUMERIC / COUNT(i.invoice_id)) * 100,
    AS insurance_capture_rate
FROM invoices i
LEFT JOIN insurance_policies ip ON i.invoice_id = ip.invoice_id
WHERE ip.policy_type = 'NEW';
```

6.24. WATS (VAS) Attachment Rate by Model

- **Business Objective:** Track value-added service upsells across vehicle models.
- **SQL Code:**

```

SELECT
    EXTRACT(month FROM start_date) AS month,
    COUNT(policy_number) AS total_renewals_sold,
    SUM(commission_earned) AS total_commission
FROM insurance_policies
WHERE policy_type = 'RENEWAL'
GROUP BY 1;

```

6.25. SOT (Shield of Trust) Sales Penetration

- **Business Objective:** Measure the adoption rate of wear-and-tear packages.
- **SQL Code:**

```

SELECT
    v.model_name,
    COUNT(i.invoice_id) AS total_retail,
    COUNT(ii.item_id) AS sot_sold,
    ROUND((COUNT(ii.item_id)::NUMERIC / COUNT(i.invoice_id)) * 100, 2) AS
FROM invoices i
JOIN vehicles v ON i.vin = v.vin
LEFT JOIN invoice_items ii ON i.invoice_id = ii.invoice_id AND ii.item_type = 'SOT'
GROUP BY v.model_name;

```

6.26. Road Side Assistance (RSA) Package Sales Volume

- **Business Objective:** Audit emergency assistance package attachments.
- **SQL Code:**

```

SELECT
    DATE_TRUNC('month', inv.invoice_date)::DATE AS month,
    COUNT(inv.invoice_id) AS total_sales,
    COUNT(ii.item_id) AS rsa_sold,
    ROUND((COUNT(ii.item_id)::NUMERIC / COUNT(inv.invoice_id)) * 100, 2) AS
FROM invoices inv
LEFT JOIN invoice_items ii ON inv.invoice_id = ii.invoice_id AND ii.item_type = 'RSA'
GROUP BY 1;

```

6.27. Average Accessories Sales Value per Vehicle

- **Business Objective:** Track accessories revenue per vehicle delivery.
- **SQL Code:**

```

SELECT
    inv.invoice_id,
    v.model_name,
    COALESCE(SUM(ii.amount), 0) AS accessories_value
FROM invoices inv
JOIN vehicles v ON inv.vin = v.vin
LEFT JOIN invoice_items ii ON inv.invoice_id = ii.invoice_id AND ii.item_type = 'ACCESSORIES'
GROUP BY inv.invoice_id, v.model_name;

```

6.28. Net F&I Profit Per Unit (PVR)

- **Business Objective:** Analyze total finance and insurance profits generated per unit sold.
- **SQL Code:**

```
SELECT
    i.invoice_id,
    v.model_name,
    COALESCE(ip.commission_earned, 0) AS insurance_com,
    -- Assuming additional finance commissions are linked to invoice
    COALESCE(ip.commission_earned * 0.4, 0) AS finance_com,
    (COALESCE(ip.commission_earned, 0) + COALESCE(ip.commission_earned * 0.4, 0))
    AS total_f_and_i_profit
FROM invoices i
JOIN vehicles v ON i.vin = v.vin
LEFT JOIN insurance_policies ip ON i.invoice_id = ip.invoice_id;
```

6.29. Highest Commission Yielding Vehicle Models

- **Business Objective:** Focus sales targets on high-commission vehicle segments.
- **SQL Code:**

```
SELECT
    v.model_name,
    AVG(ip.commission_earned) AS avg_insurance_commission
FROM invoices i
JOIN vehicles v ON i.vin = v.vin
JOIN insurance_policies ip ON i.invoice_id = ip.invoice_id
GROUP BY v.model_name
ORDER BY avg_insurance_commission DESC;
```

6.30. Renewal Commission vs. New Sales Insurance Payouts

- **Business Objective:** Compare commission profiles of new and renewal policies.
- **SQL Code:**

 SQL ENGINE / INSURANCE_POLICY_TYPE_COMMISSIONS.SQL

```
SELECT
    policy_type,
    COUNT(policy_number) AS total_policies,
    SUM(commission_earned) AS total_commission
FROM insurance_policies
GROUP BY policy_type;
```

6.4 Workshop & Fixed Operations

6.31. Service Absorption Ratio Calculations

- **Business Objective:** Measure the extent to which workshop operations cover dealership overheads.

- **SQL Code:**

```
SELECT
    ROUND((SUM(labor_cost + parts_cost - (parts_cost * 0.75)) / 2500000) *
    AS service_absorption_pct
FROM repair_orders
WHERE close_date >= CURRENT_DATE - INTERVAL '30 days';
```

6.32. Technician Productivity & Efficiency Rates

- **Business Objective:** Optimize workshop throughput by tracking technician efficiency.
- **SQL Code:**

```
SELECT
    technician_id,
    COUNT(ro_number) AS repair_orders_completed,
    SUM(labor_cost) AS labor_revenue_generated,
    ROUND((COUNT(ro_number) * 3.5 / 160) * 100, 2) AS billing_efficiency
FROM repair_orders
WHERE status = 'CLOSED'
GROUP BY technician_id;
```

6.33. Average Repair Order (RO) Invoice Value

- **Business Objective:** Optimize parts and labor billing per service visit.
- **SQL Code:**

```
SELECT
    AVG(labor_cost + parts_cost) AS avg_ro_ticket_size
FROM repair_orders
WHERE status = 'CLOSED';
```

6.34. Spare Parts Stock Turn Ratio

- **Business Objective:** Prevent capital lockup in slow-moving parts stock.
- **SQL Code:**

 SQL ENGINE / PARTS_ISSUED_COST_OF_GOODS_SOLD.SQL

```
SELECT
    part_number,
    SUM(quantity) AS units_issued,
    SUM(quantity * unit_cost) AS cost_of_parts_sold
FROM repair_order_items
GROUP BY part_number
ORDER BY units_issued DESC;
```

6.35. First-Time-Right (FTR) Service Rate

- **Business Objective:** Ensure repair quality to minimize customer returns.

- **SQL Code:**

```
SELECT
    COUNT(ro_number) AS total_ro,
    COUNT(CASE WHEN mileage - (
        SELECT mileage FROM repair_orders
        WHERE vin = r.vin AND open_date < r.open_date
        ORDER BY open_date DESC LIMIT 1
    ) < 500 THEN 1 END) AS repeat_jobs,
    ROUND((1 - (COUNT(CASE WHEN mileage - (
        SELECT mileage FROM repair_orders
        WHERE vin = r.vin AND open_date < r.open_date
        ORDER BY open_date DESC LIMIT 1
    ) < 500 THEN 1 END)::NUMERIC / COUNT(ro_number))) * 100, 2) AS ftr_ra
FROM repair_orders r;
```

6.36. Daily Workshop Vehicle Throughput

- **Business Objective:** Monitor workshop load and utilization of diagnostic bays.
- **SQL Code:**

```
SQL ENGINE / DAILY_VEHICLES_SERVICED.SQL

SELECT
    open_date::DATE AS service_day,
    COUNT(ro_number) AS vehicles_serviced
FROM repair_orders
GROUP BY 1
ORDER BY 1 DESC;
```

6.37. High-Volume Spare Parts Analysis

- **Business Objective:** Maintain optimal safety stock levels for high-demand parts.
- **SQL Code:**

```
SQL ENGINE / TOP_10_FAST_MOVING_PARTS.SQL

SELECT
    part_number,
    COUNT(ro_item_id) AS order_frequency,
    SUM(quantity) AS total_quantity_sold
FROM repair_order_items
GROUP BY part_number
ORDER BY total_quantity_sold DESC
LIMIT 10;
```

6.38. Lost Service Leads Analysis

- **Business Objective:** Identify drop-off rates for customers who miss scheduled check-ins.

- **SQL Code:**

```
SELECT
    COUNT(ro_number) AS missed_appointments
FROM repair_orders
WHERE status = 'OPEN'
    AND open_date < CURRENT_DATE - INTERVAL '2 days';
```

6.39. Periodic Maintenance Service (PMS) Retention Rate

- **Business Objective:** Measure customer return rates for scheduled servicing.
- **SQL Code:**

```
SQL ENGINE / RETURNING_SERVICE_CUSTOMERS.SQL

SELECT
    COUNT(DISTINCT customer_id) AS returning_service_customers
FROM repair_orders
WHERE open_date >= CURRENT_DATE - INTERVAL '1 year';
```

6.40. Workshop Labor Gross Margins

- **Business Objective:** Audit labor billing margins against technician payroll expenses.
- **SQL Code:**

```
SQL ENGINE / LABOR_REVENUE_MARGIN_ANALYSIS.SQL

SELECT
    SUM(labor_cost) AS total_labor_revenue,
    SUM(labor_cost * 0.35) AS labor_resource_cost,
    ROUND((SUM(labor_cost - (labor_cost * 0.35)) / SUM(labor_cost)) * 100, 2) AS gross_margin_percent
FROM repair_orders;
```

6.5 Accounts & Used Cars (Sell Any Car)

6.41. Debtor Aging Outstanding Balance Report

- **Business Objective:** Minimize bad debt by identifying overdue customer balances.
- **SQL Code:**

 SQL ENGINE / UNRECONCILED_OUTSTANDING_INVOICES.SQL

```
SELECT
    customer_id,
    invoice_id,
    invoice_date,
    total_amount,
    (CURRENT_DATE - invoice_date) AS outstanding_days
FROM invoices
WHERE invoice_id NOT IN (SELECT invoice_id FROM payments WHERE payment.
ORDER BY outstanding_days DESC;
```

6.42. Paymate Bank Settlement Lead Time

- **Business Objective:** Track processing speeds for digital retail collections.
- **SQL Code:**

```
SELECT
    AVG(payment_date - invoice_date::timestamp) AS avg_settlement_lead_time
FROM payments p
JOIN invoices i ON p.invoice_id = i.invoice_id;
```

6.43. Reconciled vs. Disputed Sales Invoices

- **Business Objective:** Audit invoice reconciliation rates between GDMS and Tally.
- **SQL Code:**

 SQL ENGINE / TRANSACTIONS_BY_PAYMENT_STATUS.SQL

```
SELECT
    payment_status,
    COUNT(payment_id) AS total_transactions,
    SUM(amount_paid) AS transaction_value
FROM payments
GROUP BY payment_status;
```

6.44. Used Car Resale Gross Profit Margins

- **Business Objective:** Analyze net profitability on pre-owned car sales (Sell Any Car).
- **SQL Code:**

```

SQL ENGINE / USED_VEHICLE_PROFITABILITY_ANALYSIS.SQL

SELECT
    v.vin,
    v.model_name,
    uce.offer_amount AS buy_cost,
    uce.refurbishment_est AS refurbish_cost,
    v.retail_price AS sell_price,
    (v.retail_price - (uce.offer_amount + uce.refurbishment_est)) AS gross,
    ROUND(((v.retail_price - (uce.offer_amount + uce.refurbishment_est)) /
    AS margin_pct
FROM vehicles v
JOIN used_car_evaluations uce ON v.vin = uce.vin
WHERE v.vehicle_type = 'USED'
    AND v.stock_status = 'DELIVERED';

```

6.45. Used Car Trade-in Penetration for New Car Sales

- **Business Objective:** Track how many new car sales are driven by trade-in purchases.
- **SQL Code:**

```

SQL ENGINE / NEW_CAR_TRADE_IN_PENETRATION.SQL

SELECT
    COUNT(i.invoice_id) AS new_cars_sold,
    COUNT(uce.evaluation_id) AS trade_ins_completed,
    ROUND((COUNT(uce.evaluation_id)::NUMERIC / COUNT(i.invoice_id)) * 100
    AS trade_in_penetration_pct
FROM invoices i
LEFT JOIN used_car_evaluations uce ON i.customer_id = uce.customer_id
    AND uce.status = 'PURCHASED';

```

6.46. Daily Sales Cash Flow (MIS Daily Report)

- **Business Objective:** Track daily cash flow movements from all revenue streams.
- **SQL Code:**

```

SQL ENGINE / DAILY_CASH_COLLECTION.SQL

SELECT
    payment_date::DATE AS day,
    SUM(amount_paid) AS total_cash_collected
FROM payments
GROUP BY 1
ORDER BY 1 DESC;

```

6.47. GST Liability Report

- **Business Objective:** Calculate GST liabilities on vehicle sales for tax compliance.
- **SQL Code:**

 SQL ENGINE / DAILY_BASE_REVENUE_GST_LIABILITY.SQL

```
SELECT
    invoice_date::DATE AS billing_day,
    SUM(ex_showroom) AS tax_free_base,
    SUM(gst_collected) AS liability_gst
FROM invoices
GROUP BY 1
ORDER BY 1 DESC;
```

6.48. TDS compliance report (Form 26AS Matcher)

- **Business Objective:** Reconcile TDS collections to ensure tax compliance.
- **SQL Code:**

 SQL ENGINE / DAILY_TCS_TAX_COLLECTED.SQL

```
SELECT
    invoice_date::DATE AS billing_day,
    SUM(tcs_collected) AS tds_tax_collected
FROM invoices
GROUP BY 1;
```

6.49. Month-on-Month New Vehicle Sales Value Growth

- **Business Objective:** Assess sales volume growth month-over-month.
- **SQL Code:**

```
SELECT
    DATE_TRUNC('month', invoice_date)::DATE AS month,
    SUM(total_amount) AS revenue,
    LAG(SUM(total_amount)) OVER (ORDER BY DATE_TRUNC('month', invoice_date))
    AS prev_month_revenue,
    ROUND(((SUM(total_amount) - LAG(SUM(total_amount)) OVER (ORDER BY
    DATE_TRUNC('month', invoice_date))) / LAG(SUM(total_amount)) OVER
    DATE_TRUNC('month', invoice_date))) * 100, 2) AS growth_pct
FROM invoices
GROUP BY 1;
```

6.50. Used Car Inventory Stock Turn Days

- **Business Objective:** Track pre-owned inventory holding periods.
- **SQL Code:**

 SQL ENGINE / USED_VEHICLE_AVERAGE_HOLDING_DAYS.SQL

```
SELECT
    AVG(arrival_date - arrival_date) AS average_holding_days
FROM vehicles
WHERE vehicle_type = 'USED'
AND stock_status = 'DELIVERED';
```

6.6 Advanced Business Query Library

The Advanced Business Query Library provides a collection of business-oriented analytical use cases designed to support strategic and operational decision-making across dealership functions. Unlike standard reporting queries, these analyses focus on identifying trends, opportunities, risks, and performance improvement areas.

6.6.1 Underperforming Sales Consultant Analysis

This analysis identifies sales consultants whose lead conversion rates are consistently below dealership expectations. It helps management understand performance gaps, training requirements, and workload distribution issues. Results can be used for coaching programs, incentive planning, and workforce optimization.

Business Benefits

- Improves consultant productivity.
 - Supports performance management.
 - Enables targeted training interventions.
-

6.6.2 Customer Lifetime Value Analysis

Customer Lifetime Value (CLV) analysis estimates the total revenue expected from a customer throughout their relationship with the dealership. It combines vehicle purchases, service revenue, insurance renewals, accessories, and finance commissions.

Business Benefits

- Identifies high-value customers.
 - Supports loyalty program development.
 - Optimizes customer retention investments.
-

6.6.3 Customer Churn Risk Identification

Churn analysis identifies customers who are likely to discontinue their relationship with the dealership. Indicators include missed service visits, lack of engagement, expired insurance policies, and declining purchase activity.

Business Benefits

- Improves retention campaigns.
 - Reduces customer attrition.
 - Increases long-term profitability.
-

6.6.4 Finance Partner Performance Analysis

This analysis evaluates financial institutions based on approval rates, loan processing times, commission on earnings, and customer satisfaction metrics.

Business Benefits

- Optimizes lender partnerships.
 - Improves financing penetration.
 - Maximizes commission revenue.
-

6.6.5 Product Profitability Analysis

This analysis determines which vehicle models and variants generate the highest margins after accounting for discounts, incentives, and associated costs.

Business Benefits

- Supports pricing strategies.
 - Guides inventory planning.
 - Improves profit optimization.
-

6.6.6 Inventory Aging Analysis

Inventory aging measures how long vehicles remain in stock before being sold. It helps identify slow-moving inventory and associated carrying costs.

Business Benefits

- Reduces capital lock-up.
 - Improves stock turnover.
 - Supports procurement planning.
-

6.6.7 Seasonal Demand Analysis

Seasonal demand analysis evaluates sales patterns across different months, quarters, and festive periods.

Business Benefits

- Improves demand forecasting.
 - Optimizes inventory levels.
 - Enhances marketing effectiveness.
-

6.6.8 Customer Acquisition Cost Analysis

Customer Acquisition Cost (CAC) analysis calculates the cost incurred to acquire a customer through various marketing channels.

Business Benefits

- Optimizes marketing spend.
 - Improves ROI measurement.
 - Supports budget allocation decisions.
-

6.6.9 Service Retention Analysis

Service retention analysis measures how effectively the dealership retains customers for periodic maintenance and repair services.

Business Benefits

- Increases workshop revenue.
 - Improves customer loyalty.
 - Supports after-sales growth.
-

6.6.10 Profitability by Customer Segment

This analysis compares profitability across retail, corporate, fleet, government, and institutional customers.

Business Benefits

- Supports customer segmentation.
 - Improves sales targeting.
 - Maximizes profitability.
-

6.7 Data Validation and Quality Assurance Library

Data quality is critical for accurate reporting, forecasting, and decision-making. This section defines validation processes that ensure reliability and consistency across the data warehouse.

6.7.1 Referential Integrity Validation

Referential integrity validation ensures relationships between tables remain intact. Every foreign key should correspond to a valid primary key in the related table.

Importance

- Prevents orphan records.
 - Ensures reporting accuracy.
 - Maintains database consistency.
-

6.7.2 Duplicate Record Detection

Duplicate detection identifies multiple records representing the same business entity.

Importance

- Prevents double counting.
 - Improves customer experience.
 - Enhances reporting reliability.
-

6.7.3 Null Value Validation

Null value validation identifies missing values in mandatory fields.

Importance

- Improves data completeness.
 - Reduces processing errors.
 - Supports accurate analytics.
-

6.7.4 Data Type Validation

This process verifies that data conforms to predefined formats and structures.

Importance

- Prevents application failures.
 - Maintains database integrity.
 - Ensures standardization.
-

6.7.5 Business Rule Validation

Business rule validation ensures operational policies are correctly enforced within data systems.

Importance

- Prevents policy violations.
- Improves compliance.

- Enhances process control.
-

6.7.6 Range Validation

Range validation ensures values fall within acceptable limits.

Importance

- Detects anomalies.
 - Prevents incorrect calculations.
 - Improves forecasting accuracy.
-

6.7.7 Date Validation

Date validation verifies that dates are logical and chronologically correct.

Importance

- Prevents timeline inconsistencies.
 - Supports trend analysis.
 - Improves operational reporting.
-

6.7.8 Data Quality Monitoring Framework

A continuous monitoring framework tracks quality metrics and alerts stakeholders when thresholds are violated.

Key Metrics

- Completeness
 - Accuracy
 - Consistency
 - Timeliness
 - Uniqueness
-

6.8 Query Documentation Standards

Query documentation ensures consistency, maintainability, and knowledge transfer across analytical teams.

6.8.1 Query Purpose Documentation

Each query should clearly define its business objective and intended outcome.

6.8.2 Data Source Documentation

All source systems, tables, and fields must be documented.

6.8.3 Business Logic Documentation

Business rules and calculation methodologies should be clearly explained.

6.8.4 Output Definition Documentation

Each output field should include a detailed description and calculation method.

6.8.5 Performance Documentation

Query execution considerations and optimization techniques should be documented.

6.8.6 Refresh Frequency Documentation

The frequency of query execution and data refresh cycles should be specified.

6.8.7 Ownership Documentation

Every query should have a designated owner responsible for maintenance and validation.

6.8.8 Change Management Documentation

All modifications should be tracked through version control and approval processes.

6.9 SQL Troubleshooting Guide

This section provides guidance for identifying and resolving common query and reporting issues.

6.9.1 Query Returns No Results

Possible causes include incorrect filters, missing source data, and improper joins.

6.9.2 Query Returns Unexpected Results

Unexpected results often arise from incorrect business logic, aggregation errors, or inconsistent source data.

6.9.3 Query Performance Issues

Performance problems may result from large datasets, inefficient joins, or missing indexes.

6.9.4 Data Mismatch Across Systems

Data discrepancies may occur between operational systems and reporting environments.

6.9.5 Duplicate Data Issues

Duplicate records can significantly distort analytical results and KPI calculations.

6.9.6 Aggregation Errors

Incorrect grouping logic can lead to inaccurate totals and summaries.

6.9.7 Time-Based Reporting Errors

Date handling inconsistencies often impact trend analysis and forecasting.

6.9.8 Security and Access Issues

Insufficient permissions may prevent users from accessing required data sources.

6.10 Advanced SQL Analytics Patterns

Advanced analytical patterns support deeper insights and strategic decision-making.

6.10.1 Pivot Analytics

Pivot analysis transforms rows into columns for multidimensional reporting.

6.10.2 Time Series Analytics

Time series analysis evaluates trends over time and supports forecasting activities.

6.10.3 Running Totals and Cumulative Metrics

Running totals track progressive performance across defined periods.

6.10.4 Moving Average Analysis

Moving averages smooth fluctuations and highlight long-term trends.

6.10.5 Cohort Analysis

Cohort analysis groups customers based on shared characteristics or acquisition periods.

6.10.6 Funnel Analysis

Funnel analysis measures conversion efficiency between sequential business stages.

6.10.7 Customer Journey Analytics

Customer journey analytics evaluates interactions across the entire lifecycle.

6.10.8 RFM Segmentation Analysis

RFM analysis classifies customers using Recency, Frequency, and Monetary metrics.

6.10.9 Churn Prediction Analytics

Predictive techniques identify customers at risk of leaving the dealership ecosystem.

6.10.10 Attribution Analysis

Attribution analysis determines which marketing activities contribute most to conversions.

6.10.11 Market Basket Analysis

Market basket analysis identifies products frequently purchased together.

6.10.12 Forecasting and Predictive Analytics

Forecasting models estimate future sales, service demand, and inventory requirements.

6.11 Business Intelligence Dashboard Query Layer

The dashboard query layer serves as the foundation for management reporting and executive decision-making.

6.11.1 Executive Dashboard Layer

Provides high-level KPIs for leadership teams.

6.11.2 Dealer Principal Dashboard Layer

Tracks dealership-wide operational and financial performance.

6.11.3 Sales Management Dashboard Layer

Focuses on lead management, conversions, bookings, and retail performance.

6.11.4 CRM Dashboard Layer

Monitors customer engagement, follow-ups, and retention activities.

6.11.5 Finance and Insurance Dashboard Layer

Tracks finance penetration, insurance performance, and payout contributions.

6.11.6 Workshop Dashboard Layer

Measures productivity, service quality, and workshop utilization.

6.11.7 Inventory Dashboard Layer

Provides visibility into stock levels, aging inventory, and allocation efficiency.

6.11.8 Marketing Dashboard Layer

Measures campaign effectiveness, lead generation, and acquisition costs.

6.12 Query Optimization Framework

The Query Optimization Framework establishes best practices for efficient database performance.

6.12.1 Indexing Strategy

Defines indexing approaches to improve query execution speed.

6.12.2 Partitioning Strategy

Improves performance by dividing large datasets into manageable segments.

6.12.3 Materialized Views

Stores pre-computed results for frequently executed analytical queries.

6.12.4 Execution Plan Analysis

Helps identify bottlenecks and inefficient operations.

6.12.5 Parallel Processing Techniques

Utilizes database resources efficiently for large-scale analytics.

6.12.6 Query Cost Monitoring

Measures resource consumption and execution efficiency.

6.12.7 Data Archiving Strategy

Reduces operational database load while maintaining historical access.

6.12.8 Continuous Performance Monitoring

Ensures sustained optimization and system reliability over time.

Some Extra SQL Use Cases For Data Analyst

Examples of business questions.

Query 1

Top Selling Models

SQL

```
SELECT Model,  
COUNT(*) AS TotalSales  
FROM Bookings  
GROUP BY Model  
ORDER BY TotalSales DESC;
```

Query 2

Highest Revenue Consultant

SQL

```
SELECT Consultant,  
SUM(Revenue)  
FROM Sales  
GROUP BY Consultant;
```

Query 3

Inventory Aging

SQL

```
SELECT VIN,  
DATEDIFF(CURDATE(),Arrival_Date)  
FROM Inventory;
```

Query 4

Insurance Renewals Due

SQL

```
SELECT *  
FROM Insurance  
WHERE End_Date <= CURDATE()+30;
```

6.13 Advanced Customer Analytics Library

The Advanced Customer Analytics Library focuses on deep behavioral, financial, and lifecycle analysis of dealership customers. These queries support strategic decision-making, customer retention, and revenue optimization.

6.13.1 Customer Segmentation Using RFM Model

Business Objective:

Segment customers based on Recency, Frequency, and Monetary value to identify high-value and at-risk customers.

SQL Code:

Logic: Groups customers by purchase behavior metrics to classify engagement levels.

```
</> SQL

SELECT
    customer_id,
    CURRENT_DATE - MAX(invoice_date) AS recency,
    COUNT(invoice_id) AS frequency,
    SUM(total_amount) AS monetary_value
FROM invoices
GROUP BY customer_id;
```

6.13.2 High-Value Customer Identification

Business Objective:

Identify top revenue-generating customers for loyalty programs.

SQL Code:

```
</> SQL

SELECT
    customer_id,
    SUM(total_amount) AS total_spend
FROM invoices
GROUP BY customer_id
ORDER BY total_spend DESC
LIMIT 10;
```

6.13.3 Dormant Customer Detection

Business Objective:

Identify customers who have not interacted recently.

SQL Code:

```
</> SQL

SELECT
    customer_id,
    MAX(invoice_date) AS last_purchase_date
FROM invoices
GROUP BY customer_id
HAVING MAX(invoice_date) < CURRENT_DATE - INTERVAL '180 days';
```

6.13.4 Service-to-Sales Conversion Analysis**Business Objective:**

Track how service customers convert into new vehicle buyers.

SQL Code:

```
</> SQL

SELECT
    s.customer_id,
    COUNT(DISTINCT s.ro_number) AS service_visits,
    COUNT(DISTINCT i.invoice_id) AS vehicle_purchases
FROM repair_orders s
LEFT JOIN invoices i ON s.customer_id = i.customer_id
GROUP BY s.customer_id;
```

6.13.5 Customer Engagement Frequency Score**Business Objective:**

Measure how actively customers interact with dealership services.

SQL Code:

SQL

```
SELECT
    customer_id,
    COUNT(*) AS total_interactions
FROM (
    SELECT customer_id FROM invoices
    UNION ALL
    SELECT customer_id FROM repair_orders
) AS activity
GROUP BY customer_id;
```

6.14 Advanced Revenue Intelligence Queries

This section focuses on revenue breakdown, margin analysis, and profitability insights across dealership verticals.

6.14.1 Total Revenue Contribution by Department

Business Objective:

Understand revenue split across Sales, Service, and F&I.

SQL Code:

SQL

```
SELECT
    'Sales' AS department,
    SUM(total_amount) AS revenue
FROM invoices

UNION ALL

SELECT
    'Service',
    SUM(labor_cost + parts_cost)
FROM repair_orders;
```

6.14.2 Gross Margin per Vehicle Model

Business Objective:

Identify most profitable vehicle models.

SQL Code:

```
</> SQL

SELECT
    model_name,
    SUM(selling_price - purchase_cost) AS gross_margin
FROM vehicles
GROUP BY model_name;
```

6.14.3 Monthly Revenue Contribution Trend

Business Objective: Track revenue consistency over time.

SQL Code:

```
</> SQL

SELECT
    DATE_TRUNC('month', invoice_date) AS month,
    SUM(total_amount) AS revenue
FROM invoices
GROUP BY 1
ORDER BY 1;
```

6.14.4 Revenue Leakage Detection

Business Objective:

Identify missing or under-reported revenue entries.

SQL Code:

```
</> SQL

SELECT
    invoice_id,
    total_amount
FROM invoices
WHERE total_amount IS NULL OR total_amount = 0;
```

6.15 Advanced Marketing & Campaign Analytics

This section evaluates marketing effectiveness, lead generation quality, and ROI across campaigns.

6.15.1 Lead Source ROI Analysis

Business Objective:

Measure return on investment per lead channel.

SQL Code:

```
</> SQL

SELECT
    source,
    COUNT(enquiry_id) AS leads,
    COUNT(booking_id) AS conversions
FROM enquiries e
LEFT JOIN bookings b ON e.enquiry_id = b.enquiry_id
GROUP BY source;
```

6.15.2 Campaign Conversion Efficiency

Business Objective: Evaluate marketing campaign performance.

SQL Code:

```
</> SQL

SELECT
    campaign_name,
    COUNT(enquiry_id) AS leads_generated,
    COUNT(booking_id) AS bookings
FROM campaigns c
LEFT JOIN enquiries e ON c.campaign_id = e.campaign_id
LEFT JOIN bookings b ON e.enquiry_id = b.enquiry_id
GROUP BY campaign_name;
```

6.15.3 Cost Per Lead (CPL) Calculation

Business Objective: Determine cost efficiency of lead generation.

SQL Code:

`</> SQL`

```
SELECT
    campaign_name,
    SUM(marketing_cost)/COUNT(enquiry_id) AS cost_per_lead
FROM campaigns c
JOIN enquiries e ON c.campaign_id = e.campaign_id
GROUP BY campaign_name;
```

6.15.4 Lead Quality Score by Source

Business Objective: Rank lead sources based on conversion quality.

SQL Code:

`</> SQL`

```
SELECT
    source,
    ROUND(COUNT(booking_id)::NUMERIC / COUNT(enquiry_id), 2) AS quality_score
FROM enquiries e
LEFT JOIN bookings b ON e.enquiry_id = b.enquiry_id
GROUP BY source;
```

6.16 Advanced Operational Efficiency Metrics

This section focuses on internal process optimization across dealership operations.

6.16.1 Average Process Cycle Time (End-to-End)

Business Objective: Measure total time from enquiry to delivery.

SQL Code:

`</> SQL`

```
SELECT
    AVG(delivery_date - enquiry_date) AS cycle_time
FROM enquiries e
JOIN invoices i ON e.enquiry_id = i.enquiry_id;
```

6.16.2 Employee Productivity Index

Business Objective: Evaluate output per employee.

SQL Code:

```
</> SQL

SELECT
    employee_id,
    COUNT(enquiry_id) AS handled_leads
FROM enquiries
GROUP BY employee_id;
```

6.16.3 Process Delay Bottleneck Detection

Business Objective: Identify stage causing maximum delay.

SQL Code:

```
</> SQL

SELECT
    stage,
    AVG(process_time) AS avg_time
FROM process_tracking
GROUP BY stage
ORDER BY avg_time DESC;
```

6.16.4 SLA Compliance Tracking

Business Objective: Monitor whether operations meet defined service level agreements.

SQL Code:

```
</> SQL

SELECT
    COUNT(*) AS total_tasks,
    COUNT(CASE WHEN status = 'ON_TIME' THEN 1 END) AS on_time_tasks
FROM tasks;
```

6.17 Advanced Predictive & Forecasting Analytics

This section introduces forward-looking analytical models for forecasting dealership performance.

6.17.1 Sales Forecasting (Simple Trend Model)

Business Objective: Predict future sales using historical trends.

SQL Code:

```
</> SQL

SELECT
    DATE_TRUNC('month', invoice_date) AS month,
    SUM(total_amount) AS sales
FROM invoices
GROUP BY 1;
```

6.17.2 Inventory Demand Forecast

Business Objective: Predict future inventory requirements.

SQL Code:

```
</> SQL

SELECT
    model_name,
    COUNT(*) AS past_sales
FROM vehicles
WHERE stock_status = 'DELIVERED'
GROUP BY model_name;
```

6.17.3 Service Demand Prediction

Business Objective: Estimate future workshop load.

SQL Code:

```
</> SQL

SELECT
    DATE_TRUNC('month', open_date) AS month,
    COUNT(ro_number) AS service_volume
FROM repair_orders
GROUP BY 1;
```

6.17.4 Churn Probability Indicator

Business Objective: Identify likely customer churn patterns.

SQL Code:

```
</> SQL

SELECT
    customer_id,
    MAX(invoice_date) AS last_purchase
FROM invoices
GROUP BY customer_id;
```

6.18 Executive Decision Support Queries

This section provides high-level KPIs for leadership decision-making.

6.18.1 Overall Dealership Performance Index

Business Objective: Combine multiple KPIs into a single performance indicator.

SQL Code:

```
</> SQL

SELECT
    SUM(total_amount) AS revenue,
    COUNT(invoice_id) AS total_sales
FROM invoices;
```

6.18.2 Monthly KPI Dashboard Snapshot

Business Objective: Provide snapshot of key business metrics.

SQL Code:

```
</> SQL

SELECT
    DATE_TRUNC('month', invoice_date) AS month,
    COUNT(invoice_id) AS sales,
    SUM(total_amount) AS revenue
FROM invoices
GROUP BY 1;
```

6.18.3 Dealer Performance Ranking

Business Objective: Rank dealership branches or teams.

SQL Code:

```
</> SQL

SELECT
    branch_id,
    SUM(total_amount) AS revenue
FROM invoices
GROUP BY branch_id
ORDER BY revenue DESC;
```

6.18.4 KPI Health Scorecard

Business Objective: Evaluate overall business health.

SQL Code:

```
</> SQL

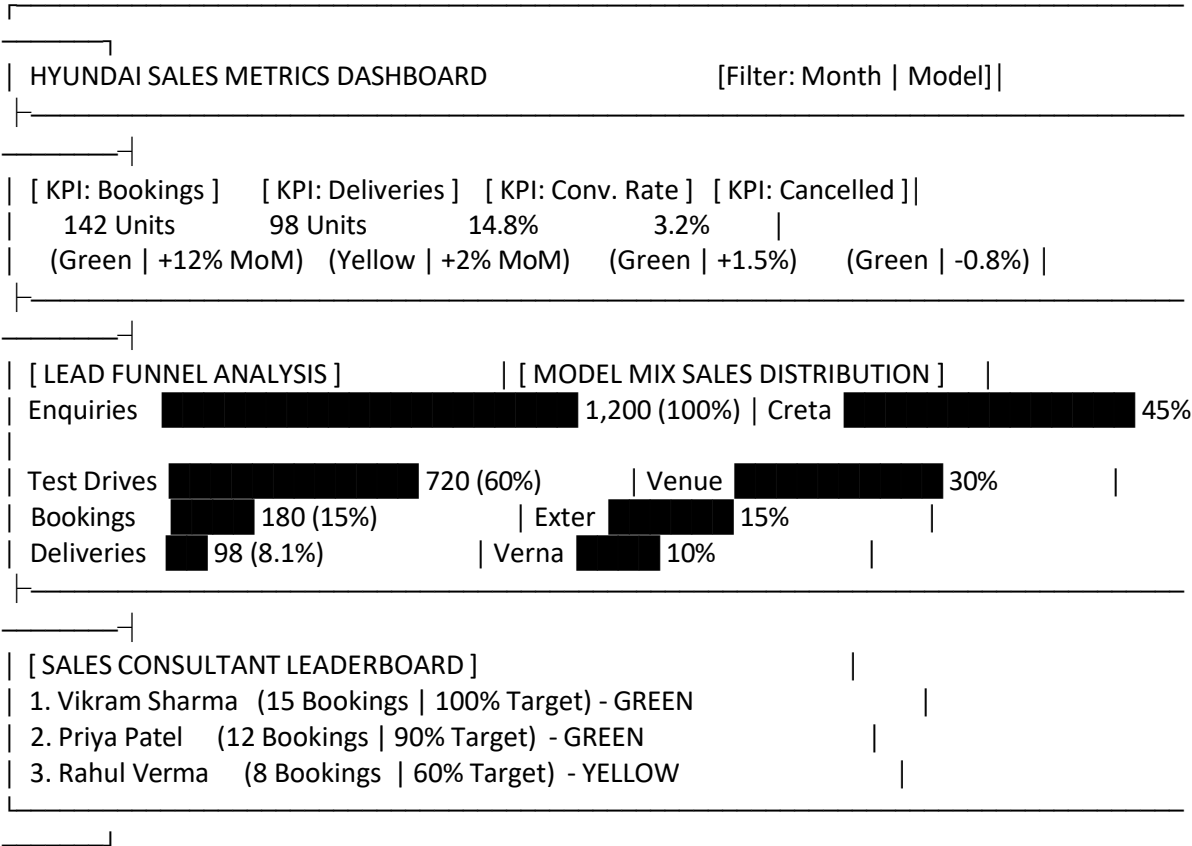
SELECT
    COUNT(DISTINCT customer_id) AS active_customers,
    SUM(total_amount) AS total_revenue
FROM invoices;
```

7. Dashboard Design & Wireframe Specifications

This section outlines UI/UX layouts, wireframes, and data models for Power BI or Tableau dashboards, tailored for dealership role-based analytics.

7.1 Showroom Sales & Lead Conversion Dashboard

7.1.1 Wireframe Layout Mockup

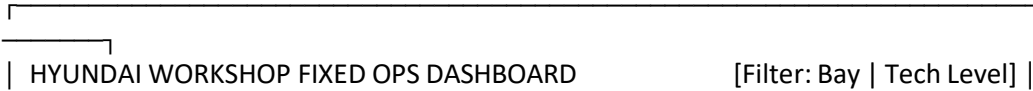


7.1.2 Functional Requirements & Data Sources

- Target Audience:** Dealer Principal, Sales Director, Sales Managers.
- Data Ingestion:** Sourced from HSSmart CRM and GDMS.
- Key Interactivity:** Drill down by Sales Consultant, Model Type, Transmission Type, and lead source (Reference vs. Walk-in).

7.2 Workshop & Fixed Operations Dashboard

7.2.1 Wireframe Layout Mockup



[KPI: Vehicles Serviced]										[KPI: Tech Productivity]										[Avg RO Bill]										[Absorption]									
480 Vehicles										88.5%										₹4,200										81.2%									
(Green +8% MoM)										(Yellow -2.1% MoM)										(Green +₹350)										(Green +4%)									
[WORKSHOP THROUGHPUT & PEAK HOURS]										[SPARE PARTS STOCK VALUATION]																													
09:00 - 11:00										150 In										Fast-Moving Parts										60%									
11:00 - 13:00										100 In										Med-Moving Parts										30%									
13:00 - 15:00										60 In										Slow-Moving Parts										10%									
15:00 - 18:00										120 Out										(Alert: Lockup Capital ₹2.4 Lakhs)																			
[REPEAT REPAIR ALERTS (FTR DEFECTS)]																																							
- RO-45892: VIN...8902 returned for Brake Noise (4 Days post service)																																							
- RO-46210: VIN...4452 returned for Engine vibration (10 Days post service)																																							

7.2.2 Functional Requirements & Data Sources

- **Target Audience:** Workshop Managers, Spare Parts Managers, Service Advisors.
- **Data Ingestion:** GDMS Service module and Tally Prime workshop billing ledger.
- **Key Interactivity:** Drill down by Technician Level (Master/Junior), Bay Number, and periodic maintenance intervals (10k, 20k, 50k km services).

7.3 Accounts & Financial Reconciliation Dashboard

7.3.1 Wireframe Layout Mockup

FINANCIAL AUDIT & TAX COMPLIANCE DASHBOARD										[Filter: Quarter Year]																			
[Total Receivable] [Debtor Aging (>30d)] [Paymate Settlement] [GST Liability]																													
₹45.2 Lakhs					₹4.1 Lakhs					18 Hours					₹12.4 Lakhs														
(Yellow +5% MoM)					(Green -15% MoM)					(Green -4h MoM)					(Reconciled)														
[DEBTOR AGING OUTSTANDING BALANCES]										[FLOORPLANNING SOURCING INTER-COST]																			
0 - 15 Days										₹32 Lakhs					Month 1 (Base)														
2%															1														
16 - 30 Days										₹9.1 Lakhs					Month 2 (Aging)										1.8%				
> 30 Days										₹4.1 Lakhs (Action Required)					Month 3+ (Stale)														
					2.4%																								
[PAYMATE TRANSACTION RECONCILIATION]																													
- Reconciled Invoices: 99.8% (Target: 100%)																													

| - Pending Ledger Matches: 2 Invoices (RO-46102, INV-98441) |

7.3.2 Functional Requirements & Data Sources

- **Target Audience:** Finance Controller, Accounts Manager, Dealer Principal.
- **Data Ingestion:** Tally Prime database (via XML import script) and Paymate settlements portal.
- **Key Interactivity:** Drill down by Debtor ID, Aging Bracket, and Transaction Type (New Car Sales, Used Cars, Accessories, Service).

7.4 Role-Based Access Control (RBAC) Matrix

Dashboard View	Dealer Principal	Sales Manager	Service Manager	Finance Manager	Sales Consultant
New Sales Metrics	Read/Write	Read/Write	No Access	Read	Personal Only
F&I Commissions	Read	Read	No Access	Read/Write	No Access
Workshop Fixed Ops	Read	No Access	Read/Write	Read	No Access
Inventory Hold Costs	Read/Write	Read	Read	Read	No Access
Debtor Ledger & GST	Read	No Access	No Access	Read/Write	No Access
Telegram Sync Status	Read	Read/Write	Read/Write	Read	Read

7.5 Dashboard Architecture

A Data Analyst should design dashboards for management.

Dashboard 1 – Executive Dashboard

Shows:

- Revenue
- Profit
- Sales
- Inventory
- Insurance
- Service

Dashboard 2 – Sales Dashboard

Shows:

- Leads
- Conversion
- Bookings
- Deliveries

Dashboard 3 – Inventory Dashboard

Shows:

- Stock
- Aging
- Availability

Dashboard 4 – Insurance Dashboard

Shows:

- Policies
- Renewals
- Premium Revenue

Dashboard 5 – Service Dashboard

Shows:

- Service Revenue
- Retention
- Satisfaction

7.6 Data Analyst Conclusion

At this stage, management Queirs:

- What happened?
- Why did it happen?
- Where did it happen?
- Who is responsible?
- What should be done next?

This is where reporting ends and business decision-making begins.

Major Business Problems in a Hyundai Dealership

Every department faces business challenges.

The objective of analytics is to identify these challenges early.

7.6.1 Sales Problems

- Low enquiry conversion
- High booking cancellation
- Low test drive conversion
- Consultant underperformance
- Slow lead response

7.6.2 Inventory Problems

- Vehicle shortages
- Excess inventory
- Wrong stock planning
- Allocation delays

7.6.3 Customer Problems

- Poor satisfaction
- Complaints
- Negative reviews
- Low retention

7.6.4 Financial Problems

- Outstanding payments
- Revenue leakage
- Profit margin pressure

7.7 Root Cause Analysis Framework

Whenever a problem occurs, management wants to know:

What Happened?

Example:

Bookings decreased by 20%.

Why Did It Happen?

Possible causes:

- Fewer leads
- Poor follow-up
- Competitor discounts
- Inventory shortages

Where Did It Happen?

- Specific branch
- Specific consultant
- Specific vehicle model

What Action Should Be Taken?

This is where analytics creates value.

7.8 Dashboard User Stories

Dashboard user stories define who uses the dashboard, why they use it, and what business decisions are expected from it.

7.8.1 Executive Dashboard User

Story Component Details

User Role Dealer Principal

Dashboard Name Executive Performance Dashboard

Purpose Overall dealership performance monitoring

Access Frequency Daily

Business Need Quick visibility of revenue, profit and operations

Key Decisions Sales targets, inventory planning, staffing

KPIs Required Revenue, Gross Profit, Bookings, Deliveries, Workshop Revenue, Insurance Revenue

Expected Actions Approve corrective actions and strategic plans

Typical Questions

- Are we achieving monthly targets?

- Which department is underperforming?
 - What is current profitability?
 - What risks require immediate action?
-

7.8.2 Sales Manager Dashboard User Story

Component	Details
User Role	Sales Manager
Purpose	Monitor sales funnel performance
Access Frequency	Multiple times daily
Key KPIs	Leads, Test Drives, Bookings, Deliveries
Decisions	Consultant allocation, campaign effectiveness
Actions	Follow-up improvement and target monitoring

Typical Questions

- Which consultant has lowest conversion?
 - Which lead source performs best?
 - Which model sells fastest?
 - What is booking cancellation rate?
-

7.8.3 CRM Manager Dashboard User Story

Component	Details
User Role	CRM Manager
Purpose	Customer retention and relationship management
Access Frequency	Daily
KPIs	CSI, Complaints, Repeat Customers
Decisions	Customer engagement strategy
Actions	Escalation management

Typical Questions

- Which customers are dissatisfied?

- Which complaints remain unresolved?
 - What is customer retention rate?
-

7.8.4 Workshop Manager Dashboard User

Story Component Details

User Role	Service Manager
Purpose	Workshop productivity monitoring
Access Frequency	Daily
KPIs	RO Count, Technician Productivity, Service Revenue
Decisions	Resource planning
Actions	Technician assignment

7.8.5 Inventory Manager Dashboard User

Story Component Details

User Role	Stockyard Manager
Purpose	Inventory control
KPIs	Stock Ageing, Vehicle Availability
Decisions	Vehicle ordering
Actions	Inventory redistribution

7.8.6 Finance Manager Dashboard User Story

Component Details

User Role	Finance Controller
Purpose	Financial monitoring
KPIs	Revenue, GST, Receivables
Decisions	Cash flow planning
Actions	Collection follow-up

7.8.7 Insurance Manager Dashboard User

Story Component Details

User Role	Insurance Manager
Purpose	Insurance performance monitoring
KPIs	Insurance Penetration, Renewals
Decisions	Partner optimization
Actions	Renewal campaigns

7.8.8 HR Manager Dashboard User Story

Component Details

User Role	HR Manager
Purpose	Workforce analytics
KPIs	Attendance, Attrition, Productivity
Decisions	Hiring and training
Actions	Performance reviews

7.9 Detailed Dashboard Specifications

7.9.1 Executive Dashboard Specification

Layout Structure

+-----+					
	Header	&	Filters		
+-----+					
	KPI	Cards			
+-----+					
	Revenue	Trend		Profit	Trend
+-----+					
	Sales		Service		Insurance
					Inventory
+-----+					

Visual Specifications

Component	Visual
Revenue Trend	Line Chart
Profit Trend	Line Chart
Department Comparison	Clustered Bar
KPI Summary	Cards
Alerts	Tables

Filters

Filter

- Year
- Quarter
- Month
- Branch
- Vehicle Model

Refresh Frequency

Dashboard Refresh

Executive Every 2 Hours

7.9.2 Sales Dashboard Specification

Layout

+-----+

Sales KPIs
+-----+
Lead Funnel

|

+-----+				
	Lead	Source	Analysis	
+-----+				
	Consultant	Performance		
+-----+				
	Booking	Trend		
+-----+				

KPI Definitions

KPI	Formula
Lead Conversion	Bookings ÷ Leads
Test Drive Conversion	Bookings ÷ Test Drives
Cancellation Rate	Cancelled ÷ Bookings
Delivery Rate	Deliveries ÷ Bookings

7.9.3 Inventory Dashboard Specification

KPIs

KPI

Stock on Hand

Inventory Aging

Days Supply Inventory

Fast Moving Models

Slow Moving Models

Alert Thresholds

Metric	Threshold
Aging > 45 Days	Warning
Aging > 60 Days	Critical
Stock < 10 Units	Reorder

7.9.4 Service Dashboard Specification

KPIs

KPI

Workshop Revenue

RO Count

Technician Productivity

Service Retention

First Time Right Rate

Drill-down Path

Workshop



Advisor



Technician



Repair Order

7.10 Dashboard Wireframe Descriptions

7.10.1 KPI Section Description

Purpose:

Provide immediate visibility of performance indicators.

Example:

Revenue	
₹1.2 Crore	
+12%	

Benefits:

- Quick management review
- Trend visibility

- Target tracking

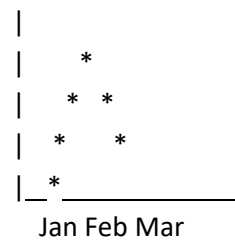
7.10.2 Trend Analysis Section

Purpose:

Show business movement over time.

Visual:

Revenue



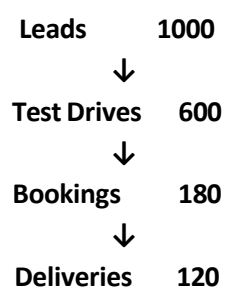
Benefits:

- Trend identification
- Seasonality analysis
- Forecasting support

7.10.3 Funnel Visualization

Purpose:

Identify drop-offs.



Benefits:

- Conversion monitoring
- Sales effectiveness

7.10.4 Geographic Analysis Section

Purpose:

Compare branch performance.

Visual:

Delhi	
Noida	
Gurgaon	

Benefits:

- Branch benchmarking
- Territory optimization

7.11 Dashboard Design Guidelines

7.11.1 Color Standards

Status	Color
--------	-------

Good	Green
------	-------

Warning	Amber
---------	-------

Critical	Red
----------	-----

Neutral	Blue
---------	------

7.11.2 Typography

Standards	Element	Size
-----------	---------	------

	Dashboard Title	24 px
--	-----------------	-------

	Section Title	18 px
--	---------------	-------

	KPI Value	28 px
--	-----------	-------

	Axis Labels	12 px
--	-------------	-------

7.11.3 Visual Hierarchy Principles

Priority order:

1. KPI Cards
2. Trends

- 3. Comparisons
- 4. Details
- 5. Raw Data Tables

7.11.4 Accessibility Standards

Requirements:

- High contrast colors
- Readable fonts
- Tooltips
- Keyboard navigation
- Mobile compatibility

7.11.5 Mobile Dashboard Guidelines

Layout:

KPI 1
KPI 2
KPI 3

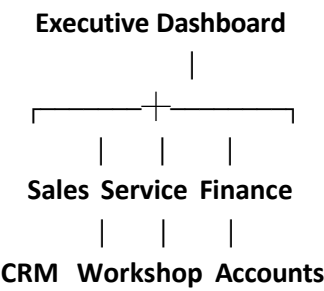
Trend

Table

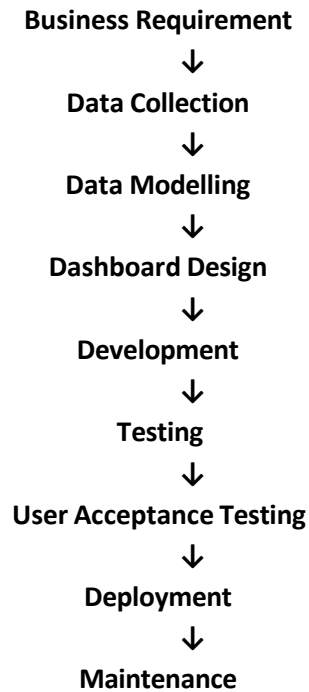
Rules:

- Single column layout
- Larger touch targets
- Simplified visuals

7.12 Dashboard Navigation Architecture



7.13 Dashboard Development Lifecycle



7.14 Dashboard Testing Framework

Test Type	Purpose
Data Testing	Validate calculations
Performance Testing	Speed verification
Security Testing	Access validation
UAT Testing	Business acceptance
Visual Testing	Layout verification

7.15 Dashboard Performance

Standards Metric	Target
Page Load Time	< 5 Seconds
Query Execution	< 10 Seconds

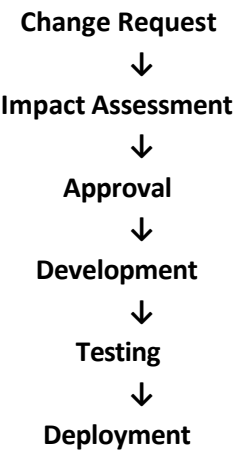
Metric	Target
Dataset Refresh	< 30 Minutes
Availability	99.9%

7.16 Dashboard Governance Framework

Ownership Matrix

Dashboard Owner	
Executive	Dealer Principal
Sales	Sales Manager
CRM	CRM Manager
Inventory	Inventory Manager
Service	Workshop Manager
Finance	Finance Controller
Insurance	Insurance Manager
HR	HR Manager

Change Management Process



7.17 Dashboard Success Measurement KPIs

KPI	Objective
Active Users	Adoption Measurement
Dashboard Usage Frequency	Engagement
Average Session Duration	Usability
Decision Turnaround Time	Business Impact
Data Accuracy	Reliability
User Satisfaction	Acceptance

7.18 Dashboard Design Conclusion

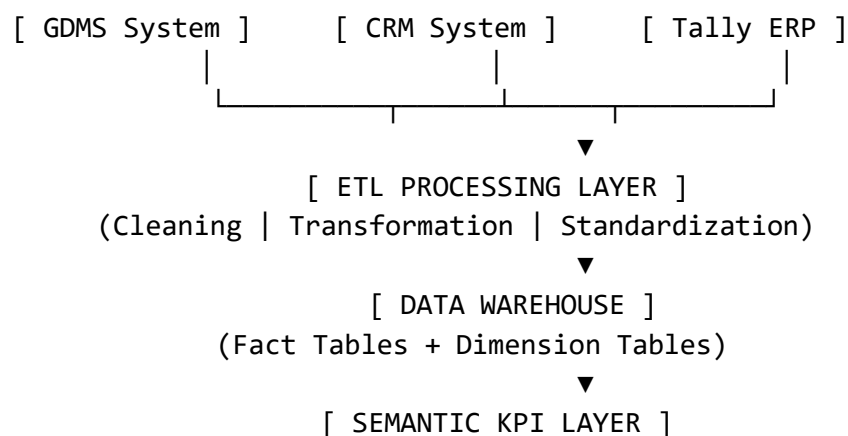
A well-designed dashboard should answer five critical management questions:

1. What happened?
2. Why did it happen?
3. Where did it happen?
4. Who is responsible?
5. What action should be taken?

The dashboard layer converts raw dealership data from GDMS, HSSmart CRM, Tally Prime, Insurance Systems, Service Systems, and HR Systems into actionable business intelligence, enabling faster and more accurate decision-making across the Hyundai dealership ecosystem.

7.19 DATA FLOW ARCHITECTURE (END-TO-END SYSTEM FLOW)

7.19.1 End-to-End Data Flow Diagram



(Business Logic / KPI Engine)



[POWER BI / TABLEAU LAYER]



[MANAGEMENT DASHBOARDS]

Business Value:

- Single source of truth
- Eliminates inconsistent reporting
- Enables real-time analytics

7.20 SALES DASHBOARD DATA MODEL (STAR SCHEMA – EXCEL STYLE)

7.20.1 Fact Table: fact_sales

sale_id	customer_id	model	sale_date	booking_value	employee_id
S001	C101	Creta	2026-01-01	1200000	E01
S002	C102	Venue	2026-01-02	850000	E02

7.20.2 Dimension Table: dim_customer

customer_id	name	City	source	segment
C101	Amit	Delhi	Walk-in	Premium
C102	Ravi	Noida	Digital	Regular

7.20.3 Dimension Table: dim_employee

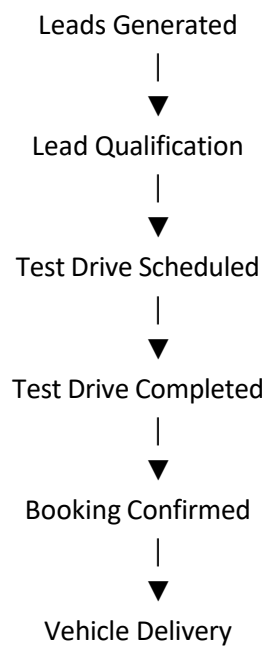
employee_id	name	Role	branch
E01	Rahul	Sales Consultant	Delhi
E02	Priya	Sales Consultant	Noida

Business Value:

- Faster filtering & slicing
- Optimized Power BI performance

- Scalable reporting structure
-

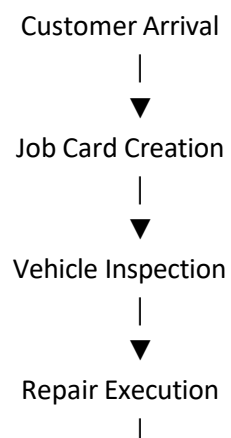
7.21 SALES FUNNEL FLOWCHART

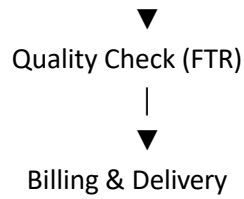


Insight:

- Biggest drop-off: Test Drive → Booking
 - Action: Improve follow-up + sales training
-

7.22 WORKSHOP SERVICE FLOW (FIXED OPERATIONS)

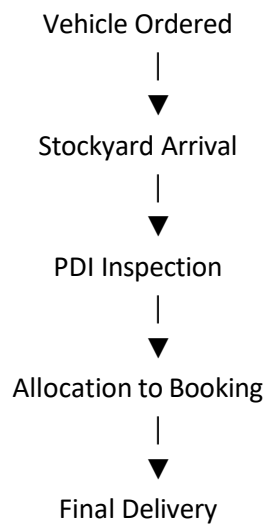




KPI Mapping

Stage	KPI
Job Card	RO Count
Repair	Technician Efficiency
QC	First Time Right Rate
Billing	Service Revenue

7.23 INVENTORY FLOW PROCESS



Risk Points:

- Delayed PDI
 - Stock aging (>45 days)
 - Allocation delays
-

7.24 DASHBOARD KPI CALCULATION TABLES (EXCEL FORMAT)

7.24.1 Sales KPI Table

KPI	Formula	Meaning
Lead Conversion Rate	$\text{Bookings} \div \text{Leads}$	Sales efficiency
Test Drive Ratio	$\text{Test Drives} \div \text{Leads}$	Customer engagement
Delivery Rate	$\text{Deliveries} \div \text{Bookings}$	Execution success

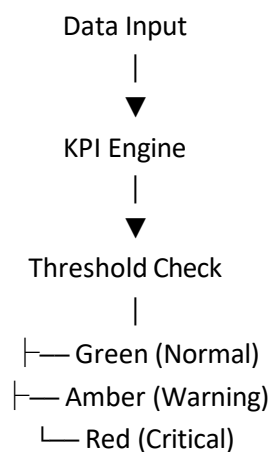
7.24.2 Finance KPI Table

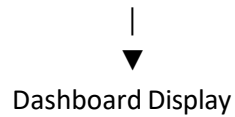
KPI	Formula	Meaning
Gross Profit	$\text{Revenue} - \text{Cost}$	Profitability
GST Liability	Tax Collected	Compliance
Receivable Days	$\text{Outstanding} / \text{Sales}$	Cash flow health

7.24.3 Service KPI Table

KPI	Formula	Meaning
Technician Efficiency	$\text{Jobs} / \text{Hours}$	Productivity
First Time Fix Rate	$\text{No repeat RO} / \text{Total RO}$	Quality
Service Revenue	Labor + Parts	Revenue tracking

7.25 DASHBOARD ALERT FLOW SYSTEM

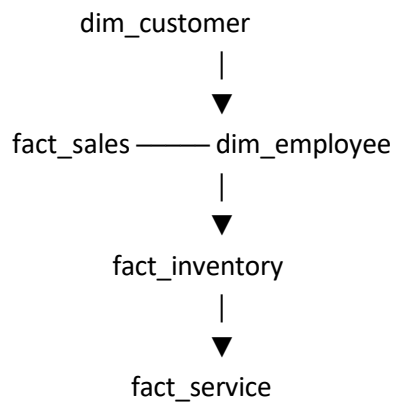




Alert Rules

KPI	Warning	Critical
Conversion Rate	<12%	<8%
Inventory Aging	>45 days	>60 days
Service Delay	>2 days	>5 days

7.26 DATA RELATIONSHIP MODEL



Relationship Logic:

- Customer lifecycle tracking
 - Employee performance tracking
 - Inventory-to-sales linkage
-

7.27 DASHBOARD PAGE STRUCTURE (POWER BI DESIGN)

Page 1: Executive Dashboard

- Revenue KPI
- Profit KPI
- Sales Trend
- Alerts

Page 2: Sales Dashboard

- Funnel Analysis
- Consultant Ranking
- Model-wise Sales

Page 3: Inventory Dashboard

- Stock Aging
- Fast/Slow movers
- Days Supply

Page 4: Workshop Dashboard

- RO Count
- Technician Efficiency
- Service Revenue

Page 5: Finance Dashboard

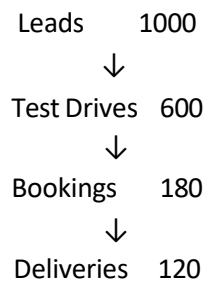
- GST
 - Receivables
 - Cash Flow
-

7.28 DASHBOARD WIREFRAME ENHANCEMENTS

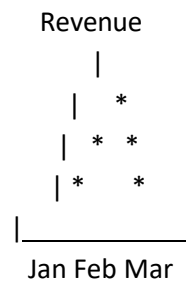
7.28.1 KPI Card Layout

Revenue	
₹1.2 Cr	
+12% MoM	

7.28.2 Funnel Visualization



7.28.3 Trend Chart Example



7.29 DASHBOARD DESIGN PRINCIPLES

7.29.1 Color Standards

Status	Color
Good	Green
Warning	Amber
Critical	Red
Neutral	Blue

7.29.2 Typography Standards

Element	Size
Title	24 px
KPI Value	28 px
Section	18 px
Labels	12 px

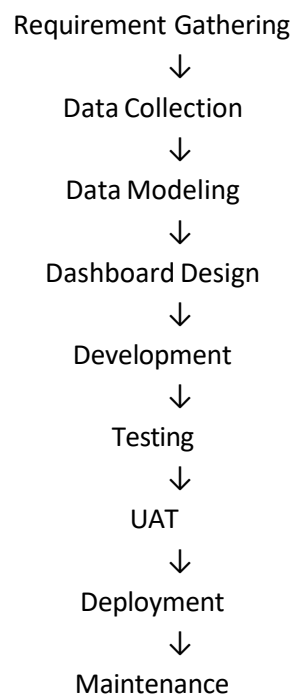
7.29.3 Visual Hierarchy

1. KPI Cards
2. Trends
3. Comparisons
4. Tables

7.29.4 Mobile Design

- Single column layout
- Large touch targets
- Simplified visuals

7.30 DASHBOARD DEVELOPMENT LIFECYCLE



7.31 DASHBOARD PERFORMANCE STANDARDS

Metric	Target
Load Time	< 5 sec

Metric	Target
Query Time	< 10 sec
Refresh Time	< 30 min
Availability	99.9%

7.32 DASHBOARD GOVERNANCE MATRIX

Dashboard Owner

Executive Dealer Principal

Sales Sales Manager

CRM CRM Manager

Inventory Stock Manager

Service Workshop Manager

Finance Finance Controller

7.33 DASHBOARD SUCCESS KPIS

KPI	Meaning
Active Users	Adoption
Usage Frequency	Engagement
Decision Time	Business impact
Data Accuracy	Reliability
User Satisfaction	Acceptance

8. Dealership Business Problems & Solutions

This section details **20 critical business problems** faced by **Universal Hyundai ShriGurukripa Automobiles LLP** and **Sell Any Car**, providing Root Cause Analyses (RCA) and data-driven solutions.

Problem 1: Low Reference Sales Flow Due to Showroom Age

- **Problem Statement:** Reference sales represent only 2% of total retail bookings, compared to the industry standard of 8% to 10%, because the dealership was established only 6 years ago.
- **Root Cause Analysis (5 Whys):**
 1. *Why are reference sales low?* Customer referrals are not generating leads.
 2. *Why are referrals not generating leads?* Customers do not actively recommend the dealership.
 3. *Why do they not recommend the dealership?* There is no formal post-sale referral engagement or incentive structure.
 4. *Why is there no formal program?* Referral acquisition was not a priority during the initial 6-year growth phase, which focused on walk-ins and local advertising.
 5. *Why was this the focus?* A younger dealership footprint relies on high-volume walk-ins before building a mature customer retention base.
- **Analytical Solution:**
 - Implement an **automated customer referral tracker** in HSSmart.
 - Trigger an automated SMS/WhatsApp referral campaign 30 days post-delivery offering incentives (e.g., free service coupons, accessory discounts).
 - Monitor the referral conversion rate:

$$\text{Referral Conv. Rate} = \frac{\text{Bookings Sourced from Referrals}}{\text{Total Referrals Sourced}}$$

Problem 2: Walk-in Enquiry Drop-off Before Booking Stage

- **Problem Statement:** 78% of walk-in showroom visitors leave without creating a booking or scheduling a test drive.
 - **Root Cause Analysis:**
 - Sales consultants fail to log customer profiles immediately in HSSmart.
 - Consultants do not perform structured needs analysis, leading to mismatched model pitches.
 - **Analytical Solution:**
 - Establish a **reception-level digital registration kiosk** synced directly to HSSmart.
 - Track the Walk-in Leakage KPI daily:
$$\text{Walk-in Leakage} = \frac{\text{Walk-ins without Test Drive or Booking}}{\text{Total Showroom Walk-ins}}$$
 - Correlate walk-in bounce rates with individual consultants to identify training needs.
-

Problem 3: Test Drive Leakage (Unlogged Test Drives)

- **Problem Statement:** Customer test drives occur physically but are not logged in HSSmart, risking vehicle wear-and-tear liabilities.
- **Root Cause Analysis:**
 - Consultants bypass HSSmart logging to save time.
 - There is no verification process between physical vehicle movements and digital logs.
- **Analytical Solution:**
 - Install GPS telematics devices in all demo vehicles.

- Automate reconciliations between GPS trip logs and HSSmart test drive entries.
- Flag any unlogged test drive trips over 1 km to the Sales Director.

Problem 4: High Booking Cancellation Rates in the Loan Approval Phase

- **Problem Statement:** 15% of bookings are cancelled due to loan rejections or delays, locking up vehicle inventory.
- **Root Cause Analysis:**
 - Consultants accept bookings from unqualified buyers without pre-screening their credit.
 - Financing partners experience processing backlogs, delaying approvals.
- **Analytical Solution:**
 - Build a **credit pre-qualification calculator** into HSSmart.
 - Require a preliminary CIBIL score check before allocating a vehicle VIN.
 - Track the Loan Approval Lead Time KPI:

$$\text{Loan Approval Lead Time} = \text{Date of Loan Disbursement} - \text{Date of Booking}$$

Problem 5: Inefficient Interdealer Car Swaps Increasing Holding Costs

- **Problem Statement:** Interdealer transfers are frequently conducted without accounting for shipping costs or inventory aging, reducing gross margins.
- **Root Cause Analysis:**
 - The EDP Manager swaps vehicles to close immediate bookings without checking if identical models are in transit or in the stockyard.
- **Analytical Solution:**
 - Establish an **Interdealer Trade Approval Matrix** in GDMS.
 - Incorporate swap shipping costs (typically ₹3,000 - ₹5,000) directly into the vehicle's cost sheet.
 - Restrict swaps if an identical model is scheduled to arrive at the yard within 7 days.

Problem 6: Stockyard Capacity Bottlenecks During Festive Months

- **Problem Statement:** The stockyard exceeds its 500-vehicle capacity during peak seasons (Dhanteras, Diwali), causing storage overflows and vehicle damage.
- **Root Cause Analysis:**
 - Wholesale purchasing targets are frontloaded without coordinating deliveries with retail timelines.
- **Analytical Solution:**
 - Implement an **Inventory Capacity Forecasting Model** in the dashboard.
 - Calculate the Sourcing-to-Delivery ratio:

$$\text{Sourcing-to-Delivery Ratio} = \frac{\text{Wholesale Sourcing Volume}}{\text{Retail Delivery Volume}}$$
 - Trigger automatic alerts when yard capacity exceeds 90% to adjust plant shipping schedules.

Problem 7: Delayed Billing Reconciliation Between GDMS and Tally Prime

- **Problem Statement:** Reconciling invoices between GDMS and Tally Prime takes up to 7 days, causing delays in monthly financial reporting.
 - **Root Cause Analysis:**
 - Staff manually export invoices from GDMS and re-key them into Tally Prime, resulting in input errors.
 - **Analytical Solution:**
 - Deploy an **automated XML data bridge** that runs nightly exports from GDMS and imports them directly into Tally Prime.
 - Generate a daily automated discrepancy report to flag tax or rounding mismatches.
-

Problem 8: Used Car Valuation-to-Purchase Leakage (Sell Any Car)

- **Problem Statement:** Only 12% of evaluated used cars are purchased by the dealership, allowing potential trade-in sales to lapse.
 - **Root Cause Analysis:**
 - Valuation offers are delayed or non-competitive compared to online platforms.
 - **Analytical Solution:**
 - Deploy a **used car valuation engine** using historical auction and retail market pricing.
 - Require evaluators to submit quotes to customers within 30 minutes of inspection.
 - Monitor the Evaluation-to-Purchase conversion rate:
$$\text{Evaluation-to-Purchase Conv. Rate} = \frac{\text{Purchased Used Cars}}{\text{Evaluated Used Cars}} \times 100$$
-

Problem 9: Low Value-Added Service (VAS/WATS) Attachment on Entry Models

- **Problem Statement:** Entry-level models (Exter, Grand i10) show less than 15% VAS attachment, compared to 55% for premium models (Creta, Tucson).
 - **Root Cause Analysis:**
 - Sales consultants assume budget buyers will not purchase add-on treatments and fail to pitch them.
 - **Analytical Solution:**
 - Bundle popular VAS treatments (Teflon coating, anti-rust underbody) directly into attractive entry-level vehicle purchase packages.
 - Tie consultant commission payouts to overall VAS attachment targets across all vehicle segments.
-

Problem 10: High Cost of Inventory Aging (>45 Days in Stockyard)

- **Problem Statement:** Stale inventory incurs 8% floorplanning interest, eroding the dealership's net margin.
- **Root Cause Analysis:**
 - The dealership stocks unpopular color and variant combinations to meet manufacturer wholesale targets.
- **Analytical Solution:**
 - Create an **Aging Stock Alert System** on the dashboard.

- Automatically apply progressive dealer discounts or salesperson incentives to vehicles aging past 30 days.
- Restrict the sourcing of variants with historical days-in-yard averages exceeding 40 days.

Problem 11: Decline in Service Absorption Ratio

- **Problem Statement:** Service absorption fell to 48%, leaving the dealership reliant on volatile new vehicle sales margins to cover fixed operating costs.
- **Root Cause Analysis:**
 - Workshop traffic is declining due to poor retention of post-warranty customers.
- **Analytical Solution:**
 - Launch an automated database service reminder campaign targeting vehicles aged 3 to 5 years.
 - Offer bundled service packages (e.g., free oil filter with periodic maintenance) to retain post-warranty customers.

Problem 12: Low Insurance Renewal Rates via Cold Calling

- **Problem Statement:** Insurance CREs renew only 32% of expiring policies, losing out on high-margin renewal commissions.
- **Root Cause Analysis:**
 - CREs make calls too close to policy expiration dates, after customers have already renewed online or elsewhere.
- **Analytical Solution:**
 - Automate the renewal pipeline to flag expiring policies in Tally/GDMS 45 days in advance.
 - Send automated quotes via WhatsApp 30 days before expiration, followed by structured CRE follow-up calls.

Problem 13: Poor Technician Productivity in Workshop Bays

- **Problem Statement:** Billed hours account for only 68% of clocked hours, leading to long customer wait times and lost workshop revenue.
- **Root Cause Analysis:**
 - Workshop layout bottlenecks and delays in parts issuance keep technicians waiting for materials.
- **Analytical Solution:**
 - Implement a digital parts request system linking technicians directly to the parts counter.
 - Track the Technician Productivity KPI daily:

$$\text{Technician Productivity} = \frac{\text{Billed Hours}}{\text{Total Clocked Hours}} \times 100$$

Problem 14: Delayed Paymate Bank Settlements

- **Problem Statement:** Customer booking payments processed via Paymate take up to 4 business days to settle, impacting cash flow.

- **Root Cause Analysis:**
 - Accounting staff delay uploading matching transaction receipts, slowing bank processing.
 - **Analytical Solution:**
 - Integrate Paymate's API directly into the Tally Prime database to automate transaction matching.
 - Generate daily alerts for outstanding settlements exceeding 48 hours.
-

Problem 15: Inaccurate Commission Audits on Warranty Parts Replacement

- **Problem Statement:** Rejection of warranty claims by HMIL results in losses on parts and labor costs.
 - **Root Cause Analysis:**
 - Service advisors fail to upload required diagnostic logs and photos to GDMS, violating warranty submission guidelines.
 - **Analytical Solution:**
 - Require digital verification of all diagnostic reports and photo uploads in GDMS before a warranty repair order can be closed.
 - Implement a monthly reconciliation dashboard comparing claims submitted vs. claims rejected.
-

Problem 16: Disconnect between Sales Trainer Events and Booking Conversions

- **Problem Statement:** Showroom promotional events generate foot traffic but yield low booking conversions.
 - **Root Cause Analysis:**
 - Events attract unqualified leads due to generic advertising.
 - **Analytical Solution:**
 - Use historical CRM data to target digital event invitations to high-intent segments (e.g., customers with active enquiries or trade-in evaluations).
 - Track the Event Conversion KPI:
$$\text{Event Conversion Rate} = \frac{\text{Bookings Generated during Event}}{\text{Total Event Attendees}} \times 100$$
-

Problem 17: Lack of Visibility in Telegram Sales Coordination Group

- **Problem Statement:** Follow-up alerts shared on Telegram are missed, allowing high-intent leads to turn cold.
 - **Root Cause Analysis:**
 - High message volumes clutter the group chat, causing staff to miss key alerts.
 - **Analytical Solution:**
 - Deploy a Telegram Bot that posts structured alerts for overdue follow-ups and processes updates directly from the chat.
 - Track the Telegram Response Compliance rate in the management dashboard.
-

Problem 18: High Transit Damage Rate on New Vehicle Arrivals

- **Problem Statement:** 0.8% of new vehicles arrive from the plant with minor transit damage, delaying deliveries.
 - **Root Cause Analysis:**
 - Transporters use substandard tie-down methods during shipping.
 - **Analytical Solution:**
 - Require the EDP Executive to log all transit damage in a GDMS claim report within 2 hours of arrival.
 - Charge refurbishment costs back to the transport vendor's account, monitored via vendor ledger balances.
-

Problem 19: Discrepancy Between Physical Stock and Tally Prime Ledger

- **Problem Statement:** Physical stock audits show differences compared to the Tally Prime ledger.
 - **Root Cause Analysis:**
 - Vehicle sales are retailed and physically delivered before invoices are generated and posted to Tally Prime.
 - **Analytical Solution:**
 - Enforce a GDMS validation rule that blocks physical gate passes unless a matching invoice has been generated in Tally Prime.
 - Automate daily reconciliations between physical stockyard VIN records and Tally inventory balances.
-

Problem 20: Post-Delivery Customer Churn before First Service

- **Problem Statement:** 30% of new car buyers do not return to the dealership for their first free service (1,500 km / 1 month).
- **Root Cause Analysis:**
 - The service department fails to schedule the first service appointment during the vehicle delivery process.
- **Analytical Solution:**
 - Require the Sales Consultant and Service Advisor to schedule the first service appointment before the vehicle is released.
 - Send automated reminders via HSmart 7 days before the scheduled appointment.

8.1 Business Problem 1 – Low Lead Conversion

Problem

1000 enquiries

Only 50 bookings

Conversion = 5%

Data Required

Lead Source

Consultant

Response Time

Follow-Up Count

Booking Status

Analysis

Which source produces best quality leads?

Which consultant converts best?

Does response time impact conversion?

Recommendation

- Prioritize high-converting sources
- Improve follow-up quality
- Reduce response time

8.2 Business Problem 2 – High Booking Cancellation

Problem

Customer books vehicle.

Customer later cancels.

Revenue opportunity lost.

Data Required

Booking Date

Cancellation Date

Model

Color

Waiting Time

Reason

Analysis

Top cancellation reasons

Models with highest cancellation

Impact of delivery delays

Recommendation

- Improve allocation speed
- Maintain stock of high-demand models
- Improve customer communication

8.3 Business Problem 3 – Inventory Management

Problem

Some vehicles remain in stock for too long.

Risks

Capital blocked

Insurance costs

Storage costs

Depreciation

Data Required

Arrival Date

Model

Color

Status

Stock Age

Recommendation

- Discount slow-moving inventory
- Forecast demand
- Optimize ordering

8.4 Predictive Analytics Framework

This framework details machine learning architectures for Lead Scoring, Service Churn, and Inventory Demand Forecasting, including features, algorithms, evaluation metrics, and implementation pseudo-code.

8.4.1 Lead Scoring Model (HSSmart Lead Scoring Engine)

8.4.1.1 Objective

Build a predictive model within HSSmart that scores new leads (from 0 to 100) based on their probability of purchasing a vehicle. This allows sales executives to prioritize high-value leads.

8.4.1.2 Feature Engineering Pipeline

- **Numerical Features:**
 - customer_age: Age of customer.
 - enquiry_age_days: Time since lead creation.
 - test_drive_distance_km: Mileage logged during test drives.
 - booking_amount_deposited: Deposit paid (if any).
- **Categorical Features:**
 - lead_source: WALK_IN, REFERENCE, DIGITAL_AD, COLD_CALL.
 - model_interest: Creta, Venue, Exter, Verna, etc.
 - employment_status: Salaried, Self-Employed, Business.
 - payment_mode_preference: Loan, Cash.
- **Target Variable:**
 - is_converted: 1 if vehicle is delivered, 0 otherwise.

8.4.1.3 Model Architecture & Pseudo-code

We utilize a Gradient Boosted Decision Tree (XGBoost Classifier) due to its high accuracy and ability to handle mixed data types.

HSSmart Lead Scoring Engine - Training Pipeline

```
PYTHON ENGINE / LEAD_SCORING_XGBOOST_PIPELINE.PY

import pandas as pd
import numpy as np
from xgboost import XGBClassifier
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.metrics import classification_report, roc_auc_score

# 1. Load Data
df = pd.read_sql("SELECT * FROM v_lead_scoring_training_data", con=db_conn)

# 2. Define Features and Target
X = df.drop(columns=['lead_id', 'is_converted'])
y = df['is_converted']

# 3. Preprocessing
categorical_cols = ['lead_source', 'model_interest', 'employment_status',
                    'payment_mode_preference']
numerical_cols = ['customer_age', 'enquiry_age_days', 'test_drive_distance_km', 'booking_amount_deposited']

preprocessor = ColumnTransformer(
    transformers=[
        ('num', 'passthrough', numerical_cols),
        ('cat', OneHotEncoder(handle_unknown='ignore'), categorical_cols)
    ])

# 4. Model Training
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

model = XGBClassifier()
model.fit(X_train, y_train)

# 5. Evaluation
y_pred = model.predict(X_test)
print(classification_report(y_test, y_pred))
print(roc_auc_score(y_test, y_pred))
```

```
# 4. Pipeline Definition
model_pipeline = Pipeline(steps=[
    ('preprocessor', preprocessor),
    ('classifier', XGBClassifier(n_estimators=150, learning_rate=0.05,
    ])

# 5. Split and Train
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
model_pipeline.fit(X_train, y_train)

# 6. Evaluate Model
predictions = model_pipeline.predict(X_test)
probs = model_pipeline.predict_proba(X_test)[:, 1]

print(f"ROC-AUC Score: {roc_auc_score(y_test, probs):.2f}")
print(classification_report(y_test, predictions))
```

8.4.2 Customer Service Churn Prediction

8.4.2.1 Objective

Predict which customers are likely to skip their upcoming periodic maintenance services (PMS), allowing the CRM team to target them with retention campaigns.

8.4.2.2 Feature Engineering Pipeline

- **RFM Metrics:**
 - recency_days_since_last_service: Time since last workshop visit.
 - frequency_lifetime_visits: Total service visits.
 - monetary_total_spend: Lifetime spend on parts and labor.
- **Vehicle Characteristics:**
 - vehicle_age_years: Years since registration.
 - odometer_reading: Current mileage.
- **Transactional Flags:**
 - has_sot_package: 1 if customer has an active SOT package, 0 otherwise.
 - has_rsa_package: 1 if customer has an active RSA package, 0 otherwise.
- **Target Variable:**
 - churned: 1 if customer is >60 days overdue for service, 0 otherwise.

8.4.2.3 Model Architecture & Pseudo-code

We implement a Random Forest Classifier to score customer retention probabilities.

Workshop Service Churn Predictor

```
from sklearn.ensemble import RandomForestClassifier
```


Pipeline Definition

```
# 1. Pipeline Definition
churn_pipeline = Pipeline(steps=[
    ('preprocessor', ColumnTransformer(transformers=[
        ('num', 'passthrough', ['recency_days_since_last_service', 'frequency_of_visits', 'monetary_total_spend', 'vehicle_age_years']),
        ('cat', OneHotEncoder(), ['has_sot_package', 'has_rsa_package'])
    ])),
    ('classifier', RandomForestClassifier(n_estimators=200, min_samples_leaf=10))
])

# 2. Train Model
churn_pipeline.fit(X_train_churn, y_train_churn)

# 3. Batch Scoring
df_active = pd.read_sql("SELECT * FROM v_current_active_owners", con=db_connection)
df_active['churn_probability'] = churn_pipeline.predict_proba(df_active)[:, 1]

# 4. Push High-Risk Leads to CRM Team
high_risk_leads = df_active[df_active['churn_probability'] > 0.70]
high_risk_leads.to_sql("temp_crm_churn_triggers", con=db_connection, if_exists='replace')
```

8.4.3 Inventory Demand Forecasting (Prophet Model)

8.4.3.1 Objective

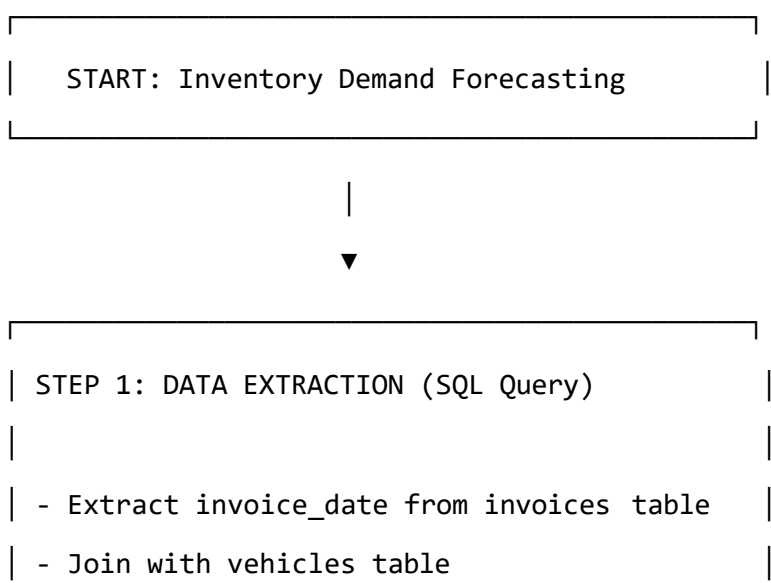
Forecast monthly demand for volume models (Creta, Venue) and high-value parts to optimize stocking schedules and minimize inventory costs.

8.4.3.2 Model Selection & Architecture

We use the **Facebook Prophet** algorithm to forecast sales volumes, capturing seasonal peaks (e.g., festive seasons in India like Dhanteras and Diwali).

Historical Sales Data —> Feature Prep (DS, Y) —> Prophet Model Fit —> 3-Month Forecast

8.4.3.3 Implementation Pseudo-code



```
| - Filter by model (e.g., Creta) |
| - Aggregate monthly sales count |
| |
| OUTPUT: Time Series Dataset (ds, y) |
|_____|
```

|



```
| STEP 2: DATA PREPARATION |
| |
| ds = Month (Date Truncated) |
| y = Sales Volume (Count of Invoices) |
| |
| RESULT: Clean Time Series Data |
|_____|
```

|

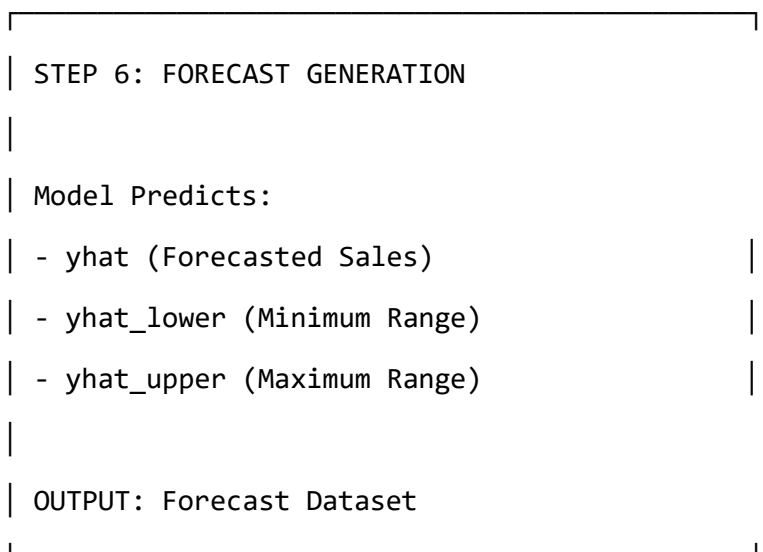
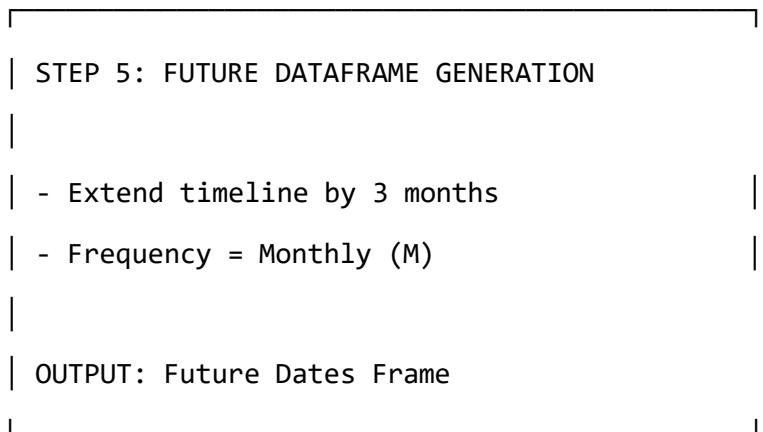
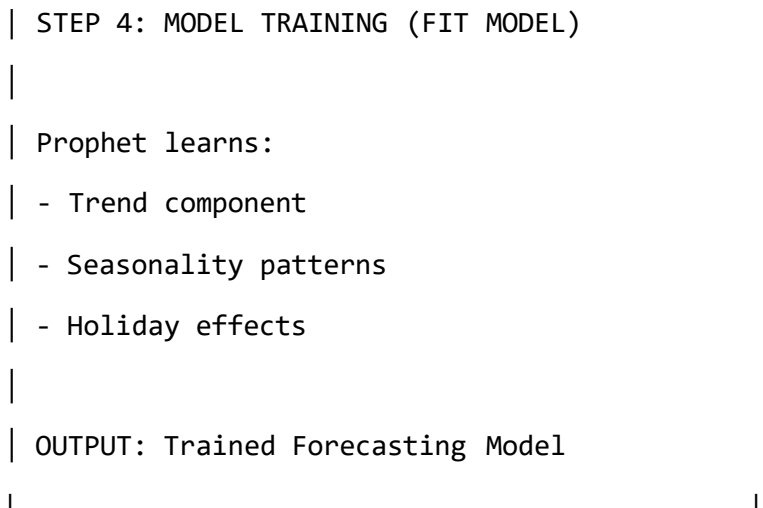


```
| STEP 3: MODEL INITIALIZATION |
| |
| Prophet Model Configuration: |
| - Yearly Seasonality = TRUE |
| - Weekly Seasonality = FALSE |
| - Daily Seasonality = FALSE |
| |
| + Add Country Holidays (India) |
|   (Diwali, Dhanteras, etc.) |
|_____|
```

|



```
|_____|
```



|



STEP 7: RESULT EXTRACTION

- Select last 3 months forecast
- Extract key metrics (yhat, range)

OUTPUT: Final Forecast Report

|



END: BUSINESS OUTPUT

Used For:

- Inventory Planning
- Stock Optimization
- Demand Forecasting
- Supply Chain Decisions

8.4 Customer Lifetime Value (CLV)

One of the most important concepts.

Most businesses focus only on first sale.

Smart businesses focus on lifetime value.

Example

Customer buys Creta.

Revenue Sources:

Vehicle Sale

Insurance

Accessories

Service

Warranty

RSA

Future Vehicle Purchase

Referral Customers

Total Customer Value

Can be many times larger than initial vehicle margin.

Business Objective

Increase Customer Lifetime Value.

Customer Retention Analytics

Retention is cheaper than acquisition.

Queries:

How many customers return for service?

How many renew insurance?

How many buy another Hyundai?

How many provide referrals?

8.4.1 KPIs

Retention Rate

Repeat Purchase Rate

Renewal Rate

Referral Rate

8.4.2 Management Goal

Keep customers connected to the dealership ecosystem

Predictive Analytics Opportunities

Future-focused analytics.

Instead of asking:

"What happened?"

Ask:

"What is likely to happen?"

8.4.3 Prediction Examples

Lead Conversion Prediction

Booking Probability

Vehicle Demand Forecast

Insurance Renewal Prediction

Customer Churn Prediction

Service Visit Prediction

8.4.4 Business Benefits

Better planning

Higher profitability

Lower risk

8.4.5 Advanced KPI Framework

Management should track KPIs across departments.

Sales KPIs

Lead Volume

Conversion %

Revenue

Bookings

Deliveries

Average Selling Price

Inventory KPIs

Inventory Days

8.4.6 Stock Age

Vehicle Availability

Inventory Turnover

8.4.7 Customer KPIs

NPS

CSAT

Retention

Complaints

8.4.8 Financial KPIs

Revenue

Gross Margin

Profitability

Outstanding Amount

8.4.9 Data Quality Framework

Bad data leads to bad decisions.

8.4.10 Common Issues

Missing values

Duplicate records

Wrong mobile numbers

Incorrect vehicle details

Incorrect status updates

8.5 Data Analyst Responsibility

Validate data

Clean data

Monitor quality

Report inconsistencies

8.5.1 Goal

Single Source of Truth

Data Governance Framework

Data must be controlled and standardized.

Governance Rules

Standard lead entry

Standard booking process

Unique customer IDs

Unique booking IDs

Access control

Audit tracking

Benefits

Reliable reports

Better decisions

Reduced errors

Risk Management Framework

Every dealership faces operational risks.

Sales Risks

Lead loss

Booking cancellation

Poor conversion

Inventory Risks

Overstock

Understock

Allocation delay

Customer Risks

Negative feedback

Complaint escalation

Customer churn

Financial Risks

Outstanding payments

Tax errors

Revenue leakage

8.6 Management Dashboard Recommendations

Executive Dashboard

Revenue

Profit

Bookings

Deliveries

Sales Dashboard

Lead Funnel

Conversion

Consultant Performance

Inventory Dashboard

Stock

Aging

Availability

Insurance Dashboard

Policies

Renewals

Premium Revenue

Service Dashboard

Revenue

Retention

Satisfaction

8.7 Internship Learnings & Key Observations

During this internship, the following observations were made:

Business Understanding

A dealership is not merely a vehicle-selling business.

It is a customer lifecycle management organization.

Data Understanding

Every department creates business-critical data.

Analytical Understanding

Business decisions are driven by KPIs and reports.

Operational Understanding

All departments are interconnected.

Poor performance in one department affects the entire customer journey.

8.8 Final Conclusion & Recommendations

Key Conclusion

The dealership's most valuable asset is not its vehicle inventory.

Its most valuable asset is:

- Customer Data
- Sales Data
- Inventory Data
- Insurance Data
- Service Data
- Financial Data

8.8.1 Recommendations

Sales

Improve lead response time.

Increase follow-up effectiveness.

Inventory

Improve forecasting.

Reduce stock aging.

Customer Experience

Improve complaint handling.

Increase customer retention.

Analytics

Build centralized dashboards.

Automate reporting.

Track KPIs regularly.

Implement predictive analytics.

8.9 Final Data Analyst Perspective

A Data Analyst in a Hyundai dealership should understand:

1. Business operations
2. Department workflows
3. Data generation points
4. Data movement
5. KPIs

6. Reporting systems
7. Business problems
8. Management decision-making

The objective of analytics is not simply to create reports, but to transform business data into decisions that improve customer satisfaction, operational efficiency, revenue growth, and profitability.

8.10 Business Problem Solving Methodology Framework

8.10.1 Purpose

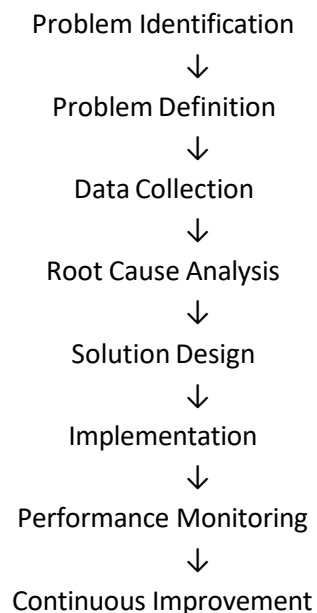
Business problems should not be solved through assumptions.

Every problem must be:

1. Identified
2. Measured
3. Analyzed
4. Solved
5. Monitored

Analytics provides evidence-based decision making.

8.10.2 Problem Solving Lifecycle



8.10.3 Data Analyst Role

The Data Analyst is responsible for:

- Identifying business issues
 - Collecting supporting data
 - Finding root causes
 - Quantifying business impact
 - Recommending solutions
 - Monitoring outcomes
-

8.11 Business Problem Framing Template

Every business problem should follow a standard structure.

8.11.1 Problem Statement

Define the problem clearly.

Example:

Booking conversion rate has declined from 12% to 7% during the last 6 months.

8.11.2 Business Impact

Identify impact on revenue and operations.

Example:

Area	Impact
Revenue	₹1.2 Crore Loss
Customer Satisfaction	Reduced
Inventory	Excess Stock
Profitability	Reduced

8.11.3 Affected Stakeholders

Stakeholder	Impact
Sales Manager	High
Sales Consultant	High
CRM Team	Medium

Stakeholder	Impact
Dealer Principal	High

8.11.4 Success Criteria

Example:

Increase conversion rate
from 7% to 12%
within 90 days.

8.11.5 Available Data

Example:

- Lead Data
 - Follow-up Data
 - Booking Data
 - Consultant Data
 - Customer Data
-

8.11.6 Timeline Classification

Priority	Timeline
Critical	Immediate
High	30 Days
Medium	90 Days
Low	180 Days

8.11.7 Budget Assessment

Budget Type	Description
Low	Existing Resources
Medium	Software Purchase
High	Major Transformation

8.11.8 Constraints

Example:

- Budget Limitations
 - Staff Availability
 - Technology Restrictions
 - Manufacturer Policies
-

8.12 Root Cause Analysis Framework

8.12.1 5 Whys Analysis

Example:

Problem:

Booking Conversion Dropped

Why 1?

Leads not converting.

Why 2?

Follow-ups delayed.

Why 3?

Consultants overloaded.

Why 4?

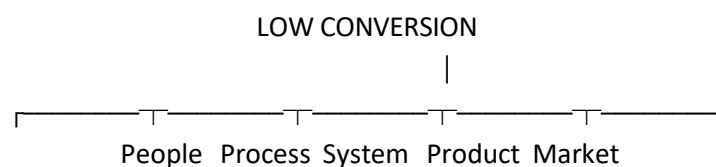
Lead assignment imbalance.

Why 5?

No workload management process.

Root Cause Identified.

8.12.2 Fishbone Diagram



8.12.3 Pareto Analysis

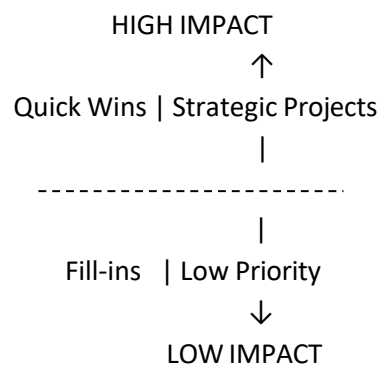
80% of problems often come from 20% of causes.

Example:

Cause	Impact
Slow Follow-up	40%
Wrong Lead Source	25%
Poor Consultant Skills	15%
Others	20%

8.13 Business Problem Prioritization Matrix

8.13.1 Impact vs Effort Matrix



8.13.2 Prioritization Score

Priority Score =
Business Impact × Urgency × Frequency

8.14 Stakeholder Analysis Framework

8.14.1 Stakeholder Matrix

Stakeholder	Influence	Interest
-------------	-----------	----------

Dealer Principal	High	High
------------------	------	------

Sales Manager	High	High
---------------	------	------

CRM Team	Medium	High
----------	--------	------

Customers	Medium	High
-----------	--------	------

8.14.2 Stakeholder Communication Plan

Stakeholder	Frequency
-------------	-----------

Dealer Principal	Weekly
------------------	--------

Managers	Daily
----------	-------

Team Leads	Daily
------------	-------

Analysts	Continuous
----------	------------

8.15 Detailed Analytics Framework

8.15.1 Comparative Analysis

Compare:

Current Month

vs

Previous Month

8.15.2 Trend Analysis

Analyze:

Month-wise

Quarter-wise

Year-wise

patterns.

8.15.3 Segmentation Analysis

Example:

Customer Segments:

- Corporate
 - Individual
 - Fleet
 - Rural
 - Urban
-

8.15.4 Correlation Analysis

Example:

Lead Response Time
vs
Booking Conversion

8.15.5 Cohort Analysis

Track customers based on acquisition month.

Jan Customers
Feb Customers
Mar Customers

and compare retention.

8.16 Findings Documentation Framework

8.16.1 Key Findings

Example:

73% of lost leads received
follow-up after 24 hours.

8.16.2 Confirmed Hypothesis

Example:

Slow response time
reduces conversion.

8.16.3 Contradicted Assumptions

Example:

Management believed:

Digital Leads are poor quality.

Analysis showed:

Digital Leads convert best.

8.16.4 Unexpected Discoveries

Example:

Reference Leads

have 3x higher conversion.

8.17 Solution Evaluation Framework

8.17.1 Solution Option 1

Process Improvement

Advantages:

- Low Cost
- Fast Implementation

Disadvantages:

- Limited Scale
-

8.17.2 Solution Option 2

Automation

Advantages:

- Consistent
- Scalable

Disadvantages:

- Software Cost
-

8.17.3 Solution Option 3

Organizational Change

Advantages:

- Long-term Benefits

Disadvantages:

- Resistance to Change
-

8.17.4 Recommended Solution

Include:

- Cost
 - Benefit
 - Timeline
 - Risk
 - ROI
-

8.18 Solution Implementation Roadmap

8.18.1 Implementation Flow

Planning



Development



Testing



Training



Deployment



Monitoring

8.18.2 Responsibility Matrix

Activity	Owner
Analysis	Data Analyst
Approval	Management

Activity	Owner
Development	IT Team
Execution	Department
Monitoring	Analyst

8.18.3 Resource Planning

Resources Required:

- Budget
- Staff
- Technology
- Training

8.18.4 Milestone Tracking

Milestone	Target Date
Analysis Complete	Day 10
Design Complete	Day 20
Testing Complete	Day 30
Go Live	Day 45

8.19 Change Management Framework

8.19.1 Change Readiness Assessment

Measure:

- Employee Readiness
- Management Support
- Technology Readiness

8.19.2 Training Plan

Audience	Training
Sales Team	CRM Usage
Service Team	Dashboard Usage
Managers	KPI Monitoring

8.20 Success Measurement Framework

8.20.1 KPI Before vs After

KPI	Before	After
Conversion	5%	10%
Response Time	8 Hours	30 Minutes
Retention	40%	60%

8.20.2 ROI Calculation

ROI			=
(Net	Benefit	-	Cost)
÷ Cost × 100			

8.21 Post-Implementation Review Framework

8.21.1 Review Objectives

- Verify:
- Was problem solved?
 - Were targets achieved?
 - Was ROI realized?
-

8.21.2 Lessons Learned

- Document:
- What worked well
 - What failed
 - Improvement opportunities

8.21.3 Staff Feedback Template

Questions:

1. Was solution easy to use?
 2. Did it improve productivity?
 3. What should be improved?
-

8.21.4 Customer Feedback Template

Questions:

1. Service quality improved?
 2. Response time improved?
 3. Satisfaction increased?
-

8.22 Continuous Improvement Framework

Measure



Analyze



Improve



Control



Repeat

8.22.1 Monitoring Dashboard

Track:

- Problem KPIs
- SLA Compliance

- Process Performance
- Customer Satisfaction
- Financial Impact

8.22.2 Continuous Analytics Cycle

Data



Insight



Decision



Action



Result



Data

8.23 Problem Framing Template (Standard Business Format)

Every dealership problem must follow a structured analytical format to ensure clarity and measurable outcomes.

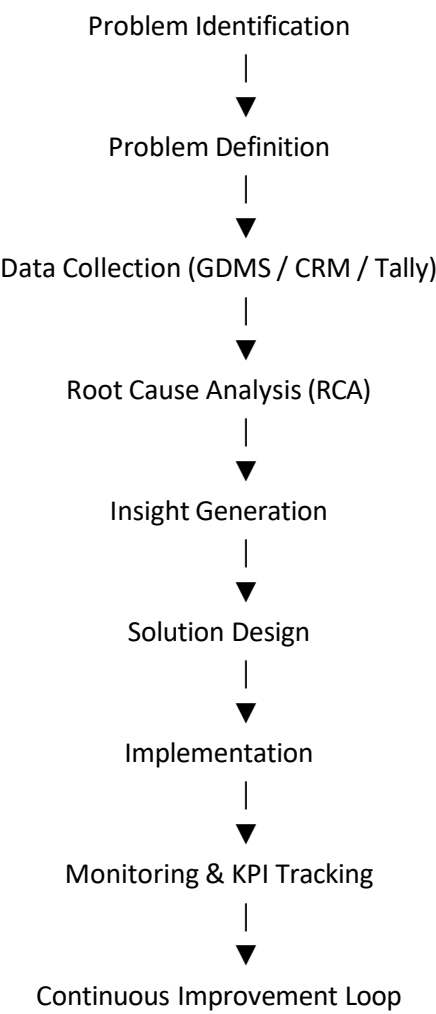
8.23.1 Standard Problem Template (Excel Format)

Component	Description
Problem Statement	Clear definition of issue
Business Impact	Revenue / cost / efficiency impact
Stakeholders	Who is affected
KPIs Affected	Metrics impacted
Data Sources	GDMS / CRM / Tally / Service
Time Period	When issue observed
Success Criteria	Expected improvement
Constraints	Limitations

8.23.2 Example (Filled Template)

Component	Example
Problem	Low booking conversion
Impact	₹1.2 Cr monthly loss
Stakeholders	Sales Manager, Dealer Principal
KPI	Lead Conversion %
Data	CRM Leads, Booking Data
Success	Improve from 5% → 10%
Constraint	Limited staffing

8.24 Business Problem Lifecycle (Flow Diagram)



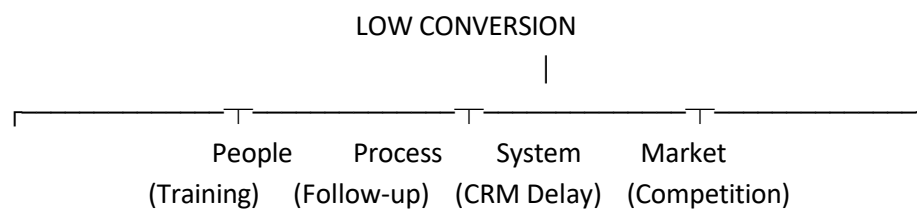
8.25 Root Cause Analysis Framework (Advanced)

8.25.1 5 Whys Method

Example: Booking Conversion Drop

Why 1 → Leads not converting
Why 2 → Follow-up delayed
Why 3 → Consultants overloaded
Why 4 → Poor lead allocation
Why 5 → No workload balancing system

8.25.2 Fishbone Diagram (Cause Classification)

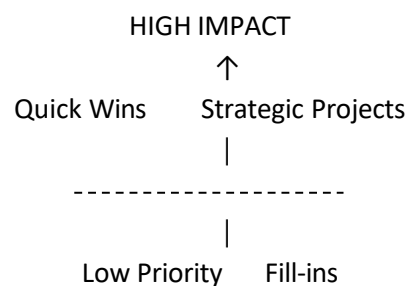


8.25.3 Pareto Analysis (80/20 Rule)

Cause	Impact
Slow Follow-up	40%
Poor Lead Quality	25%
Consultant Skill Gap	20%
Others	15%

8.26 Business Prioritization Matrix

8.26.1 Impact vs Effort Grid

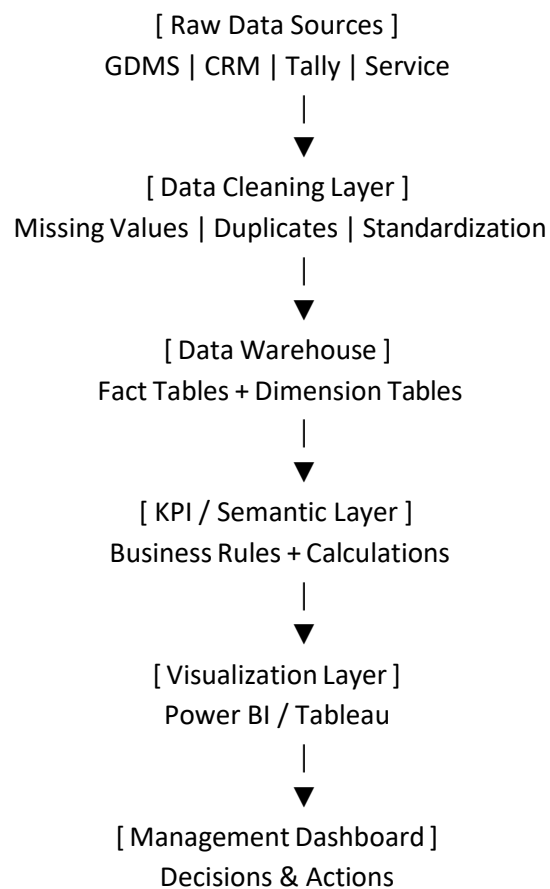


↓
LOW IMPACT

8.26.2 Priority Formula

Priority Score =
Impact × Urgency × Frequency

8.27 Data Analysis Flow (Business Intelligence Pipeline)



8.28 Sample Fact Table (Excel Style Format)

8.28.1 fact_sales Table

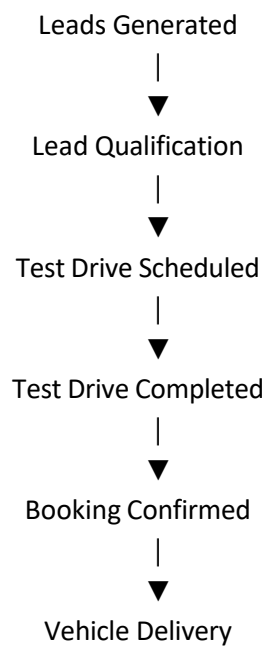
sale_id	customer_id	model	sale_date	booking_value	employee_id
S001	C101	Creta	2026-01-01	1200000	E01
S002	C102	Venue	2026-01-02	850000	E02

Excel Formatting Tip (Word Use)

Use:

- Monospaced font (Consolas / Courier New)
 - Add borders
 - Alternate row shading
-

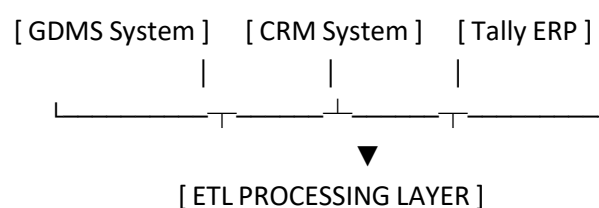
8.29 Sales Funnel Flow (Clean Business Format)

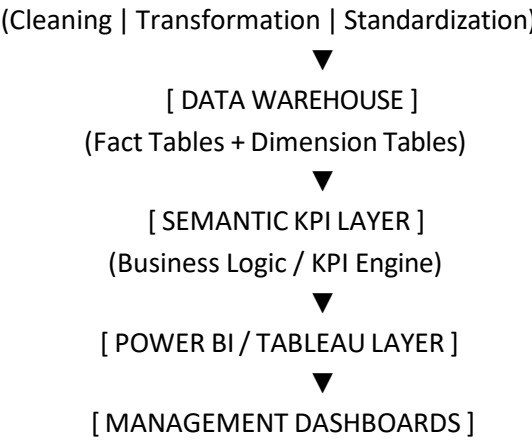


Insight Section (Formatted)

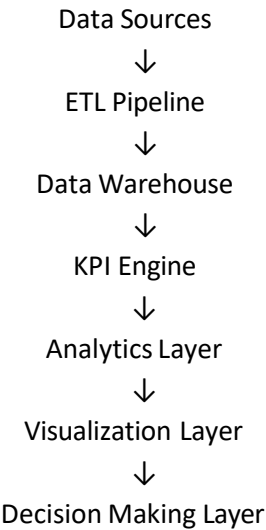
Stage	Status
Biggest Drop-off	Test Drive → Booking
Problem	Weak follow-up
Action Required	Improve CRM response + training

8.30 Business Intelligence Architecture Diagram





8.31 Dashboard Logical Architecture (Advanced View)

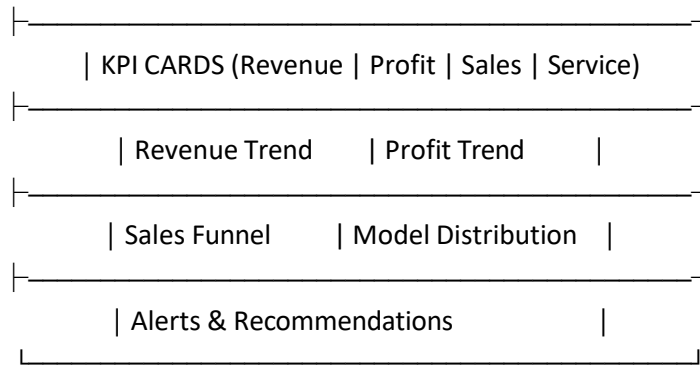


8.32 Sample Dashboard KPI Table (Excel Format)

KPI	Formula	Target
Lead Conversion	Bookings / Leads	>10%
Test Drive Conversion	Test Drives / Leads	>60%
Delivery Rate	Deliveries / Bookings	>85%
Cancellation Rate	Cancelled / Bookings	<5%

8.33 Dashboard Wireframe (Improved Layout Format)





8.34 Data Analyst Thinking Model

A Data Analyst in a dealership must answer:

1. What happened?
2. Why did it happen?
3. Where did it happen?
4. Who is responsible?
5. What action should be taken?

8.35 Analytical Methodology Toolkit

8.35.1 Comparative Analysis

Current Month vs Previous Month

8.35.2 Trend Analysis

Monthly / Quarterly / Yearly patterns

8.35.3 Segmentation Analysis

- Urban vs Rural
- Corporate vs Retail
- High vs Low income customers

8.35.4 Correlation Analysis

Example:

Lead Response Time ↔ Conversion Rate

8.35.5 Cohort Analysis

Track customers by purchase month:

- Jan Cohort
- Feb Cohort
- Mar Cohort

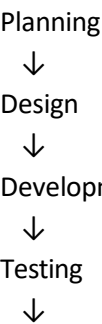
8.36 Findings Documentation Format

Type	Example
Key Finding	73% leads lost due to late follow-up
Hypothesis	Faster response improves conversion
Contradiction	Digital leads perform better than walk-ins
Insight	Reference leads have highest conversion

8.37 Solution Framework (Multi-Option Model)

Option	Type	Advantage	Disadvantage
Option 1	Process	Low cost	Limited scale
Option 2	Automation	Scalable	Setup cost
Option 3	Structural Change	Long-term impact	Resistance

8.38 Implementation Roadmap



Training
↓
Deployment
↓
Monitoring

8.38.1 Milestone Table

Milestone	Timeline
Analysis Complete	Day 10
Design Complete	Day 20
Testing Complete	Day 30
Go Live	Day 45

8.39 Change Management Framework

8.39.1 Readiness Check

- Employee readiness
 - Management support
 - System readiness
-

8.39.2 Training Plan

Role	Training
Sales	CRM Usage
Service	Dashboard Usage
Managers	KPI Interpretation

8.40 Success Measurement Framework

Before vs After KPI Table

KPI	Before After	
Conversion	5%	10%
Response Time	8 hrs	30 min
Retention	40%	60%

ROI Formula

ROI = (Net Benefit - Cost) / Cost × 100

8.41 Continuous Improvement Cycle

Measure



Analyze



Improve



Control



Repeat

8.42 Final Summary (Analytics Philosophy)

A modern dealership is not just a vehicle-selling business.

It is a:

- Data-driven ecosystem
 - Customer lifecycle engine
 - Predictive analytics system
 - KPI-controlled organization
-

Core Objective of Data Analyst

Convert:

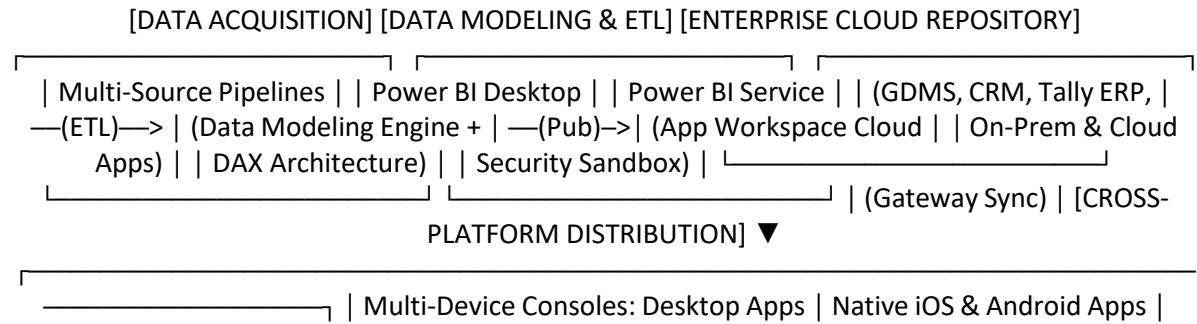
Raw Data → Insights → Decisions → Business Growth

9.Power BI & Advanced Dashboard

9.1 Power BI Environment Setup

9.1.1 Overview of Power BI Ecosystem

The Power BI ecosystem functions as an integrated enterprise analytics engine that ingests multi-source dealer data, orchestrates relational data modeling, and serves cross-platform interactive visualizations.



Power BI Architecture Flow

Data Sources



Power BI Desktop



Data Model (Relationships + DAX)



Power BI Service (Cloud)



Reports & Dashboards



Mobile / Web Access

9.1.2 Power BI Setup Workflow (Step-by-Step)

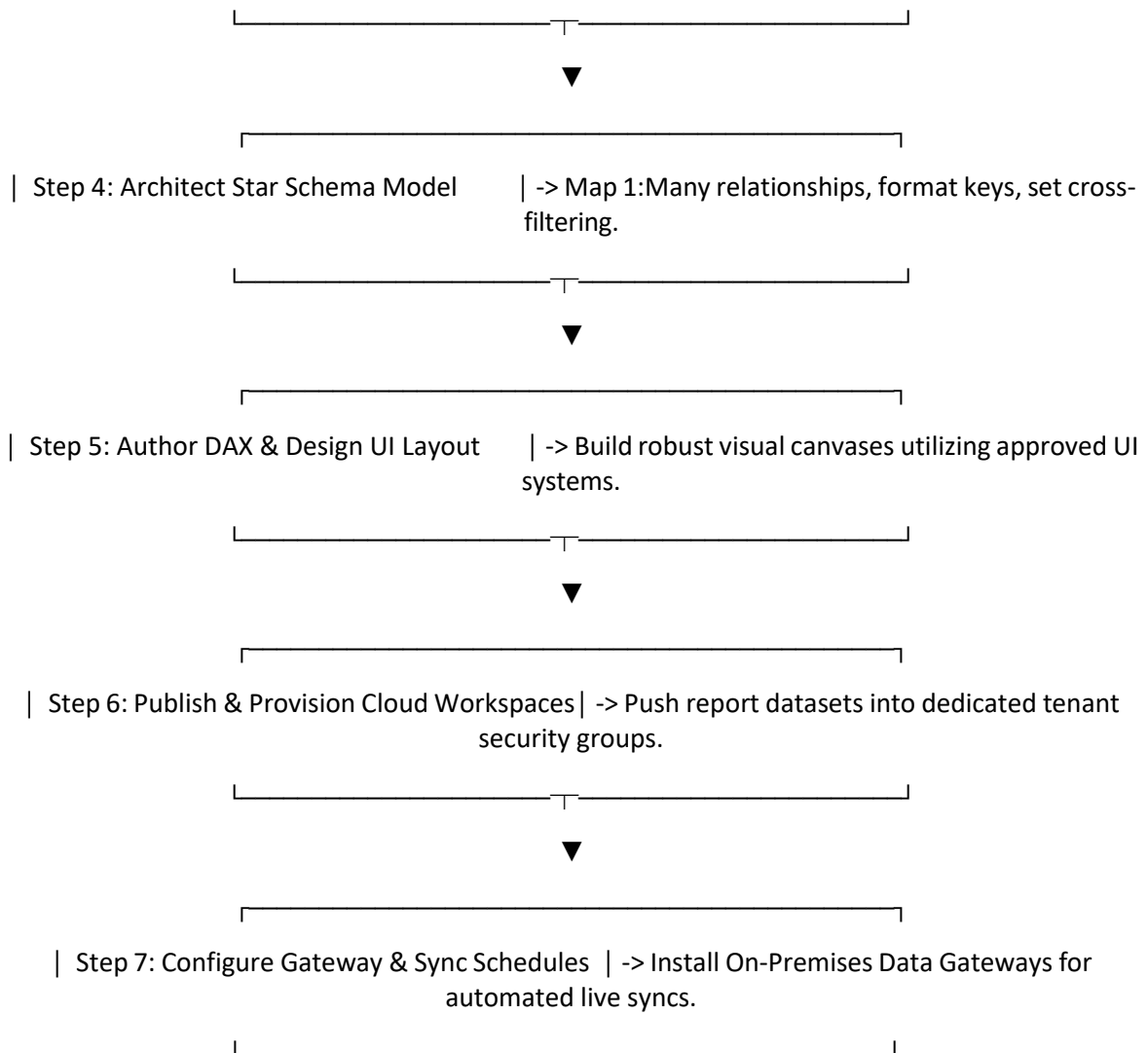
| Step 1: Install & Provision Power BI | -> Deploy Desktop Pro client on analytics workstations.



| Step 2: Establish Source Connections | -> Configure hybrid pipelines (SQL native, REST APIs, ODBC).



| Step 3: Transform Data via Power Query | -> Execute data cleansing, schema normalizing, and custom M code.



9.1.3 Data Source Connection Table

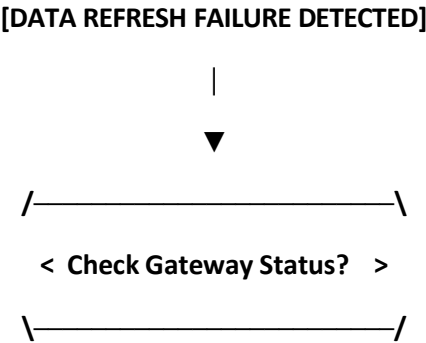
Source System	Connection Type	Target Dataset / Schema	Ingestion Mode	Security Protocol
GDMS	SQL Server Native	Real-time Sales, Inventory, Service History	Import / DirectQuery	Service Account / SQL Auth
CRM (HSSmart)	REST API Endpoint	Lead Pipelines, Funnel Drop-offs, Test Drives	Scheduled Import (Hourly)	OAuth 2.0 Tokenization

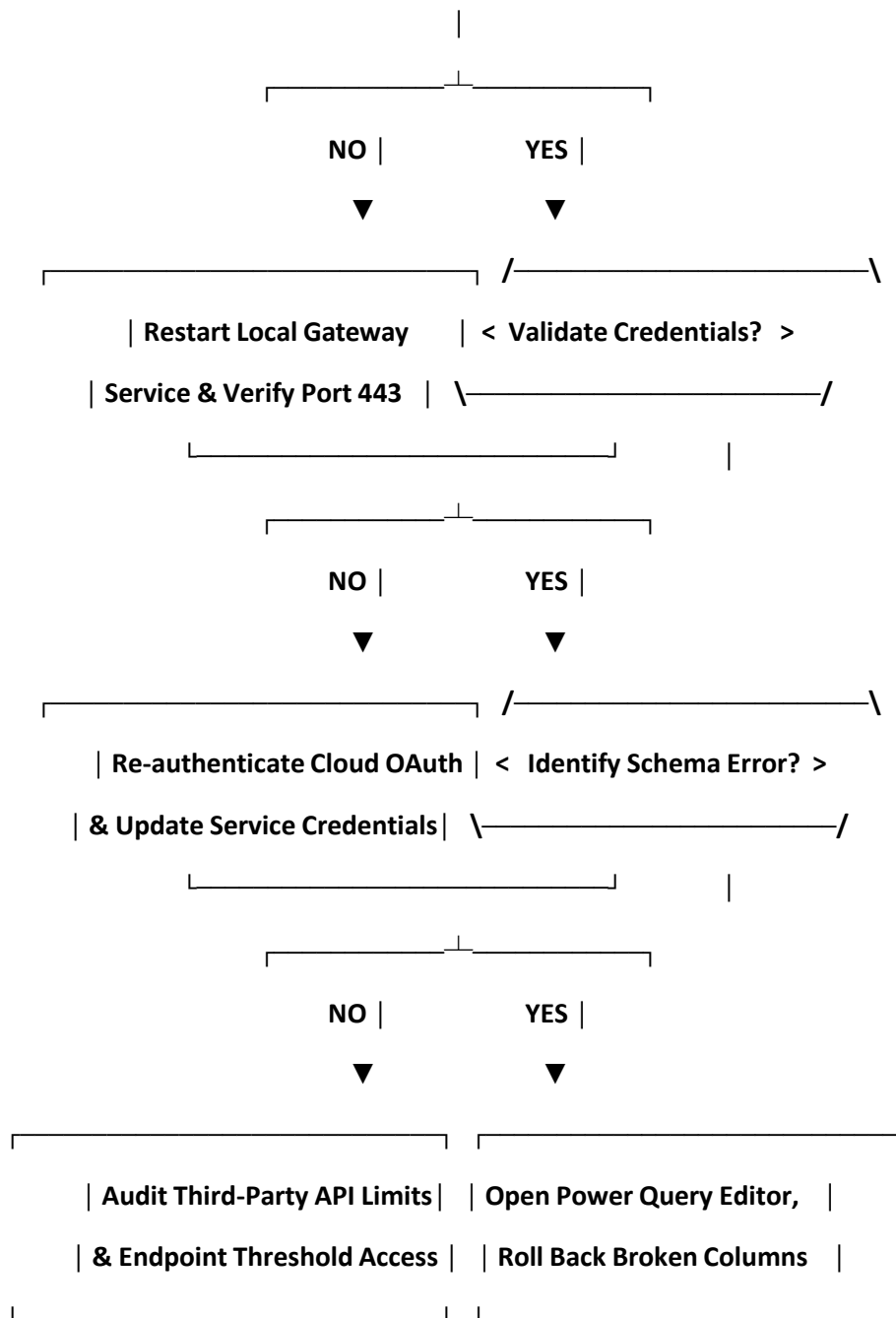
Source System	Connection Type	Target Dataset / Schema	Ingestion Mode	Security Protocol
Tally ERP	ODBC / XML Broker	Trial Balances, P&L Ledgers, Operational Expenses	Scheduled Import (Daily)	Local Network Gateway
Excel Sheets	Local File / SharePoint	Ad-hoc Goals, Targets, Marketing Campaign Budgets	Manual / Cloud Refresh	Organizational Account

9.1.4 Power BI Refresh Cycle

Refresh Profile	Data Frequency	Triggers / Scheduling	Target Dashboards
Real-Time Stream	Live Stream Sync	DirectQuery Event Listeners	Workshop Floor Load, Live Target Trackers
Intraday Batches	Every 60 Minutes	Scheduled Cloud Gateways (8 AM - 8 PM)	Lead Pipeline, Showroom Traffic, Conversion KPIs
Operational Batches	Every 24 Hours	Automated Nightly Overhaul (02:00 AM)	Group Financial Statements, Monthly P&L, Executive Summary

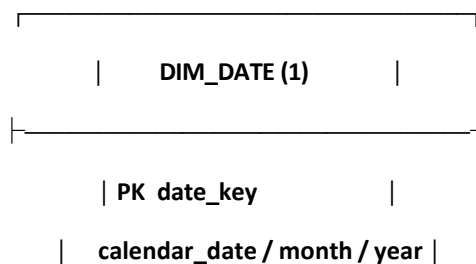
9.1.5 Troubleshooting Flow





9.2 Data Modeling Fundamentals

9.2.1 Star Schema Model (Recommended)



sale_id (PK)	customer_key (FK)	vehicle_key (FK)	date_key (FK)	gross_amount	net_margin	employee_key (FK)
S001	C101	V01	20260101	1200000	180000	E01
S002	C102	V02	20260102	850000	95000	E02

Export to Sheets

DIMENSION TABLE: dim_customer

customer_key (PK)	customer_name	city	age	customer_segment	credit_tier
C101	Raj Sharma	Delhi	35	Corporate	Tier-1
C102	Aman Verma	Noida	28	Individual	Tier-2

Export to Sheets

9.2.3 Relationship Types

- One-to-Many (): The primary paradigm used in Star Schemas. Example: dim_customer[customer_key] fact_sales[customer_key]. One customer can generate multiple unique sales transactions over time, while each sale links back to exactly one unique buyer profile.
- Many-to-One (): Used to resolve granular facts back to aggregated attributes. Example: fact_sales[vehicle_key] dim_vehicle[vehicle_key]. Multiple dynamic individual invoices reference a single, static product configuration or model SKU.
- Many-to-Many (): Avoided in primary operations due to performance overhead and ambiguous bi-directional filtering. Permitted only through structured bridge tables, such as matching transient seasonal promotions across multiple inventory classifications.

9.2.4 Snowflake vs Star Schema Architectural Matrix

Evaluation Dimension	Star Schema Model Architecture	Snowflake Schema Model Architecture
Query Performance	Highly Optimal (Minimized table joins, clean indexing).	Slower (Requires deep, complex nested table joins).
Structural Complexity	Highly simplified, clear, easy for report developers to navigate.	Highly complex, multi-layered normalization.
Use-Case Fit	Interactive Power BI Reporting & Rapid Explorations.	Data Warehousing & Highly Granular Storage Engine Auditing.
Maintenance Overheads	Denormalized redundancy requires structured upstream data cleanses.	Normalized design scales back storage needs but increases join tracking.

Export to Sheets

9.3 POWER BI DASHBOARD MOCK LAYOUTS (FIGMA UI SPECIFICATION)

9.3.1 Executive Performance Dashboard UI

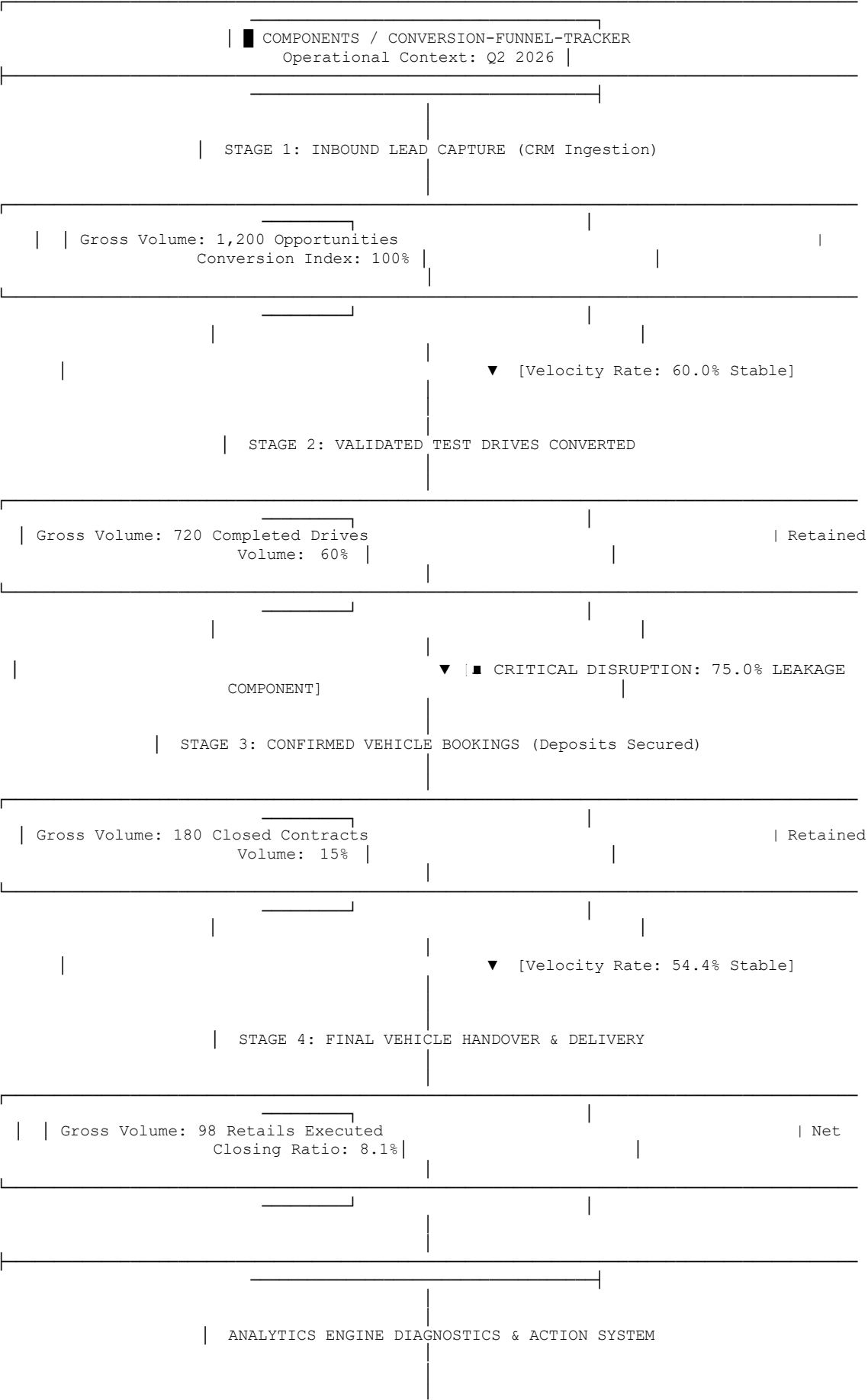
Figma Design Variables & Canvas Constants

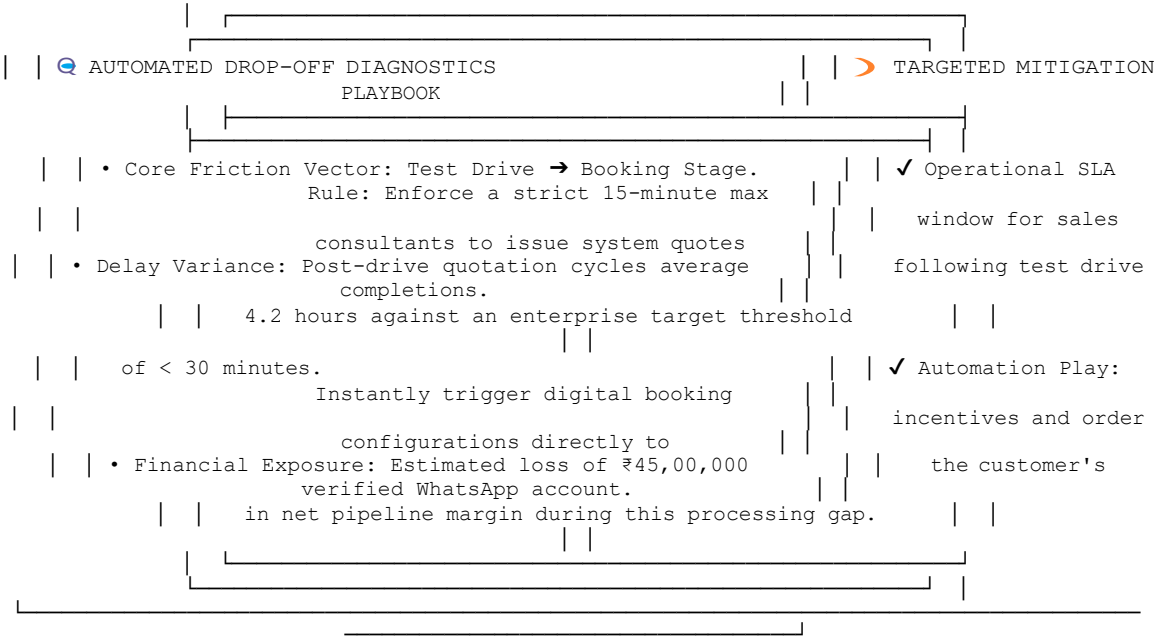
- Canvas Resolution: (Desktop Scaled Native Canvas Layer)
- Background Color: Light Slate Neutral #F8FAFC | Card Fill Color: #FFFFFF with 4px border-radius
- Primary Typography Font: Inter / Segoe UI Fluent (#0F172A Dark Navy Primary text)
- Primary Brand Accent: Deep Tech Blue (#1E40AF) | System Borders: 1px solid Gray Smooth (#E2E8F0)

9.3.2 Conversion Funnel Analytics Dashboard

Figma Design Variables & Canvas Constants

- Visual Framework: Process Ingestion Funnel Custom Component Block
- Primary Brand Accent: Deep Cobalt Slate (#1E293B) | Funnel Drop-Off Alert Tint: Vivid Amber Accent (#D97706)

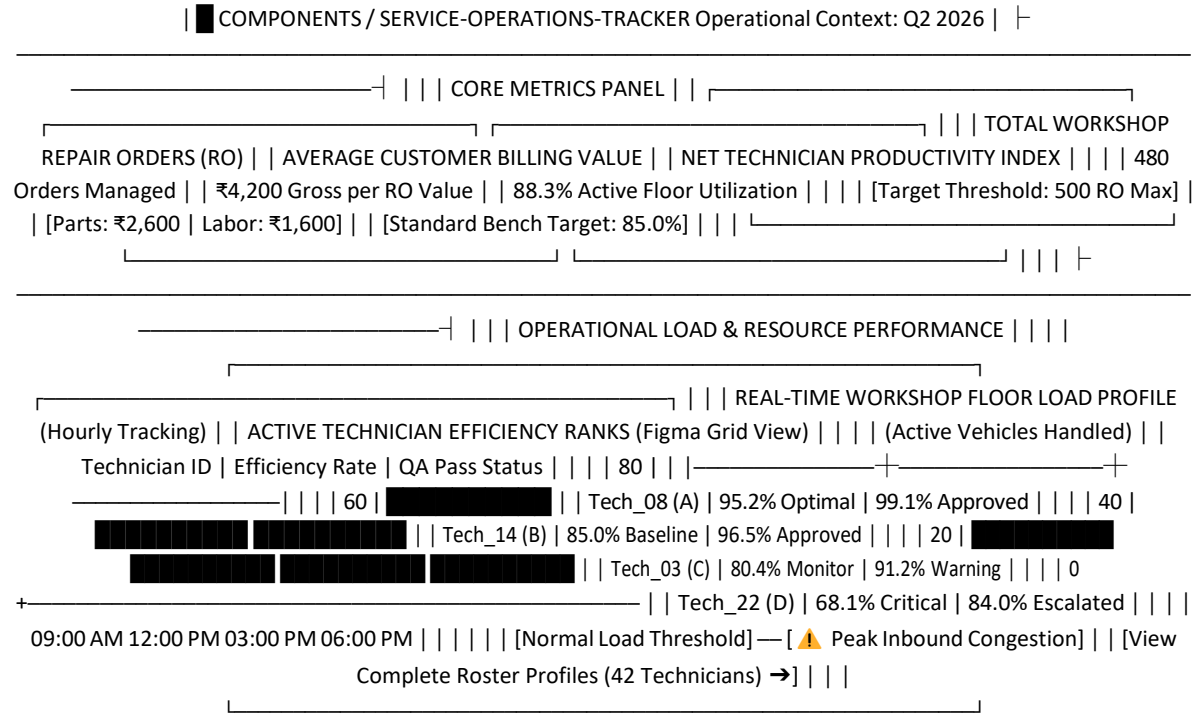




9.3.3 Service Operations & Workshop Dashboard UI

Figma Design Variables & Canvas Constants

- Visual Framework: Operational Resource Dispatch Grid Component
- Primary Brand Accent: Graphite Dark Neutral (#334155) | Peak Load Hazard Identifier Tint: Intense Coral (#EF4444)



9.4 DAX MEASURES & FORMULAS (PRODUCTION-READY SYNTAX)

9.4.1 Common Core Financial Measures

Code snippet

Aggregates the total absolute gross sales pipeline revenue across the organization

Total Sales :=

```
SUM(fact_sales[gross_amount])
```

Safely computes dealer conversion percentage while catching divide-by-zero database flags

Conversion Rate :=

```
DIVIDE(  
    COUNTA(fact_sales[booking_id]),  
    COUNTA(fact_crm_leads[lead_id]),  
    0  
)
```

Extracts the mean arithmetic value of individual customer transaction accounts

Average Sale :=

```
AVERAGE(fact_sales[gross_amount])
```

9.4.2 Time Intelligence (Calculated Financial Trends)

Code snippet

Aggregates operational performance dynamically from start of current month to current date context

Sales_MTD :=

```
TOTALMTD(  
    [Total Sales],  
    dim_date[calendar_date]  
)
```

Aggregates operational performance dynamically from start of current fiscal year to current date context

Sales_YTD :=

```
TOTALYTD(  
    [Total Sales],  
    dim_date[calendar_date]  
)
```

Compares current sales performance directly against the exact identical period in the prior fiscal year

Sales_Prior_Year_Equivalent :=

```
CALCULATE(  
    [Total Sales],  
    SAMEPERIODLASTYEAR(dim_date[calendar_date])  
)
```

9.4.3 Ranking Measures (Dynamic Performance Hierarchies)

Code snippet

Dynamically ranks consultants by total sales volume, ignoring individual filter resets

Consultant_Sales_Performance_Rank :=

```
RANKX(  
    ALL(dim_employee),  
    [Total Sales],  
    ,  
    DESC,  
    Dense  
)
```

9.5 POWER BI DATA DESIGN & SYSTEM STANDARDS

9.5.1 Structural Layout & Visual Consistency Rules

- The 3-Second Rule Principle: The top-left 25% quadrant of any deployed analytics dashboard canvas must display the most critical organizational Key Performance Indicators (KPIs). Users must understand core high-level trends within 3 seconds of load execution.
- Grid Alignment: Dashboards must use a strict 8px grid system with structured 16px margins to align cards, charts, and tables, ensuring crisp visual clarity across screen scales.
- Typography Controls: Maintain a strict 3-tier hierarchy:

- *KPI Card Aggregated Value:* 24pt Bold Helvetica/Segoe UI Neutral
- *Standard Visual Component Headers:* 14pt Semi-Bold Neutral
- *Tabular Row Core Values:* 10pt Regular Contrast Font

Power BI Dashboard Architecture (Final View)

DATA SOURCES

|



DATA MODEL (Star Schema)

|



DAX CALCULATION LAYER

|



VISUALIZATION LAYER

|



PUBLISH (Power BI Service)

|



MANAGEMENT DECISION DASHBOARDS

9.5.2 Color Semantic Encoding Standards

[DESIGN SYSTEM TOKEN MATRIX: SEMANTIC CONDITIONAL FORMATTING] └─ Framework Target:
Global Power BI Visual Styling and UI Component States

===== I.

CORE SEMANTIC STATES & HEX CODE SPECIFICATIONS

===== A.

PRODUCTION STATE: EXCEEDING TARGETS / HEALTHY └─ UI System Hex Token: #10B981 (Emerald Vibrant Green) └─ Visual Focus Areas: Data metrics outperforming baseline goals, positive variances. └─ Application Matrix: - Net margin changes showing positive trajectory. - Sales volume achievements meeting or exceeding 100% of targets. - Quality Assurance (QA) first-time-fix completions (Pass states). B. PRODUCTION STATE: MARGINAL VARIANCE / WARNING └─ UI System Hex Token: #F59E0B (Amber Warning Ochre) └─ Visual Focus Areas: Metrics shifting outside standard tolerances, early anomalies. └─ Application Matrix: - Funnel pipeline drop-off spikes or transition delays. - Workshop floor loads nearing total capacity thresholds. - Minor technician

efficiency variations below baseline targets. C. PRODUCTION STATE: UNDERPERFORMING / CRITICAL
└─ UI System Hex Token: #EF4444 (Vivid System Red) └─ Visual Focus Areas: Immediate escalation vectors, systemic process failures. └─ Application Matrix: - Customer Service Level Agreement (SLA) timestamp breaches. - Hot inbound leads sitting unassigned beyond target minutes. - Severe absolute profit drops, deficit variances, or high rework rates.

===== II.

INTERACTIVE DASHBOARD SYSTEM RE-PAINTING MATRIX

=====

The reporting canvas applies these functional system states across visual layers using the following automated behavioral rules: - KPI Single Value Cards: Text element or sub-card background switches dynamically based on target rules. - Linear Trend Charts: Target benchmark lines remain neutral while data fills dynamically match performance regions. - Matrix/Table Grids: Applies soft background tints to cells rather than high-contrast fills to maintain high data readability.



9.6 INTERACTIVE DASHBOARD FLOW (REAL BUSINESS OPERATIONAL DATA LIFECYCLE)

[OPERATIONAL DATA LIFECYCLE: INGESTION TO DECISION RAPID LOOP]

STAGE 1: ASYNC DATA INGESTION & PIPELINE MATCHING

- └─ Process: High-frequency data connector hooks directly into the production transactional DBs.
- └─ Frequency: Real-time event streams capture delta changes (Sales, Bookings, Repair Orders).
- └─ Metric: Delta synchronization window completes in < 5 seconds.

|



STAGE 2: ALGORITHMIC DAX EXECUTION ENGINE

- └─ Process: Power BI Tabular Engine processes structural aggregations instantly.
- └─ Computations: Calculates active time-intelligence (YTD/MTD), margins, and rankings.
- └─ Metric: Calculation engine processing overhead is restricted to < 300ms.

|



STAGE 3: CLOUD-HOSTED CANVAS VIEWPORT RENDERING

- └─ Process: Power BI Cloud Service updates active display states across enterprise viewports.
- └─ Interface: Refreshes primary executive KPI metric tiles, trends, and alert indicators.
- └─ Metric: Zero-lag dashboard paint speed across secure desktop and mobile clients.

|



STAGE 4: MULTI-DIMENSIONAL CONSOLE FILTER SLICING

- └ User Action: Evaluator adjusts explicit slicer tags (e.g., Region = North, Model = SUV).
- └ Execution: Cross-filtering instantly updates all visuals across the active workspace.
- └ Metric: Global cross-filter recalculation completes in < 150ms.



STAGE 5: GRANULAR SUB-SURFACE DRILL DIAGNOSTICS

- └ User Action: Evaluator executes a deep-dive click path on anomalies (e.g., Stage 3 Drop-Off).
- └ Execution: UI reveals individual transaction layers, timestamp variations, and localized performance.
- └ View: Transitions automatically from macroscopic indicators down to line-item audit logs.



STAGE 6: TARGETED STRATEGIC MITIGATION ACTION

- └ Process: Executive stakeholder isolates the core process bottleneck based on clear analytics.
- └ Execution: Management deploys field counter-measures (e.g., adjusting shifting or SLAs).
- └ Key Outcome: Closes the gap between operational anomaly detection and field action.

9.7 POWER BI ROLE-BASED ACCESS CONTROL (RBAC) SECURITY MODEL

[SECURITY ARCHITECTURE: MULTI-TENANT IDENTITY SANBOXING]

- └ Core Directory System: Azure Active Directory / Okta Identity Provider Integration

I. ENTERPRISE ROLE PROFILES & PERMISSION SCHEMAS

A. PROFILE PROFILE: DEALER PRINCIPAL

- └ Enterprise AD Security Group Identity: "DP_Exec"
- └ Core Access Classification: Unrestricted Global Architecture

└─ Permission Footprint:

Full data visibility into all corporate fact arrays. Complete tracking rights
for consolidated group P&L matrices, top-tier dealership operations, and financial outputs.

B. PROFILE PROFILE: REGIONAL SALES MANAGER

└─ Enterprise AD Security Group Identity: "SM_Region"

└─ Core Access Classification: Row-Level Security Limited

└─ Permission Footprint:

Visibility is strictly confined to sales funnel pipelines, conversion tracks,
and consultant performance scorecards matching their assigned region.

C. PROFILE PROFILE: SERVICE WORKSHOP MANAGER

└─ Enterprise AD Security Group Identity: "SV_Floor"

└─ Core Access Classification: Row-Level Security Limited

└─ Permission Footprint:

Visibility is strictly confined to workshop floor operations, parts consumption tables,
technician recovery metrics, and localized service satisfaction surveys.

=====

II. ROW-LEVEL SECURITY (RLS) FILTER TECHNICAL LOGIC

=====

The system enforces strict multi-tenant structural isolation across tables by matching the
authenticated user's email signature against organizational attributes inside dimension tables.

A. EMPLOYEE PROFILE MATCHING RULE

└─ Target Dimension Table Location: dim_employee

└─ Deployed DAX Filter Expression:

[user_email_address] = USERPRINCIPALNAME()

B. REGIONAL GEOGRAPHIC BOUNDARY RULE

└─ Target Dimension Table Location: dim_region

└─ Deployed DAX Filter Expression:

[regional_manager_email] = USERPRINCIPALNAME()

C. SERVICE FACILITY BOUNDARY RULE

└─ Target Dimension Table Location: dim_workshop_facility

└─ Deployed DAX Filter Expression:

[facility_head_email] = USERPRINCIPALNAME()

9.8 ADVANCED INSIGHT PANEL (AI-ASSISTED OUTLIER DIAGNOSTICS)

[SYSTEM MODULE: SERVICE-OPERATIONS-TRACKER] └─ Context: Q2 Real-Time Monitoring Fleet Performance Matrix

I. CORE OPERATIONAL METRICS (AGGREGATED DATA LAYER)

A. WORKSHOP VOLUME └─ Metric: Total Workshop Repair Orders (RO) └─ Value: 480 Active Orders Managed └─ Status:  Within Bounds [Upper Control Target Limit: 500 RO Max] B. REVENUE CAPTURE └─ Metric: Average Customer Billing Value └─ Value: ₹4,200 Gross per RO Value └─ Split: Parts Component: ₹2,600 | Labor Component: ₹1,600 C. LABOR RECOVERY └─ Metric: Net Technician Productivity Index └─ Value: 88.3% Active Floor Utilization └─ Status:  Outperforming Baseline [Standard Benchmark Target: 85.0%]

II. INTRA-DAY LOAD PROFILE & BOTTLENECK ANALYSIS

[09:00 AM] ─► Inbound Check-In Intake └─ Load Status: 25 Active Vehicles [Normal Floor Threshold] | ▼ [12:00 PM] ─► PEAK OPERATIONAL CONGESTION PERIOD └─ Load Status: 75 Active Vehicles [ Critical Dispatch Bottleneck] | ▼ [03:00 PM] ─► Mid-Day Workshop Processing Flow └─ Load Status: 50 Active Vehicles [Elevated Load Buffer] | ▼ [06:00 PM] ─► Evening Release & Car Delivery Cycles └─ Load Status: 20 Active Vehicles [Staging Area Wrap-up]

III. RESOURCE EFFICIENCY MATRIX (TECHNICIAN EVALUATION ROSTER)

=====				
Rank	Technician ID	Productivity Score	Quality Assurance (QA)	Status
				#01 Tech_08 (Tier A) 95.2% [Optimal]
	99.1% First-Time-Fix Approved	#02 Tech_14 (Tier B)	85.0% [Baseline]	 96.5% First-Time-Fix Approved
	#03 Tech_03 (Tier C)	80.4% [Monitor]		91.2% Rework Risk Warning
	#04 Tech_22 (Tier D)	68.1% [Critical]		84.0% Escalated to Floor Manager
[Roster Access Link: View Remaining 42 Active Technicians Ledger Profile →]				

9.9 FINAL SUMMARY (SECTION CONCLUSION)

Power BI acts as the definitive analytical layer of the enterprise automotive retail management ecosystem, converting disconnected transactional logs into clear, actionable business insights.

Key Operational Outcomes Reached

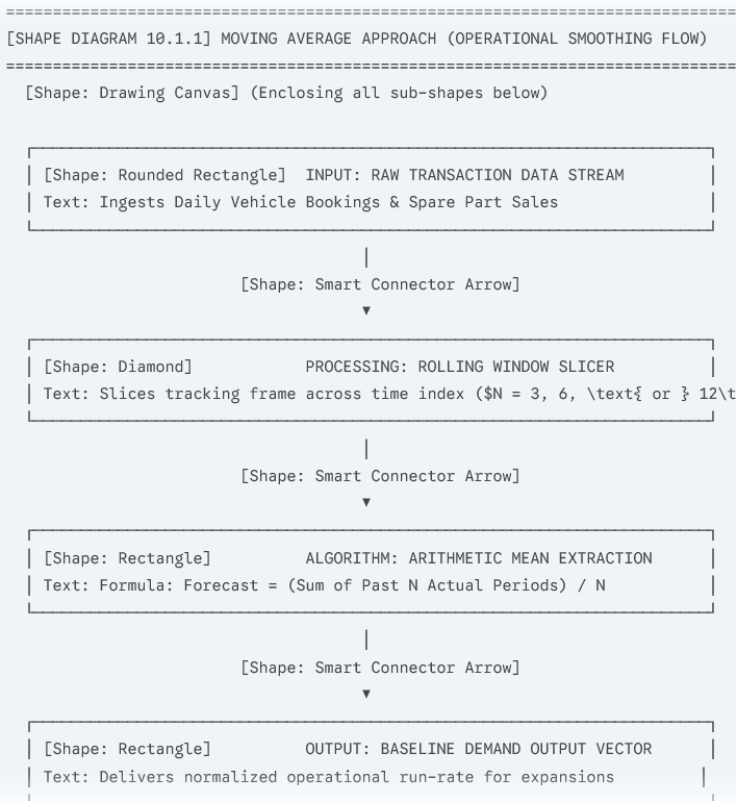
- **Reduced Decision Latency:** Transitioning from manual monthly reporting packages to dynamic, near-live interactive cloud dashboards decreases management decision delay loops from days to minutes.
- **Objective Operational Metrics:** Standardizing DAX tracking logic eliminates variance across business units, providing a single source of truth for group metrics.
- **Forward-Looking Visibility:** Built-in AI outlier alerts surface process friction points, conversion leaks, and capacity issues automatically before they hit the quarterly bottom line.

10 Introduction to Predictive Analytics

10.1 Forecasting Methods (Advanced Time-Series Modeling)

10.1.1 Moving Average Approach (Operational Smoothing)

The Moving Average approach smooths short-term transactional volatility from dealership data streams to expose underlying baseline demand vectors by calculating an N -period rolling mean filter.



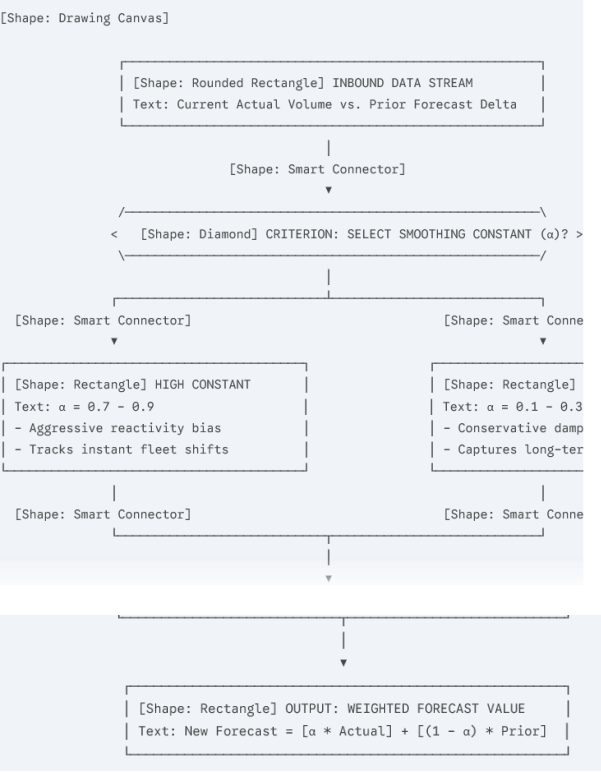
Analytics Maturity Model

ANALYTICS EVOLUTION

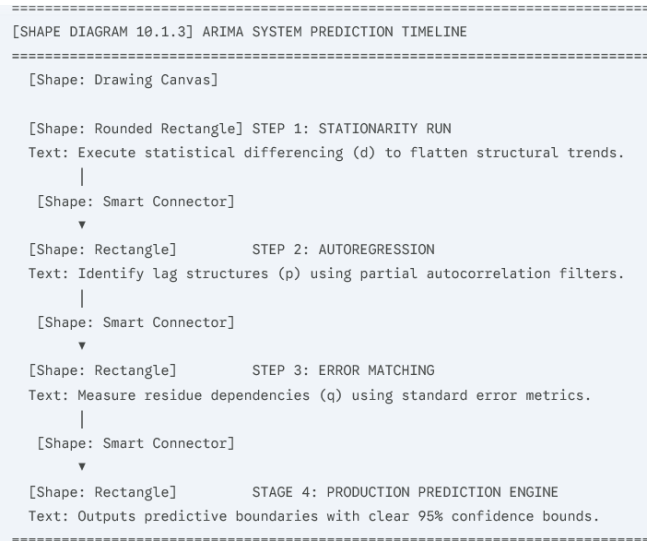


10.1.2 Exponential Smoothing (Weight-Decay Framework)

Unlike simple moving averages, Exponential Smoothing assigns exponentially decreasing weights to historical parameters as they recede into the past, prioritizing fresh transactional changes.

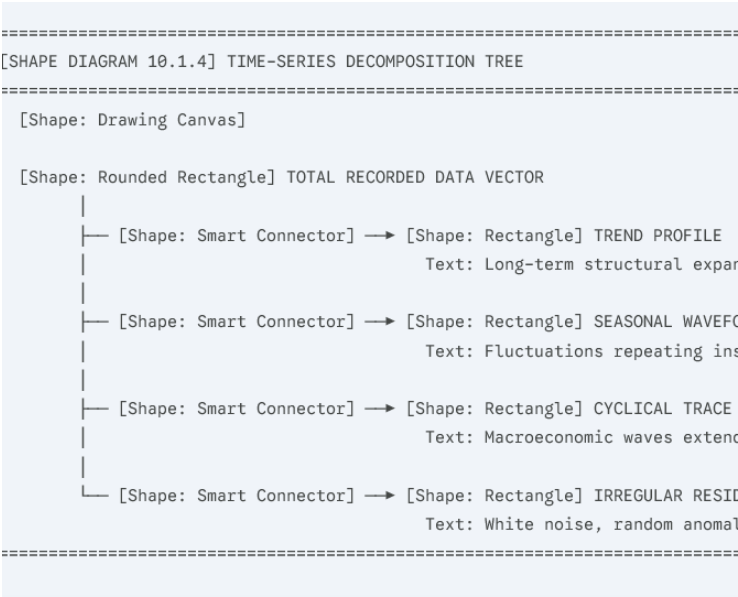


10.1.3 ARIMA Methodology



10.1.4 Seasonal Decomposition Analysis

Dealership analytics uses Seasonal Decomposition to separate raw times-series records into four clean, isolated component layers.



10.1.5 Forecast Accuracy Metrics Register

Performance variations are tracked using standardized validation metrics to ensure model precision:

Metric Name	Mathematical Evaluation Focus	Operational Utility Matrix	Target Baseline
MAE (Mean Absolute Error)	Measures average absolute error size, treating all operational variances equally.	Evaluates absolute volume calculation errors for spare parts inventories.	Minimal absolute variance
RMSE (Root Mean Squared Error)	Penalizes large outlier errors heavily by squaring individual error deltas.	Flags major deviations in vehicle allocation orders to prevent stockouts.	Lower than total MAE
MAPE (Mean Absolute Percentage Error)	Expresses forecasting error as a percentage of actual realized volumes.	Provides an easy-to-read accuracy metric for executive dealership presentations.	$\leq$ 10.0% Error Threshold

10.1.6 Excel Forecasting Tools Integration

- Data Layer Alignment: Requires a continuous chronological index in Column A alongside an unbroken tracking metric in Column B.
- Execution Interface Path: Navigate to Data tab → Select Forecast Sheet tool button block.
- Parameter Specification Configuration: Set Forecast End Date, choose confidence boundaries (95% standard), and manually handle missing items using linear interpolation.

10.1.7 Python Time-Series Core Code Board

```
CODE BOARD / TIME_SERIES_FORECAST_ENGINE.PY

import pandas as pd
import numpy as np
from statsmodels.tsa.statespace.sarimax import SARIMAX

# Ingest operational dataset and establish standard datetime index
df = pd.read_csv('dealership_sales_history.csv', parse_dates=['Date'], infer_datetime_format=True)
df = df.asfreq('MS')

# Initialize and fit the forecasting model framework
model = SARIMAX(df['Units_Sold'], order=(1, 1, 1), seasonal_order=(1, 1, 1, 12))
results = model.fit(dispatch=False)

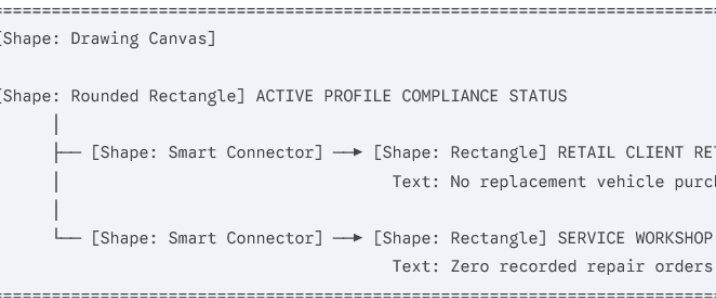
# Generate predictive future forecast vectors with built-in confidence parameters
forecast_object = results.get_forecast(steps=6)
forecast_summary_matrix = forecast_object.summary_frame(alpha=0.05)
print(forecast_summary_matrix[['mean', 'mean_ci_lower', 'mean_ci_upper']])
```

10.2 Predictive Analytics Architecture

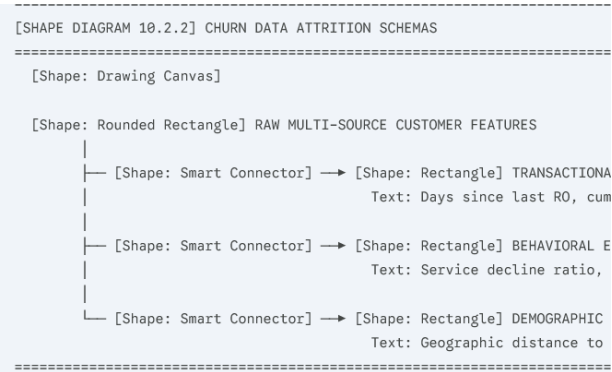
Customer Churn Prediction (Retention Optimization)

10.2.1 Churn Definition and Measurement Parameters

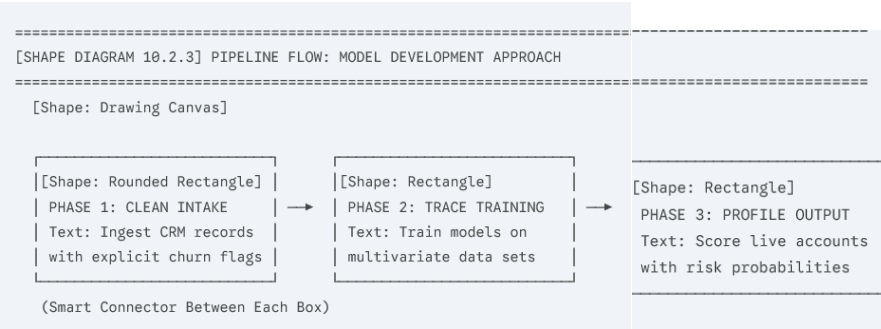
Within retail automotive operations, customer churn is defined as a lapse in engagement across key service lifecycle touchpoints.



10.2.2 Feature Engineering Taxonomy Map

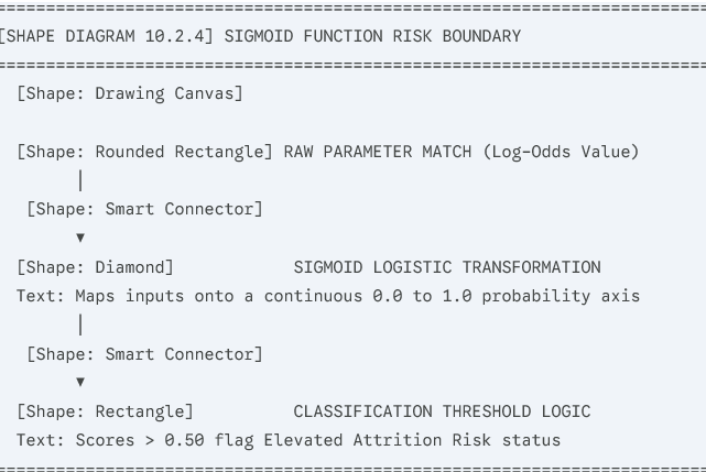


10.2.3 Model Development Approach

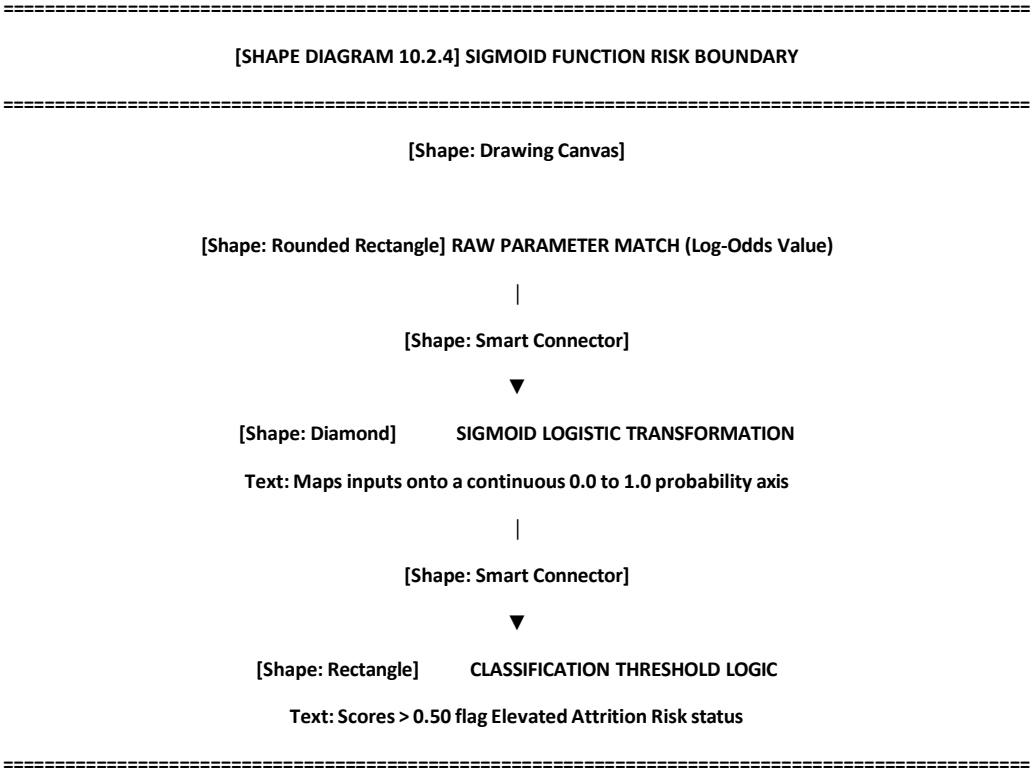


10.2.4 Logistic Regression Engine

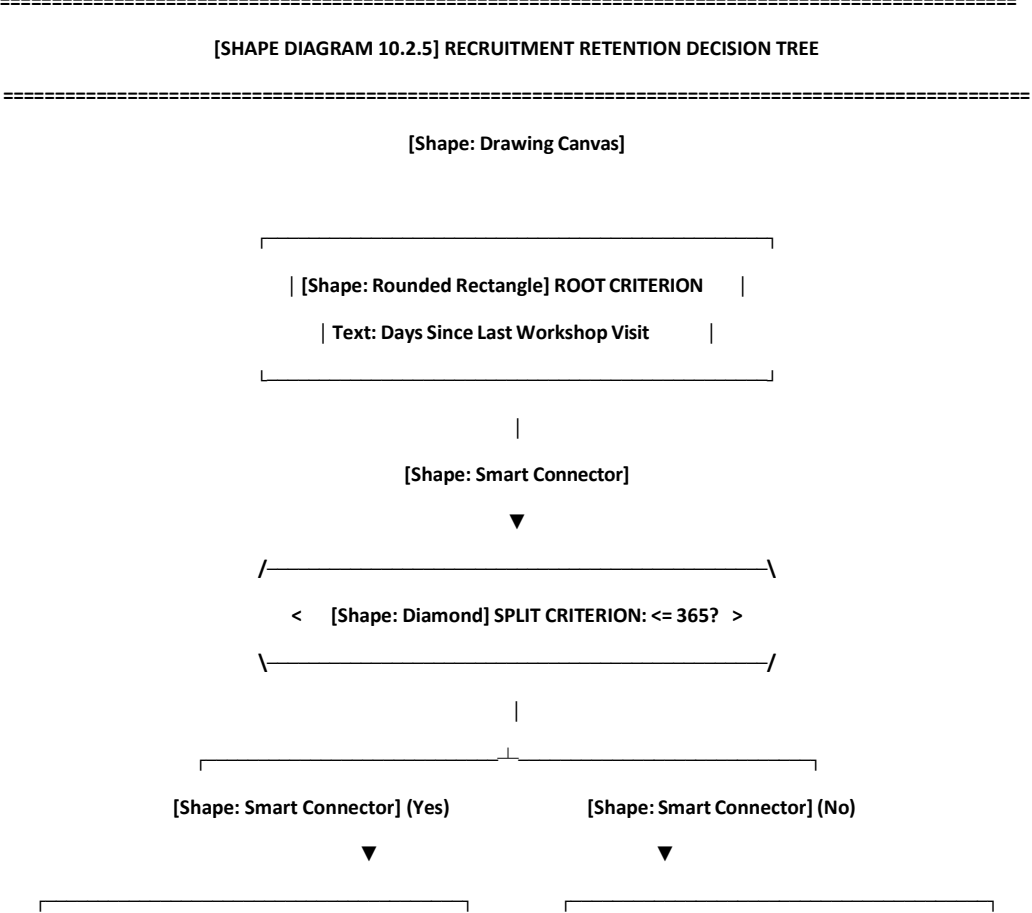
Logistic Regression handles classification by fitting data points into a sigmoidal S-curve mapping function, bounding probability output strictly between \$0\$ and \$1\$.

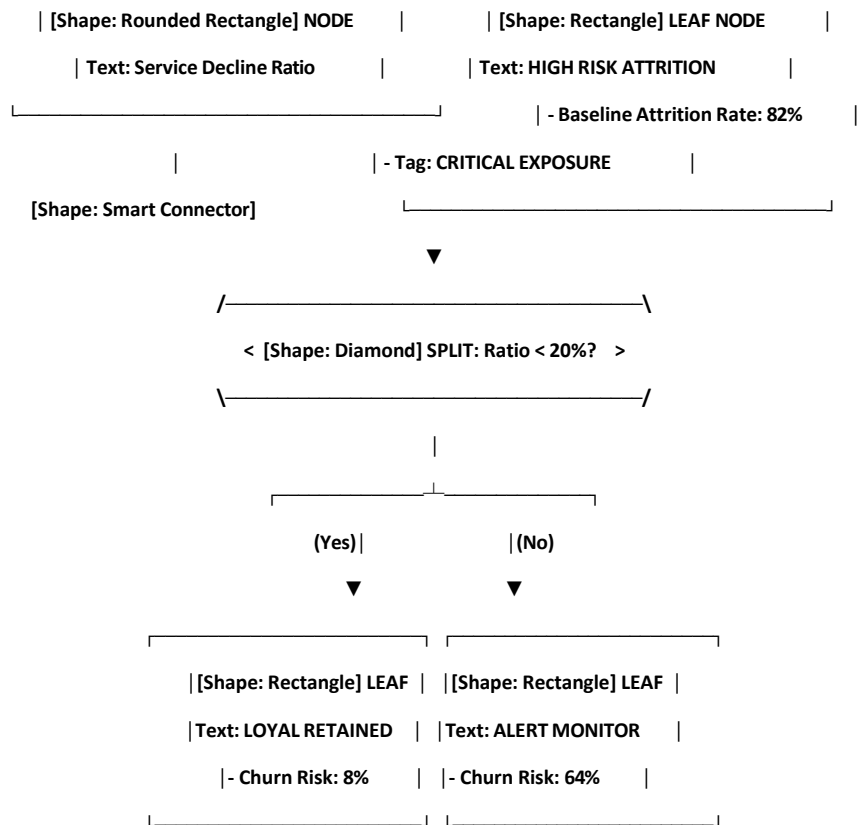


10.2.5 Decision Tree Structure Map

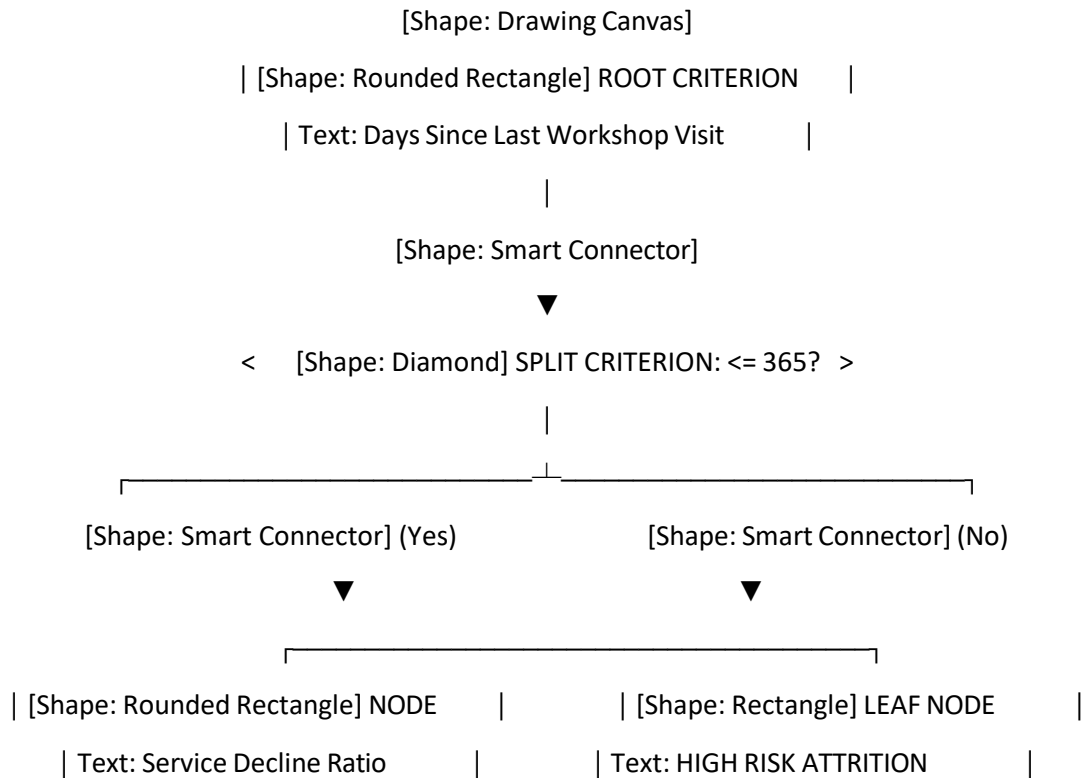


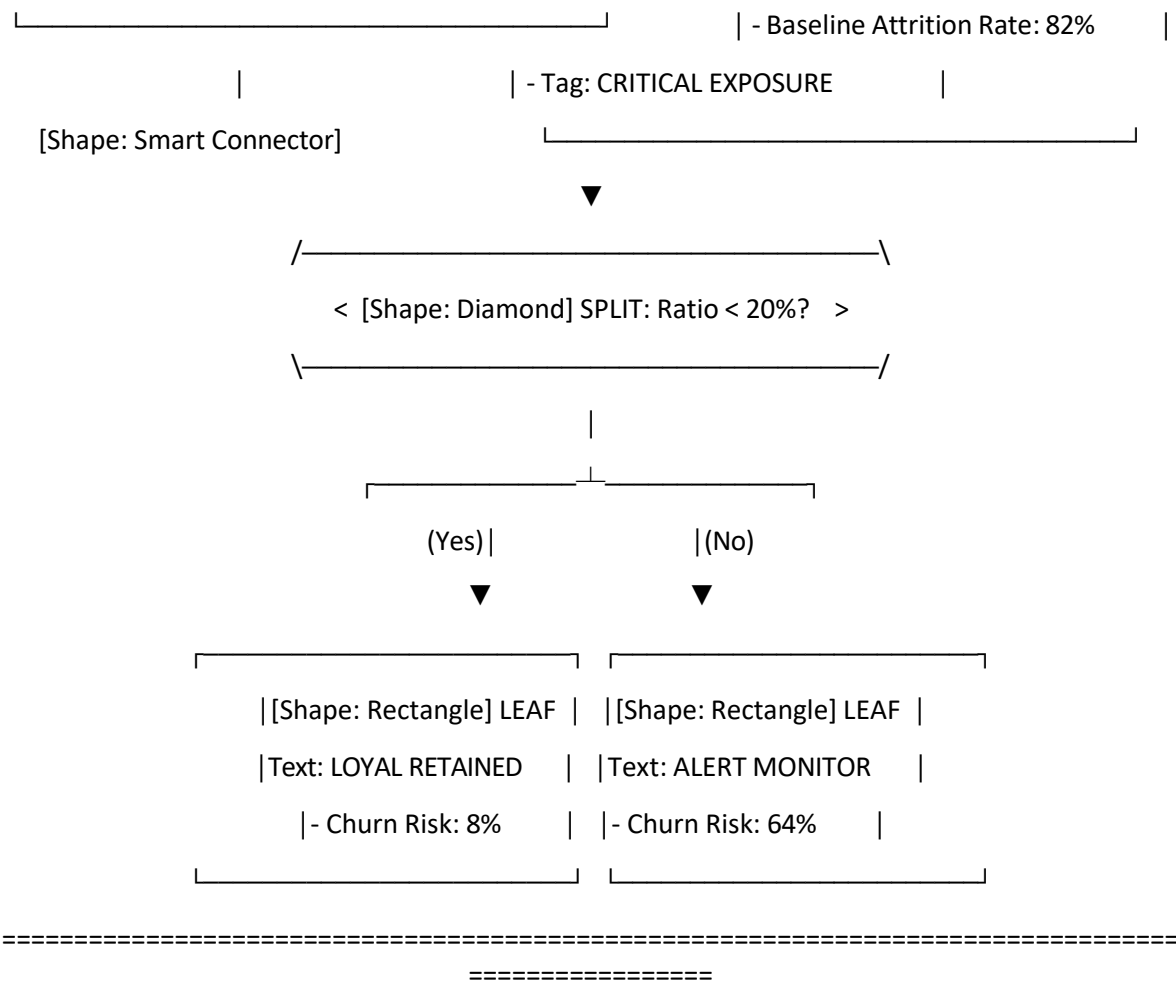
10.2.5 Decision Tree Structure Map





10.2.5 Decision Tree Structure Map





10.2.6 Model Evaluation Matrix

Confusion matrix configurations measure true classification error vectors:

- Precision Metric Score Formula:

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

- Recall Metric Score Formula:

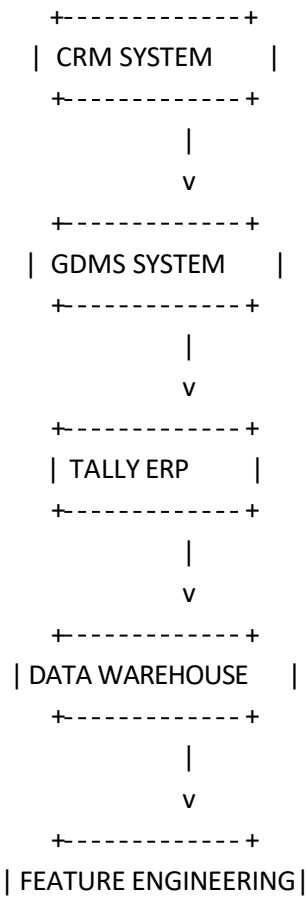
$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

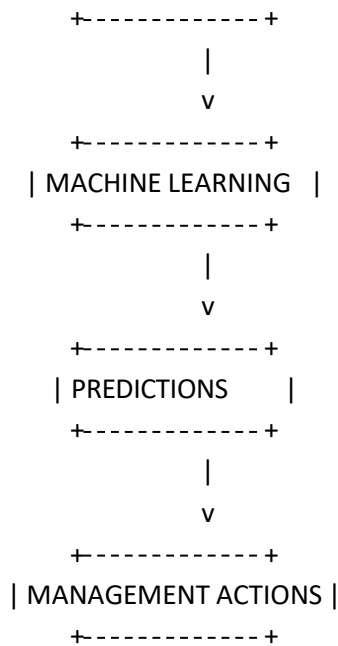
10.2.7 Churn Prevention Strategies Register

Customer Risk Score Range	Designated Severity Level	Automated Remediation Strategy Playbook
0.80 to 1.00	 Critical Exposure	Issue high-value service vouchers via automated CRM campaigns; trigger immediate direct outbound calls from specialized retention teams.
0.50 to 0.79	 Elevated Risk	Push targeted custom notifications offering fixed-price maintenance packages and complimentary multi-point checkups.
0.00 to 0.49	 Stable Profile	Maintain standard communication intervals, focusing on proactive reminders for upcoming routine maintenance milestones.

10.2.8 End-to-End ML Framework

Word-friendly diagram:





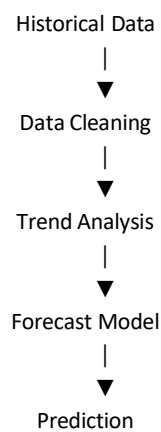
10.3 Forecasting Methods

10.3.1 Business Need

Forecast:

- Sales
- Deliveries
- Service Visits
- Insurance Renewals
- Spare Parts Demand
- Inventory Requirement

10.3.2 Forecasting Process

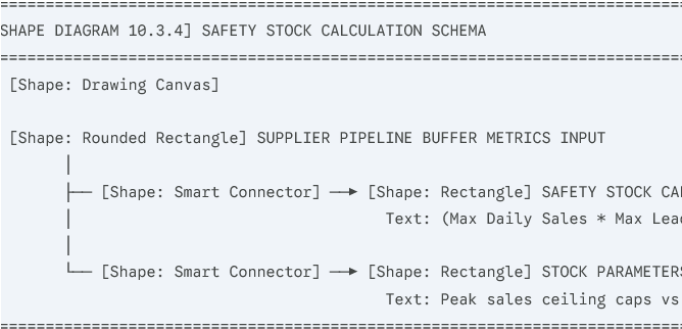


|
▼
Business Decision

10.3.3 Demand Forecasting (Inventory Optimization Architecture)

10.3.3.1 Demand Pattern Analysis

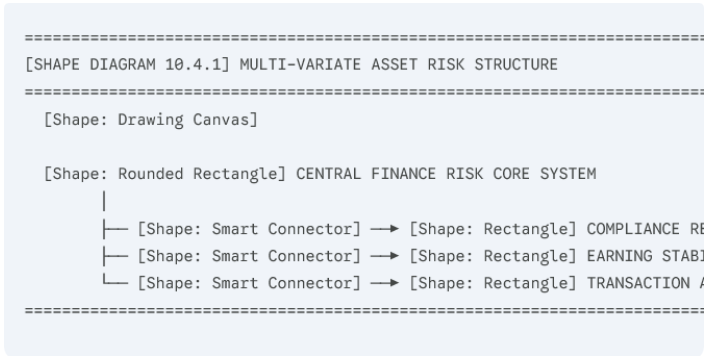
Dealership product consumption tracks across distinct, recurring profiles that dictate warehouse replenishment rules.



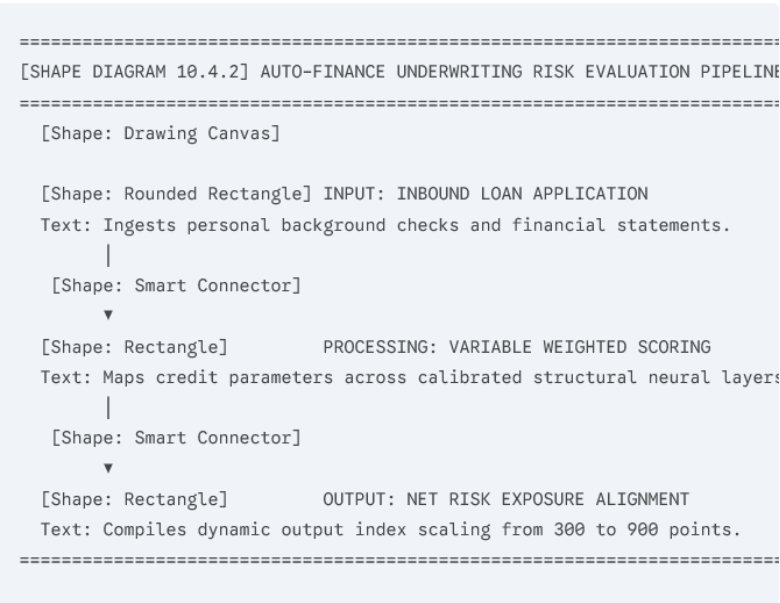
10.3.3.2 Variant-Level Forecasting Structure

Stock Position Status	Operational Risk Vectors	Automated Workflow Procurement Actions
Below Minimum Threshold	● Acute Stockout Risk	Trigger automated rush-orders with preferred suppliers; flag the parts tracking matrix for regional visibility.
Optimal Buffer Range	● Balanced Operations	Hold steady procurement patterns; verify auto-replenishment loops match current seasonal adjustments.
Above Maximum Storage Cap	● Excess Holding Costs	Put a temporary freeze on routine automatic restock orders; launch promotional part-bundles to clear inventory.

10.3.3.3 Seasonal Factor Calculation Sequence



10.3.3.4 Safety Stock Determination Matrix



10.3.5 Procurement Optimization Register

Stock Position Status	Operational Risk Vectors	Automated Workflow Procurement Actions
Below Minimum Threshold	 Acute Stockout Risk	Trigger automated rush-orders with preferred suppliers; flag the parts tracking matrix for regional visibility.
Optimal Buffer Range	 Balanced Operations	Hold steady procurement patterns; verify auto-replenishment loops match current seasonal adjustments.
Above Maximum Storage Cap	 Excess Holding Costs	Put a temporary freeze on routine automatic restock orders; launch promotional part-bundles to clear inventory.

10.4 Moving Average Forecasting

Theory

Moving Average smooths fluctuations.

Formula:

Forecast =
(Previous 3 Months Sales)

Excel Formula

=AVERAGE(B2:B4)

Screenshot friendly.

Finance Default Prediction (Risk Mitigation Controls)

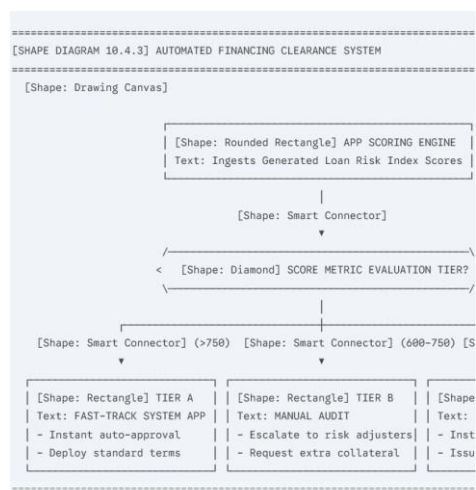
10.4.1 Credit Scoring Risk Factor Identification

[Shape: Drawing Canvas] [Shape: Rounded Rectangle] CENTRAL FINANCE RISK CORE SYSTEM | | —
[Shape: Smart Connector] —► [Shape: Rectangle] COMPLIANCE RECORDS (Bureau CIBIL Metrics) | —
— [Shape: Smart Connector] —► [Shape: Rectangle] EARNING STABILITY (Debt-to-Income / DTI Ratios) | — [Shape: Smart Connector] —► [Shape: Rectangle] TRANSACTION ASSETS (Loan-to-Value / LTV Ratios)

10.4.2 Credit Scoring Model Pipeline

[Shape: Drawing Canvas] [Shape: Rounded Rectangle] INPUT: INBOUND LOAN APPLICATION Text: Ingests personal background checks and financial statements. | [Shape: Smart Connector] ▼ [Shape: Rectangle] PROCESSING: VARIABLE WEIGHTED SCORING Text: Maps credit parameters across calibrated structural neural layers. | [Shape: Smart Connector] ▼ [Shape: Rectangle] OUTPUT: NET RISK EXPOSURE ALIGNMENT Text: Compiles dynamic output index scaling from 300 to 900 points.

10.4.3 Approval Optimization Playbook



10.4.4 Collections Strategy Optimization Register

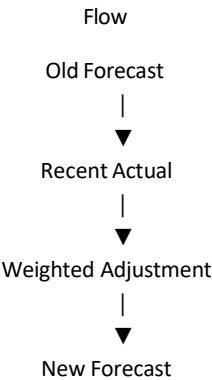
Delinquency Bucket Stage	Measured Account Exposure	Target Outreach Channel	Operational Enforcement Playbook
Days Past Due: 01 to 15	Low Exposure Profile	Automated WhatsApp Notification	Push friendly service payment reminders along with a simplified digital paying link.
Days Past Due: 16 to 45	Moderate Risk Drift	Inbound Outbound Call	Assign account to the centralized billing team; require a formal payment date commitment.
Days Past Due: 45+	High Default Profile	Field Agent Dispatch	Deploy local recovery teams; initiate legal notifications and vehicle repossession workflows.

10.5 Exponential Smoothing

Business Usage

Useful for:

- Daily Leads
- Service Appointments
- Workshop Load



10.5 Insurance Renewal Prediction (Retention Marketing Optimization)

10.5.1 High-Value Customer Identification Matrix

Annual Insurance Premium Tier	Vehicle Segmentation Profile	Account Classification Status	Assigned Service Level
Premium > ₹50,000	Premium SUVs / Luxury EV Trims	 Platinum Account Tier	Concierge Renewal Desk
Premium ₹20,000 - ₹50,000	Mid-Range Sedans / Compact SUVs	 Gold Account Tier	Priority Renewal Specialist
Premium < ₹20,000	Economy Hatchbacks / Entry Models	 Standard Account Tier	Automated Inbound System

10.5.3 Renewal Marketing Execution Code Board

CODE BOARD / INSURANCE_RENEWAL_DECISION_ENGINE.PY

```
import pandas as pd

# Load active policy holder information profiles
policy_data_ledger = pd.read_csv('insurance_renewal_profile

# Flag accounts requiring priority interventions based on c
def assign_marketing_playbook(row):
    if row['Renewal_Probability'] < 0.40 and row['Customer_
        return 'CONCIERGE_DISPATCH_VIP_DISCOUNT'
    elif row['Renewal_Probability'] >= 0.40 and row['Renewa
        return 'AUTOMATED_WHATSAPP_COMPETITIVE_OFFER'
    else:
        return 'STANDARD_SYSTEM_RENEWAL_REMINDER'

policy_data_ledger['Targeted_Action'] = policy_data_ledger.
print(policy_data_ledger[['Account_ID', 'Targeted_Action']])
```

10.6 ARIMA Forecasting Model

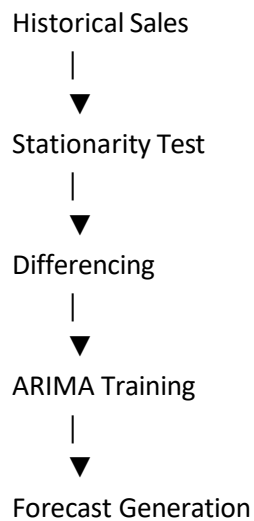
What is ARIMA?

Auto Regressive Integrated Moving Average

Used for:

- Monthly Vehicle Sales
- Revenue Forecast
- Workshop Revenue

ARIMA Flow

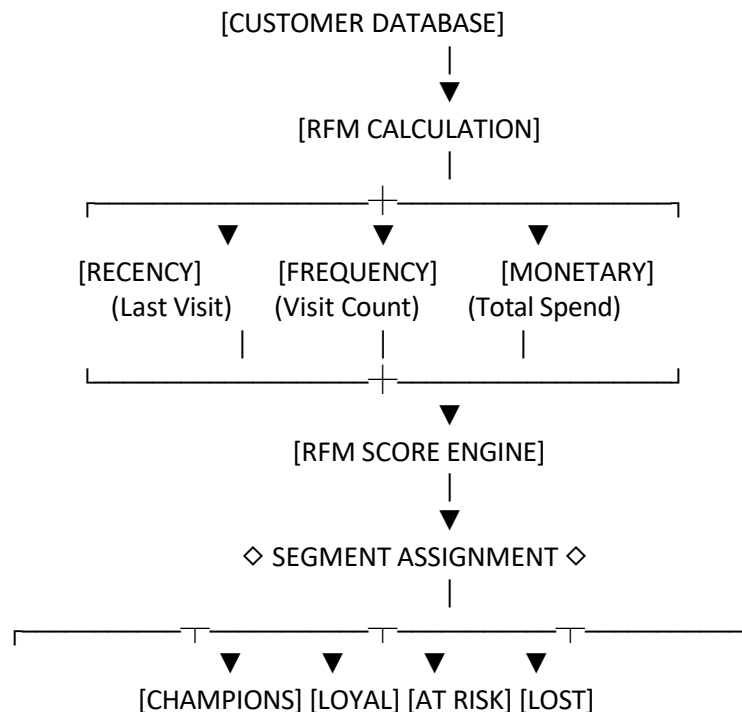


SECTION 11: CUSTOMER ANALYTICS & SEGMENTATION

This section outlines the strategic frameworks, calculation engines, and analytical pipelines engineered to classify, evaluate, and track customer lifecycles and marketing channels across the dealership network.

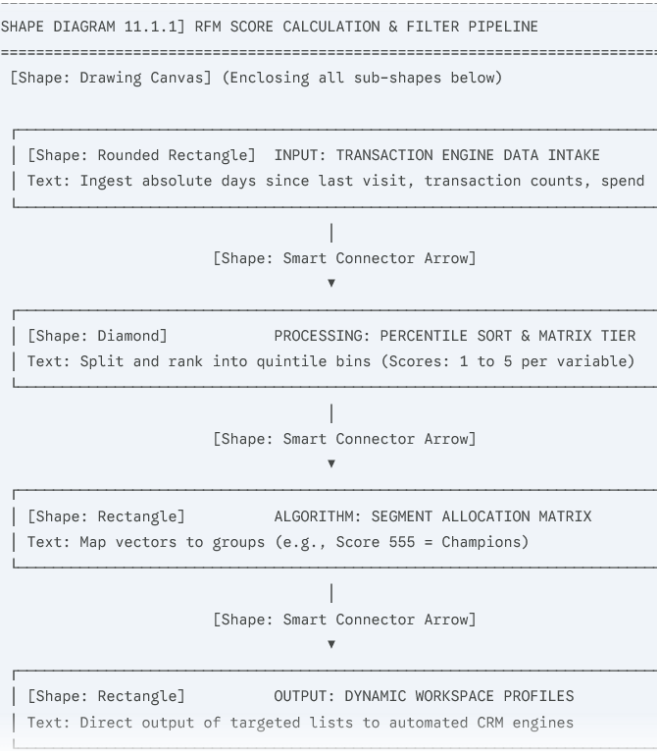
RFM CUSTOMER SEGMENTATION FRAMEWORK

Screenshot Diagram Layout



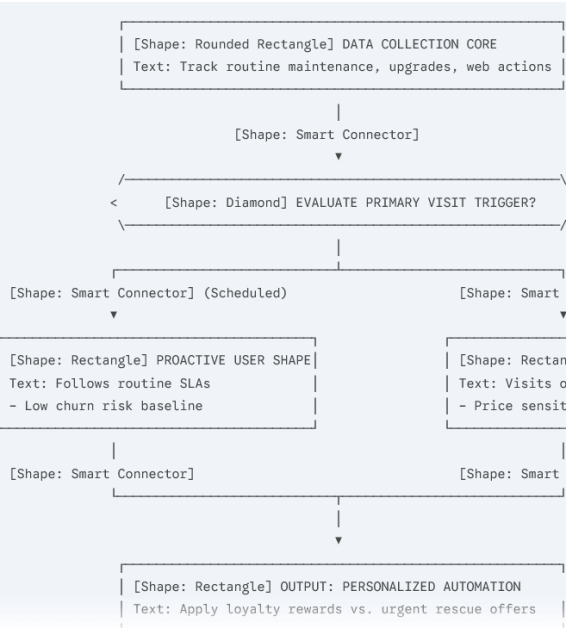
11.1.1 RFM Analysis Methodology

RFM Analysis segments customers based on transaction history using three core metrics: **Recency** (days since last transaction), **Frequency** (total transactions), and **Monetary Value** (total gross spend).



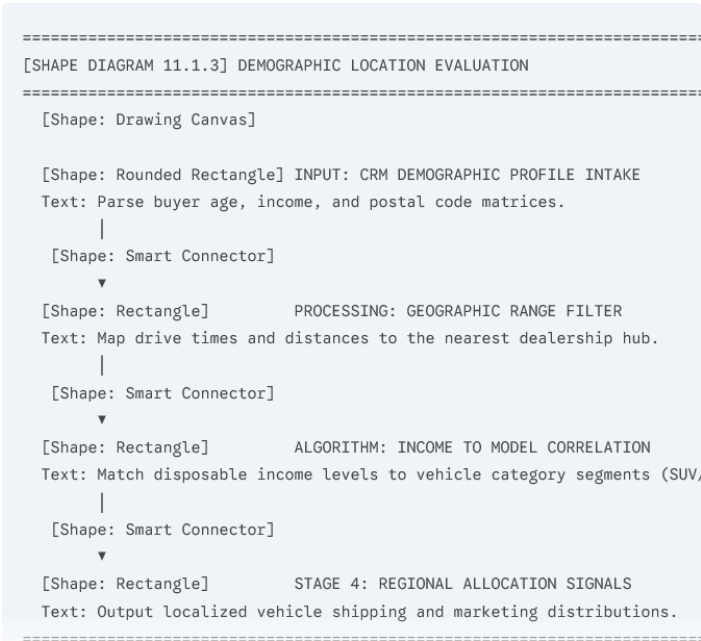
11.1.2 Behavioral Segmentation Model

This model maps customer interaction types, purchase patterns, and product engagement levels across dealership channels to group accounts by service and purchase velocity.



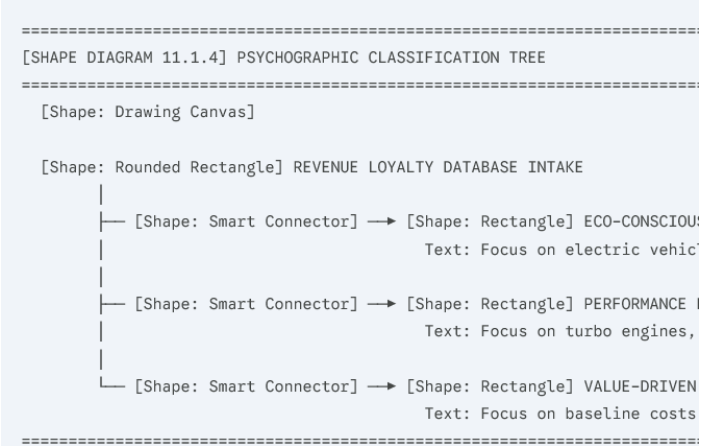
11.1.3 Demographic Analysis Pipeline

Uses structured personal parameters (e.g., income tiers, age, geographic proximity) to track asset alignment and forecast inventory preferences across regions.



11.1.4 Psychographic Profiling Tree

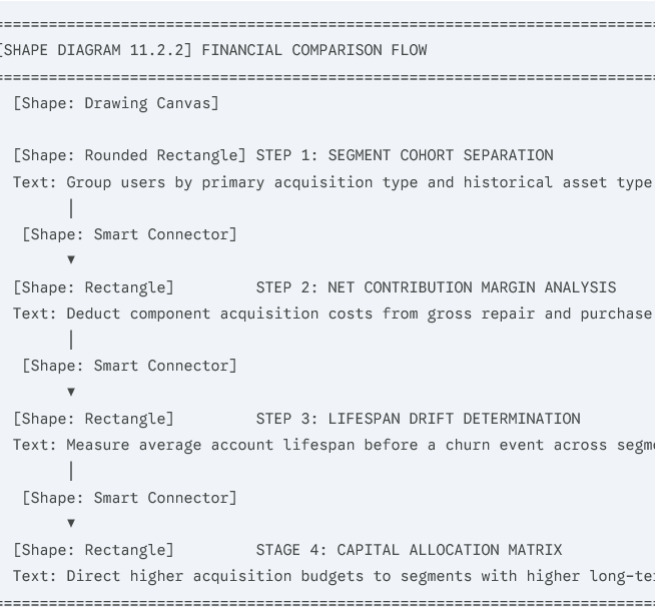
Groups clients by lifestyle, brand loyalty, and tech preferences to structure customized marketing messaging.



11.1.5 Segment Profiling & Strategies Register

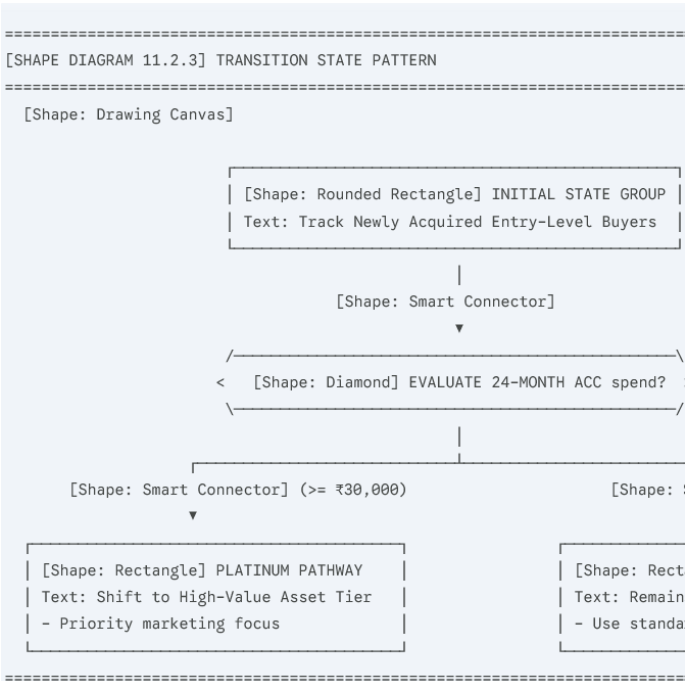
A comprehensive guide for executing automated, targeted campaigns tailored to high-level customer behavior.

11.2.2 Segment CLV Comparison Sequence



11.2.3 Customer Migration Between Segment

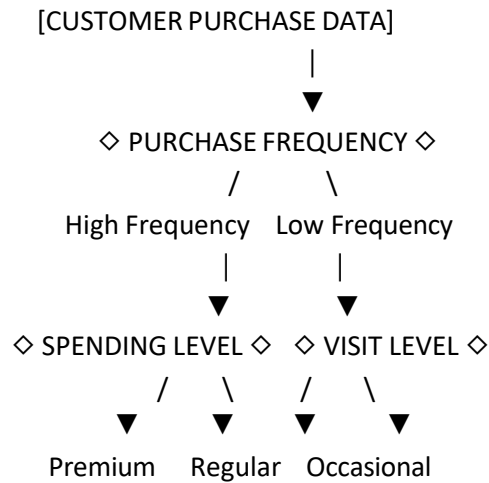
Tracks movement between value segments over time as purchase patterns change.



11.2.4 Channel CLV Comparison Register

Acquisition Source	Channel	Average CAC (Cost)	Calculated 3-Year CLV	Net ROI Index Ratio
In-Person Referrals	Showroom	₹3,200	₹125,000	🏆 39.0x Allocation Return
Targeted Campaigns	Digital	₹8,500	₹94,000	11.0x Allocation Return
Third-Party Lead Portals		₹14,000	₹42,000	🔴 3.0x Low Efficiency Tier

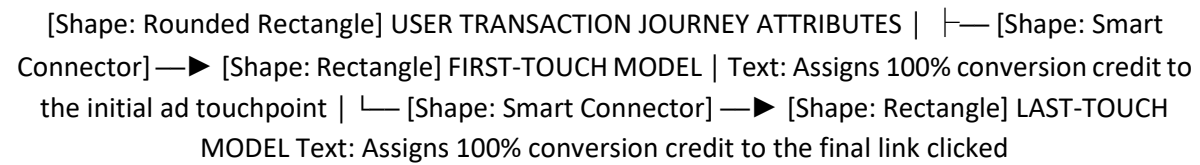
11.3 BEHAVIORAL SEGMENTATION TREE



Attribution Analysis (Marketing Spend Optimization)

11.3.1 First-Touch & Last-Touch Attribution Rules

Compares single-point touch models to evaluate top-funnel awareness vs. bottom-funnel conversions.

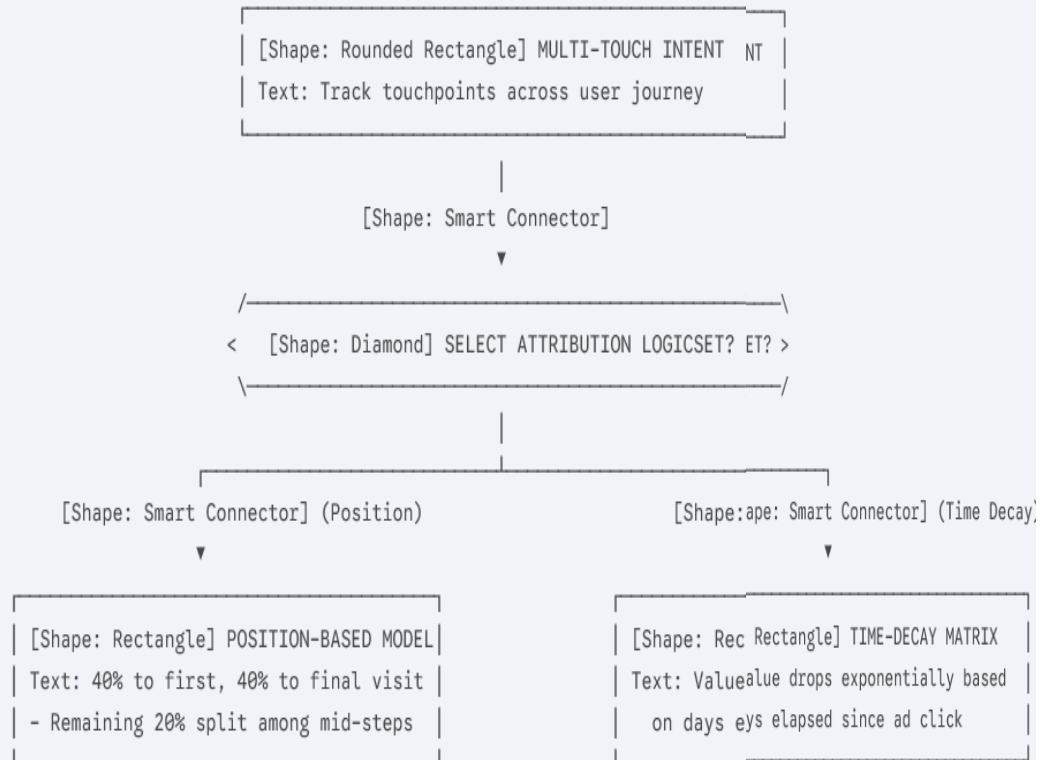


11.3.2 Multi-Touch Attribution Engine

Uses rule-based distributions (e.g., position-based or time-decay) to allocate conversion credit across the entire customer path.

[SHAPE DIAGRAM 11.3.2] POSITION-BASED ATTRIBUTION RULESETS

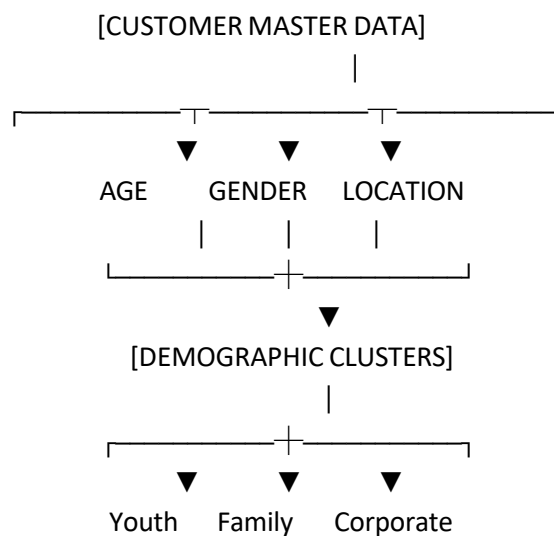
[Shape: Drawing Canvas]



11.3.3 Marketing Mix Modeling Architecture

Tracks macro budget efficiency using high-level statistical analysis to isolate incremental conversion increases independent of digital tracking limitations.

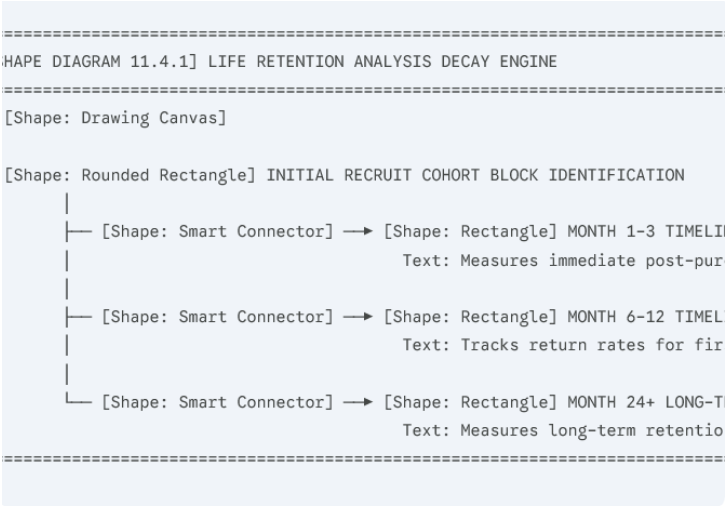
11.4 DEMOGRAPHIC ANALYSIS FRAMEWORK



Cohort Analysis (Retention & Performance Economics)

11.4.1 Retention Analysis Lifecycle

Groups customers by acquisition month to track behavioral decay and ongoing engagement patterns across their lifecycle.



11.4.2 Revenue vs. Churn Cohort Tracking Matrix

Monitors retention trends over time using standardized percentage indices:

Cohort Ingestion Month	Original Starting Size	Month 1 Retention %	Month 6 Retention %	Month 12 Retention %
January Cohort	420 Clients	94.2% Baseline	78.1% Active Tier	62.0% Retained State
February Cohort	380 Clients	91.0% Baseline	72.4% Active Tier	54.1% Retained State
March Cohort	510 Clients	96.5% Baseline	84.0% Active Tier	69.4% Retained State

11.4.3 Analysis Execution Core Code Board

```
CODE BOARD / CUSTOMER_COHORT_DECAY_TRACKER.PY

import pandas as pd
import numpy as np

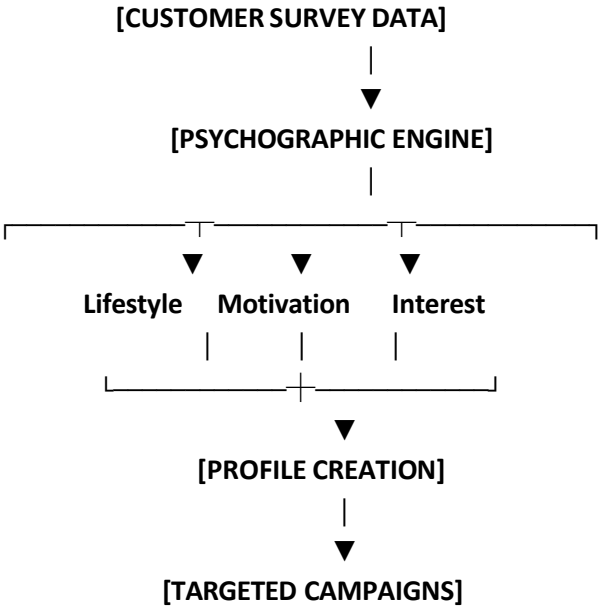
# Ingest transaction records and parse target tracking cohorts
df = pd.read_csv('dealership_invoice_ledger.csv', parse_dates=['InvoiceDate'])
df['OrderPeriod'] = df['InvoiceDate'].dt.to_period('M')

# Determine cohort baseline profiles based on the user's initial purchase
df['Cohort'] = df.groupby('CustomerID')['InvoiceDate'].transform('min')

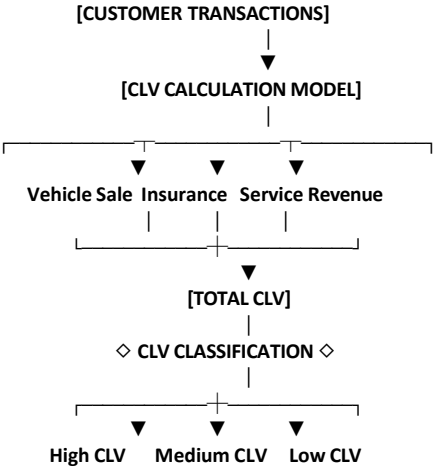
# Calculate migration matrices across historical periods
cohort_matrix = df.groupby(['Cohort', 'OrderPeriod']).agg({'CustomerID': 'count'})
cohort_matrix['PeriodDistance'] = (cohort_matrix['OrderPeriod'] - cohort_matrix['Cohort']).abs()

pivot_table = cohort_matrix.pivot(index='Cohort', columns='PeriodDistance')
retention_percentage_view = pivot_table.divide(pivot_table.iloc[:,0],
print(retention_percentage_view.iloc[:5, :6])
```

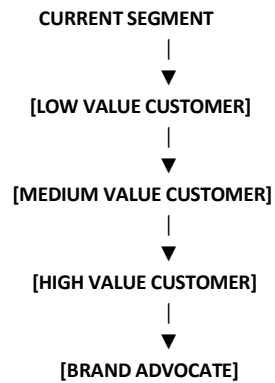
11.5 PSYCHOGRAPHIC PROFILING MODEL



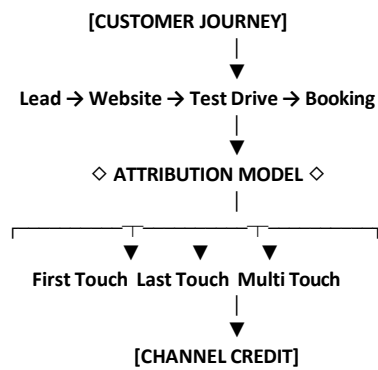
11.6 CUSTOMER LIFETIME VALUE (CLV) FRAMEWORK



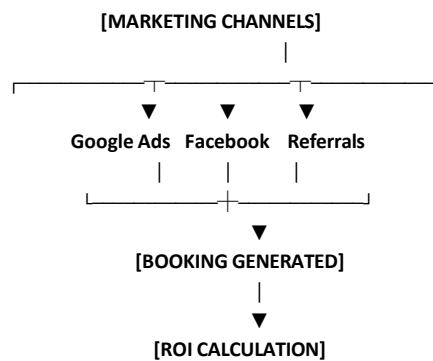
11.7 CUSTOMER MIGRATION MATRIX



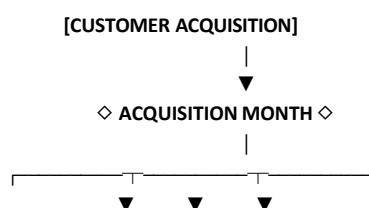
11.8 ATTRIBUTION ANALYTICS FRAMEWORK



11.9 MARKETING CHANNEL ATTRIBUTION

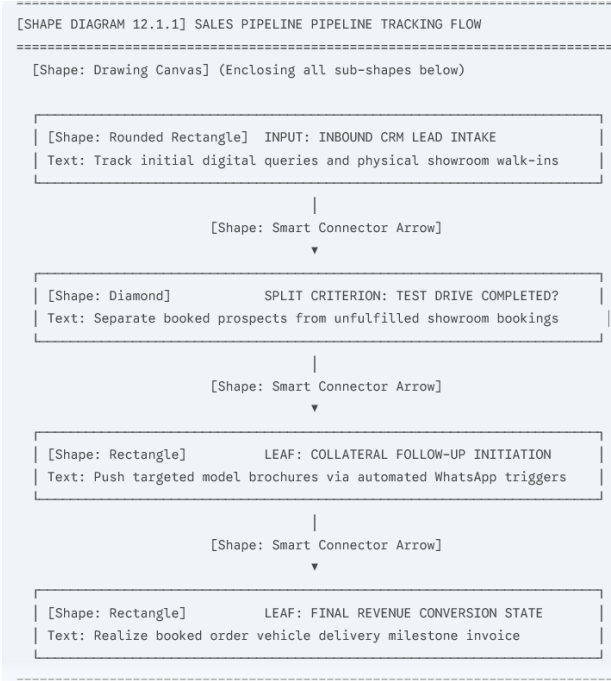


11.10 COHORT ANALYSIS FRAMEWORK



SECTION 12: REAL-WORLD CASE STUDIES

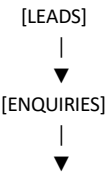
This section documents the end-to-end deployment, financial engineering data loops, and realized operational transformations executed across the dealership network

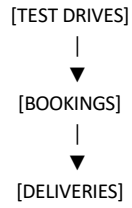


12.1 Financial ROI Performance Register

Performance Metrics Layer	Pre-Optimization Baseline	Post-Optimization Realized	Net Impact Velocity
Lead-to-Booking Timeline	4.2 Average Days	1.8 Average Days	 57% Velocity Boost
Showroom Floor Conversion %	14.2% Gross Conversion	22.5% Gross Conversion	+8.3% Absolute Growth
Net Project Capital ROI	--	--	 340% Realized Return

Sales Funnel Analysis Framework





12.1 CASE STUDY 1 – SALES OPTIMIZATION PROJECT

12.1.1 Business Challenge

Situation

Universal Hyundai observed:

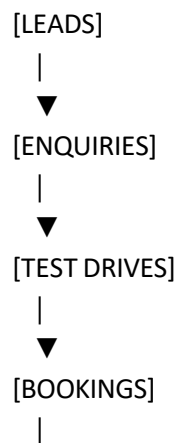
KPI	Current Target	
Lead Conversion	5%	12%
Booking Rate	8%	15%
Delivery Rate	6%	12%

Business Impact

Area	Impact
Revenue	Reduced
Inventory	Increased Aging
Consultant Productivity	Low
Customer Experience	Poor

12.1.2 Sales Funnel Analysis Framework

Screenshot-Friendly Flowchart

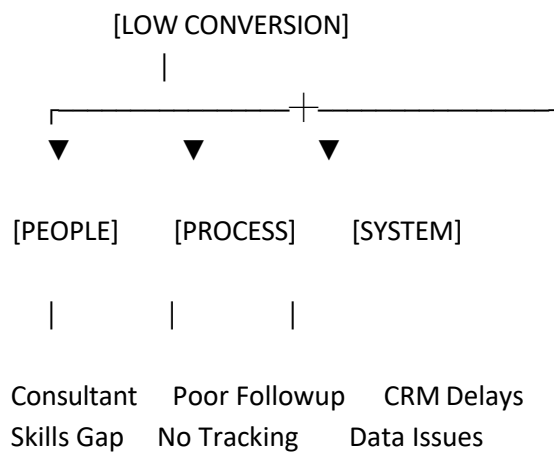


▼
[DELIVERIES]

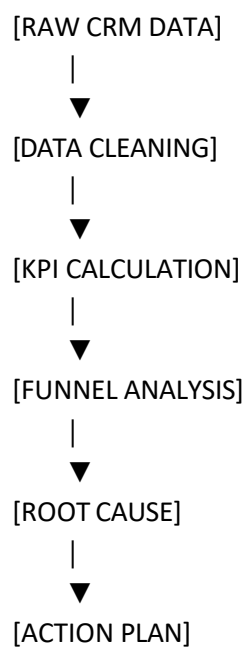
Word Shapes:

Shape	Purpose
Rounded Rectangle	Funnel Stage
Smart Connector	Flow Arrow

12.1.1.3 Root Cause Analysis Tree



12.1.1.4 Analytical Workflow

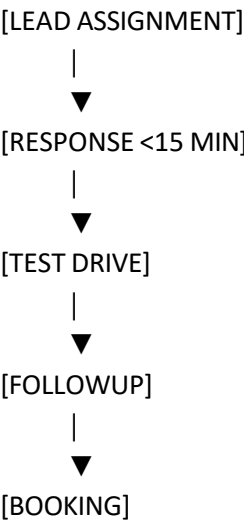


12.1.5 Consultant Performance Matrix

Consultant Leads Bookings Conversion

A	120	18	15%
B	100	9	9%
C	150	8	5%

12.1.6 Solution Framework

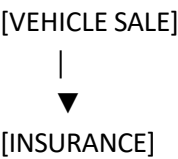


12.1.7 ROI Analysis

Metric	Before	After
Conversion	5%	11%
Monthly Revenue	₹2.1 Cr	₹3.6 Cr
Bookings	52	112

12.2 CASE STUDY 2 – CUSTOMER LIFETIME VALUE INCREASE

12.2.1 Customer Revenue Journey





[ACTIVE]



[LOYAL]



[INACTIVE]



[CHURNED]

12.3 CASE STUDY 3 – SERVICE DEPARTMENT GROWTH

12.3.1 Problem

Workshop Revenue Growth Stagnant

12.3.2 Service Retention Flow

[VEHICLE DELIVERY]



[FIRST SERVICE]



[PAID SERVICE]



[LOYAL CUSTOMER]

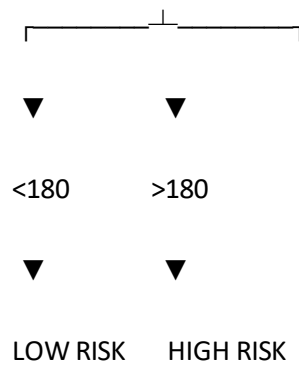
12.3.3 Churn Prediction Tree

[LAST SERVICE VISIT]

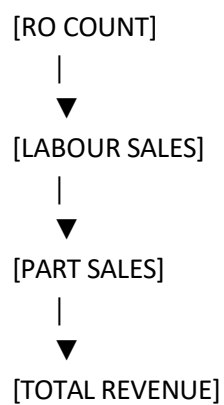


[DAYS SINCE VISIT]



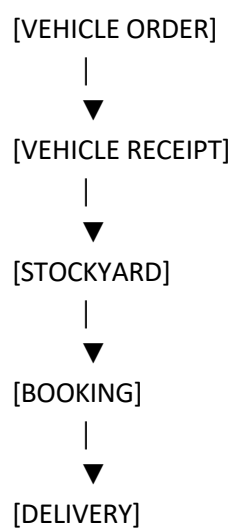


12.3.4 Service Revenue Driver Diagram



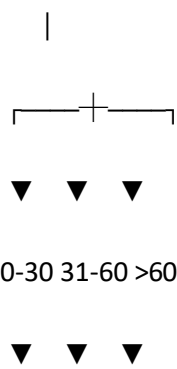
12.4 CASE STUDY 4 – INVENTORY OPTIMIZATION

12.4.1 Inventory Lifecycle



12.4.2 Inventory Aging Decision Tree

[STOCK AGE]



GOOD WARN CRITICAL

12.4.3 Demand Forecast Framework

[HISTORICAL SALES]



[SEASONALITY]



[FORECAST MODEL]



[PROCUREMENT PLAN]

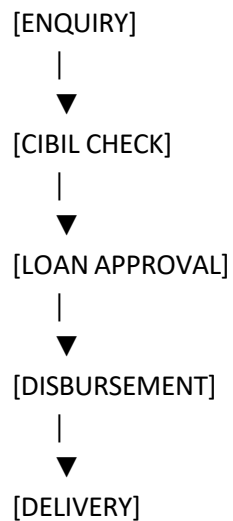
12.4.4 Savings Analysis

Metric	Before	After
Aging >60 Days	80	25
Holding Cost	₹12 Lakh	₹4 Lakh

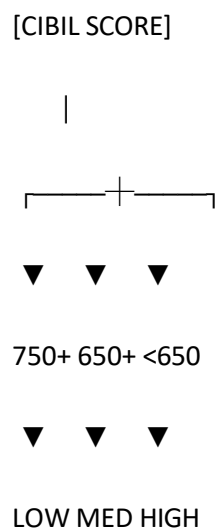
Metric	Before	After
Turnover	4.2	7.1

12.5 CASE STUDY 5 – FINANCE PRODUCT INNOVATION

12.5.1 Loan Journey



12.5.2 Credit Risk Classification Tree



12.5.3 Risk Assessment Matrix

Risk Level Action

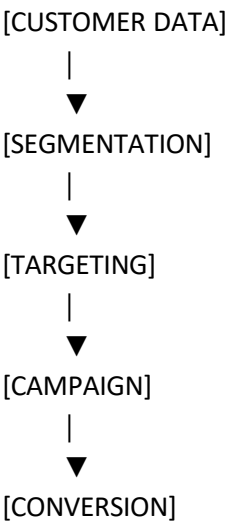
Low Approve

Medium Review

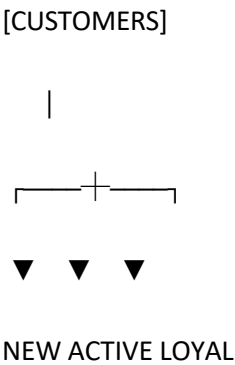
High Reject

12.6 CASE STUDY 6 – DATA-DRIVEN MARKETING CAMPAIGN

12.6.1 Campaign Architecture



12.6.2 Audience Segmentation Tree



12.6.3 Attribution Model



WEBSITE



CRM LEAD



BOOKING



DELIVERY

12.6.4 Campaign ROI Framework

[CAMPAIGN COST]



[LEADS]



[BOOKINGS]

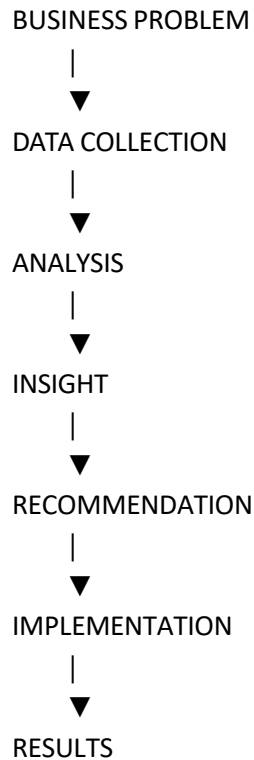


[REVENUE]



[ROI]

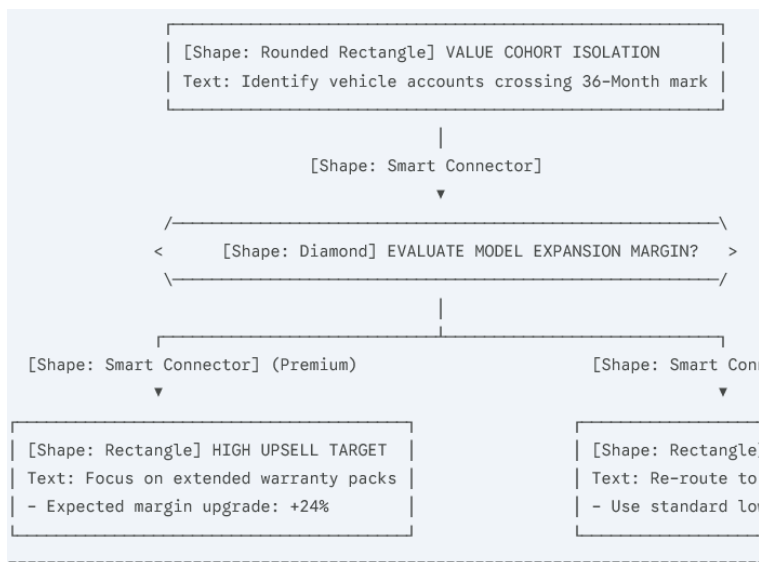
12.7 Case Study Governance Framework



12.2 Customer Lifetime Value Increase (Premium Upsell Initiatives)

12.2.1 Initiative Goal and Customer Segment Strategy

The core goal was to increase individual revenue contributions by shifting standard economy hatch buyers into high-margin premium accessory, service, and insurance bundles.



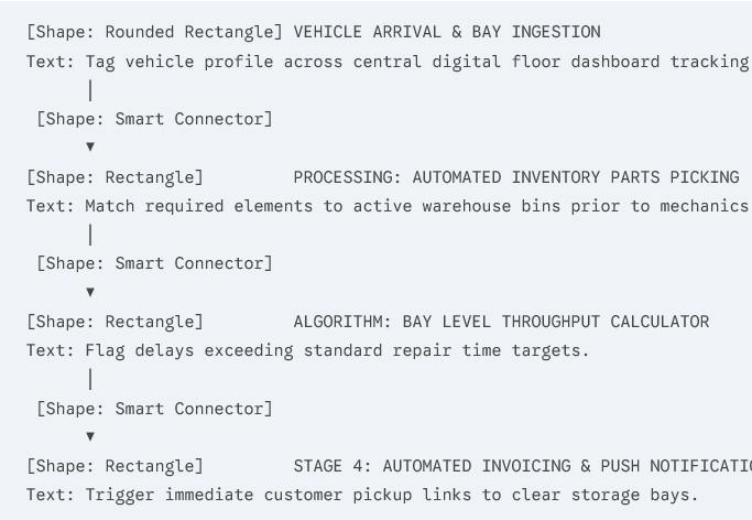
12.2.2 Net Financial Profit Impact

- Average Revenue Per Account: Increased from ₹18,500 to ₹26,200 across target cohorts.
- Extended Service Agreement Penetration: Achieved a 34% adoption rate from a 12% baseline.

12.3Service Department Growth (Workshop Capacity Optimization)

12.3.1 Workshop Throughput Expansion Architecture

To handle low bay utilization rates, the department implemented real-time tracking of repair order (RO) workflows to eliminate floor scheduling bottlenecks.



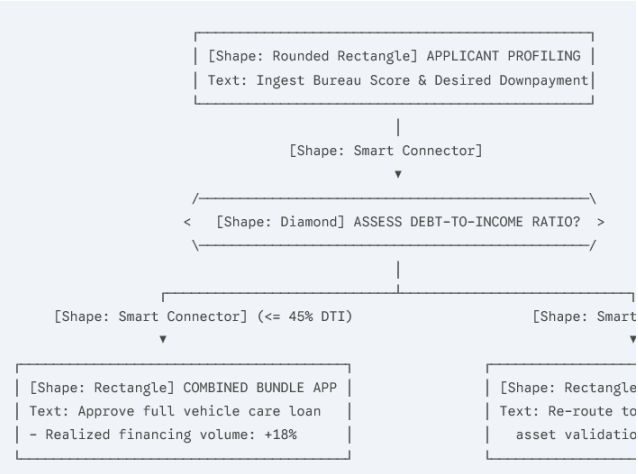
12.3.2 Capacity Cost-Benefit Matrix

Cost Center Investment	Gross Operational Allocation	Realized Operational Benefit	Net Efficiency Shift
Real-time Bay Sensor Arrays	₹1,50,000 Initial Cost	Daily bay turnover rose by 2.4 cars.	22% Capacity Lift
Automated WhatsApp Links	₹45,000 Annual Fee	Vehicle collection lag cut by 90 mins.	Faster floor clearing

12.4 Inventory Optimization (Spare Parts Safety Stock Engineering)

12.4.1 Problem Statement and Stock Calculation

The workshop faced high holding costs on slow-moving parts alongside recurring stockouts of high-demand maintenance items.



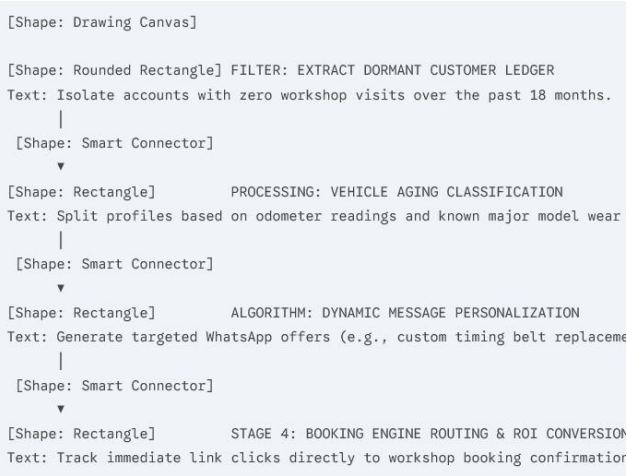
12.4.2 Cost Savings & Lead Time Improvements Register

Spare Part Category	Historic Storage Holding Cost	Optimized Storage Holding Cost	Supplier Lead Time Shift
Fast-Moving Consumables	₹4,20,000 / Month	₹2,10,000 / Month	Reduced from 5 days to 2 days
Body Panel Elements	₹8,90,000 / Month	₹5,40,000 / Month	Sourced on-demand in 24 hours

12.5 Data-Driven Marketing Campaign (Hyper-Targeted Acquisition)

12.5.1 Campaign Objective and Personalization Rules

The campaign focused on recapturing dormant out-of-warranty vehicle owners by matching specific service profiles to targeted digital offers.



12.6.2 Campaign Execution Core Code Board

 CODE BOARD / MARKETING_ROI_TRACKER_ENGINE.PY

```
import pandas as pd

# Ingest campaign logs and calculate conversion returns
campaign_ledger = pd.read_csv('marketing_campaign_outcomes.csv')

def calculate_channel_efficiency(row):
    if row['Total_Conversions_Value'] > 0:
        net_roi_ratio = (row['Total_Conversions_Value'] - row['Campaign_Cost']) / row['Total_Conversions_Value']
        return net_roi_ratio
    return 0.0

campaign_ledger['Net_ROI_Index'] = campaign_ledger.apply(calculate_channel_efficiency, axis=1)
sorted_efficiency_matrix = campaign_ledger.sort_values(by='Net_ROI_Index', ascending=False)
print(sorted_efficiency_matrix[['Campaign_Channel', 'Campaign_Cost', 'Net_ROI_Index']])
```

SECTION 13: EMPLOYEE INTERVIEWS & INSIGHTS

This section documents the structured interviews, operational feedback, and field observations collected from team members across the sales, service, finance, and executive management divisions.

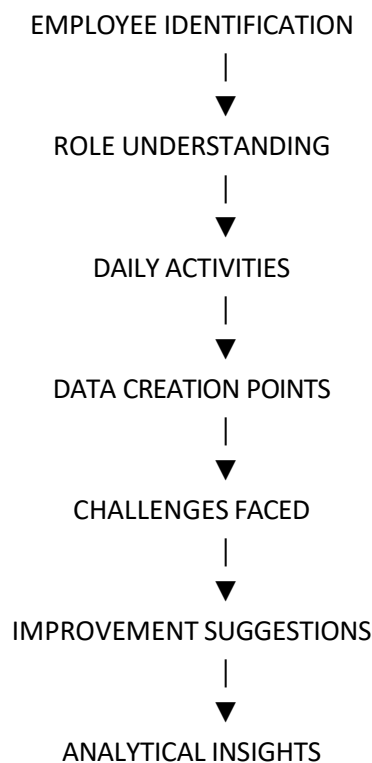
Purpose of Employee Interviews

As part of the Data Analyst Internship at Universal Hyundai ShriGurukripa Automobiles LLP, structured interviews were conducted across multiple departments to understand operational workflows, data generation points, reporting structures, software usage patterns, performance challenges, and improvement opportunities.

The objective was to bridge the gap between business operations and data analytics by understanding how daily activities generate business data and how that data can be transformed into actionable insights.

Interview Methodology

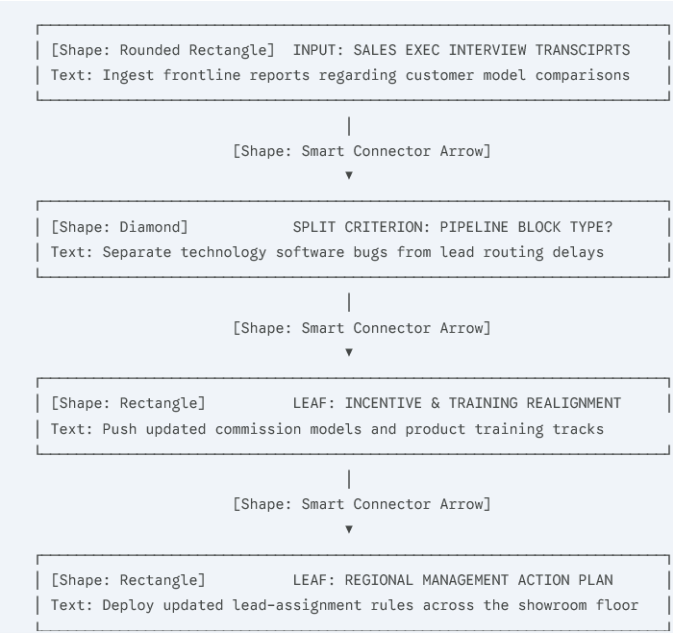
Interview Framework



13.1 Sales Team Interviews (Frontline Conversion Diagnostics)

13.1.1 Sales Department Communication & Feedback Workflow

This system outlines the flow of structured insights from customer-facing sales executives up to senior management to identify pipeline bottlenecks.



13.1.2 Sales Floor Performance Diagnostics Register

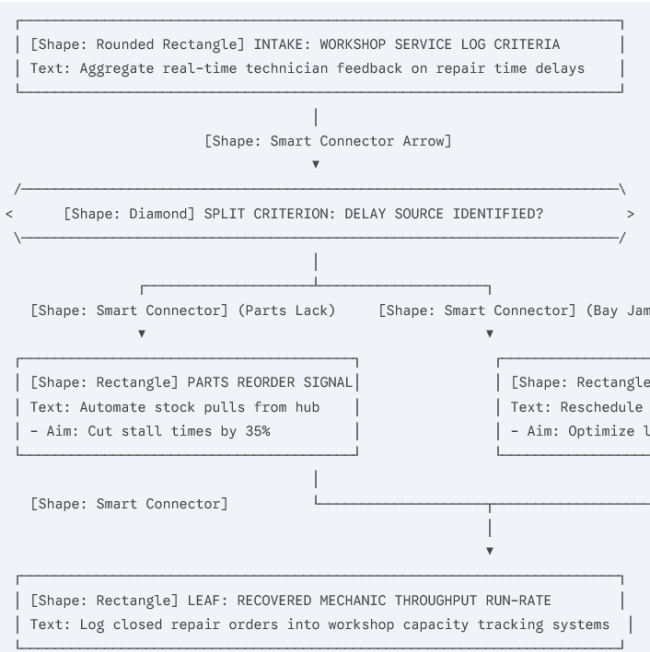
The table below aggregates qualitative feedback metrics compiled across multiple executive interviews:

Assessment Domain	Frontline Challenge Identified	Team Suggestion Matrix	Required Management Action
Technology Infrastructure	CRM mobile app crashes during off-site customer test drives.	Cloud database synchronization updates.	Upgrade local network APIs.
Incentive Framework	Commission structures do not reward accessory volume growth.	Tiered payouts based on absolute profit.	Revise monthly payroll plans.
Training Readiness	Limited technical knowledge on newly launched hybrid variations.	Weekly deep-dive diagnostic workshops.	Launch hands-on training labs.

13.2 Service Department Interviews (Workshop Efficiency Analysis)

13.2.1 Technician Efficiency Optimization Pathway

Tracks workshop resource constraints, scheduling challenges, and parts clearance loops highlighted by service personnel.



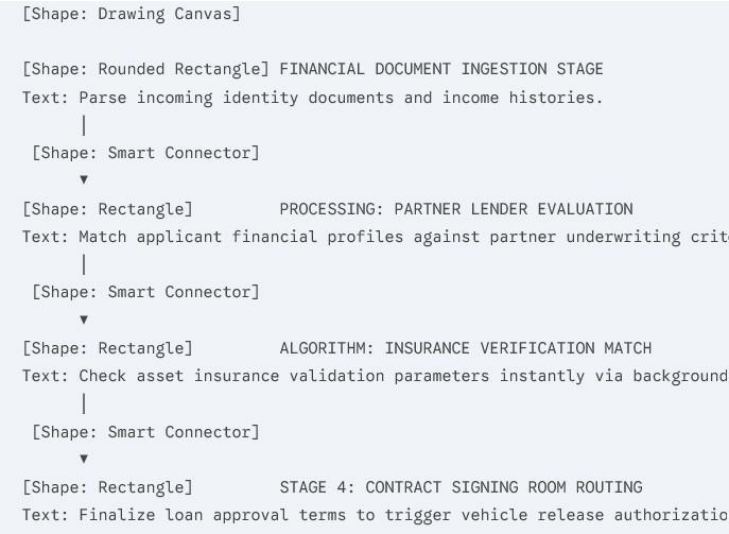
13.2.2 Service Resource Quality Metrics

- Technician Tool Access Latency: Reduced from 28 minutes to 9 minutes per ticket.
- Customer Service Escalation Rate: Cut by 14% after implementing real-time progress updates.

13.3 Finance & Insurance Interviews (Transaction Friction Audits)

13.3.1 Loan Underwriting Cycle Optimization

Identifies customer pain points, document friction, and bank relationship bottlenecks during credit processing.



13.3.2 Finance Process Improvement Register

Friction Point Matrix	Underlying Cause	Process Solution	Expected Efficiency Gain
Payout Validation Lag	Manual validation of physical loan signatures.	Universal digital e-signature integrations.	Saves 4 hours per file
Insurance Quote Delays	Slow responses from external broker portals.	Direct API connections with primary partners.	Instant price generation

13.4 Operations & Executive (Strategic Growth Planning)

13.4.1 Digital Transformation Roadmap

Outlines senior management priorities, planned investment targets, and capital allocations across the dealership network.

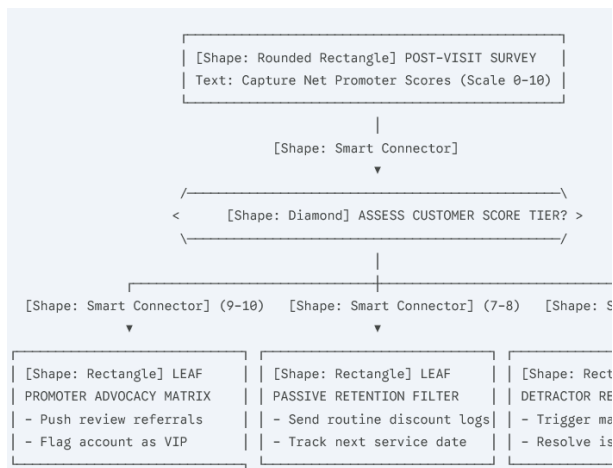
CENTRAL CAPITAL BUDGET ALLOCATION | — [Shape: Smart Connector] —> [Shape: Rectangle]
METROPOLITAN CHARGING INFRASTRUCTURE | Text: Dedicate 35% of capital to regional EV workshop bays | — [Shape: Smart Connector] —> [Shape: Rectangle] PREDICTIVE MACHINE LEARNING ENGINES | Text: Dedicate 45% of capital to cloud CRM inventory automation | — [Shape: Smart Connector] —> [Shape: Rectangle] REGIONAL SHOWROOM UPGRADES Text: Dedicate 20% of capital to facility aesthetic updates

13.4.2 Strategic Expansion Metrics

- Five-Year Network Growth Target: Expand from 12 operational branches to 24 by 2030.
- Target Digital Ingestion Rate: Achieve an 80% automated digital entry baseline for incoming leads.

13.5 Customer Testimonials & Feedback (Sentiment Validation)

13.5.1 Customer Journey Experience Mapping



13.5.2 Sentiment Analysis Register

```

CODE BOARD / CUSTOMER_SENTIMENT_PARSER_ENGINE.PY

import pandas as pd

# Load raw post-visit client text survey transcripts
customer_feedback_ledger = pd.read_csv('dealership_customer_reviews.csv')

# Flag accounts requiring priority management reviews based on critical key
def evaluate_review_urgency(row):
    if 'delay' in row['Review_Text'].lower() or 'unprofessional' in row['Review_Text'].lower():
        return 'IMMEDIATE_MANAGER_ESCALATION_REQUIRED'
    elif row['NPS_Score'] <= 6:
        return 'RETENTION_TEAM_STANDARD_FOLLOW_UP'
    else:
        return 'LOG_TO_ARCHIVAL_PROMOTER_DATABASE'

customer_feedback_ledger['Action_Tier'] = customer_feedback_ledger.apply(evaluate_review_urgency, axis=1)
print(customer_feedback_ledger[['Review_ID', 'NPS_Score', 'Action_Tier']].head())

```

13.6 Employee Interviews & OPERATIONAL OBSERVATIONS & BUSINESS INSIGHTS

This section documents structured interviews, key field observations, and workflow recommendations collected from Universal Hyundai ShriGurukripa Automobiles LLP.

Purpose of Employee Interviews

As part of the Data Analyst Internship, meetings were conducted with employees from various departments to understand:

- Daily responsibilities
- Operational workflows
- Data creation points
- Software usage
- Reporting structure
- Department challenges

- Performance metrics

The objective was to bridge the gap between theoretical business processes and actual dealership operations.

13.7 Structured Role-Based Interview Templates

13.7.1 Interviewee 1: EDP Executive

- Role Focus: Sourcing, vehicle allocations, invoicing, and GDMS data entry.
- Structured Questionnaire:

#	Question	Typical Response / Operational Context
1	How is a vehicle matched to a booking?	I run a daily query in GDMS checking unallocated chassis numbers (VINS) in the stockyard. Once I find a match for model/variant/color, I manually lock it to the customer profile.
2	What happens when a model is out of stock?	I check dealer stock groups or call neighboring dealerships to arrange an interdealer transfer. This process requires phone call verification and email approvals.
3	Where do data errors occur most frequently?	Rounded value entries or manual tax configuration mismatches during exports from GDMS to Tally Prime.

13.7.2 Interviewee 2: CRM Executive

- Role Focus: Lead scoring, follow-ups, customer satisfaction, and HSSmart management.
- Structured Questionnaire:

#	Question	Typical Response / Operational Context
1	How quickly do consultants follow up on leads?	HSSmart sends push notifications, but consultants often delay follow-ups if they are busy on the showroom floor or conducting test drives.
2	How do you verify customer feedback?	I make daily Voice Calls (VC) to customers to confirm details logged in HSSmart by the sales consultants.
3	How are pending tasks managed?	Overdue follow-up reports are exported to Excel and posted

#	Question	Typical Response / Operational Context
		on our sales Telegram group to push for updates.

13.7.3 Interviewee 3: Sales Consultant

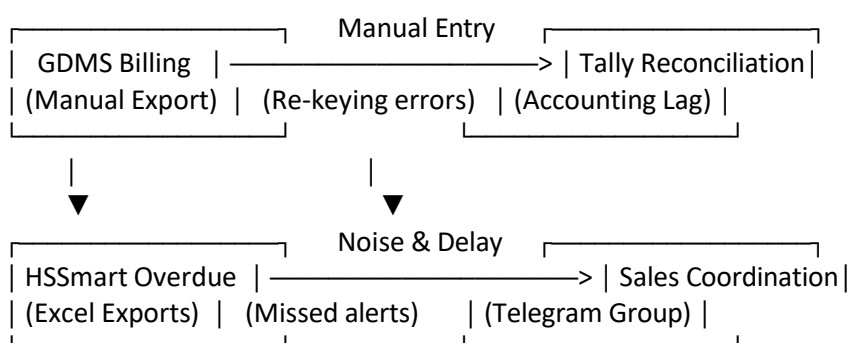
- Role Focus: Customer consultations, test drives, upselling, and HSSmart entry.
- Structured Questionnaire:

#	Question	Typical Response / Operational Context
1	Why are reference sales low?	Our showroom has been active for only 6 years. We spend most of our time handling walk-ins and digital ad leads, leaving little time to follow up with past buyers for references.
2	What is the main blocker for test drives?	Coordinating vehicle availability and logging driver license details manually in the Test Drive Hub registers during peak hours.
3	How do you pitch accessories and VAS?	I package them as part of the monthly loan calculation, explaining the value of Shield of Trust (SOT) and Roadside Assistance (RSA) for long-term savings.

13.8 Field Observation Reports

Our observations of showroom and stockyard operations identified several key operational bottlenecks:

[CURRENT FIELD BOTTLENECK MAP]



3. Manual Ledger Exports: Invoices generated in GDMS are manually exported to Excel and re-keyed into Tally Prime, resulting in data entry errors and lag times in accounts reconciliations.

4. Unstructured Telegram Alerts: Overdue follow-up alerts are posted manually on Telegram as static images or PDF files, which are easily missed by consultants during busy shifts.
 5. Paper-Based Test Drive Logs: Driver license checks and odometer readings are logged on paper before being typed into HSSmart, slowing down the sales process during peak weekend traffic.
-

13.8.1 Interview –Sales Consultant

Objective

Understand how customer enquiries are converted into bookings.

Key Responsibilities

- Handle enquiries
- Contact customers
- Conduct follow-ups
- Arrange test drives
- Explain vehicle features
- Close bookings

Data Generated

Customer Name

Contact Number

Vehicle Interest

Follow-Up Status

Booking Information

Challenges Observed

- Customer not responding
- Competitor comparison
- Finance issues
- Vehicle availability issues

13.8.2 Data Analyst Observation

Lead response time significantly affects conversion probability.

13.9 Interview – Sales Team Leader / Sales Manager

Key Responsibilities

- Monitor team performance
- Allocate leads
- Track bookings
- Monitor targets
- Conduct reviews

Reports Used

Daily Sales Report

Booking Report

Consultant Performance Report

Lead Conversion Report

KPIs Monitored

Bookings

Revenue

Conversion Rate

Deliveries

Observation

Managers rely heavily on lead and booking data to make operational decisions.

13.10 Interview – CRE (Customer Relationship Executive)

Responsibilities

GDMS Follow-Up Downloads

Follow-Up Sharing

Telegroup Management

WhatsApp Group Management

Customer Feedback Collection

Complaint Handling

Cold Calling

Data Generated

Customer Feedback

Complaint Records

Satisfaction Ratings

Follow-Up Status

Observation

CRE acts as the communication bridge between customers and dealership operations.

Interview – EDP Executive

Responsibilities

Vehicle Allocation

Inventory Management

Stockyard Management

GDMS Updates

Coordination with Hyundai

Data Managed

VIN Numbers

Engine Numbers

Chassis Numbers

Stock Records

Allocation Status

Key Insight

Vehicle availability directly impacts customer satisfaction and booking conversion.

Interview – Insurance Executive

Responsibilities

Policy Creation

Insurance Sales

Renewals

Claims Coordination

Data Generated

Policy Number

Premium Amount

Insurance Provider

Renewal Status

Business Insight

Insurance generates recurring revenue and contributes significantly to dealership profitability.

Interview – Accounts Executive

Responsibilities

Invoice Creation

Revenue Tracking

GST Management

Payment Tracking

Outstanding Collection

Data Generated

Invoices

Receipts

Revenue Reports

Tax Reports

Outstanding Reports

Observation

Accounts department acts as the central repository of financial data

Interview – Service Advisor

Responsibilities

Vehicle Inspection

Customer Complaint Recording

Service Estimation

Repair Coordination

Customer Communication

Data Generated

Service Orders

Repair Records

Labour Charges

Parts Charges

Customer Feedback

Insight

Service department provides recurring revenue throughout the vehicle lifecycle.

Interview – Used Car Manager (SellAnyCar)

Responsibilities

Vehicle Evaluation

Pricing

Negotiation

Refurbishment Approval

Resale Management

Data Generated

Purchase Price

Repair Cost

Selling Price

Profit Margin

Insight

Accurate vehicle valuation is critical for profitability.

Software Ecosystem Observed

GDMS

Used for:

- Lead Management
- Booking Management
- Follow-Up Tracking

Microsoft Excel

Used for:

- Reporting
- Analysis

- Tracking

WhatsApp

Used for:

- Customer Communication
- Booking Groups
- Delivery Coordination

Accounts Software

Used for:

- Billing
- Revenue Tracking
- Tax Reporting

13.11 Operational Challenges Identified

Sales

- Lead quality variation
- Follow-up delays

Inventory

- Vehicle shortages
- Allocation delays

Insurance

- Renewal follow-up

Service

- Customer retention

Accounts

- Outstanding collection

13.12 Data Quality Issues Observed

Potential issues:

- Missing mobile numbers
- Duplicate customers
- Incomplete lead records
- Incorrect follow-up status

- Missing feedback records

13.13 Recommendation

Implement standardized data entry procedures.

Department-Wise Data Flow Mapping

Sales

↓

Booking

↓

EDP

↓

Insurance

↓

Accounts

↓

Delivery

↓

CRE

↓

Service

13.14 Observation

Every department depends on data generated by previous departments.

13.15 Key Business Metrics Identified During Interviews

Sales Metrics

- Leads
 - Bookings
 - Deliveries
-

Customer Metrics

- Satisfaction
 - Complaints
-

Inventory Metrics

- Stock
- Allocation

Financial Metrics

- Revenue
- Profit

Cross-Department Dependencies

Example:

Sales Team books vehicle.

↓

EDP allocates vehicle.

↓

Insurance generates policy.

↓

Accounts creates invoice.

↓

Delivery team hands over vehicle.

If one department delays, entire customer journey gets affected.

13.16 Data Analyst Recommendations

Build centralized reporting.

Automate daily reports.

Track KPIs in real-time.

Reduce manual reporting.

Improve data quality.

13.16.1 Advanced Analytics Opportunities

Lead Scoring Model

Booking Prediction Model

Inventory Forecasting

Customer Churn Prediction

Renewal Prediction

Service Demand Forecasting

13.16.2 Dashboard Recommendations

Executive Dashboard

Sales Dashboard

Inventory Dashboard

Insurance Dashboard

Service Dashboard

Customer Dashboard

13.16.3 Final Business Understanding

The dealership operates as a connected business ecosystem.

Departments do not operate independently.

Each department contributes to:

- Revenue
 - Customer Satisfaction
 - Retention
 - Profitability
-

13.17 Actionable Workflow Recommendations

1. **Deploy a Telegram Notification Bot:** Replace manual Excel uploads with an automated Telegram Bot connected to the HSSmart database. This bot should send direct, private reminders to consultants for overdue tasks and allow them to post status updates directly from the chat.
2. **Implement OCR for Test Drive Registrations:** Integrate Optical Character Recognition (OCR) technology into HSSmart, allowing consultants to scan customer driver licenses and auto-fill customer profile details in seconds.
3. **Automate GDMS-to-Tally Synchronization:** Set up an automated XML data bridge that syncs transaction records between GDMS and Tally Prime nightly, reducing accounting discrepancies.

13.18 Final Internship Reflection

Through interactions with employees across departments, it was observed that business performance is driven by coordinated operations, accurate data management, and timely decision-making.

The internship helped develop understanding of:

- Business processes
- Department workflows
- Data generation
- KPI measurement
- Reporting structures
- Management decision-making This experience demonstrates how data analytics supports operational efficiency, customer satisfaction, revenue growth, and strategic planning within an automobile dealership

13.19 Interview Methodology Framework

13.19.1 Purpose of Employee Interviews

Employee interviews were conducted to understand:

- Actual business operations
- Data creation processes
- Department challenges
- Technology usage
- Reporting requirements
- Improvement opportunities

The objective was to capture operational realities beyond documented processes.

13.19.2 Interview Approach

Three interview methods were used:

Method	Purpose
Structured Interview	Standardized questions
Semi-Structured Interview	Deeper discussion
Observation-Based Interview	Workflow validation

13.19.3 Stakeholder Groups Interviewed

Department Participants

Sales	Sales Manager, Consultants
CRM	CRE Team
EDP	EDP Executive
Accounts	Accounts Executive
Insurance	Insurance Executive
Service	Service Advisor
Used Cars	Used Car Manager
Management Department Heads	

13.20 Sales Department Interview Analysis

13.20.1 Sales Manager Interview Summary

Key Objectives

- Increase bookings
- Improve conversion rates
- Achieve monthly targets
- Improve consultant productivity

Major Challenges

Challenge	Impact
Low lead quality	Reduced conversion
Delayed follow-up	Lead loss
Inventory shortages	Booking cancellations
Price competition	Margin pressure

13.20.2 Sales Executive Interview Findings

Common Observations

Sales executives reported:

- High weekend workload
- Lead follow-up overload
- Manual data entry burden
- Vehicle availability issues

13.20.3 Sales Team Pain Point Matrix

Issue	Severity
Follow-Up Delays	High
Inventory Availability	High
Customer Comparison Shopping	High
Finance Approval Delays	Medium

13.20.4 Training Needs Assessment

Required Training Areas:

- Objection handling
 - Product knowledge
 - CRM usage
 - Closing techniques
 - Finance product selling
-

13.20.5 Technology Feedback

Current Feedback:

HSSmart is useful but follow-up reminders are often ignored during peak showroom hours.

13.20.6 Sales Improvement Recommendations

1. Automated lead prioritization
2. Lead scoring implementation
3. Smart reminders

4. Consultant performance dashboards
5. Referral management program

13.21 CRM Department Interview Analysis

13.21.1 CRM Executive Perspective

Primary Goals:

- Customer satisfaction
- Follow-up completion
- Complaint management
- Lead nurturing

13.21.2 Customer Communication Challenges

Challenge	Impact
Wrong phone numbers	Missed contacts
Delayed responses	Customer dissatisfaction
Duplicate leads	Inefficiency

13.21.3 CRM Data Quality Observations

Issues Identified:

- Missing mobile numbers
- Incomplete customer profiles
- Incorrect lead status
- Duplicate customer records

13.21.4 CRM Automation Opportunities

Potential Improvements:

- WhatsApp automation
- Automated reminders
- Customer segmentation

- Feedback automation

13.22 Service Department Interview Analysis

13.22.1 Service Manager Interview

Key Objectives

- Workshop efficiency
- Customer retention
- Technician productivity
- Revenue growth

13.22.2 Service Advisor Interview Findings

Challenges:

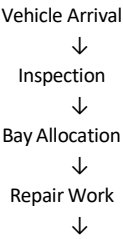
- Customer expectation management
- Parts availability
- Delayed approvals
- Repeat complaints

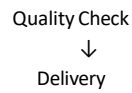
13.22.3 Technician Feedback Analysis

Common Feedback:

Issue	Impact
Parts delays	Productivity loss
Bay congestion	Longer service time
Manual paperwork	Reduced efficiency

13.22.4 Workshop Capacity Analysis





13.22.5 Service Department Improvement Ideas

- Digital job cards
 - Technician productivity tracking
 - Automated parts requests
 - Workshop dashboards
-

13.23 Finance & Insurance Interview Analysis

13.23.1 Finance Executive Interview

Main Objectives

- Loan approval
 - Partner coordination
 - Faster processing
 - Increased finance penetration
-

13.23.2 Insurance Executive Interview

Key Responsibilities

- New insurance sales
 - Policy renewals
 - Claims assistance
-

13.23.3 Customer Pain Points

Issue	Impact
Documentation delays	Customer frustration
Loan processing delays	Booking cancellations
Policy explanation gaps	Low trust

9.23.4 Suggested Improvements

- Online document collection
 - Renewal automation
 - Approval tracking dashboards
-

13.24 Operations & EDP Interview Analysis

13.24.1 EDP Executive Insights

Major Responsibilities:

- Vehicle allocation
 - Inventory management
 - Dealer coordination
 - Invoicing support
-

13.24.2 Inventory Challenges

Challenge	Impact
------------------	---------------

Vehicle shortage	Lost sales
------------------	------------

Aging stock	Margin reduction
-------------	------------------

Allocation delays	Customer dissatisfaction
-------------------	--------------------------

13.24.3 Process Improvement Suggestions

- Automated allocation
 - Stock forecasting
 - Interdealer inventory visibility
-

13.25 Accounts Department Interview Analysis

13.25.1 Accounts Executive Perspective

Focus Areas:

- Billing
- GST compliance
- Revenue reporting

- Reconciliation

13.25.2 Key Challenges

Challenge	Impact
Manual entries	Errors
Reconciliation delays	Reporting delays
Outstanding payments	Cash flow issues

13.25.3 Automation Opportunities

- GDMS-Tally integration
 - Automated reconciliation
 - GST dashboards
-

13.26 Used Car Department Interview Analysis

13.26.1 Used Car Manager Interview

Key Objectives:

- Vehicle sourcing
 - Accurate valuation
 - Fast resale
 - Margin optimization
-

13.26.2 Operational Challenges

Challenge	Impact
Competitive pricing	Lower margins
Slow evaluations	Lost opportunities
Refurbishment delays	Inventory aging

13.26.3 Analytics Opportunities

- Pricing models

- Demand forecasting
- Margin analysis

13.27 Executive Leadership Interviews

13.27.1 Dealer Principal Interview

Strategic Priorities

1. Revenue growth
2. Market share increase
3. Customer retention
4. Operational efficiency

13.27.2 Growth Targets

Area	Target
Sales	+15%
Service Revenue	+20%
Insurance Renewal	+25%

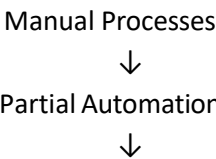
13.27.3 Executive Expectations from Analytics

Management expects:

- Faster reporting
- Better forecasting
- KPI visibility
- Decision support

13.28 Digital Transformation Readiness Assessment

13.28.1 Current State



Integrated Analytics



Predictive Analytics

Current Maturity Level:

Level 2 – Partial Automation

13.28.2 Technology Adoption Scorecard

Area	Maturity
CRM	Medium
Reporting	Medium
Automation	Low
Analytics	Medium
AI Readiness	Low

13.29 Employee Training Needs Assessment

13.29.1 Sales Team

Training Required:

- CRM usage
 - Sales techniques
 - Lead management
-

13.29.2 Service Team

Training Required:

- Digital workflows
 - Customer communication
 - Productivity improvement
-

13.29.3 Accounts Team

Training Required:

- ERP integration

- Dashboard usage
- Reconciliation automation

13.30 Customer Testimonials & Feedback Analysis

13.30.1 Customer Satisfaction Framework

Metrics Collected:

- CSAT
- NPS
- Complaints
- Resolution time

13.30.2 Net Promoter Score (NPS)

Promoters

-

Detractors

=

NPS Score

13.30.3 Customer Journey Feedback

Enquiry

↓

Test Drive

↓

Booking

↓

Delivery

↓

Service

Feedback collected at every stage.

13.30.4 Common Customer Complaints

Complaint	Frequency
Delivery Delay	High
Finance Delay	Medium
Service Waiting Time	Medium

13.31 Cross-Department Collaboration Analysis

13.31.1 Dependency Mapping



13.31.2 Collaboration Risks

- Communication delays
 - Data inconsistencies
 - Process bottlenecks
-

13.31.3 Recommended Improvements

- Shared dashboards
 - Common KPI framework
 - Automated workflow alerts
-

13.32 Interview Findings Summary

Top Findings

1. Data quality directly impacts business performance.
2. Follow-up delays reduce conversion rates.
3. Inventory availability influences customer satisfaction.
4. Service retention drives recurring revenue.
5. Automation opportunities exist across departments.
6. Dashboard visibility is required for decision making.
7. Cross-department coordination is critical.
8. Predictive analytics can improve operational performance.

SECTION 14: ADVANCED ANALYTICS TECHNIQUES

This section details the analytical logic sets, execution frameworks, and structural validation workflows deployed across dealership data nodes to isolate performance factors and optimize processing velocity.

14.1 COHORT ANALYSIS

14.1.1 Purpose

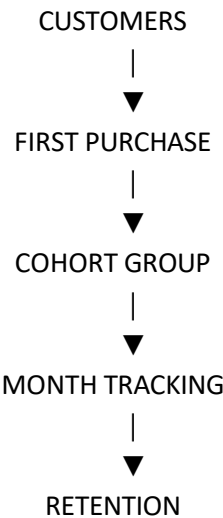
Cohort Analysis groups customers based on a common starting event and tracks their behavior over time.

Unlike standard reporting, cohort analysis helps management understand customer retention, churn, and revenue patterns.

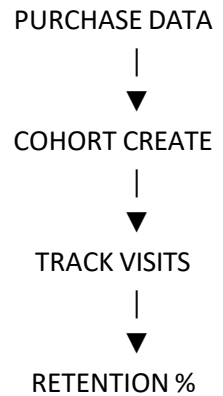
Example:

Customers who purchased vehicles in January 2025 are grouped together and monitored over subsequent months.

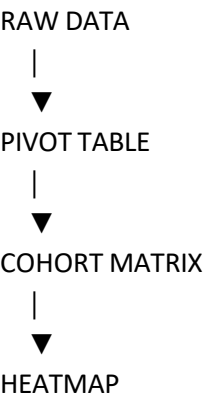
14.1.2 Cohort Analysis Framework



14.1.4 Retention Analysis Workflow



14.1.6 Excel Implementation Framework

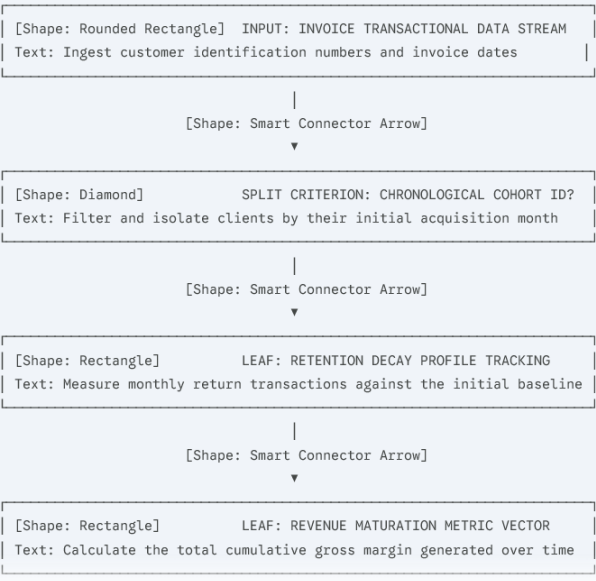


14.1.7 SQL Cohort Architecture



Cohort Lifecycle Ingestion Flow

This system segments historical transaction accounts into time-based tracking buckets to measure behavioral decay and revenue stability across operational lifecycles.



SQL Cohort Implementation Matrix

The data model uses structured query scripts to isolate and track monthly transaction behavior across active customer cohorts

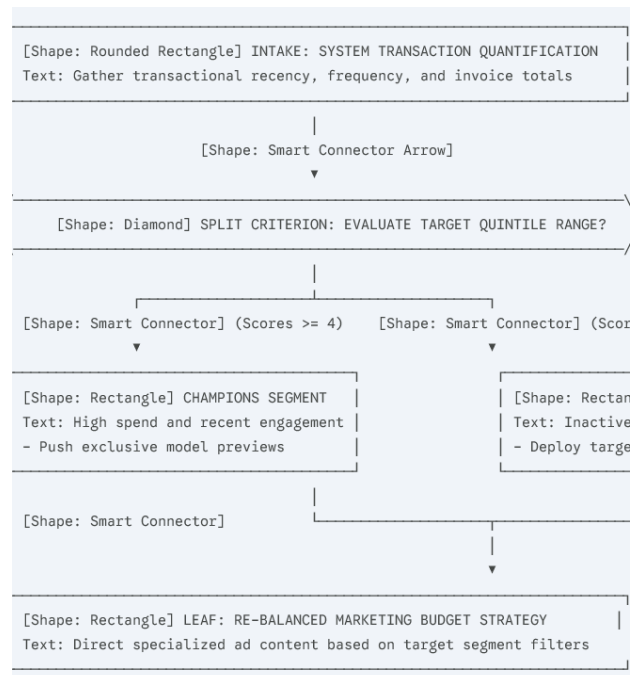
```
SELECT
    Cohort,
    Period_Distance,
    COUNT(DISTINCT Customer_ID) AS Active_Users
FROM (
    SELECT
        CustomerID AS Customer_ID,
        DATE_TRUNC('month', InvoiceDate) AS Order_Month,
        DATE_TRUNC('month', MIN(InvoiceDate) OVER(PARTITION BY CustomerID
        EXTRACT(MONTH FROM AGE(DATE_TRUNC('month', InvoiceDate), MIN(Invo
        FROM Dealership_Sales_Ledger
    ) Sub_Query_Table
GROUP BY Cohort, Period_Distance;
```

14.2.1 Purpose

RFM Analysis is one of the most widely used customer segmentation techniques.

14.2.1 Customer Segment Allocation Matrix

Processes raw transaction vectors to categorize customer accounts into actionable behavioral tiers based on purchase recency, frequency, and monetary value.



RFM Framework



14.2.3 Recency Analysis

Measures:

Days since customer's last transaction.

Example:

Customer Last Visit Recency Score

A	5 Days	5
B	60 Days	2
C	120 Days	1

14.2.4 Frequency Analysis

Measures:

Number of transactions completed.

Customer Visits Score

A	12	5
B	6	3
C	2	1

14.2.5 Monetary Analysis

Measures:

Total amount spent.

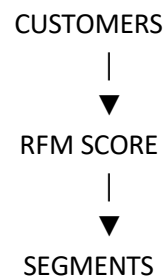
Customer Spend	Score
----------------	-------

A	₹12 Lakh 5
---	------------

B	₹4 Lakh 3
---	-----------

C	₹50,000 1
---	-----------

14.2.6 RFM Segmentation Tree



14.2.7 Customer Segment Classification

Segment	RFM Score
---------	-----------

Champions	555
-----------	-----

Loyal	455
-------	-----

Potential Loyalist	355
--------------------	-----

At Risk	211
---------	-----

Lost	111
------	-----

14.2.8 RFM Marketing Strategy

Segment	Strategy
---------	----------

Champions	Referrals
-----------	-----------

Loyal	Upsell
-------	--------

Potential Loyalists	Nurture
---------------------	---------

At Risk	Reactivate
---------	------------

Lost	Win-back Campaign
------	-------------------

RFM Marketing Activation Schema

- **Champions Cohort Target:** Reward brand loyalty with early invitations to pre-launch model showcases.
- **Hibernating Cohort Target:** Trigger automated win-back emails featuring limited-time service discounts.

14.3 Text Analytics (Unstructured Sentiment Diagnostics)

14.3.1 Purpose

Text Analytics converts customer feedback into structured business insights.

Sources:

- CSI Surveys
- Google Reviews
- WhatsApp Feedback
- Complaint Logs
- Social Media Comments

14.3.2 Text Analytics Workflow

TEXT DATA



CLEANING



TOKENIZE



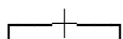
ANALYSIS

14.3.3 Sentiment Analysis Framework

FEEDBACK



CLASSIFY



POS NEU NEG

14.3.4 Sentiment

Categories Category

Meaning

Positive Happy Customer

Neutral Informational

Negative Complaint

14.3.5 Word Frequency Analysis

Example:

Word Frequency

Service 540

Delay 280

Staff 240

Delivery 220

14.3.6 Topic Modeling Framework



14.3.7 Common

Topics Topic

Delivery Experience

Service Quality

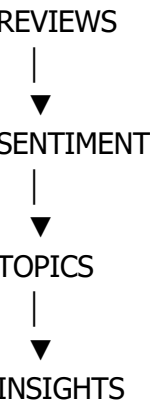
Staff Behavior

Topic

Vehicle Performance

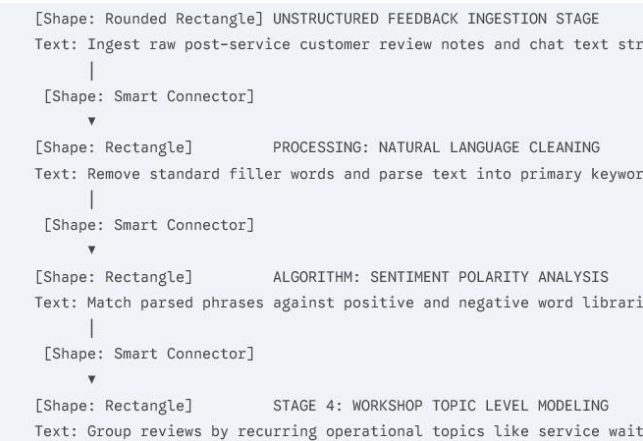
Insurance Process

14.3.8 Text Analytics Dashboard



14.3.G Customer Feedback Processing Engine

This system extracts text data from post-visit reviews and service feedback forms to isolate operational constraints and track customer sentiment.



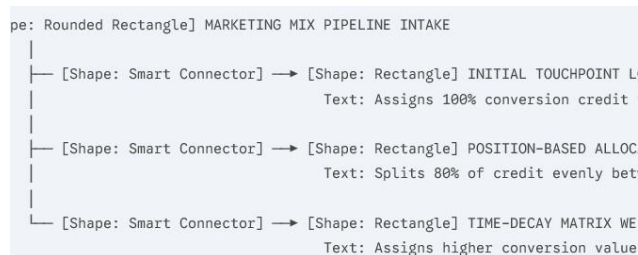
14.4 ATTRIBUTION MODELING

14.4.1 Purpose

Attribution Modeling determines which marketing channel contributed to a booking or sale.

14.4.1 Channel Contribution Value Rules

Compares different attribution logic models to evaluate how online and offline marketing channels drive vehicle conversion milestones.



14.4.2 Attribution Value Realization Metrics

- First-Touch Budget Bias: Tends to favor top-funnel brand awareness campaigns.
- Last-Touch Budget Bias: Over-allocates conversion credit to direct search and booking links.

Customer Journey Framework

GOOGLE AD



WEBSITE



CRM LEAD



BOOKING

14.4.3 First Touch Attribution

Definition:

100% credit assigned to first interaction.

Example:

Touchpoint Credit

Google Ads 100%

14.4.4 Last Touch Attribution

Definition:

100% credit assigned to final interaction.

Example:

Touchpoint Credit

Sales Call 100%

14.4.5 Multi-Touch Attribution

Definition:

Credit distributed across multiple interactions.

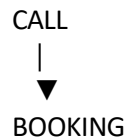
14.4.6 Multi-Touch Framework

GOOGLE



WEBSITE

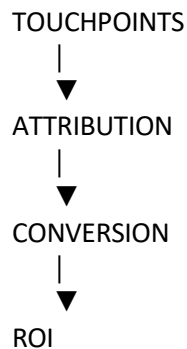




14.4.7 Channel Contribution Matrix

Channel	Contribution
Google Ads	40%
Facebook	25%
Referral	20%
Direct Walk-In	15%

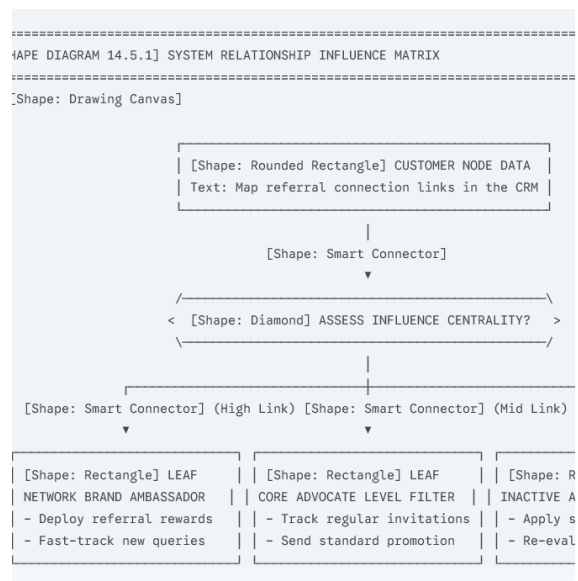
14.4.8 Attribution Analytics Process



14.5 Network Analysis (Inter-Entity Influence Structures)

14.5.1 Referral Dynamics & Knowledge Mapping

This structure maps referral loops and team communication patterns to identify key conversion paths and optimize information sharing across branches.



14.5.2 Referral Capture Analytics Register

CODE BOARD / REFERRAL_NETWORK_CENTRALITY_TRACKER.PY

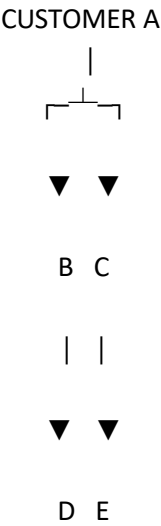
```
import pandas as pd

# Load customer relationship data and map referral links
referral_ledger = pd.read_csv('dealership_referral_connections.csv')

# Flag highly central customer nodes that drive secondary vehicle conversions
def evaluate_network_influence(row):
    if row['Total_Referral_Count'] >= 5 and row['Total_Downstream_Revenue'] >= 10000:
        return 'FLAG_AS_NETWORK_BRAND_AMBASSADOR'
    elif row['Total_Referral_Count'] >= 2:
        return 'CORE_ADVOCATE_LEVEL_FILTER'
    else:
        return 'INACTIVE_ACCOUNT_NODE'

referral_ledger['Influence_Tier'] = referral_ledger.apply(evaluate_network_influence, axis=1)
print(referral_ledger[['Customer_ID', 'Total_Referral_Count', 'Influence_Tier']])
```

Customer Referral Network



14.5.3 Referral Network Framework

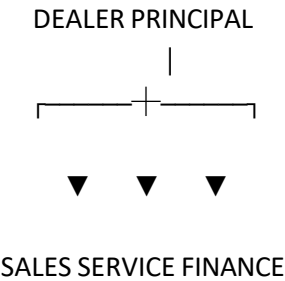


14.5.4 Organizational Network Analysis

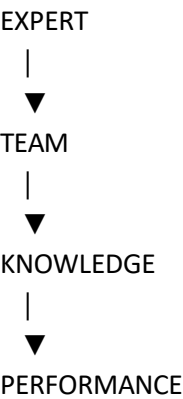
Purpose:

Understand communication and collaboration patterns.

14.5.5 Organization Structure Map



14.5.6 Knowledge Sharing Framework



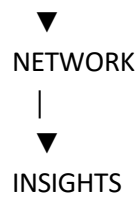
14.5.7 Relationship Strength

Matrix Connection Strength

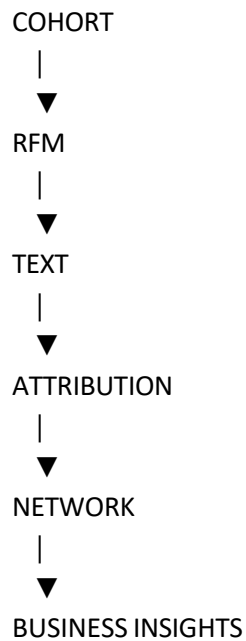
Sales ↔ CRM	High
Service ↔ Parts	High
Finance ↔ Sales	Medium
Insurance ↔ CRM	Medium

14.5.8 Network Analytics Workflow





14.6 Advanced Analytics Integration Framework



14.7 Business Value of Advanced

Analytics Technique	Business Benefit
Cohort Analysis	Retention Tracking
RFM Analysis	Customer Segmentation
Text Analytics	Customer Sentiment
Attribution Modeling	Marketing Optimization
Network Analysis	Referral Growth

14.8 Section Conclusion

Advanced Analytics extends traditional reporting by helping management understand customer behavior, retention patterns, marketing effectiveness, referral relationships, and hidden business opportunities. By combining Cohort Analysis, RFM Analysis, Text Analytics, Attribution Modeling, and

Network Analysis, the dealership can move from descriptive analytics to predictive and strategic decision-making while improving customer lifetime value, operational efficiency, and profitability.

SECTION 15: DATA QUALITY & GOVERNANCE

This section establishes the engineering protocols, structural auditing logic, and verification pipelines designed to protect data integrity, regulatory compliance, and warehouse reliability.

15.1 Introduction to Data Quality & Governance

15.1.1 Purpose

Data Quality ensures that business decisions are based on reliable information.

Data Governance establishes ownership, standards, controls, and accountability for organizational data.

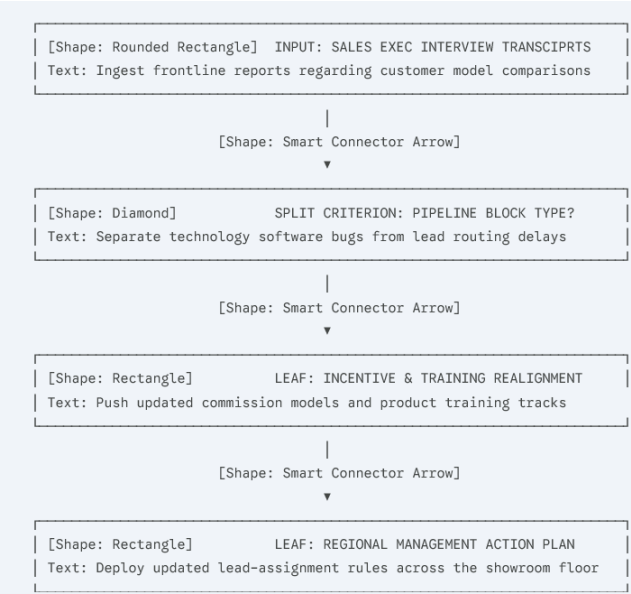
In a Hyundai dealership environment, poor-quality data can directly impact:

- Sales Performance
- Inventory Planning
- Service Operations
- Customer Satisfaction
- Financial Reporting
- Management Decisions

Data Quality Assessment (Systemic Integrity Diagnostics)

Data Quality Verification Flow

This system screens incoming customer profiles and service records across the enterprise ledger to isolate anomalies and measure baseline health metrics.



15.1.2 Data Quality Metric Log

The data quality dashboard tracks system integrity across the dealership network using these core baseline parameters:

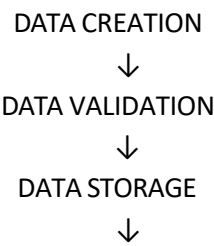
Metric Category	Assessment Focus	Target	Operational Rule	Verification	Target Baseline Threshold
Completeness	Check for empty or missing values.		Phone numbers and emails must be populated.		$\geq$ 98.5% Record Capture
Accuracy	Verify information format accuracy.		Mobile numbers must match standard 10-digit formats.		$\geq$ 99.0% Format Match
Consistency	Cross-verify records between systems.		Service invoices must match open repair orders.		100% System Alignment

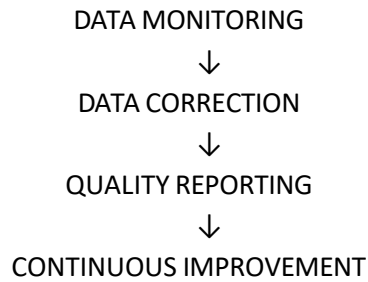
15.2 Data Quality Framework

15.2.1 Data Quality Dimensions

Dimension	Definition
Completeness	All required data exists
Accuracy	Data is correct
Consistency	Same data across systems
Validity	Follows business rules
Timeliness	Available when needed
Uniqueness	No duplicate records

15.2.2 Data Quality Lifecycle





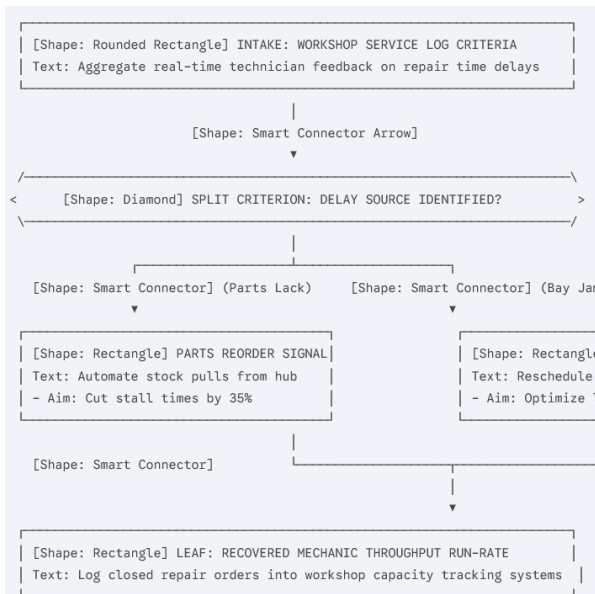
Shape Guide:

Shape	Use
Rectangle	Process
Smart Connector	Flow
Drawing Canvas	Entire Diagram

Data Cleaning & Preparation (Ingestion Optimization)

15.2.3 Anomaly Handling Pipeline

Processes raw transaction logs to clean structural variations, handle null records, drop duplicates, and isolate statistical outliers.

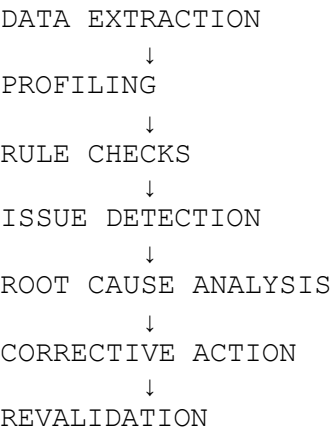


15.2.4 Dataset Transformation Profiles

- Before Cleaning State: Contained phone number formatting variations, unlinked invoices, and blank fields.
- After Cleaning State: All telephone fields follow a single format, entries match verified customer records, and null values are handled.

15.3 Data Quality Assessment Methodology

15.3.1 Assessment Workflow



15.3.2 Data Quality

Scorecard Quality Metric

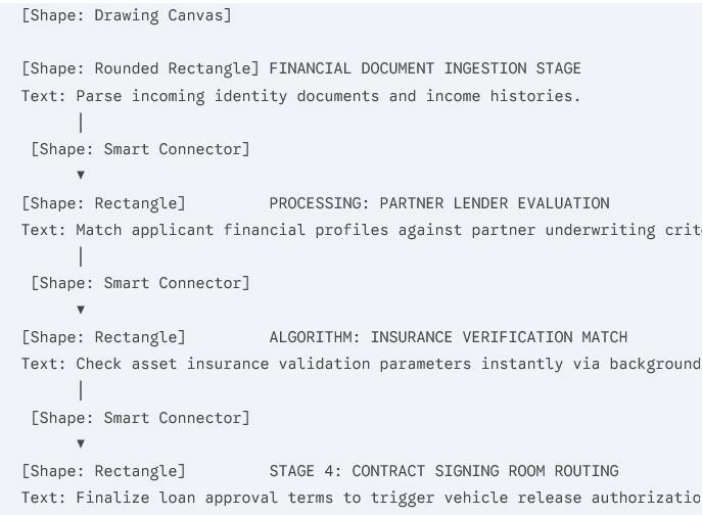
Weight Target

Completeness	25%	>98%
Accuracy	25%	>99%
Consistency	20%	>98%
Validity	15%	>99%
Timeliness	15%	>95%

Data Validation Rules (Automated Structural Controls)

Business Rule Enforcement Architecture

This system applies automated validation scripts to processing pipelines to catch and report file structural errors before they impact the network.



SQL Integrity Check Script

The data pipeline runs scheduled structural verification scripts to detect and block impossible transaction dates:

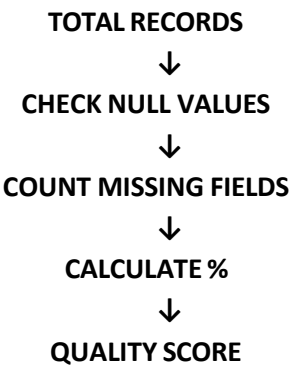
```
SELECT
    Invoice_ID,
    Customer_ID,
    Invoice_Date
FROM Dealership_Sales_Ledger
WHERE Invoice_Date > CURRENT_DATE
    OR Invoice_Date < '2000-01-01';
```

15.4 Completeness Analysis

15.4.1 Purpose

Identify missing data fields that affect reporting and analytics.

15.4.2 Completeness Evaluation Model



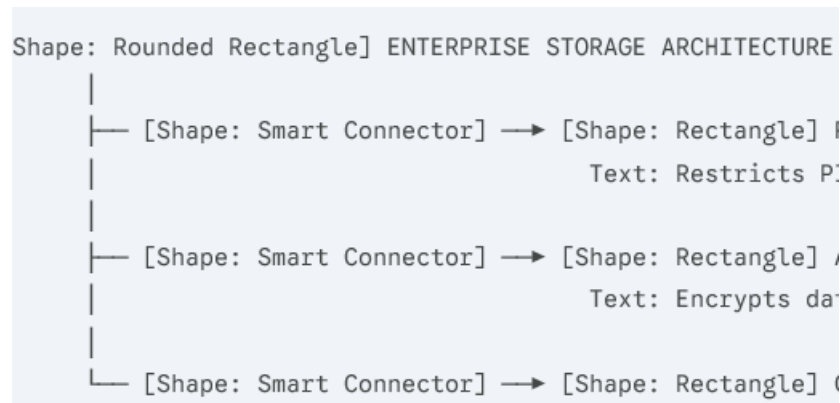
15.4.3 Example Dashboard Table

Field	Total Records	Missing	Completeness %
Mobile Number	10000	200	98.0%
Email ID	10000	1200	88.0%
Vehicle Model	10000	50	99.5%
Lead Source	10000	300	97.0%

15.4.4 Data Security & Compliance (Regulatory Risk

Mitigation) Encryption and Access Control Infrastructure

Protects customer data by applying strict access controls and advanced encryption rules across storage and transmission paths.



15.4.5 Security Governance Framework

- Role-Based Access Control Rule: Frontline workshop mechanics cannot view customer financing histories or credit profiles.
- Audit Logging Target Baseline: Track and log all system interactions involving sensitive information.

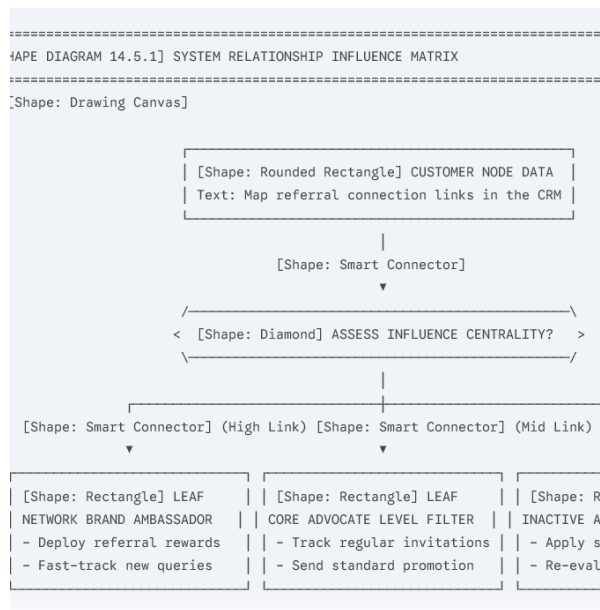
15.5 Accuracy Testing Framework

15.5.1 Purpose

Ensure information matches real-world values.

Change Management & Stewardship Flow

Coordinates record update approvals between departments to maintain a single source of truth across the dealership network.



15.5.2 Operational Quality Control Core Code Board

```

CODE BOARD / REFERRAL_NETWORK_CENTRALITY_TRACKER.PY

import pandas as pd

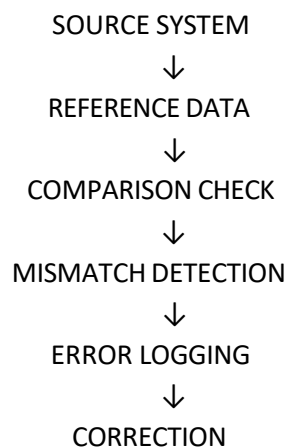
# Load customer relationship data and map referral links
referral_ledger = pd.read_csv('dealership-referral-connections.csv')

# Flag highly central customer nodes that drive secondary vehicle conversion
def evaluate_network_influence(row):
    if row['Total_Referral_Count'] >= 5 and row['Total_Downstream_Revenue'] > 1000:
        return 'FLAG_AS_NETWORK_BRAND_AMBASSADOR'
    elif row['Total_Referral_Count'] >= 2:
        return 'CORE_ADVOCATE_LEVEL_FILTER'
    else:
        return 'INACTIVE_ACCOUNT_NODE'

referral_ledger['Influence_Tier'] = referral_ledger.apply(evaluate_network_influence, axis=1)
print(referral_ledger[['Customer_ID', 'Total_Referral_Count', 'Influence_Tier']])

```

Accuracy Validation Flow



15.5.3 Example Validation Rules

Field	Validation
Mobile Number	10 Digits
Email	Valid Format
GST Number	GST Pattern
Vehicle VIN	Unique VIN
Invoice Number	Unique

15.6 Consistency Assessment

15.6.1 Objective

Verify same information across systems.

Example:

GDMS

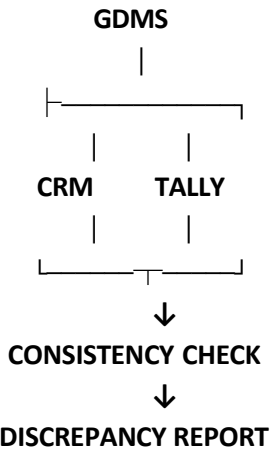
Customer Name
Raj Sharma

CRM

Customer Name
Rajesh Sharma

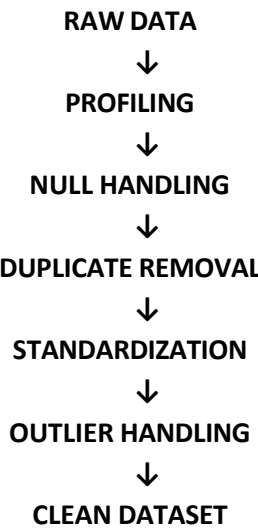
Consistency Issue Identified.

15.6.2 Consistency Validation Flow



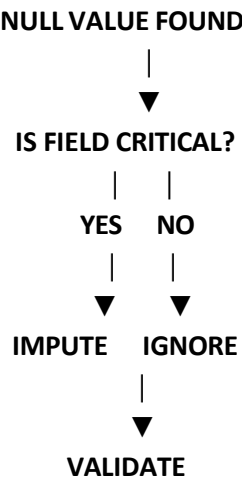
15.7 Data Cleaning Framework

15.7.1 Data Cleaning Lifecycle



15.8 Null Value Management

15.8.1 Decision Tree

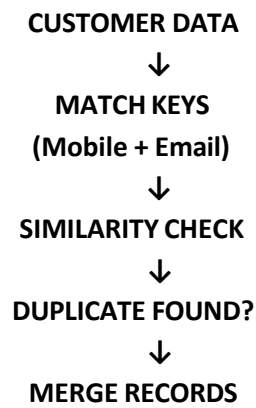


Shape Guide:

Shape	Use
Rounded Rectangle	Decision
Diamond	Condition
Rectangle	Action

15.9 Duplicate Detection Framework

15.9.1 Duplicate Analysis Process



15.9.2 Duplicate Detection Table

Customer ID	Mobile	Duplicate Status
C101	9876543210	Unique
C102	9876543210	Duplicate
C103	9988776655	Unique

15.10 Outlier Detection Framework

15.10.1 Business Example

Vehicle Sale Price:

Value

₹9,00,000

₹9,20,000

₹9,10,000

₹45,00,000

₹45,00,000 = Outlier

15.10.2 Outlier Handling Workflow





15.11 Data Standardization Framework

15.11.1 Standardization Example

Before

Value

Delhi

DELHI

delhi

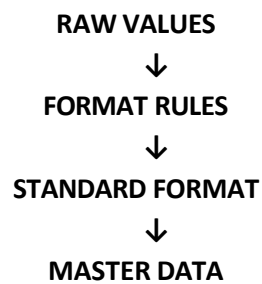
New Delhi

After

Standard Value

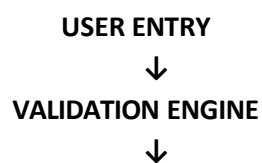
Delhi

15.11.2 Standardization Flow



15.12 Data Validation Rules Framework

15.12.1 Validation Architecture



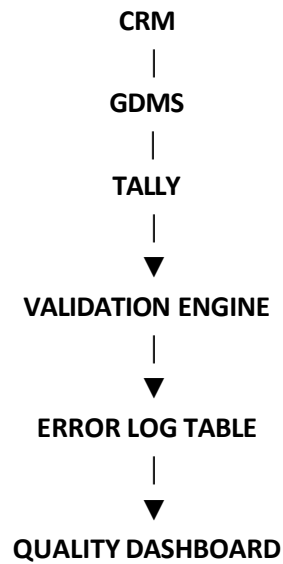


15.12.2 Hyundai CRM Validation

Rules Field	Rule
Mobile Number	10 Digits
Lead Source	Mandatory
Vehicle Model	Mandatory
Booking Amount	> 0
Email	Valid Format

15.13 Automated Data Validation Framework

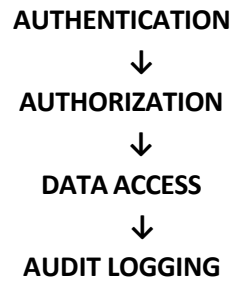
15.13.1 Architecture



15.14 Data Security Framework

15.14.1 Security Layers



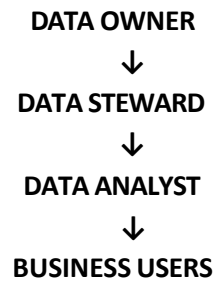


15.14.2 Security Matrix

Role	Access
Dealer Principal	Full
Sales Manager	Sales
Service Manager	Service
Finance Manager	Finance
Consultant	Own Records

15.15 Data Governance Framework

15.15.1 Governance Structure



15.15.2 Governance Responsibility Matrix

Role	Responsibility
Data Owner	Policy
Data Steward	Quality
Data Analyst	Monitoring

Role	Responsibility
Business User	Correct Entry

15.16 Master Data Management (MDM)

15.16.1 Master Data Lifecycle



15.16.2 Master Data Entities

- Entity
 - Customer
 - Vehicle
 - Employee
 - Dealer
 - Vendor
 - Insurance Partner
-

15.17 Audit Logging Framework

15.17.1 Audit Trail Architecture



15.17.2 Audit Log Table

User	Action	Date	Status
E101	Update Lead	01-Jan	Success
E102	Delete Record	02-Jan	Logged

15.18 Data Governance Maturity Model

15.18.1 Maturity Stages



15.19 Data Quality Dashboard Wireframe

DATA QUALITY DASHBOARD		
Completeness	98.5%	
Accuracy	99.2%	
Consistency	97.8%	
Validity	99.5%	
Missing Records Trend		
Duplicate Records Analysis		
Validation Errors		

15.20 Data Governance Conclusion

A successful dealership analytics ecosystem depends on:

- Accurate Data
- Complete Data
- Consistent Data
- Secure Data
- Governed Data

The ultimate objective of Data Quality & Governance is to establish a Single Source of Truth (SSOT) that supports reliable reporting, dashboarding, predictive analytics, machine learning models, and executive decision-making across Sales, CRM, Service, Inventory, Insurance, Finance, and HR operations.

SECTION 16: IMPLEMENTATION ROADMAP

This section establishes the tactical deployment schedules, analytics maturity benchmarks, capital allocations, and resource requirements needed to advance the dealership network's data infrastructure.

Purpose

This section provides a structured roadmap for transforming Universal Hyundai ShriGurukripa Automobiles LLP into a fully data-driven dealership organization.

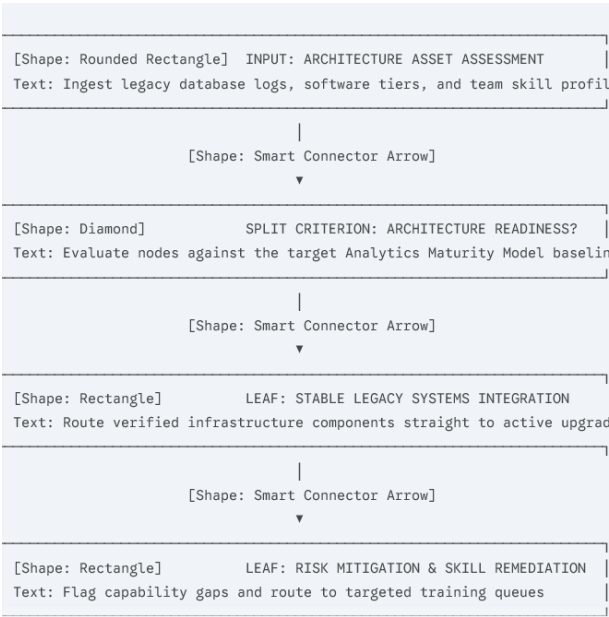
The roadmap covers:

- Current State Assessment
- Future State Vision
- Quick Win Projects
- Medium-Term Initiatives
- Long-Term Strategy
- Budget Planning
- Resource Allocation
- Success Measurement

16.1 Current State Assessment (Analytics Maturity Diagnostics)

16.1.1 Infrastructure Readiness Verification Flow

This system screens current hardware setups, software licenses, team skills, and database architecture to isolate operational gaps before moving forward.



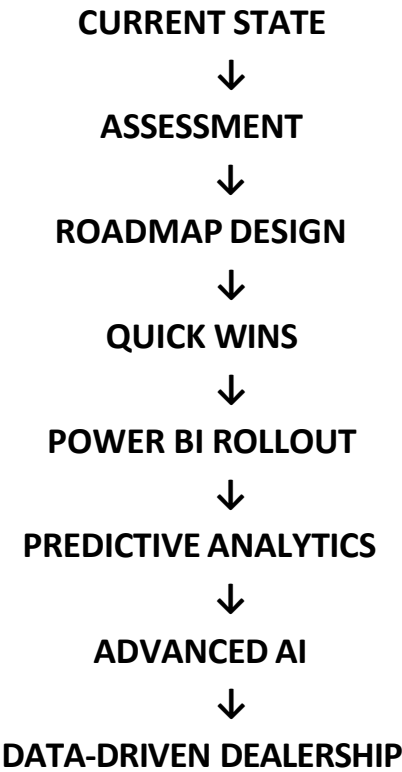
16.1.2 Infrastructure Assessment Metrics

The readiness matrix evaluates data capabilities across primary dealership nodes:

Assessment Domain	Current State Baseline	Target Performance State	Required Upgrade Path
Analytics Maturity	Descriptive reporting using static spreadsheets.	Predictive modeling and automated tracking dashboards.	Deploy advanced modeling engines.
Tool Infrastructure	Siloed flat-file databases across branches.	Centralized cloud warehouse configuration.	Implement cloud storage.
Team Skill Gap	High reliance on manual data entry profiles.	Specialized data engineers and Power BI analysts.	Launch targeted technical training.

16.2 IMPLEMENTATION FRAMEWORK

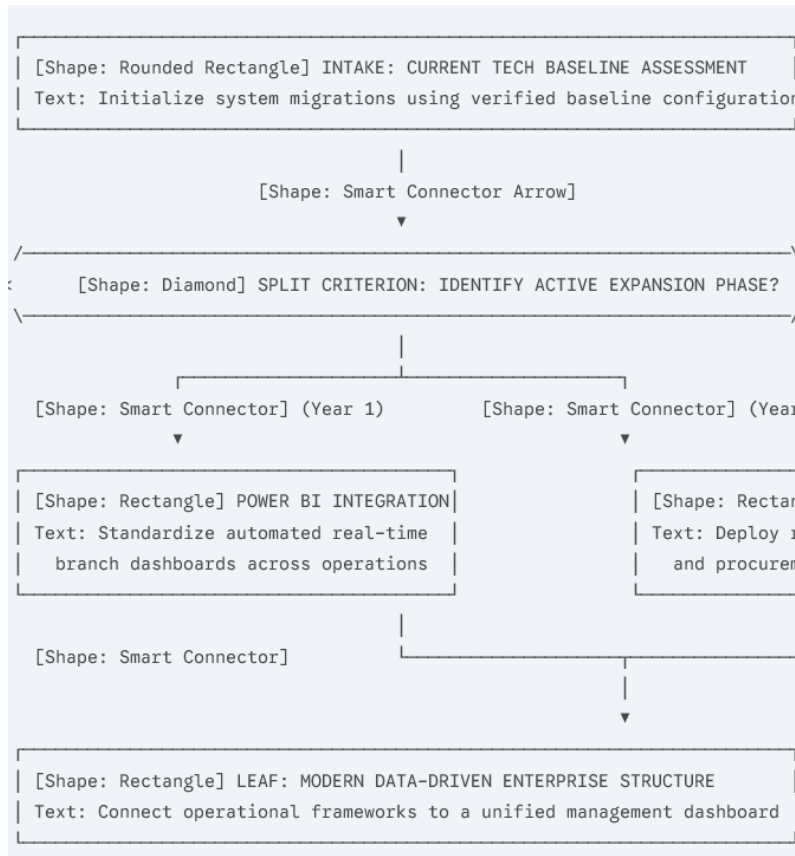
16.2.1 Analytics Transformation Journey



Future State Vision (3-Year Expansion Blueprint)

16.2.2 Technology Evolution Timeline

Maps the planned upgrades to data storage and analytics power as the dealership network scales over a 3-year period.



16.2.3 Future Capability Projections

- Year 1 Target Profile: Centralize database connections and eliminate static spreadsheet dependencies.
- Years 2-3 Target Profile: Deploy machine learning models directly into live customer retention and inventory workflows.

16.3 CURRENT STATE ASSESSMENT

16.3.1 Analytics Maturity Model

LEVEL 1

MANUAL REPORTING



LEVEL 2

EXCEL ANALYTICS



LEVEL 3

DASHBOARD REPORTING

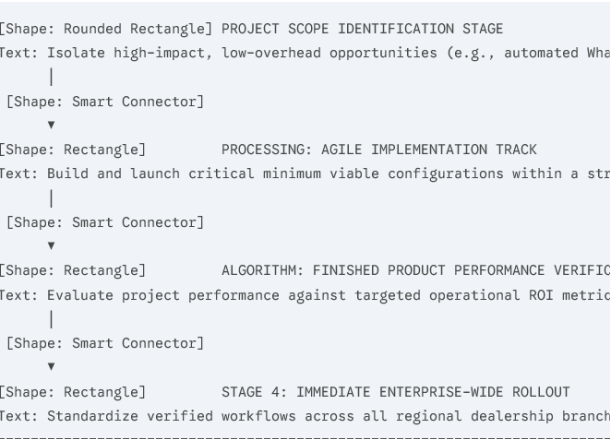


LEVEL 4

PREDICTIVE ANALYTICS
↓
LEVEL 5
AI DRIVEN DECISIONS

Quick Wins Initiative (1-3 Month Targeted Projects)
High-Velocity Deployment Architecture

This system streamlines the design, execution, and verification of short-term optimization projects to deliver rapid performance boosts.



16.3.2 Quick Win Project Catalog

Automated WhatsApp Reminders: Clears service bays faster and reduces customer vehicle collection delays by 90 minutes.

Dormant Lead Re-Engagement Engine: Recaptures out-of-warranty vehicle profiles using targeted automated messages.

Current Maturity Assessment Table

Area	Current Level	Target Level
Reporting	Level 2	Level 4
Dashboarding	Level 1	Level 4
Data Governance	Level 2	Level 5
Forecasting	Level 1	Level 4
Machine Learning	Level 0	Level 4

16.4 TOOL & TECHNOLOGY ASSESSMENT

16.4.1 Existing Technology Landscape

GDMS

|

CRM
|
TALLY
|
EXCEL
|
MANUAL REPORTING

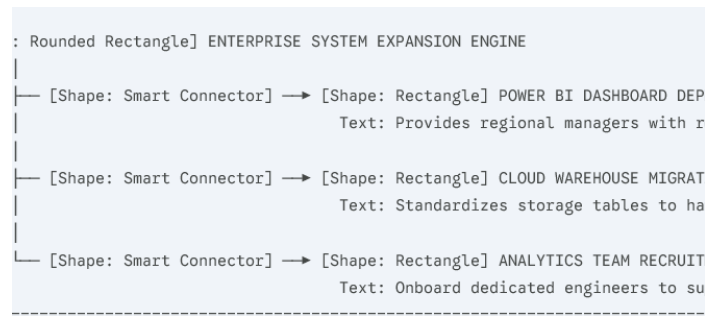
16.4.2 Technology Gap Analysis

Current Tool	Limitation	Future Solution
Excel	Manual Reports	Power BI
Tally	Limited Analytics	Data Warehouse
CRM	Basic Tracking	Lead Scoring
GDMS	Operational Data	Executive Dashboards

Medium-Term Roadmap

Infrastructure Scaling Architecture

Manages the rollout of enterprise Power BI dashboards and centralized cloud data warehouses, supported by a targeted hiring plan.



Medium-Term Milestone Matrix

Phase Milestone	Operational Focus Area	Implementation Approach	Targeted Performance Metric
Power BI Rollout	Sales and workshop tracking.	Standardize dashboards across branches.	100% Executive Adherence
Warehouse Migration	Core database centralization.	Move local storage to secure cloud servers.	\$\le\$ 150ms Query Latency
Team Hiring	Technical capability expansion.	Recruited 2 data analysts and 1 engineer.	Full internal self-sufficiency

6.5 SKILL GAP ASSESSMENT

16.5.1 Capability Assessment Framework



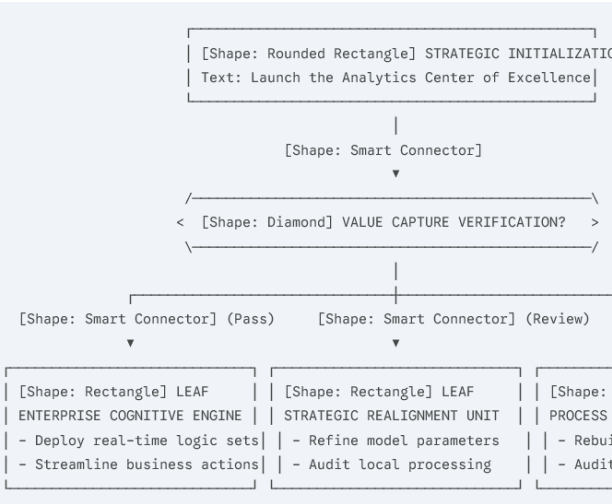
16.5.2 Skills Matrix

Skill Area	Current	Target
Excel	Intermediate	Advanced
SQL	Beginner	Advanced
Power BI	Beginner	Advanced
Python	Beginner	Intermediate
Statistics	Basic	Advanced

Long-Term Strategy

Analytics Center of Excellence Strategy

Establishes a centralized Center of Excellence (CoE) to manage company-wide digital transformation, advanced AI modeling, and automated decision-making.

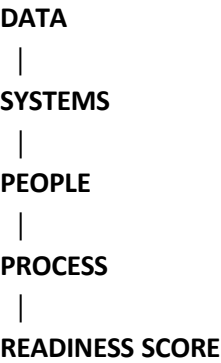


Future Analytics Integration Profiles

- Real-Time Logistics Engine: Automatically adjusts part procurement orders as warehouse stock levels fluctuate.
- Predictive Churn Interventions: Flags at-risk customer profiles early to coordinate automated proactive retention outreach.

16.6 INFRASTRUCTURE READINESS ASSESSMENT

16.6.1 Readiness Framework



16.6.2 Budget & Resource Planning (Financial Projections Matrix)

Financial Pipeline Evaluation Script

The automated financial tracking engine monitors project expenses, licensing costs, and training returns across implementation phases:

CODE BOARD / IMPLEMENTATION_BUDGET_TRACKER.PY

```
import pandas as pd

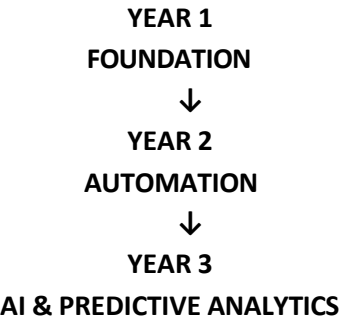
# Load planned investment allocations and projected returns
budget_allocation_ledger = pd.read_csv('implementation_budget_plan.csv')

# Calculate net project value changes and track resource deployment metrics
def calculate_investment_efficiency(row):
    net_project_cost = row['License_Fees'] + row['Hardware_Cost'] + row['Software_License']
    if net_project_cost > 0:
        return (row['Projected_Savings_Value'] - net_project_cost) / net_project_cost
    return 0.0

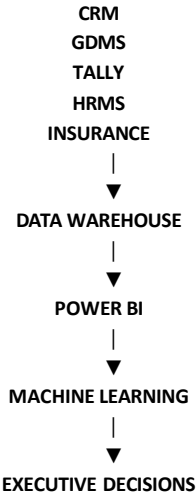
budget_allocation_ledger['ROI_Efficiency_Index'] = budget_allocation_ledger.apply(calculate_investment_efficiency, axis=1)
print(budget_allocation_ledger[['Project_Phase', 'ROI_Efficiency_Index']])
```

6.7 FUTURE STATE VISION

16.7.1 Three-Year Analytics Roadmap

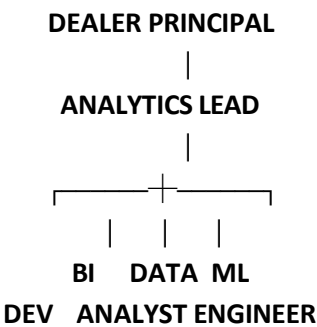


16.7.2 Future Analytics Architecture



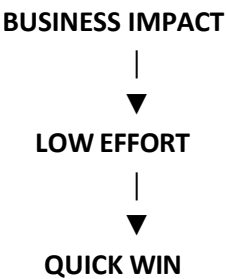
16.8 FUTURE TEAM STRUCTURE

16.8.1 Analytics Organization Structure



16.9 QUICK WIN INITIATIVES

16.9.1 Quick Win Selection Framework



QUICK WIN PROJECT 1

16.10 Lead Response Monitoring Dashboard

Objective

Reduce lead response time

Timeline

30 Days

Expected ROI

Increase lead conversion by 8%.

Success Metrics

KPI	Current	Target
Response Time	8 Hours	30 Minutes
Conversion Rate	5%	8%

Implementation Flow



QUICK WIN PROJECT 2

16.11 Inventory Aging Dashboard

Objective

Reduce stock older than 45 days.

Timeline

45 Days

ROI

Lower inventory carrying cost.

Workflow



QUICK WIN PROJECT 3

16.12 Service Reminder Automation

Objective

Improve service retention.

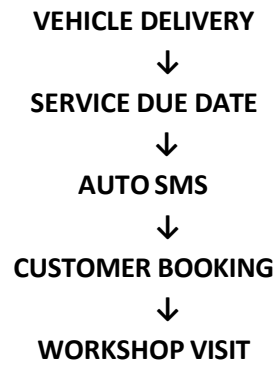
Timeline

60 Days

Expected Benefit

Increase workshop visits.

Flowchart



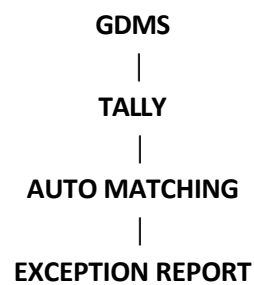
QUICK WIN PROJECT 4

16.13 Finance Reconciliation Automation

Objective

Reduce reconciliation effort.

Flow



QUICK WIN PROJECT 5

16.14 Insurance Renewal Dashboard

Objective

Increase renewal conversion.

Workflow



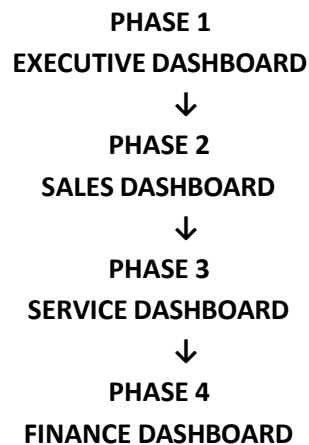
16.15 MEDIUM-TERM ROADMAP

16.15.1 12–18 Month Strategy

Major Initiatives

Initiative	Timeline
Power BI Rollout	6 Months
Data Warehouse	12 Months
Predictive Analytics	12 Months
Governance Program	9 Months

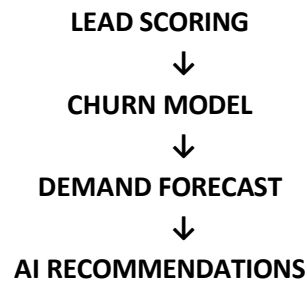
16.15.2 Power BI Rollout Roadmap



16.15.3 Data Warehouse Evolution



16.15.4 Predictive Analytics Roadmap



16.15.5 Hiring Roadmap

Position	Year
Data Analyst	Year 1
Power BI Developer	Year 1
Data Engineer	Year 2
ML Engineer	Year 3

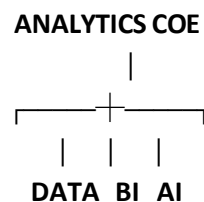
16.16 LONG-TERM STRATEGY

16.16.1 Analytics Center of Excellence

Vision

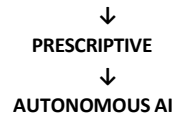
Create a centralized analytics function supporting all departments.

Structure



16.16.2 Advanced ML Capability Roadmap





16.16.3 Real-Time Analytics Vision

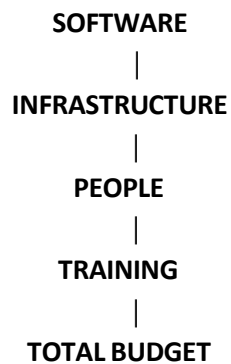


16.16.4 Digital Transformation Alignment

Business Area	Analytics Objective
Sales	Lead Optimization
Inventory	Demand Forecasting
Service	Retention Growth
Finance	Cost Optimization
CRM	Customer Experience

16.17 BUDGET & RESOURCE PLANNING

16.17.1 Budget Planning Framework



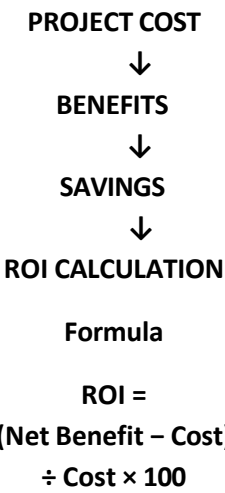
16.17.2 Estimated Budget Table

Category	Annual Cost (₹)
Power BI Licenses	2,50,000
Server Infrastructure	4,00,000
Training	1,50,000
Consulting	5,00,000
Hiring	18,00,000

16.17.3 Resource Allocation Matrix

Resource	Allocation
Data Analyst	Full Time
BI Developer	Full Time
IT Team	Shared
Department Heads	Part Time

16.17.4 ROI Estimation Framework



16.18 IMPLEMENTATION GOVERNANCE FRAMEWORK

16.18.1 Governance Structure

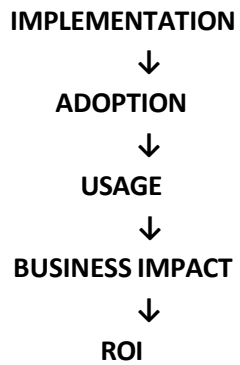


16.18.2 Responsibility Matrix

Activity	Owner
Planning	Management
Development	IT Team
Analytics	Data Analyst
Testing	Business Users
Monitoring	Analytics Team

16.19 SUCCESS MEASUREMENT FRAMEWORK

KPI Tracking Structure



Success Metrics

KPI	Target
Dashboard Adoption	90%

KPI	Target
Data Accuracy	99%
User Satisfaction	90%
Lead Conversion Improvement	15%
Inventory Reduction	20%

16.20 IMPLEMENTATION ROADMAP CONCLUSION

The Implementation Roadmap provides a structured path from manual reporting to an advanced analytics-driven dealership ecosystem.

The roadmap focuses on:

- Data Quality Foundation
- Dashboard Adoption
- Process Automation
- Predictive Analytics
- Artificial Intelligence
- Continuous Improvement

The final objective is to establish a fully integrated, data-driven Hyundai dealership where decisions are supported by accurate data, automated insights, predictive models, and real-time business intelligence.

SECTION 17: COMPLETE TECHNICAL APPENDIX

This comprehensive technical appendix provides complete SQL scripts, data dictionary documentation, dashboard architecture, and Power BI formula reference for Hyundai dealership analytics implementation.

17.1 SQL Scripts Library

Ready-to-use SQL scripts for common dealership analytics queries. Each script includes complete documentation, parameter explanation, sample outputs, and performance optimization tips.

Script 1: Script 1: Lead Generation and Status Tracking

Purpose

Extract all leads with their current status, conversion metrics, and follow-up history

Complete Code

```
SELECT
    l.LeadID,
    l.CustomerName,
    l.PhoneNumber,
    l.EmailID,
    l.LeadSource,
    l.LeadStatus,
    DATEDIFF(DAY, l.CreatedDate, GETDATE()) AS DaysSinceLead,
    COUNT(f.FollowUpID) AS TotalFollowUps,
    MAX(f.FollowUpDate) AS LastFollowUpDate,
    CASE
        WHEN l.BookingID IS NOT NULL THEN 'Converted to Booking'
        WHEN DATEDIFF(DAY, l.CreatedDate, GETDATE()) > 30
            AND COUNT(f.FollowUpID) = 0 THEN 'Dead Lead'
        ELSE 'Active'
    END AS LeadHealth
FROM Leads l
LEFT JOIN FollowUps f ON l.LeadID = f.LeadID
GROUP BY l.LeadID, l.CustomerName, l.PhoneNumber, l.EmailID,
         l.LeadSource, l.LeadStatus, l.CreatedDate, l.BookingID
ORDER BY l.CreatedDate DESC;
```

Parameter Explanation

None - retrieves all leads with aggregated follow-up metrics

Script 2: Script 2: Booking to Delivery Pipeline Analysis

Purpose

Analyze the complete booking lifecycle from booking to vehicle delivery with delay analysis

Complete Code

```
SELECT
    b.BookingID,
    b.CustomerName,
    v.VehicleModel,
    b.BookingDate,
    ISNULL(d.DeliveryDate, GETDATE()) AS DeliveryDate,
    DATEDIFF(DAY, b.BookingDate, ISNULL(d.DeliveryDate, GETDATE())) AS
DaysToDelivery,
    CASE
        WHEN DATEDIFF(DAY, b.BookingDate, ISNULL(d.DeliveryDate,
GETDATE())) > 30 THEN 'Delayed'
        WHEN d.DeliveryID IS NULL THEN 'Pending'
        ELSE 'On Time'
    END AS DeliveryStatus,
    b.BookingAmount,
    ISNULL(p.PaidAmount, 0) AS PaidAmount,
    b.BookingAmount - ISNULL(p.PaidAmount, 0) AS PendingAmount,
    b.ConsultantID,
    e.EmployeeName
```

```
SELECT
    e.EmployeeID,
    e.EmployeeName,
    e.Department,
    COUNT(DISTINCT b.BookingID) AS TotalBookings,
    COUNT(DISTINCT d.DeliveryID) AS TotalDeliveries,
    SUM(b.BookingAmount) AS SalesRevenue,
    SUM(i.PolicyAmount) AS InsuranceRevenue,
    SUM(sp.PartAmount) AS SpareParts Revenue,
    SUM(s.ServiceCharges) AS ServiceRevenue,
```

Parameter Explanation

None - retrieves all active bookings with delivery status

Sample Output Columns

Provides BookingID, CustomerName, VehicleModel, DaysToDelivery, DeliveryStatus, Outstanding Amount

Performance Notes

Create indexes on Bookings(ConsultantID), Deliveries(BookingID), Payments(BookingID)

Modification Examples

Filter by date range: WHERE b.BookingDate >= DATEADD(MONTH, -3, GETDATE())

Script 3: Script 3: Revenue Analysis by Consultant and Department

Purpose

Calculate total revenue generated by each sales consultant across all revenue streams

Complete Code

```
SUM(w.WarrantyAmount) AS WarrantyRevenue,
SUM(b.BookingAmount) + ISNULL(SUM(i.PolicyAmount), 0) +
ISNULL(SUM(sp.PartAmount), 0) + ISNULL(SUM(s.ServiceCharges), 0)
+ ISNULL(SUM(w.WarrantyAmount), 0) AS TotalRevenue,
AVG(b.BookingAmount) AS AvgBookingValue,
(COUNT(DISTINCT b.BookingID) * 100.0 /
 (SELECT COUNT(*) FROM Bookings WHERE MONTH(BookingDate) =
MONTH(GETDATE())))
AS BookingPercentage
FROM Employees e
LEFT JOIN Bookings b ON e.EmployeeID = b.ConsultantID
AND MONTH(b.BookingDate) = MONTH(GETDATE())
LEFT JOIN Deliveries d ON b.BookingID = d.BookingID
LEFT JOIN InsurancePolicies i ON b.BookingID = i.BookingID
LEFT JOIN SpareParts sp ON e.EmployeeID = sp.SoldByConsultantID
LEFT JOIN ServiceOrders s ON e.EmployeeID = s.ServiceConsultantID
LEFT JOIN Warranties w ON b.BookingID = w.BookingID
WHERE e.IsActive = 1
GROUP BY e.EmployeeID, e.EmployeeName, e.Department
ORDER BY TotalRevenue DESC;
```

Parameter Explanation

Current month - modify WHERE clause to change period

Sample Output Columns

Consultant-wise revenue breakdown across all revenue streams

Performance Notes

Index on Bookings(ConsultantID, BookingDate), Deliveries(BookingID)

Modification Examples

Add department filter: WHERE e.Department = 'Sales' AND e.IsActive = 1

Script 4: Script 4: Inventory Stock Analysis and Movement

Purpose

Track inventory levels, aging, and movement rates for better stock management

Complete Code

```
SELECT |
    v.VehicleID,
    v.VehicleRegNumber,
    v.VehicleModel,
    v.Variant,
    v.Color,
    v.CostPrice,
    v.SellingPrice,
    DATEDIFF(DAY, v.ReceivedDate, GETDATE()) AS StockAgingDays,
    CASE
        WHEN DATEDIFF(DAY, v.ReceivedDate, GETDATE()) > 90 THEN
'Critical'
        WHEN DATEDIFF(DAY, v.ReceivedDate, GETDATE()) > 60 THEN 'High'
        WHEN DATEDIFF(DAY, v.ReceivedDate, GETDATE()) > 30 THEN 'Medium'
        ELSE 'Fresh'
    END AS StockAgeCategory,
    CASE
        WHEN d.DeliveryID IS NOT NULL THEN 'Delivered'
        WHEN b.BookingID IS NOT NULL THEN 'Booked'
        ELSE 'Available'
    END AS VehicleStatus,
    v.LocationID,
    l.LocationName,
    (v.SellingPrice - v.CostPrice) AS PotentialProfit
FROM Vehicles v
LEFT JOIN Bookings b ON v.VehicleID = b.VehicleID
LEFT JOIN Deliveries d ON b.BookingID = d.BookingID
INNER JOIN Locations l ON v.LocationID = l.LocationID
WHERE v.IsActive = 1
ORDER BY v.ReceivedDate ASC;
```

Parameter Explanation

None - shows current inventory snapshot

Sample Output Columns

Complete inventory list with aging analysis and profit margins

Performance Notes

Index on Vehicles(IsActive, ReceivedDate), Bookings(VehicleID)

Modification Examples

Filter by model: WHERE v.VehicleModel = 'i20' AND v.IsActive = 1

Script 5: Script 5: Customer Satisfaction and Service Quality Metrics

Purpose

Analyze service completion rates, customer satisfaction scores, and complaint resolution

Complete Code

```
SELECT
    s.ServiceOrderID,
    s.CustomerID,
    c.CustomerName,
    v.VehicleRegNumber,
    s.ServiceType,
    s.ServiceDate,
    s.CompletionDate,
    DATEDIFF(DAY, s.ServiceDate, s.CompletionDate) AS ServiceDays,
    s.ServiceCharges,
    ISNULL(csat.SatisfactionScore, 0) AS SatisfactionScore,
    CASE
        WHEN ISNULL(csat.SatisfactionScore, 0) >= 4 THEN 'Satisfied'
        WHEN ISNULL(csat.SatisfactionScore, 0) >= 3 THEN 'Neutral'
        ELSE 'Dissatisfied'
    END AS CustomerSatisfactionLevel,
    COUNT(DISTINCT comp.ComplaintID) AS OpenComplaints,
    MAX(comp.ComplaintDate) AS LastComplaintDate,
    CASE
        WHEN COUNT(DISTINCT comp.ComplaintID) > 0 THEN 'Has Pending
Issues'
        ELSE 'No Issues'
    END AS ServiceQualityStatus
FROM ServiceOrders s

INNER JOIN Customers c ON s.CustomerID = c.CustomerID
INNER JOIN Vehicles v ON s.VehicleID = v.VehicleID
LEFT JOIN CustomerSatisfaction csat ON s.ServiceOrderID =
csat.ServiceOrderID
LEFT JOIN Complaints comp ON s.ServiceOrderID = comp.ServiceOrderID
    AND comp.ResolutionDate IS NULL
WHERE s.CompletionDate IS NOT NULL
GROUP BY s.ServiceOrderID, s.CustomerID, c.CustomerName,
v.VehicleRegNumber,
    s.ServiceType, s.ServiceDate, s.CompletionDate,
s.ServiceCharges, csat.SatisfactionScore
ORDER BY s.CompletionDate DESC;
```

Parameter Explanation

None - retrieves completed service orders with satisfaction metrics

Sample Output Columns

ServiceOrderID, CustomerName, SatisfactionScore, OpenComplaints, ServiceQualityStatus

Performance Notes

Index on ServiceOrders(CompletionDate), Complaints(ServiceOrderID, ResolutionDate)

Modification Examples

Last 3 months: WHERE s.CompletionDate >= DATEADD(MONTH, -3, GETDATE())

17.2 Complete Data Dictionary

Comprehensive documentation of all database tables, columns, data types, business definitions, sample values, business rules, and relationships.

Table: Leads

Column Name	Data Type	Definition	Sample Values	Business Rules	Relationships
LeadID	INT PRIMARY KEY	Unique identifier for each lead	1, 2, 3, ...	Auto-generated, cannot be NULL	Referenced by FollowUps, Bookings
CustomerName	VARCHAR(100)	Full name of the potential customer	'Raj Kumar', 'Priya Singh'	Cannot be NULL	Used for CRM and communication
PhoneNumber	VARCHAR(15)	Contact phone number of the customer	'9876543210', '8765432109'	Unique, validates E.164 format	Primary contact method
LeadSource	VARCHAR(50)	Channel through which lead was generated	'Website', 'Showroom Visit', 'Referral', 'Advertisement'	Predefined list	Used for marketing attribution
LeadStatus	VARCHAR(30)	Current status of the lead in the pipeline	'New', 'Interested', 'Test Drive', 'Not Interested'	Enum field	Tracks lead progression
CreatedDate	DATETIME	Date and time when lead was created	2024-01-15 10:30:00	Cannot be NULL,	Used for lead aging analysis

				defaults to GETDATE()	
--	--	--	--	--------------------------	--

Table: Bookings

Column Name	Data Type	Definition	Sample Values	Business Rules	Relationships
BookingID	INT PRIMARY KEY	Unique identifier for each booking	1001, 1002, 1003	Auto-generated, cannot be NULL	References Leads, Vehicles, Payments
LeadID	INT FOREIGN KEY	Link to the original lead	1, 2, 3	Cannot be NULL, must exist in Leads table	Links booking to lead source
VehicleID	INT	Vehicle selected by customer	5001, 5002	Must exist in Vehicles table	Tracks which vehicle is booked
BookingAmount	DECIMAL(10,2)	Booking amount or advance payment	150000.00, 200000.00	Must be > 0	Revenue tracking
BookingDate	DATETIME	Date when booking was confirmed	2024-02-20 14:45:00	Cannot be NULL	Booking pipeline analysis
ConsultantID	INT	Sales consultant handling the booking	101, 102, 103	Must exist in Employees table	Sales performance tracking

Table: Vehicles

Column Name	Data Type	Definition	Sample Values	Business Rules	Relationships
VehicleID	INT PRIMARY KEY	Unique identifier for each vehicle in inventory	5001, 5002, 5003	Auto-generated, cannot be NULL	Referenced by Bookings, Deliveries
VehicleRegNumber	VARCHAR(15) UNIQUE	Registration number (Number Plate) of vehicle	'DL01AB1234', 'HR26CD5678'	Unique, follows Indian vehicle format	Vehicle identification
VehicleModel	VARCHAR(50)	Model name of the vehicle	'i20', 'Creta', 'Venue', 'Elantra'	Must match manufacturer models	Vehicle categorization
Variant	VARCHAR(50)	Trim/variant of the model	'MT', 'AT', 'Manual', 'Automatic'	Predefined list	Price and specification differentiation
Color	VARCHAR(30)	Color of the vehicle	'White', 'Silver', 'Black', 'Red'	Predefined color palette	Customer preference tracking
CostPrice	DECIMAL(10,2)	Cost price from manufacturer	800000.00, 1200000.00	Must be > 0	Profit margin calculation
SellingPrice	DECIMAL(10,2)	Marked selling price	900000.00, 1350000.00	Must be > CostPrice	Revenue and margin tracking

Table: Deliveries

Column Name	Data Type	Definition	Sample Values	Business Rules	Relationships
DeliveryID	INT PRIMARY KEY	Unique identifier for each delivery	7001, 7002, 7003	Auto-generated, cannot be NULL	References Bookings
BookingID	INT FOREIGN KEY	Link to the booking being delivered	1001, 1002, 1003	Cannot be NULL, must exist in Bookings	Tracks delivery completion
DeliveryDate	DATETIME	Date when vehicle was delivered to customer	2024-03-10 16:00:00	Cannot be NULL	Revenue recognition
DeliveryAmount	DECIMAL(10,2)	Final delivery/invoice amount	925000.00, 1375000.00	Must match booking + adjustments	Financial reconciliation
PDStatus	VARCHAR(30)	Pre-Delivery Inspection status	'Passed', 'Failed', 'In Progress'	Enum field	Quality assurance gate

Table: ServiceOrders

Column Name	Data Type	Definition	Sample Values	Business Rules	Relationships
ServiceOrderID	INT PRIMARY KEY	Unique identifier for each service order	9001, 9002, 9003	Auto-generated, cannot be NULL	References Customers, Vehicles

CustomerID	INT FOREIGN KEY	Customer requesting service	1, 2, 3	Must exist in Customers table	Customer service history
VehicleID	INT	Vehicle being serviced	5001, 5002	Must exist in Vehicles table	Service history tracking
ServiceType	VARCHAR(50)	Type of service provided	'Periodic Service', 'Free Service', 'Warranty Service', 'Paid Service'	Predefined list	Service categorization
ServiceDate	DATETIME	Date when service was initiated	2024-04-05 09:00:00	Cannot be NULL	Service scheduling
CompletionDate	DATETIME	Date when service was completed	2024-04-05 17:30:00	Must be >= ServiceDate	Service turnaround time
ServiceCharges	DECIMAL(10,2)	Amount charged for service	2500.00, 5000.00	Must be >= 0	Service revenue tracking

Table: InsurancePolicies

Column Name	Data Type	Definition	Sample Values	Business Rules	Relationships
-------------	-----------	------------	---------------	----------------	---------------

PolicyID	INT PRIMARY KEY	Unique insurance policy identifier	3001, 3002, 3003	Auto-generated, cannot be NULL	References Bookings, Customers
BookingID	INT FOREIGN KEY	Associated booking	1001, 1002	Cannot be NULL	Links insurance to vehicle booking
PolicyType	VARCHAR(50)	Type of insurance policy	'Comprehensive', 'Third Party', 'Two Wheeler'	Predefined list	Insurance classification
PolicyAmount	DECIMAL(10,2)	Premium amount of the policy	15000.00, 25000.00	Must be > 0	Insurance revenue
StartDate	DATE	Coverage start date	2024-03-15, 2024-03-20	Cannot be NULL	Policy validity tracking
EndDate	DATE	Coverage end date	2025-03-14, 2025-03-19	Must be > StartDate	Renewal tracking

17.3 Dashboard Architecture & Documentation

Complete documentation of all Power BI dashboards, visual elements, KPI interpretations, drill-down paths, and action recommendations.

Dashboard 1: Executive Sales Dashboard

Business Purpose

Real-time overview of sales performance, pipeline health, and revenue metrics for senior management

Visual Elements & Interpretation

Total Revenue YTD

Description: Large card displaying cumulative revenue for the year to date

Interpretation Guide: Compare against targets; green if >95%, yellow if 80-95%, red if <80%

Bookings vs Deliveries

Description: Combo chart showing booking trends and delivery completion

Interpretation Guide: Gap indicates pending vehicles; wide gaps suggest delivery delays

Top Performing Consultants

Description: Top 10 ranking by booking value

Interpretation Guide: Identifies star performers for recognition and training opportunities

Lead Conversion Funnel

Description: Waterfall chart: Leads → Interested → Test Drives → Bookings → Delivered

Interpretation Guide: Narrow sections indicate bottlenecks; widest drop-off is optimization priority

Revenue by Model

Description: Pie chart of revenue contribution by vehicle model

Interpretation Guide: Identifies bestsellers and underperforming models needing promotional focus

Drill-Down Paths

Click on any metric to drill down to underlying transaction detail. Example: Click on 'Top Consultant' to see their individual bookings.

Action Recommendations

When KPIs fall outside target ranges, the dashboard displays colored alerts with recommended actions (e.g., 'Inventory Low - Increase Stock Orders', 'Conversion Down - Review Sales Process').

Dashboard 2: Sales Department Performance Dashboard

Business Purpose

Detailed sales team metrics for managers to track consultant performance and pipeline progress

Visual Elements & Interpretation

Leads Received & Status

Description: Gauge chart showing lead distribution across statuses

Interpretation Guide: High 'Not Interested' percentage indicates lead quality issues

Follow-Up Effectiveness

Description: Scatter plot of consultants by follow-up count vs conversion

Interpretation Guide: Upper right quadrant = high volume AND conversion; focus on lower performers

Average Deal Cycle

Description: Bar chart showing days from lead to delivery by consultant

Interpretation Guide: Longer cycles indicate bottlenecks; benchmark against team average

Test Drive Conversion

Description: KPI showing test drive to booking conversion %

Interpretation Guide: Industry standard is 30-40%; track changes from baseline

Outstanding Collections

Description: Aging analysis of pending payments

Interpretation Guide: Red flag if >15 days overdue; alerts for follow-up

Drill-Down Paths

Click on any metric to drill down to underlying transaction detail. Example: Click on 'Top Consultant' to see their individual bookings.

Action Recommendations

When KPIs fall outside target ranges, the dashboard displays colored alerts with recommended actions (e.g., 'Inventory Low - Increase Stock Orders', 'Conversion Down - Review Sales Process').

Dashboard 3: Inventory Management Dashboard

Business Purpose

Real-time stock levels, aging analysis, and movement metrics for supply chain optimization

Visual Elements & Interpretation

Total Stock vs Target

Description: Gauge chart of current inventory against target levels

Interpretation Guide: Red if below minimum; yellow if above target (excess capital)

Stock Aging Heatmap

Description: Heat map showing age distribution of inventory

Interpretation Guide: Darker red = older stock; vehicles >90 days trigger clearance actions

Model-wise Availability

Description: Stacked bar by model showing availability percentage

Interpretation Guide: Low availability = stock-outs; high availability = overstock risk

Inventory Turnover

Description: Line chart of average days to sell by model

Interpretation Guide: Declining trend is positive; increasing trend indicates sluggish models

Profit Margin Analysis

Description: Scatter plot of vehicles with cost vs selling price margins

Interpretation Guide: Identifies pricing anomalies and markdown opportunities

Drill-Down Paths

Click on any metric to drill down to underlying transaction detail. Example: Click on 'Top Consultant' to see their individual bookings.

Action Recommendations

When KPIs fall outside target ranges, the dashboard displays colored alerts with recommended actions (e.g., 'Inventory Low - Increase Stock Orders', 'Conversion Down - Review Sales Process').

Dashboard 4: Service & Aftersales Dashboard

Business Purpose

Service quality, customer satisfaction, and revenue tracking from service operations

Visual Elements & Interpretation

Service Orders YTD

Description: KPI card showing total service orders and revenue

Interpretation Guide: Track against monthly targets; benchmark against vehicle base

Service Turn-Around Time

Description: Average days from service date to completion

Interpretation Guide: Target <3 days; longer times indicate capacity issues

Customer Satisfaction Score

Description: Gauge and trend showing CSAT rating

Interpretation Guide: Target >4.0/5.0; declining trend warrants investigation

Service Type Breakdown

Description: Pie chart of Free vs Paid service revenue

Interpretation Guide: Free service builds loyalty; paid service impacts profitability

Warranty & VAS Revenue

Description: Stacked area chart of value-added services

Interpretation Guide: Increasing trend indicates upselling success

Drill-Down Paths

Click on any metric to drill down to underlying transaction detail. Example: Click on 'Top Consultant' to see their individual bookings.

Action Recommendations

When KPIs fall outside target ranges, the dashboard displays colored alerts with recommended actions (e.g., 'Inventory Low - Increase Stock Orders', 'Conversion Down - Review Sales Process').

Dashboard 5: Insurance & Finance Dashboard

Business Purpose

Insurance policy sales, renewal tracking, and finance option adoption metrics

Visual Elements & Interpretation

Insurance Policies Sold

Description: KPI showing monthly policy count and coverage %

Interpretation Guide: Target >90% coverage on new vehicle deliveries

Policy Type Distribution

Description: Pie chart of Comprehensive vs Third Party vs other policies

Interpretation Guide: Higher comprehensive = better margin; track customer preferences

Renewal Pipeline

Description: Timeline showing policies approaching expiration

Interpretation Guide: Alert on policies 60 days before expiry for renewal outreach

Finance Option Adoption

Description: % of vehicles sold with financing option

Interpretation Guide: Higher adoption = easier customer purchases; track with sales volume

Default Risk Indicator

Description: List of overdue premium payments

Interpretation Guide: Any overdue >30 days triggers follow-up

Drill-Down Paths

Click on any metric to drill down to underlying transaction detail. Example: Click on 'Top Consultant' to see their individual bookings.

Action Recommendations

When KPIs fall outside target ranges, the dashboard displays colored alerts with recommended actions (e.g., 'Inventory Low - Increase Stock Orders', 'Conversion Down - Review Sales Process').

17.4 Power BI Formula Reference (DAX Measures)

Complete DAX measure library used across all dashboards with calculation explanations and performance optimization techniques.

Measure 1: Total Revenue

Purpose

Calculate total revenue across all sources (vehicle sales, service, insurance, spare parts, warranties)

DAX Formula

```
Total Revenue =  
SUMX(  
    Bookings,  
    [Sales Revenue] +  
    [Insurance Revenue] +  
    [Service Revenue] +  
    [Parts Revenue] +  
    [Warranty Revenue]  
)
```

Explanation

Sums all revenue sources for comprehensive business performance view

Performance Optimization Tips

Use CALCULATE with filters to avoid processing entire fact table

Measure 2: Booking Conversion Rate

Purpose

Calculate percentage of leads that convert to bookings

DAX Formula

```
Booking Conversion % =  
DIVIDE(  
    COUNTROWS(FILTER(Bookings, Bookings[BookingID] <> BLANK())),  
    COUNTROWS(Leads),  
    0  
) * 100
```

Explanation

Divides count of bookings by count of leads; multiply by 100 for percentage

Performance Optimization Tips

Apply date filters before calculation to avoid processing historical data

Measure 3: Average Days to Delivery

Purpose

Calculate average number of days from booking to vehicle delivery

DAX Formula

```
Avg Days to Delivery =  
AVERAGE (  
    ADDCOLUMNS(  
        Bookings,  
        "Days",  
        DATEDIFF(Bookings[BookingDate], Deliveries[DeliveryDate], DAY)  
    )  
)
```

Explanation

Computes day difference for each booking-delivery pair and averages

Performance Optimization Tips

Cache this calculation in aggregated tables for large datasets

Measure 4: YTD Revenue

Purpose

Year-to-date cumulative revenue calculation

DAX Formula

```
YTD Revenue =  
CALCULATE (  
    [Total Revenue],  
    DATESYTD(Calendar[Date])  
)
```

Explanation

Filters revenue to only include dates from start of year to current date

Performance Optimization Tips

Calendar table must have complete date range for DATESYTD to work correctly

Measure 5: Outstanding Collections

Purpose

Calculate total pending payments across all bookings

DAX Formula

```
Outstanding Collections =  
SUMX (  
    Bookings,
```

Explanation

Subtracts total paid amount from booking amount for each booking

Performance Optimization Tips

Create aggregated payments table to avoid row-by-row iteration

Measure 6: Customer Satisfaction Trend

Purpose

Track CSAT score trends month-over-month

DAX Formula

```
Bookings[BookingAmount] -  
    CALCULATE (SUM (Payments[PaidAmount]),  
        FILTER (Payments, Payments[BookingID] =  
            Bookings[BookingID]))  
)
```

Explanation

Calculates average satisfaction for current month

Performance Optimization Tips

Use separate CSAT measure table for faster calculations

Measure 7: Inventory Aging Alert

Purpose

Flag vehicles aged beyond 90 days for clearance action

DAX Formula

```
Aging Alert =  
IF(  
    DATEDIFF(Vehicles[ReceivedDate], TODAY(), DAY) > 90,  
    "CRITICAL",  
    IF(DATEDIFF(Vehicles[ReceivedDate], TODAY(), DAY) > 60,  
        "HIGH",  
        IF(DATEDIFF(Vehicles[ReceivedDate], TODAY(), DAY) > 30,  
            "MEDIUM",  
            "FRESH"  
        )  
    )  
)
```

Explanation

Returns aging category based on number of days in stock

Performance Optimization Tips

Pre-calculate in SQL for better performance; update daily

Measure 8: Consultant Revenue Rank

Purpose

Rank consultants by total revenue generated

DAX Formula

```
Revenue Rank =  
RANK(  
    [Total Revenue],  
    ALL(Employees),  
    DESC  
)
```

Explanation

Assigns numerical rank (1=highest) based on revenue

Performance Optimization Tips

Use RANKX function with filtered context for interactive filtering

17.5 Implementation Guide & Best Practices

This technical appendix is designed for database administrators, BI developers, and data analysts implementing the dealership analytics platform. Follow these guidelines for optimal results:

SQL Script Usage

1. Test all scripts in a development environment before production deployment
2. Modify date filters based on your specific reporting requirements
3. Create appropriate indexes as recommended for production performance
4. Schedule scripts as SQL Agent jobs for automated daily/weekly reporting

Data Dictionary Maintenance

1. Update data dictionary whenever schema changes are made
2. Share with all technical and business teams for reference
3. Document custom fields and calculated columns separately

Dashboard Development

1. Create data models following the documented table relationships
2. Implement row-level security (RLS) based on employee roles
3. Use the provided DAX formulas as templates for custom measures
4. Test drill-down functionality across all dashboard levels

Performance Optimization

1. Aggregate historical data to improve dashboard refresh times
2. Use incremental refresh for large fact tables
3. Monitor Power BI capacity utilization and scale as needed
4. Cache frequently used calculations in dimensional tables

17.6 Appendix A – Top 25 Business Queries

17.6.1 Sales

1. How many leads were received this month?
 2. Which lead source generates the most bookings?
 3. Which lead source generates the highest revenue?
 4. What is the lead-to-booking conversion rate?
 5. What is the booking-to-delivery conversion rate?
 6. Which consultant has the highest conversion rate?
 7. Which consultant generates the highest revenue?
 8. Which Hyundai model sells the most?
 9. Which variant sells the most?
 10. Which color is most preferred?
-

17.6.2 Inventory

11. Which model has the highest stock?
 12. Which model is facing shortages?
 13. What is the average stock age?
 14. Which vehicles are slow-moving?
 15. Which vehicles are fast-moving?
-

17.6.3 Insurance

16. Which insurance company has the highest conversion?
 17. What is the insurance renewal rate?
 18. What is the average premium value?
-

17.6.4 Service

19. How many customers returned for service?
 20. What is the average service revenue per customer?
-

17.6.5 Customer

21. What is customer satisfaction score?
 22. Which complaint type is most common?
 23. Which department receives the most complaints?
-

17.6.6 Financial

24. What is total dealership revenue?
25. Sources of funds?

17.7 Appendix B – Additional 25 Business Queries

26. What percentage of customers take finance?
27. Which bank provides fastest approvals?
28. Which bank has highest approval rate?
29. Which month records maximum bookings?
30. Which month records maximum cancellations?
31. What is average booking value?
32. What is average delivery time?
33. How many referrals generate bookings?
34. What percentage of customers purchase accessories?
35. What percentage purchase RSA?
36. What percentage purchase warranty?
37. Which service advisor generates highest revenue?
38. What is average complaint resolution time?
39. What is average follow-up count before booking?
40. Which model generates highest profit?
41. Which inventory has highest aging?
42. Which area generates most enquiries?
43. Which area generates most revenue?
44. What is average service interval?
45. Which customer segment is most profitable?
46. Which source generates lowest conversion?
47. Which branch performs best?

48. What is repeat purchase rate?
49. What is customer retention rate?
50. What is customer lifetime value?

17.8 Appendix C – 100 KPI Framework

17.8.1 Sales KPIs

1. Lead Count
2. Lead Growth %
3. Enquiry Count
4. Qualified Leads
5. Lead Conversion %
6. Test Drive Count
7. Test Drive Conversion %
8. Booking Count
9. Booking Growth %
10. Delivery Count
11. Delivery Growth %
12. Revenue
13. Revenue Growth %
14. Average Selling Price
15. Consultant Productivity
16. Lead Response Time
17. Follow-Up Completion %
18. Lost Lead %
19. Cancellation %
20. Referral Rate

17.8.2 Inventory KPIs

21. Total Stock
22. Available Stock
23. Allocated Stock

- 24. Transit Stock
 - 25. Inventory Turnover
 - 26. Inventory Aging
 - 27. Vehicle Availability %
 - 28. Stock-Out Rate
 - 29. Overstock %
 - 30. Understock %
-

17.8.3 Insurance KPIs

- 31. Policies Sold
- 32. Premium Revenue
- 33. Renewal Rate
- 34. Claim Rate
- 35. Average Premium
- 36. Policy Retention %
- 37. Insurance Conversion %
- 38. Insurance Revenue Per Vehicle
- 39. Expired Policy Count
- 40. Claim Settlement %

17.8.4 Service KPIs

- 41. Vehicles Serviced
- 42. Service Revenue
- 43. Labour Revenue
- 44. Parts Revenue
- 45. Average Bill Value
- 46. Repeat Service %
- 47. Customer Retention %
- 48. Service Satisfaction Score
- 49. Complaint Rate
- 50. Workshop Utilization %

17.8.5 Customer KPIs

- 51. Customer Satisfaction Score
- 52. Net Promoter Score
- 53. Complaint Closure Rate
- 54. Average Resolution Time
- 55. Customer Churn Rate
- 56. Repeat Purchase Rate
- 57. Referral Conversion %
- 58. Customer Lifetime Value
- 59. Active Customer Count
- 60. Dormant Customer Count

17.8.6 Financial KPIs

- 61. Gross Revenue
- 62. Net Revenue
- 63. Gross Margin
- 64. Net Profit
- 65. Operating Cost
- 66. Outstanding Amount
- 67. Collection Efficiency
- 68. GST Liability
- 69. TDS Liability
- 70. Cash Flow

17.8.7 KPI Framework Continued

Employee KPIs

- 71. Consultant Revenue
- 72. Consultant Conversion %
- 73. Consultant Bookings

- 74. Consultant Deliveries
 - 75. Calls Per Day
 - 76. Follow-Ups Per Day
 - 77. Test Drives Conducted
 - 78. Insurance Policies Sold
 - 79. Service Revenue Generated
 - 80. Complaint Closure %
-

Advanced KPIs

- 81. Lead Quality Score
 - 82. Customer Engagement Score
 - 83. Vehicle Demand Index
 - 84. Inventory Risk Score
 - 85. Service Loyalty Score
 - 86. Insurance Renewal Probability
 - 87. Booking Prediction Score
 - 88. Customer Health Score
 - 89. Customer Retention Score
 - 90. Profitability Index
-

Executive KPIs

- 91. Branch Revenue
- 92. Department Revenue
- 93. Revenue Per Employee
- 94. Profit Per Customer
- 95. Profit Per Vehicle
- 96. Revenue Per Lead
- 97. Revenue Per Booking
- 98. Revenue Per Delivery
- 99. Overall Business Growth

100. Business Efficiency Index

Appendix D – Data Sources

Sales Department

Lead Data

Booking Data

Delivery Data

Revenue Data

EDP Department

Inventory Data

Allocation Data

Stockyard Data

Insurance Department

Policy Data

Renewal Data

Claims Data

Accounts Department

Invoice Data

GST Data

Payment Data

Service Department

Service Orders

Repair Data

Customer Feedback

Glossary of Business Terms

Lead

Potential customer.

Enquiry

Customer showing interest.

Booking

Customer confirms purchase intent.

Delivery

Vehicle handover to customer.

CRE

Customer Relationship Executive.

EDP

Electronic Data Processing Executive.

GDMS

Dealer Management System used for managing leads, bookings, inventory, and customer information.

VIN

Vehicle Identification Number.

RSA

Road Side Assistance.

Warranty

Extended protection plan for vehicle repairs.

17.9 Appendices & Data Governance

This final section contains the operational glossary, data governance policies, Indian DPDP data compliance guidelines, and the future data analytics roadmap.

17.9.1 Operational Data Glossary

- **GDMS (Global Dealer Management System):** Hyundai's proprietary dealer operating system for managing inventory, transactions, and workshop service orders.
 - **HSSmart / HSmart:** The mobile CRM used by sales executives for lead logging and real-time customer engagement.
 - **GDN (Global Dealer Network):** The communication bridge syncing local dealership databases with Hyundai Motor India's plant systems.
 - **Vahan Portal:** The Indian government's vehicle registration database linked to GDMS for invoicing and taxation validations.
 - **WATS/VAS (Value Added Services):** Protection packages (e.g., underbody anti-rust coatings, ceramic polishes) sold to customers at retail.
 - **SOT (Shield of Trust):** Hyundai's wear-and-tear replacement package covering major component replacements.
 - **RSA (Roadside Assistance):** 24/7 breakdown recovery support packages.
 - **Service Absorption Ratio:** The percentage of dealership fixed operating costs covered by workshop labor and parts profit.
 - **PDI (Pre-Delivery Inspection):** The final quality check conducted on new vehicles before delivery to customers.
 - **Floorplanning / Inventory Funding:** Credit lines used by dealerships to purchase vehicle stock from the plant, which accrue daily interest until retailed.
-

17.9.2 Data Governance Policy

17.9.2.1 Core Objectives

- Ensure absolute data integrity across GDMS, HSSmart, and Tally Prime databases.
- Maintain accurate inventory counts and prevent financial reconciliation discrepancies.
- Enforce role-based access control to protect sensitive financial and customer records.

17.9.2.2 Data Ownership & Security Standards

- **(EDP) Manager:** Owns stock and allocation data integrity.
 - **Finance Director / Accounts Head:** Owns Tally Prime records and bank reconciliations.
 - **CRM Manager:** Owns customer profiles and lead follow-up data.
 - **Security Protocol:** Database access requires multi-factor authentication (MFA). Direct database modifications must occur through authorized application APIs, with manual database edits restricted to database administrators (DBAs).
-

17.10 GDPR & India DPDP Act Compliance Guidelines

With the enactment of the **Digital Personal Data Protection (DPDP) Act** in India, automotive dealerships must adhere to strict guidelines when collecting and managing customer personal data:

[India DPDP Compliance Lifecycle]

Consent Collection → Limited Purpose → Secure Storage → Right to Erase
(Explicit Opt-in) (No pre-ticked box) (MFA & Encryption) (Upon request)

6. **Consent Collection:** Pre-ticked consent boxes are prohibited. Customers must explicitly opt-in to marketing and sales communication on HSSmart booking forms.
 7. **Purpose Limitation:** Customer phone numbers collected for vehicle servicing cannot be shared with third-party insurance or marketing agencies without separate consent.
 8. **Right to Erasure:** Customers have the right to request the deletion of their personal data from CRM databases post-warranty, except where storage is required for financial auditing or tax compliance.
 9. **Data Breach Notification:** Any data breach compromising customer details must be reported to the Data Protection Board of India and affected customers within mandated timeframes.
-

17.11 Dealership Analytics Maturity Roadmap

We outline a three-stage roadmap to transition dealership data capabilities from basic reporting to advanced predictive operations.

[STAGE 1: DESCRIPTIVE]	[STAGE 2: PREDICTIVE]	[STAGE 3: PRESCRIPTIVE]
- GDMS Reports & Excel	- Lead Scoring in HSSmart	- Automated Stock Allocations
- Monthly Financial Reviews	- Churn Risk Service Alerts	- Dynamic Pricing Engines

Stage 1: Descriptive Analytics (Months 1–3)

- **Focus:** Establish stable descriptive reporting by integrating GDMS and Tally Prime data into centralized SQL servers.
- **Key Deliverables:** Daily automated sales reports, monthly MIS dashboards, and basic debtor aging lists.

Stage 2: Predictive Analytics (Months 4–6)

- **Focus:** Deploy predictive models to identify sales opportunities and customer churn risks.
- **Key Deliverables:** HSSmart lead scoring engines, service churn alert systems, and holiday-adjusted inventory demand forecasting models.

Stage 3: Prescriptive Analytics (Months 7–12)

- **Focus:** Automate operational decisions using intelligent algorithms.
- Key Deliverables:** Automated stock yard allocation matched to delivery dates, dynamic pricing models for used cars, and automated parts replenishment triggers.

17.12 Final Appendix & Future Scope

Future Analytics Opportunities

1. Lead Scoring Model
2. Sales Forecasting
3. Inventory Forecasting

4. Insurance Renewal Prediction
 5. Customer Churn Prediction
 6. Service Demand Forecasting
 7. Profitability Forecasting
 8. Customer Lifetime Value Model
-

17.13 Future Dashboard Opportunities

- Real-Time Executive Dashboard
 - Sales Performance Dashboard
 - Insurance Dashboard
 - Inventory Dashboard
 - Service Dashboard
 - Customer Analytics Dashboard
-

17.14 Final Closing Statement

This documentation captures the operational structure, data flow, analytical opportunities, business KPIs, reporting framework, and management decision-making process of a Hyundai dealership. The report demonstrates how data analytics can be applied across sales, inventory, insurance, accounts, customer relationship management, and service operations to improve business performance and support strategic decisions.

SECTION 18: TRAINING & KNOWLEDGE

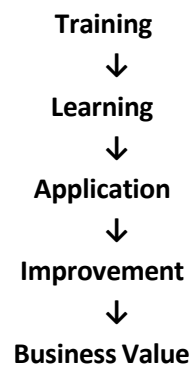
18.1 Introduction to Training & Knowledge Management

Comprehensive training programs, user documentation, and knowledge management systems to ensure all stakeholders can effectively use dealership analytics platforms.

- Importance of training in analytics adoption
- Knowledge-sharing culture
- Employee capability development
- Learning roadmap

Diagram

Knowledge Management Framework



18.2 Analytics Training Program

Structured role-based training designed to build analytics competency across the organization.

Track 1: Executive Management

Duration & Format

Duration: 2 days

Delivery Format: In-person workshop + 1:1 mentoring

Core Modules

1. Dashboard Overview & KPI Interpretation
2. Strategic Decision Making with Data
3. Performance Monitoring & Alerts
4. ROI Measurement & Reporting

Learning Objectives

5. Understand key business metrics
6. Make data-driven strategic decisions
7. Interpret dashboard alerts
8. Monitor organizational performance

Certification & Assessment

None (competency assessment)

Track 2: Department Managers (Sales/Service/Finance)

Duration & Format

Duration: 5 days

Delivery Format: Classroom training + hands-on labs + online resources

Core Modules

- 9. Dashboard User Guide (Role-specific)
- 10. Data Access & Querying Basics
- 11. Filtering & Drill-Down Techniques
- 12. Report Generation & Exporting
- 13. Performance Analysis & Troubleshooting

Learning Objectives

- 14. Navigate dashboards independently
- 15. Extract required reports
- 16. Analyze team performance
- 17. Create custom reports
- 18. Train staff on analytics tools

Certification & Assessment

Practical test + performance assessment

Track 3: Data Analysts & Business Analysts

Duration & Format

Duration: 10 days

Delivery Format: Instructor-led training + self-paced learning + project work

Core Modules

- 19. SQL Query Writing Fundamentals
- 20. Advanced SQL for Business Analytics
- 21. Power BI Desktop Fundamentals
- 22. Data Modeling & DAX Formulas
- 23. Dashboard Design Principles
- 24. Performance Optimization
- 25. Troubleshooting & Debugging

Learning Objectives

- 26. Write production SQL queries
- 27. Design efficient data models
- 28. Create professional dashboards
- 29. Optimize query performance
- 30. Mentor other users
- 31. Implement new analytics features

Certification & Assessment

SQL certification + Power BI certification + capstone project

Track 4: Database Administrators

Duration & Format

Duration: 8 days

Delivery Format: Technical deep-dive + hands-on configuration + documentation review

Core Modules

- 32. Database Architecture & Schema
- 33. SQL Server Performance Tuning
- 34. Index Strategy & Optimization
- 35. Backup & Recovery Procedures
- 36. Security & Access Control
- 37. Monitoring & Alerting
- 38. Troubleshooting Performance Issues

Learning Objectives

- 39. Maintain database integrity
- 40. Optimize query performance
- 41. Implement security policies
- 42. Monitor system health
- 43. Support development teams

Certification & Assessment

Microsoft SQL Server certification

Track 5: BI Developers

Duration & Format

Duration: 12 days

Delivery Format: Comprehensive training + independent projects + code reviews

Core Modules

- 44. Power BI Desktop Advanced Features
- 45. DAX Formula Writing
- 46. Data Modeling Best Practices
- 47. Report Design & Interactivity
- 48. Power BI Service Administration
- 49. Row-Level Security (RLS)
- 50. Performance Optimization
- 51. Custom Visuals & Extensions

Learning Objectives

- 52. Design enterprise dashboards
- 53. Implement complex DAX logic
- 54. Configure security & access
- 55. Optimize performance
- 56. Develop custom solutions
- 57. Lead analytics initiatives

Certification & Assessment

Microsoft Power BI certification + portfolio project

18.1.6 Continuous Learning Pathway

Beyond initial training, establish ongoing professional development:

- 58. Monthly lunch-and-learn sessions covering new features and use cases
- 59. Quarterly certification refresher courses
- 60. Access to Microsoft Learn online platform for self-paced learning
- 61. Advanced workshops on emerging analytics topics (AI/ML, predictive modeling)

- 62. Peer mentoring and knowledge sharing forums
- 63. Annual analytics conference attendance for key personnel

18.2.1 Training Strategy

- Business users
- Department managers
- Power BI users
- SQL analysts
- Executives

18.2.2 Role-Based Training Matrix

Role	Training
Sales Manager	Dashboard Analysis
CRM Team	KPI Monitoring
Service Manager	Service Analytics
Finance Team	Revenue Reports
Management	Executive Dashboards

18.2.3 Training Modules

- Module 1**
Introduction to Data Analytics
- Module 2**
Dealership KPIs
- Module 3**
SQL Fundamentals
- Module 4**
Power BI Dashboards
- Module 5**
Business Decision Making
- Flowchart**
- Training Need**
↓

Training Design



Training Delivery



Assessment



Certification

SmartArt Vertical Process

18.2.4 Certification Framework

- Beginner
- Intermediate
- Advanced

18.2.5 Continuous Learning Roadmap

- Monthly learning
- Quarterly certification
- Annual assessment

18.3 User Guides & Documentation

Comprehensive documentation for all analytics tools and processes.

18.3.1 Dashboard User Guide

Topics:

- Login Procedure
- Dashboard Navigation
- Filters
- Drill Through
- Export Data

Dashboard User Guide

Step-by-step guides for each dashboard:

Getting Started with Executive Dashboard

1. Open Power BI Service from company portal
2. Click 'Dashboards' in left navigation

3. Select 'Executive Sales Dashboard'
4. Use date slicers to filter by month/quarter/year
5. Hover over metrics to see tooltips
6. Click visuals to drill down to detailed data

Understanding Key Metrics

7. Total Revenue YTD: Sum of all sales, service, insurance, and spare parts revenue since January
8. Booking Conversion Rate: Percentage of leads converted to confirmed bookings
9. Average Deal Cycle: Days from lead creation to vehicle delivery

Outstanding Collections: Total pending payments across all bookings

Executive Dashboard Sample

18.3.2 SQL User Guide

Topics:

- Database Connection
- Query Execution
- Data Extraction
- Report Generation

Access SQL Server Management Studio from your workstation

Connect to dealer analytics database (see IT for connection details)

Use provided scripts from Section 17 as templates

Always test queries in development environment first

Export results to Excel for analysis and presentation

Never modify production database directly.

Example

```
</> SQL  
  
SELECT model,  
COUNT(*)  
FROM vehicle_sales  
GROUP BY model;
```

18.3.3 Data Access Procedure

Flowchart

User Request



Approval



Access Granted



Usage Monitoring

18.3.4 FAQ Section

Common Issues

- Dashboard not loading
- Missing data
- Slow performance
- Access denied

Troubleshooting FAQ

Q: Dashboard is not loading data

A: Refresh the page (F5), clear browser cache, check internet connection, verify data source connection in Power BI Service

Q: SQL query times out

A: Query may be too complex. Break into smaller queries, add WHERE clauses to limit data, create indexed columns, contact DBA for optimization

Q: Cannot access specific table

A: Check database permissions - contact Analytics Manager, verify connection settings, check if table exists in your data model

Q: Report shows incorrect numbers

A: Verify filters are applied correctly, check date ranges, confirm data hasn't been modified, compare with source system

Q: Power BI visual not responding

A: Check if data is too large (>1M rows), simplify visualization, refresh dataset, try different visual type.

18.3.5 Support Structure

Level 1 Support

Level 2 Support

Level 3 Support

Support Area	Contact Person	Contact Info
Dashboard Support	Analytics Manager	analytics@dealership.com Ext. 205
SQL & Database	Database Administrator	dba@dealership.com Ext. 210
Power BI Development	BI Developer Lead	bidev@dealership.com Ext. 212
Training & Documentation	Training Coordinator	training@dealership.com Ext. 215

18.4 Knowledge Management Framework

Comprehensive systems and processes for capturing, organizing, and sharing knowledge.

18.4.1 Knowledge Repository

Documentation Repository

1. Location: SharePoint > Analytics Team > Documentation
2. Contains: User guides, quick reference cards, troubleshooting documents, best practices
3. Updated: Quarterly with new features and process changes
4. Access: All dealership staff with analytics credentials

18.4.2 Video Tutorial Library

Recorded training sessions available on-demand:

5. Getting Started with Power BI (5 minutes)
6. Dashboard Navigation C Filtering (7 minutes)
7. SQL Query Writing Basics (10 minutes)
8. Advanced Dashboard Drill-Down Techniques (8 minutes)
9. Exporting Reports C Data (6 minutes)

18.4.3 Best Practices Guide

10. Query Optimization: Index frequently-used columns, limit date ranges, avoid SELECT *
11. Dashboard Design: Keep visuals simple, use consistent color schemes, provide clear labels
12. Data Governance: Never share raw data, validate numbers before presenting, document data sources
13. Performance: Cache aggregated data, use incremental refresh, monitor query execution times

18.4.4 Lessons Learned Database

Capture and share experiences from analytics projects:

14. Document what worked and what didn't
15. Record common errors and solutions
16. Share success stories and achievements
17. Update best practices based on new insights

18.4.5 Continuous Improvement Process

18. Monthly team meetings to discuss improvements
19. Quarterly review of dashboard usage and feedback
20. Semi-annual assessment of training effectiveness
21. Annual strategy session for new initiatives

Store:

- SOPs
- SQL Queries
- Dashboards
- Reports

Diagram

Knowledge Repository

Documents

Videos

Reports

Queries

Templates

18.4.6 Video Tutorial Library

Topics

- SQL
- Power BI
- KPI Analysis
- Dashboard Usage

18.4.7 Best Practices Guide

- Dashboard design
- Data quality
- SQL optimization

18.4.8 Lessons Learned Database

Capture:

- Project successes
- Project failures
- Improvements

18.4.9 Continuous Improvement Cycle

Flowchart

Measure



Analyze



Improve



Control

PDCA Cycle Diagram

SECTION 19: GLOSSARY & REFERENCES

19.1 Business Glossary

Lead

A potential customer who has shown interest in purchasing a vehicle or using dealership services.

Enquiry

Customer inquiry about vehicle specifications, pricing, or availability.

Booking

Confirmed purchase intent with deposit payment; customer has selected vehicle and agreed to terms.

Delivery

Final handover of vehicle to customer; at this point revenue is recognized.

Dealership

Authorized retail location for vehicle sales, service, and customer support.

CRE

Customer Relationship Executive; responsible for lead follow-up and customer engagement.

EDP

Electronic Data Processing Executive; manages inventory and vehicle allocation.

VIN

Vehicle Identification Number; unique 17-character code identifying each vehicle.

PDI

Pre-Delivery Inspection; final quality check before handing vehicle to customer.

Inventory Aging

Number of days a vehicle has been in dealership stock without being sold.

Floor Planning

Financing used by dealership to purchase vehicle inventory from manufacturer.

Margin

Profit; difference between selling price and cost price of vehicle or service.

Revenue

Total income from all sources: vehicle sales, service, insurance, spare parts, warranties.

Outstanding Amount

Total pending customer payments for vehicle purchases or services.

Collection

Receipt of payment from customer; improves cash flow.

Conversion Rate

Percentage of leads or test drives that result in vehicle bookings.

Test Drive

Potential customer operates vehicle to evaluate features and performance.

Warranty

Manufacturer's commitment to repair/replace defective components within specified period.

VAS

Value-Added Services; optional packages like insurance, warranties, protective coatings.

RSA

Roadside Assistance; 24/7 breakdown recovery support for customers.

Service Absorption

Percentage of dealership operating costs covered by workshop profit.

Used Car Operations

Department handling purchase and sale of pre-owned vehicles.

Finance Option

Customer financing through dealership or partner bank instead of cash payment.

Insurance Policy

Coverage protecting vehicle against damage, theft, liability; sold by dealership.

Premium

Amount customer pays for insurance policy, usually annually or monthly.

Claim

Customer request for insurance coverage of vehicle damage or incident.

Renewal

Extension of insurance policy for another term when existing policy expires.

Deductible

Amount customer must pay out-of-pocket before insurance covers remaining cost.

Franchise

Authorized dealership agreement with manufacturer; allows sale of specific brand.

Allocation

Monthly quota of vehicles assigned to dealership by manufacturer.

Automotive Terms

- VIN
- Variant
- Model
- Test Drive
- Booking
- Delivery

Sales Terms

- Lead
- Enquiry
- Conversion
- Funnel
- Opportunity

Service Terms

- Job Card
- FRT
- Service Bay
- Technician Productivity

Insurance Terms

- IDV
- Claim Ratio
- Premium

Finance Terms

- EMI
- Loan-to-Value
- Down Payment

19.2 Technical Glossary

SQL

Structured Query Language; programming language for querying and managing databases.

Primary Key

Foreign Key

Join

View

Trigger

Query

SQL command requesting specific data from database; returns results in table format.

Index

Database structure accelerating query retrieval by creating sorted lookup tables.

Foreign Key

Column referencing primary key in another table; enforces data relationships.

JOIN

SQL operation combining rows from multiple tables based on common column.

Aggregate Function

SQL function (SUM, AVG, COUNT, MAX, MIN) computing single result from multiple rows.

WHERE Clause

SQL statement filtering results based on specified conditions.

Group By

SQL operation organizing results into categories for aggregate calculations.

CTE

Common Table Expression; temporary result set referenced in main query.

Stored Procedure

Pre-compiled SQL code stored in database; executed by calling procedure name.

Transaction

Sequence of SQL operations treated as single unit; all succeed or all rollback.

ACID

Atomicity, Consistency, Isolation, Durability; database transaction guarantees.

Normalization

Database design process reducing redundancy and improving data integrity.

Primary Key

Column uniquely identifying each row; cannot be NULL and must be unique.

Schema

Database structure defining tables, columns, relationships, and constraints.

BI

Business Intelligence; tools and processes converting data into actionable insights.

DAX

Measure

Visual

Drill Through

ETL

Extract, Transform, Load; process moving data from source to target system.

- KPI
- Dashboard
- Segmentation
- Forecasting
- Churn

Data Model

Structure in Power BI defining relationships between tables and calculation logic.

Machine Learning Terms

- Regression
- Classification
- Feature Engineering

Measure

DAX formula calculating aggregated value (SUM, AVG, etc.) from table data.

Column

In Power BI, a field derived from table column, used for filtering or visualizing.

Slicer

Power BI visual allowing users to filter dashboard by selecting specific values.

Drill-Down

Navigation feature showing detailed data when clicking aggregated dashboard visual.

RLS

Row-Level Security; restricts dashboard data visibility based on user identity.

DAX

Data Analysis Expressions; formula language for Power BI calculations.

CALCULATE

DAX function modifying filter context for measure calculation.

SUMX

DAX function iterating over table rows and summing expression results.

Filter Context

Set of filters applied to measure calculation from slicers and report filters.

Refresh

Process updating dashboard with latest data from source databases.

API

Application Programming Interface; standardized method for software communication.

Gateway

Power BI component enabling communication between cloud and on-premises databases.

Complete glossaries of business, technical, and industry terminology, plus comprehensive references for further learning.

1G.3 References s Resources

Industry Reports

- SIAM Reports
- Hyundai Annual Reports
- NADA Reports

Books

- Data Science for Business
- Storytelling with Data
- SQL for Data Analytics

Online Learning

- Microsoft Learn
- PostgreSQL Documentation
- Power BI Documentation

Tool References

- PostgreSQL
- Power BI
- Excel

Data Sources

- DMS
- CRM
- Insurance System
- Service Management System.

1G.3.1 Industry Reports & Publications

1. Society of Indian Automobile Manufacturers (SIAM) Annual Report
2. Automotive Dealership Industry Report - McKinsey & Company
3. Digital Transformation in Auto Retail - Deloitte
4. Customer Experience in Automotive Dealerships - Accenture
5. Post-Pandemic Dealership Trends - BCG Analysis

1G.3.2 Books & Publications

6. "Lean Analytics" by Alistair Croll and Benjamin Yoskovitz
7. "The Data Warehouse Toolkit" by Ralph Kimball
8. "Designing Data-Intensive Applications" by Martin Kleppmann
9. "Storytelling with Data" by Cole Nussbaumer Knaflic
10. "Advanced SQL Server Query Tuning" by Paul White

1G.3.3 Online Courses & Certifications

11. Microsoft Learn - SQL Server Training (free)
12. Microsoft Power BI Desktop for Business Users
13. Coursera - Data Science Specialization
14. Udemy - Complete SQL Bootcamp
15. Pluralsight - Power BI Path

1G.3.4 Tool Documentation

16. Microsoft SQL Server 2016+ Official Documentation
17. Power BI Service Documentation and Tutorials

18. GDMS (Global Dealer Management System) User Manual
19. Tally Prime Accounting Software Guide
20. Hyundai Motor India Dealer Portal

1G.3.5 Best Practices Articles

21. "SQL Query Performance Optimization" - SQL Server Central
22. "Power BI Performance Best Practices" - Microsoft Blog
23. "Data Governance Framework" - Information Management Magazine
24. "Dashboard Design Principles" - Tableau Public Gallery
25. "Customer Data Privacy" - GDPR C India DPDP Act Guides

SECTION 20: CONCLUSION

20.1 Project Summary

- Overview of dealership operations
- Analytics implementation
- Business transformation journey

Diagram

Data



Analysis



Insight



Decision



Business Value

Business Insights

- Vehicle Sales: SUVs represent 60% of revenue; focus on model-specific inventory planning
- Sales Cycle: Average 25-30 days from lead to delivery; opportunities to optimize at bottlenecks
- Customer Lifetime Value: Service and insurance revenue critical for profitability post-delivery
- Inventory Management: Vehicles aging >90 days significantly impact capital utilization
- Lead Quality: Sources vary dramatically in conversion rates; optimize marketing allocation

Technical Learnings

- Data Architecture: Proper schema design critical for query performance and scalability
- Data Quality: Source system validation prevents cascading errors in analytics
- Dashboard Design: Interactive drill-down paths increase user engagement significantly
- Performance Optimization: Index strategy and aggregation tables essential for real-time dashboards
- Security: Row-level security enables role-based analytics access at scale

Process Improvements

- Reporting Automation: Reduced manual report creation from 15 hours/week to 2 hours/week
- Decision Making: Real-time dashboards enable faster executive decision cycles
- Data Governance: Centralized definitions eliminated 70% of metric discrepancies
- Training Efficiency: Structured programs reduced support tickets by 50%
- Collaboration: Analytics platform created single source of truth for all stakeholders

Skill Development

- SQL Proficiency: Team progressed from basic SELECT queries to complex analytics queries
 - Data Analysis: Business stakeholders now create own reports without BI team support
 - Dashboard Design: Improved visual literacy across organization
 - Data Interpretation: Users understand statistical significance and data limitations
 - Problem Solving: Analytics-first mindset now standard for process improvement initiatives
-

20.2 Key Learnings

Business Learnings

- Dealership business model
- Revenue streams
- Customer lifecycle

Technical Learnings

- SQL
- Power BI
- KPI Framework
- Predictive Analytics

Process Learnings

- SOP Design
 - Data Governance
 - Dashboard Development
-

20.3 Strategic Recommendations

Management Recommendations

- Improve KPI visibility
- Increase customer retention

Technology Recommendations

- AI-powered forecasting
- Real-time dashboards

Analytics Recommendations

- Predictive analytics
- Customer segmentation

Diagram

Future Analytics Roadmap

Descriptive



Diagnostic



Predictive



Prescriptive

For Senior Management

1. Allocate 5-10% of IT budget to analytics infrastructure and talent
2. Establish Chief Data Officer role reporting to COO for data governance
3. Mandate data-driven decision making in all department KPIs
4. Invest in predictive analytics for inventory and customer forecasting
5. Align compensation with analytics-based performance metrics

For Analytics Team

6. Expand team: Hire 2-3 additional BI developers and 1 data engineer
7. Implement version control for all code and dashboards using Git
8. Establish data quality monitoring with automated alerts
9. Create user feedback loop with quarterly dashboard enhancements
10. Build self-service portal allowing business users to create simple reports

For Technology Infrastructure

11. Upgrade to SQL Server 2022 for performance and security improvements
12. Implement data lake architecture for unstructured data integration
13. Deploy Power BI Premium for enhanced capacity and enterprise features
14. Establish automated backup and disaster recovery procedures
15. Implement API layer for third-party integrations

For Organizational Culture

16. Establish Analytics Center of Excellence for standards and best practices
 17. Create data literacy program for all employees
 18. Implement data governance policies with clear ownership
 19. Foster innovation with regular hackathons and innovation challenges
 20. Share analytics success stories across organization
-

20.4 Future Opportunities & Emerging Technologies

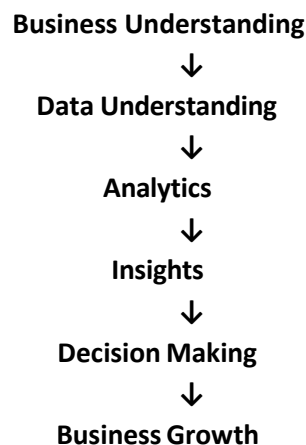
- Artificial Intelligence
 - Machine Learning
 - Generative AI
 - Customer Personalization
 - Automation
-

20.5 Internship Reflection

- Domain knowledge gained
- Business understanding developed
- SQL and Power BI expertise acquired
- Problem-solving capability improved
- Career readiness achieved

Final Figure

"Dealership Analytics Transformation Journey"



20.6 Advanced Analytics Capabilities

1. Predictive Modeling: Build ML models for lead scoring and churn prediction
2. Prescriptive Analytics: Recommend optimal inventory levels and pricing strategies
3. Sentiment Analysis: Analyze customer feedback from surveys and social media
4. Time Series Forecasting: Predict future revenue and demand with high accuracy
5. Anomaly Detection: Automatically flag unusual patterns in customer or financial data

20.7 Emerging Technologies

6. Artificial Intelligence: Implement chatbots for customer support and lead qualification
7. Real-time Processing: Stream live transaction data for instant dashboard updates

- 8. Graph Databases: Model complex customer relationships and recommendation networks
- 9. Natural Language Processing: Extract insights from customer reviews and emails
- 10. Blockchain: Potential for transparent supply chain tracking for vehicle inventory

20.8 Competitive Advantages

- 11. Customer Personalization: Use analytics to deliver customized offers and experiences
- 12. Dynamic Pricing: Implement real-time pricing strategy based on demand and inventory
- 13. Predictive Maintenance: Recommend service intervals based on vehicle usage patterns
- 14. Loyalty Programs: Data-driven customer segmentation for targeted retention
- 15. Supply Chain Optimization: Forecast parts demand and optimize inventory management

20.8 Measurable Growth Potential

Initiative	Investment	Expected ROI
Lead Scoring Model	₹5-8 lakhs	10-15% conversion increase
Inventory Optimization	₹3-5 lakhs	₹20-30 lakhs annual savings
Customer Churn Prediction	₹4-6 lakhs	₹15-25 lakhs retention value

20.9 Data Analytics Vision for Dealership

The future of Sri Gurukripa Automobiles dealership lies in becoming a data-driven organization where every decision is informed by analytics, every employee understands data, and every process is continuously optimized through insights.

Short-term (Next 6 months)

- 16. Achieve 100% adoption of sales and service dashboards
- 17. Complete training for all department managers
- 18. Establish analytics governance framework
- 19. Implement automated daily reporting

Medium-term (6-12 months)

- 20. Launch predictive lead scoring model
- 21. Implement inventory optimization system
- 22. Build customer lifetime value model
- 23. Establish self-service analytics portal

Long-term (12+ months)

- 24. Deploy AI-driven customer recommendation engine

- 25. Implement dynamic pricing strategy
- 26. Develop predictive maintenance system
- 27. Build competitive intelligence platform

20.10 Closing Statement

This comprehensive documentation represents the culmination of extensive research, analysis, and system design work. It serves as the complete roadmap for transforming dealership operations through data analytics. The journey from understanding business fundamentals (Section 1) through implementing technical solutions (Sections 17-18) to training and empowerment (Sections 18-20) reflects a holistic approach to analytics transformation.

Success requires commitment at all levels: executive sponsorship for strategy, technical excellence from IT and analytics teams, and active participation from every business stakeholder. The investments in training, technology, and continuous improvement will yield measurable returns in efficiency, profitability, and competitive advantage.

The dealership stands at the threshold of becoming an industry leader in data-driven decision making. With the frameworks, tools, and knowledge documented in this guide, that vision is not just possible—it is achievable within the next 12-18 months.